Algorithmic Non-commutative Class Field Theory

Volume III: Higher-Rank Buildings, Non-Equilibrium Topology and Higher-Order Games

Ruqing Chen

GUT Geoservice Inc., Rivière-Beaudette, Québec, Canada

ruqing@hotmail.com

September 2026

doi:10.5281/zenodo.23109523

Contents

1 Higher-Rank Buildings: The Cubic Trace Identity, Garland Rigidity, and the Hecke Pencil **1**

1.1 Flags of PG$(2, q)$ and the counts of an $\widetilde{A}_2$ complex 2
1.2 The $\mathbb{Z}/3$ grading and the cubic trace identity 3
1.3 The first nonzero trace in every rank 5
1.4 Garland rigidity by the local spectral gap 8
1.5 What rigidity removes: Brillouin zones, abelian towers, and the non-commutative turn 10
1.6 The Hecke–Laplacian and the spectrum of the cubic pencil 12
1.7 Worked examples: Y_{21}, Y_{24} and Y_{42} 14
1.8 Verification battery 15
1.9 Scope, residue, corrections 15

2 Simplicial Sandpiles on $\widetilde{A}_2$ Complexes: Sign Obstructions, Hodge Factorization, and Avalanche Statistics **17**

2.1 Toppling matrices and the sign obstruction 18
2.2 Critical groups and the Hodge factorization 20
2.3 The vertex sandpile: exact avalanche statistics 21
2.4 Scaling: what can and cannot be claimed 23
2.5 The example Y_{21} 23
2.6 Verification battery 25
2.7 Scope, residue, corrections 25

3 Phase Oscillators on Buildings: Chiral Connections, Frustrated Ground States, and Transverse Bifurcations **27**

3.1 The model and its gauge structure 28
3.2 Symmetries and the gradient sector 29
3.3 Twisted states and the triadic lag 30
3.4 Mean field and Ott–Antonsen reduction 32
3.5 Frequency shift and geometric phase 34
3.6 Why a tree cannot carry the mechanism 34
3.7 The example Y_{21} 35
3.8 Transverse Hopf bifurcations on Y_{168} 35
3.9 Verification battery 39
3.10 Scope, residue, corrections 39

4 The Geodesic Edge Operator: Unitary Remainders, Ramanujan Equivalence, and Quantum Spectral Chaos **41**
4.1 The geodesic edge operator . . . 42
4.2 The unitary remainder . . . 43
4.3 Factorization and the Ramanujan equivalence . . . 44
4.4 Level statistics: what is provable . . . 45
4.5 The example Y_{21} . . . 46
4.6 Breaking the arithmetic degeneracy: the crossover to GUE . . . 47
4.7 Verification battery . . . 57
4.8 Scope, residue, corrections . . . 57

5 Chamber Games and Higher-Order Interactions: Pairwise Reductions, Interfaces, and Entrainment **59**
5.1 The model and its pairwise reduction . . . 60
5.2 Zero noise: pairs, clusters, cages . . . 61
5.3 Finite noise: no topological catalysis . . . 62
5.4 Cyclic dominance and oscillations . . . 62
5.5 Nonlinear chamber games . . . 63
5.6 Chiral games, interfaces, and an economic reading . . . 66
5.7 Clocks, entrainment and anchors . . . 69
5.8 Verification battery . . . 71
5.9 Scope, residue, corrections . . . 71

6 Hecke Algebras on Chambers, Iwasawa Modules, and the 2-adic Valuation Tower **73**
6.1 Hecke algebras . . . 74
6.2 Localization and the absence of a $\mathbb{Z}_p$-direction . . . 75
6.3 The Iwasawa module of Hecke Jacobians . . . 76
6.4 The Cartwright–Steger model and the canonical 2-tower . . . 77
6.5 Growth of Hecke Jacobians . . . 79
6.6 The five-step scheme . . . 80
6.7 Volume III in retrospect, and Volume IV . . . 80
6.8 The 2-part along the canonical tower . . . 81
6.9 Localization over the non-analytic group $\mathbb{Z}_2[[P]]$ . . . 88
6.10 Verification battery . . . 91
6.11 Corrections . . . 91

State of Volume III **93**

Chapter 1

Higher-Rank Buildings: The Cubic Trace Identity, Garland Rigidity, and the Hecke Pencil

This chapter sets up the topology that the rest of Volume III uses on $\widetilde{A}_2$ complexes, with full proofs where earlier chapters quoted results. Four things are established. First, the trace mechanism. A diagonal conjugation $DC^{(3)}D^{-1} = \omega C^{(3)}$ comes from the $\mathbb{Z}/3$ type grading, and it forces $\operatorname{Tr}(C^{(3)})^k = 0$ for $3 \nmid k$. Reading off the diagonal blocks of $(C^{(3)})^3$, each as a count of closed walks, gives $\operatorname{Tr}(C^{(3)})^3 = \operatorname{Tr} A_1^3 - 3q|E| + 3nq^3$. Under the flag condition this equals $3|F| - 3q|E| + 3nq^3 = n(q-1)^2(q+1)$, and the counts $|E|$ and $|F|$ are read off the point–line flags of $\mathrm{PG}(2,q)$ in the vertex links. In every rank d the same mechanism, with the partial flags of $\mathrm{PG}(d,q)$ counted by q-multinomials, gives $\operatorname{Tr}(C^{(d+1)})^{d+1} = n\prod_{i=1}^{d}(1-q^i)$, whose sign is $(-1)^d$. Second, Garland rigidity in *local* form. If every vertex link of a finite two-dimensional complex is the incidence graph of $\mathrm{PG}(2,q)$ with $q \geq 2$, then $H^1(Y;\mathcal{E}) = 0$ for *every* unitary local system $\mathcal{E}$. The proof is self-contained and uses only the spectral gap $1 - \sqrt{q}/(q+1) > \frac{1}{2}$ of the link. It needs no building, no free action and no flag condition, and it therefore covers the complexes of Vol. I–II, Chapter 16 that fail the flag condition. Three consequences follow: $b_1 = 0$, the group $H_1(Y;\mathbb{Z})$ is finite, and every unitary representation of $\pi_1(Y)$ is infinitesimally rigid. So no Brillouin zone exists, abelian or not. Every ℓ-adic analytic Galois tower over Y has a perfect Lie algebra of dimension ≥ 3; commutative Iwasawa theory has no rank-two sector, and the smallest possible Iwasawa algebra is non-commutative. Third, the Hecke pencil. $\det(I - uC^{(3)}) = \det P(u)$ holds on the nose. The nontrivial eigenvalues of Δ^{H} lie in the open right half-plane, which removes the hypothesis of Vol. I–II, Theorem 16.3.2(ii). The factor $(1-u^3)$ occurs with multiplicity exactly $b_0 = 1$, and the exact derivative identity is $\frac{d}{du}\det(I - uC^{(3)})\big|_{u=1} = (-1)^n\, n(q-1)^2(q+1)\,|\operatorname{Jac}^{\mathrm{H}}(Y)|$. Under the flag condition this reads $(-1)^n \operatorname{Tr}(C^{(3)})^3 \cdot |\operatorname{Jac}^{\mathrm{H}}(Y)|$. The factorization $(1-u^3)^{b_2-b_1}P(u)$ proposed in the proposed programme is false, by a degree count, and we say where Betti numbers actually enter. Fourth, the complexes Y_{21} and Y_{42} are rebuilt from the triangle presentation and every number is checked exactly. The flag part of $\operatorname{Tr}(C^{(3)})^3$ is 63 and 126, but because the flag condition fails, the true traces are 651 and 630.

Status of the results

Every numbered statement is proved. The only external inputs, each marked where it is used, are Lazard's theory of uniform pro-ℓ groups (Proposition 1.5.3(iii)) and Weil's identification of the tangent space of a representation variety (Proposition 1.5.1(ii)). Nothing in Sections 1.1–1.4 or 1.6 depends on either. The Garland vanishing theorem, quoted in [Vol. I–II, Ch. 18, Thm. 18.1.1] and [Vol. I–II, Ch. 22, Thm. 22.6.1] from [8, 9], is proved here in the local form needed (Theorem 1.4.3). All finite identities were machine-checked exactly on complexes rebuilt from scratch (Table 1.2). Section 1.9 records what is open and the corrections to the proposed programme.

1.1 Flags of $\mathrm{PG}(2,q)$ and the counts of an $\widetilde{A}_2$ complex

We keep Vol. I–II, Definition 16.1.1: Y has vertex set V ($n = |V|$), edge set E and triangle set F, a type map $\tau\colon V \to \mathbb{Z}/3$, the type-increasing adjacency A_1 and $A_2 = A_1^t$ with $A_1A_2 = A_2A_1$. The *link* $\mathrm{Lk}(v)$ is the graph whose vertices are the neighbours of v, with $w \sim x$ when $\{v, w, x\} \in F$. We shall often use the following purely local hypothesis in place of (ii)–(iii):

(L) every vertex link $\mathrm{Lk}(v)$ is isomorphic to the point–line incidence graph $\mathcal{I}(q)$ of $\mathrm{PG}(2,q)$, the neighbours of type $\tau(v)+1$ going to points and those of type $\tau(v)+2$ to lines.

This is the situation for quotients of thick $\widetilde{A}_2$ buildings, and it was verified on every complex of [Vol. I–II, Ch. 16] (Heawood links, $q = 2$).

Lemma 1.1.1 (Flag counts in $\mathrm{PG}(2,q)$)**.** $\mathrm{PG}(2,q)$ *has* q^2+q+1 *points and* q^2+q+1 *lines. Every line carries* $q+1$ *points, every point lies on* $q+1$ *lines, and two distinct points lie on exactly one line. The number of flags (incident point–line pairs) is* $(q^2+q+1)(q+1)$*. Consequently* $\mathcal{I}(q)$ *is a bipartite* $(q+1)$*-regular graph with* $2(q^2+q+1)$ *vertices and* $(q^2+q+1)(q+1)$ *edges. For* $q = 2$ *it is the Heawood graph, with* 14 *vertices,* 21 *edges and girth* 6*.*

Proof. Points and lines are the 1- and 2-dimensional subspaces of $\mathbb{F}_q^3$. There are $(q^3-1)/(q-1) = q^2+q+1$ lines through the origin, and by duality ($W \mapsto W^\perp$ for the standard bilinear form) the same number of planes. A plane $W \cong \mathbb{F}_q^2$ contains $(q^2-1)/(q-1) = q+1$ lines through the origin, and dually a line lies in $q+1$ planes. Two distinct lines through the origin span exactly one plane. Counting flags by points gives $(q^2+q+1)(q+1)$. The girth of $\mathcal{I}(q)$ is 6: there is no 4-cycle, because two points share at most one line, and every triangle of the plane gives a 6-cycle. □

Lemma 1.1.2 (Counts from the links)**.** *Assume* (L)*. Then:*

(a) $|E| = n(q^2+q+1)$*, each edge lies in exactly* $q+1$ *triangles, and* $|F| = \frac{1}{3}\,n(q^2+q+1)(q+1)$*;*

(b) $\mathrm{Tr}(A_1A_2) = |E|$*;*

(c) $\mathrm{Tr}\,A_1^3 = 3|F| + 3s$*, where* $s \geq 0$ *is the number of* 3*-cycles* $x \to y \to z \to x$ *of type-increasing edges whose vertex set is* not *a triangle. So* $\mathrm{Tr}\,A_1^3 \geq 3|F|$*, with equality if and only if the flag condition* (iv) *of Vol. I–II, Definition 16.1.1 holds.*

Proof. (a) Count vertex–edge incidences. The edges at v are the vertices of $\mathrm{Lk}(v)$, and there are $2(q^2+q+1)$ of them by Lemma 1.1.1, so $2|E| = 2n(q^2+q+1)$. The triangles through an edge $\{v, w\}$ are the edges of $\mathrm{Lk}(v)$ at w, and there are $q+1$ of them. The triangles at v are the edges of $\mathrm{Lk}(v)$, i.e. the flags of $\mathrm{PG}(2,q)$, so counting vertex–triangle incidences gives $3|F| = n(q^2+q+1)(q+1)$.

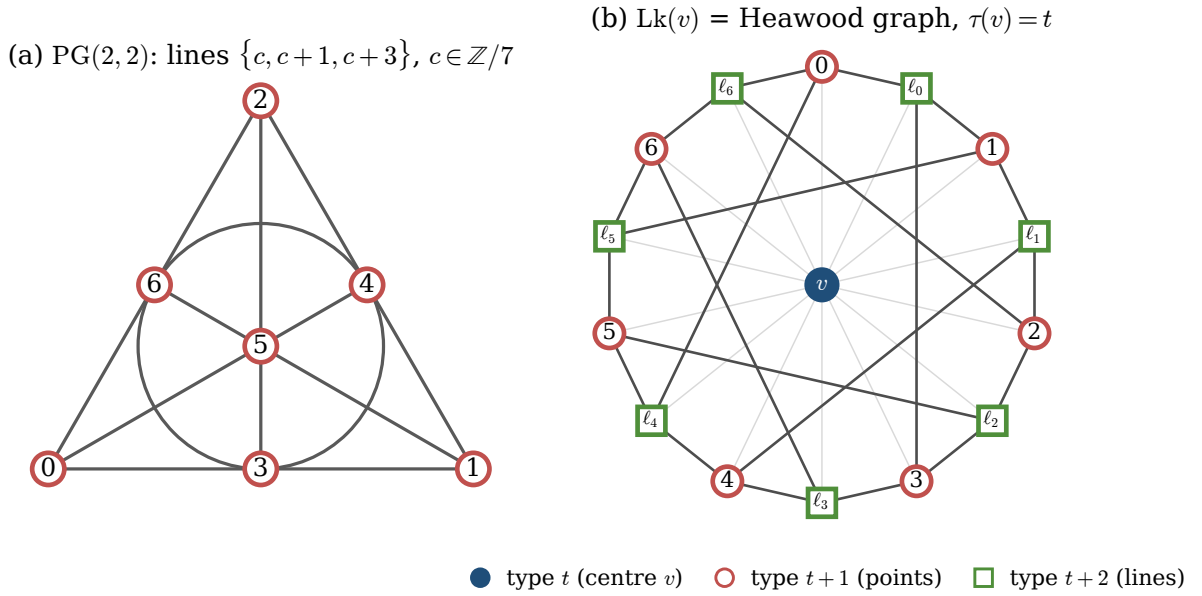


Figure 1.1: The local geometry of the complexes of Vol. I–II for $q = 2$. (a) The Fano plane PG(2, 2), labelled as in the triangle presentation: its seven lines are the translates $\{c, c+1, c+3\}$, $c \in \mathbb{Z}/7$, of the Singer difference set $\{0, 1, 3\}$. (b) The link of a vertex v of type t, which by (L) is the incidence graph of (a), the Heawood graph: its 7 vertices of type $t+1$ are the points, its 7 vertices of type $t+2$ are the lines $\ell_c = \{c, c+1, c+3\}$, and its 21 edges are the flags, i.e. the chambers through v (faint spokes). The three colours are the three classes of the $\mathbb{Z}/3$ type grading used in Proposition 1.2.1. Drawn by `code/make_figures.py`.

(b) Each edge $\{x, y\}$ joins vertices of different types, and exactly one of its two orientations is type-increasing: if $\tau(y) = \tau(x) + 1$ it is $x \to y$, and if $\tau(y) = \tau(x) + 2$ then $\tau(x) = \tau(y) + 1$ and it is $y \to x$. Hence A_1 has exactly $|E|$ entries equal to 1, and $\operatorname{Tr}(A_1 A_1^t) = \sum_{x,y} A_1(x, y)^2 = |E|$.

(c) $\operatorname{Tr} A_1^3$ counts closed walks $x \to y \to z \to x$ of type-increasing edges. Such a walk visits the three types, so x, y, z are distinct. A triangle $\{a, b, c\}$ with $\tau = (t, t+1, t+2)$ carries exactly one such cycle, $a \to b \to c \to a$ (the reverse is type-decreasing), and that cycle contributes three closed walks, one for each starting point. The remaining closed walks come in cyclic classes of three, one class for each non-triangle 3-cycle. □

For $q = 2$: $|E| = |F| = 7n$, and each vertex sees 14 edges and 21 triangles, which are the 14 vertices and 21 edges of the Heawood graph (Figure 1.1).

1.2 The $\mathbb{Z}/3$ grading and the cubic trace identity

The rank-one obstruction. For a $(q+1)$-regular graph, the companion $C = \left(\begin{smallmatrix} T_p & -qI \\ I & 0 \end{smallmatrix}\right)$ of the Ihara pencil $1 - uT_p + qu^2$ satisfies $\operatorname{Tr} C = 0$ and

$$\operatorname{Tr} C^2 = \operatorname{Tr} T_p^2 - 2qn = n(q+1) - 2qn = -n(q-1) < 0$$

[Vol. I–II, Ch. 3, Lem. 3.1.1]. The positive term $\operatorname{Tr} T_p^2 = 2|E|$ counts closed walks that return by reversing an edge, and it is beaten by the $-qI$ block, which is hit twice. By [Vol. I–II, Ch. 3,

Thm. 3.1.5] no non-negative matrix of any size carries this spectrum.

In rank two the companion of the cubic pencil $P(u) = I - uA_1 + qu^2A_2 - q^3u^3I$ is

$$C^{(3)} = \begin{pmatrix} A_1 & -qA_2 & q^3I \\ I & 0 & 0 \\ 0 & I & 0 \end{pmatrix} \in M_{3n}(\mathbb{Z}). \tag{1.2.1}$$

Proposition 1.2.1 (Grading conjugation)**.** *Let $\omega = e^{2\pi i/3}$, $D_1 = \operatorname{diag}(\omega^{-\tau(v)})_{v\in V}$ and $D = \operatorname{diag}(D_1,\ \omega D_1,\ \omega^2 D_1)$. Then*

$$D_1A_1D_1^{-1} = \omega A_1, \qquad D_1A_2D_1^{-1} = \omega^2 A_2, \qquad D\,C^{(3)}D^{-1} = \omega\,C^{(3)}.$$

Consequently $\operatorname{Tr}(C^{(3)})^k = 0$ for every k with $3 \nmid k$, the spectrum of $C^{(3)}$ (with multiplicities) is invariant under multiplication by ω, and $\det(I - uC^{(3)}) \in \mathbb{Z}[u^3]$.

Proof. $(D_1A_1D_1^{-1})_{xy} = \omega^{\tau(y)-\tau(x)}A_1(x,y) = \omega A_1(x,y)$, because $A_1(x,y) \neq 0$ forces $\tau(y) = \tau(x)+1$. Transposing gives $\omega^{-1} = \omega^2$ for A_2. Write $D = \operatorname{diag}(d_1D_1, d_2D_1, d_3D_1)$ with $(d_1,d_2,d_3) = (1,\omega,\omega^2)$. Block (i,j) of $DC^{(3)}D^{-1}$ is $d_id_j^{-1}D_1C^{(3)}_{ij}D_1^{-1}$:

$$(1,1)\colon\ \omega A_1; \quad (1,2)\colon\ \omega^{-1}\cdot\omega^2(-qA_2) = \omega(-qA_2); \quad (1,3)\colon\ \omega^{-2}q^3I = \omega q^3 I; \quad (2,1),(3,2)\colon\ \omega I.$$

So $DC^{(3)}D^{-1} = \omega C^{(3)}$, and therefore $\operatorname{Tr}(C^{(3)})^k = \omega^k \operatorname{Tr}(C^{(3)})^k$, which vanishes when $\omega^k \neq 1$. Similarity of $C^{(3)}$ and $\omega C^{(3)}$ gives the invariance of the spectrum. Then $\det(I - uC^{(3)}) = \det(I - \omega uC^{(3)})$, so only powers of u^3 occur. □

Remark 1.2.2 (Walk form of the grading). Every entry of $(C^{(3)})^k$ is a signed, q-weighted sum over words in A_1 (one type step up), A_2 (one type step down, i.e. two steps up modulo 3) and the identity blocks, which shift the block index but not the vertex. A closed contribution to the trace must return to its starting vertex, and hence to its starting type. With the block bookkeeping, the net type change of a word of length k is $\equiv k \pmod 3$. This is the combinatorial content of $DC^{(3)}D^{-1} = \omega C^{(3)}$.

Theorem 1.2.3 (Cubic trace identity)**.** *For every $\widetilde{A}_2$ complex of order q, with or without the flag condition,*

$$\operatorname{Tr}(C^{(3)})^3 = \operatorname{Tr} A_1^3 - 3q\ \operatorname{Tr}(A_1A_2) + 3nq^3 = \operatorname{Tr} A_1^3 - 3q|E| + 3nq^3. \tag{1.2.2}$$

Under (L) *this equals $n(q-1)^2(q+1) + 3s$, with s as in Lemma 1.1.2(c). Under the flag condition,*

$$\operatorname{Tr}(C^{(3)})^3 = 3|F| - 3q|E| + 3nq^3 = n(q-1)^2(q+1)\ \geq 0, \tag{1.2.3}$$

with equality if and only if $q = 1$.

Proof. Write $C^{(3)}$ by columns: $\mathrm{col}_1 = (A_1, I, 0)^t$, $\mathrm{col}_2 = (-qA_2, 0, I)^t$, $\mathrm{col}_3 = (q^3I, 0, 0)^t$. The rows of $C^{(3)}$ are $r_1 = (A_1, -qA_2, q^3I)$, $r_2 = (I,0,0)$ and $r_3 = (0,I,0)$. Hence the rows of $(C^{(3)})^2$ are

$$r_1C^{(3)} = (A_1^2 - qA_2,\ -qA_1A_2 + q^3I,\ q^3A_1), \qquad r_2C^{(3)} = r_1, \qquad r_3C^{(3)} = r_2.$$

The diagonal blocks of $(C^{(3)})^3 = (C^{(3)})^2C^{(3)}$ are

$$\begin{aligned}
\big((C^{(3)})^3\big)_{11} &= (A_1^2 - qA_2)A_1 + (-qA_1A_2 + q^3I)\cdot I + q^3A_1\cdot 0 \\
&= A_1^3 - qA_2A_1 - qA_1A_2 + q^3I, \\
\big((C^{(3)})^3\big)_{22} &= r_1\cdot \mathrm{col}_2 = A_1(-qA_2) + (-qA_2)\cdot 0 + q^3I\cdot I = -qA_1A_2 + q^3I, \\
\big((C^{(3)})^3\big)_{33} &= r_2\cdot \mathrm{col}_3 = q^3I.
\end{aligned}$$

Summing traces and using $\operatorname{Tr}(A_2A_1) = \operatorname{Tr}(A_1A_2)$ gives (1.2.2). Each term has a walk meaning. $\operatorname{Tr} A_1^3$ counts closed type-increasing 3-walks. $\operatorname{Tr}(A_1A_2) = |E|$ counts the backtracking 2-walks $x \to y \leftarrow x$; it enters three times, once for each of the three cyclic positions at which the backtrack can sit inside a 3-step return, each with weight $-q$. The term q^3 is the pure constant term, entering once per block. Under (L), Lemma 1.1.2 gives $\operatorname{Tr} A_1^3 = 3|F| + 3s$, and

$$\begin{aligned} 3|F| - 3q|E| + 3nq^3 &= n\big[(q^2+q+1)(q+1) - 3q(q^2+q+1) + 3q^3\big] \\ &= n\big[(q^3+2q^2+2q+1) - (3q^3+3q^2+3q) + 3q^3\big] \\ &= n\,(q^3 - q^2 - q + 1) = n(q-1)^2(q+1). \end{aligned}$$

Under the flag condition $s = 0$. The factor $(q-1)^2(q+1)$ vanishes only at $q = 1$. □

Remark 1.2.4 (Parity versus $\mathbb{Z}/3$). In rank one a closed walk can return in two steps, so the negative block $-qI$ is felt already at $k = 2$, and it is felt twice. In rank two the grading makes $\operatorname{Tr}(C^{(3)})^k$ blind to everything for $k = 1, 2$. At $k = 3$ the chamber term $3|F| = n(q^2+q+1)(q+1)$, the flag count of $\mathrm{PG}(2,q)$ times n, dominates the backtracking loss $3q|E| - 3nq^3$. The difference is exactly $n(q-1)^2(q+1)$. This is the proof behind [Vol. I–II, Ch. 16, Thm. 16.4.2], written out in full, and its consequence for realizability is recorded in [Vol. I–II, Ch. 16, Thm. 16.4.5].

1.3 The first nonzero trace in every rank

This section settles open problem (a) of Remark 1.9.1: the value of the first nonzero trace $\operatorname{Tr}(C^{(d+1)})^{d+1}$ on $\widetilde{A}_d$ complexes. The answer is $n\prod_{i=1}^{d}(1-q^i)$ under the $\widetilde{A}_d$ flag condition, which contains the rank-one value $-n(q-1)$ and Theorem 1.2.3 as the cases $d = 1, 2$.

Setting. An $\widetilde{A}_d$ complex Y has a type map $\tau\colon V \to \mathbb{Z}/(d+1)$, and for $1 \le j \le d$ we let A_j be the 0/1 matrix with $A_j(x,y) = 1$ iff $\{x,y\}$ is an edge and $\tau(y) = \tau(x) + j$; put $A_{d+1} = I$. The local hypothesis replacing (L) is

(L_d) the link of every vertex v is isomorphic to the flag complex of $\mathrm{PG}(d,q)$, the neighbours of type $\tau(v) + i$ going to the subspaces of codimension i in $\mathbb{F}_q^{d+1}$ ($1 \le i \le d$), adjacency in the link being containment. By the duality $U \mapsto U^{\perp}$ one may equally send them to the subspaces of dimension i; for $d = 2$ that is (L).

A *closed walk of increments* $\mathbf{j} = (j_1, \dots, j_r)$, $j_i \ge 1$, is a sequence $x_0 \to x_1 \to \cdots \to x_r = x_0$ with $A_{j_i}(x_{i-1}, x_i) = 1$ (the identity step for $j_i = d+1$). The *flag condition* of rank d is

(F_d) every closed walk whose increments sum to $d+1$ has all its vertices in one simplex.

For $d = 2$ the only increments with sum 3 are $(1,1,1)$, $(1,2)$, $(2,1)$ and (3), and (F_2) is the flag condition of Lemma 1.1.2(c). The companion is

$$C^{(d+1)} = \begin{pmatrix} R_1 & R_2 & \cdots & R_d & R_{d+1} \\ I & 0 & \cdots & 0 & 0 \\ & \ddots & & & \vdots \\ 0 & \cdots & & I & 0 \end{pmatrix}, \qquad R_j = (-1)^{j-1} q^{\binom{j}{2}} A_j,$$

which is (1.2.1) for $d = 2$. When the A_j commute, $\det(xI - C^{(d+1)}) = \sum_{j=0}^{d+1} (-1)^j q^{\binom{j}{2}} A_j x^{d+1-j}$, the local Hecke polynomial of PGL_{d+1}; the companion of [Vol. I–II, Ch. 21, Thm. 21.4.1] has the

same excursion weights R_j (see the proof of Lemma 1.3.1), hence the same traces of all powers. With $D_1 = \mathrm{diag}(\zeta^{-\tau})$, $\zeta = e^{2\pi i/(d+1)}$, the proof of Proposition 1.2.1 gives $DC^{(d+1)}D^{-1} = \zeta C^{(d+1)}$ verbatim, so $\mathrm{Tr}(C^{(d+1)})^k = 0$ for $(d+1) \nmid k$.

Lemma 1.3.1 (Excursion expansion)**.** *For every $k \geq 1$, with no commutativity assumption,*

$$\mathrm{Tr}\big(C^{(d+1)}\big)^k = \sum_{\mathbf{j}} j_1\,(-1)^{k-r}\,q^{\sum_i \binom{j_i}{2}}\,\mathrm{Tr}(A_{j_1}A_{j_2}\cdots A_{j_r}),$$

the sum over the compositions $\mathbf{j} = (j_1, \ldots, j_r)$ of k with all parts $\leq d+1$.

Proof. $\big((C^{(d+1)})^k\big)_{bb}$ is a sum over closed walks of length k in the graph on the block indices $1, \ldots, d+1$ whose arcs are $1 \to j$ (weight R_j) and $i+1 \to i$ (weight I). From block 1, the arc $1 \to j$ must be followed by $j \to j-1 \to \cdots \to 1$, so every return to block 1 is an *excursion* of length j and weight R_j. A closed walk of length k is therefore a cyclic sequence of excursions of lengths $j_1, \ldots, j_r$ with $\sum j_i = k$, together with the time at which it starts. Label the excursions so that the walk starts inside the first one; there are j_1 such starting times, and each contributes the trace of a cyclic rotation of $R_{j_1}\cdots R_{j_r}$, which is $\mathrm{Tr}(R_{j_1}\cdots R_{j_r})$. Finally $\prod_i (-1)^{j_i-1} = (-1)^{k-r}$. For the companion of [Vol. I–II, Ch. 21], whose row 0 carries $q^i A_{i+1}$ and whose subdiagonal carries $-q^i I$, an excursion of length j has weight $q^{j-1}A_j \prod_{l=0}^{j-2}(-q^l) = (-1)^{j-1}q^{\binom{j}{2}}A_j = R_j$. Line (W) checks the expansion as an identity of non-commutative polynomials for $d \leq 6$, $k \leq d+1$. □

Lemma 1.3.2 (Closed walks are flags)**.** *Let $\mathbf{j}$ be a composition of $d+1$ and let $\binom{d+1}{\mathbf{j}}_q = [d+1]_q!/\prod_i [j_i]_q!$ be the q-multinomial coefficient, $[k]_q = 1 + q + \cdots + q^{k-1}$.*

(i) Under (L_d), $\mathrm{Tr}(A_{j_1}\cdots A_{j_r}) = n\binom{d+1}{\mathbf{j}}_q + s_{\mathbf{j}}$, *where $s_{\mathbf{j}} \geq 0$ is the number of closed walks of increments $\mathbf{j}$ whose vertices are not contained in one simplex. Under* (F_d), $s_{\mathbf{j}} = 0$.

(ii) The Bruhat–Tits building of PGL_{d+1} *over a non-archimedean local field with residue field $\mathbb{F}_q$ satisfies* (L_d) *and* (F_d).

Proof. (i) A closed walk of increments $\mathbf{j}$ starting at x_0 whose vertices lie in a simplex is determined by $x_1, \ldots, x_{r-1}$, which have types $\tau(x_0) + j_1 + \cdots + j_i$ and form a simplex of $\mathrm{Lk}(x_0)$, i.e. by (L_d) a chain $\mathbb{F}_q^{d+1} \supsetneq U_1 \supsetneq \cdots \supsetneq U_{r-1} \supsetneq 0$ with $\mathrm{codim}(U_i \subset U_{i-1}) = j_i$. Conversely every such chain gives such a walk, the last step $x_{r-1} \to x_0$ having increment j_r. The number of chains is the q-multinomial coefficient (choose U_{r-1}, then $U_{r-2} \supset U_{r-1}$, and so on), whatever the order of the parts. The remaining closed walks are counted by $s_{\mathbf{j}}$.

(ii) The vertices are the homothety classes of lattices $L \subset K^{d+1}$, with uniformizer π; $[L] \to [L']$ has increment j iff there are representatives with $L \supset L' \supset \pi L$ and $\dim L/L' = j$, and the simplices are the chains of such lattices. The link of $[L]$ is the flag complex of $L/\pi L \cong \mathbb{F}_q^{d+1}$. Given a closed walk $x_0 \to \cdots \to x_r = x_0$ with increments summing to $d+1$, choose L_0 in x_0 and inductively the unique representative L_i of x_i with $L_{i-1} \supset L_i \supset \pi L_{i-1}$. Then $[L_0 : L_r] = q^{d+1}$ and L_r is homothetic to L_0, so $L_r = \pi L_0$. Hence $L_0 \supset L_1 \supset \cdots \supset L_{r-1} \supset \pi L_0$ is a chain, and the vertices of the walk span a simplex. □

Theorem 1.3.3 (First nonzero trace in rank d)**.** *Let Y be a finite $\widetilde{A}_d$ complex satisfying* (L_d), *with n vertices. Then $\mathrm{Tr}(C^{(d+1)})^k = 0$ for $(d+1) \nmid k$, and*

$$\mathrm{Tr}\big(C^{(d+1)}\big)^{d+1} = n\prod_{i=1}^{d}(1-q^i) + \sum_{\mathbf{j}} j_1(-1)^{d+1-r}q^{\sum_i \binom{j_i}{2}}s_{\mathbf{j}}.$$

Under the flag condition (F_d),

$$\operatorname{Tr}\big(C^{(d+1)}\big)^{d+1} = n\prod_{i=1}^{d}(1-q^i) = (-1)^d\, n\,(q-1)^d\,[d]_q!\,.$$

For $d = 1, 2, 3, 4$ *this is* $-n(q-1)$, $n(q-1)^2(q+1)$, $-n(q-1)^3(q+1)(q^2+q+1)$ *and* $n(q-1)^4(q+1)^2(q^2+1)(q^2+q+1)$.

Proof. By Lemmas 1.3.1 and 1.3.2 it remains to show, with $m = d+1$,

$$T_m := \sum_{\mathbf{j}\models m} j_1(-1)^{m-r} q^{\sum\binom{j_i}{2}} \binom{m}{\mathbf{j}}_q = \prod_{i=1}^{m-1}(1-q^i).$$

Write $\binom{m}{\mathbf{j}}_q = [m]_q!\prod_i E_{j_i}$ with $E_j = q^{\binom{j}{2}}/[j]_q!$ and work in $\mathbb{Z}[[q]][[t]]$. Put $F(t) = \sum_{j\ge1}(-1)^{j-1}E_j t^j$. Expanding $tF'(t)/(1-F(t))$ by the geometric series gives $\sum_{\mathbf{j}} j_1 \prod_i (-1)^{j_i-1}E_{j_i}t^{\sum j_i}$, the j_1 coming from the derivative applied to the first factor; hence $T_m = [m]_q!\,[t^m](-t\frac{d}{dt}\log(1-F(t)))$. Now $1-F(t) = \sum_{j\ge0} q^{\binom{j}{2}}(-t)^j/[j]_q!$, and since $[j]_q! = (q;q)_j/(1-q)^j$, Euler's identity $\sum_j q^{\binom{j}{2}}z^j/(q;q)_j = \prod_{i\ge0}(1+zq^i)$ with $z = -(1-q)t$ gives $1-F(t) = \prod_{i\ge0}(1-(1-q)q^it)$. Therefore

$$-t\frac{d}{dt}\log(1-F(t)) = \sum_{i\ge0}\frac{(1-q)q^it}{1-(1-q)q^it} = \sum_{k\ge1} t^k\,(1-q)^k\sum_{i\ge0}q^{ik} = \sum_{k\ge1}t^k\frac{(1-q)^k}{1-q^k},$$

and $T_m = [m]_q!\,(1-q)^m/(1-q^m) = \prod_{i=1}^{m}(1-q^i)/(1-q^m) = \prod_{i=1}^{m-1}(1-q^i)$. Both sides are polynomials in q, so the identity in $\mathbb{Z}[[q]]$ is an identity of polynomials. The last form uses $1-q^i = -(q-1)[i]_q$. □

Corollary 1.3.4. *Assume* (L_d) *and* (F_d).

(i) As a polynomial in q, $\operatorname{Tr}(C^{(d+1)})^{d+1}/n$ *vanishes at* $q=1$ *to order exactly* d. *A thin complex* $(q=1)$ *satisfying* (F_d) *has* $\operatorname{Tr}(C^{(d+1)})^{d+1} = 0$.

(ii) For $q \ge 2$ *the sign of* $\operatorname{Tr}(C^{(d+1)})^{d+1}$ *is* $(-1)^d$. *For odd* d *it is negative, so no non-negative matrix, of any size, has the nonzero spectrum of* $C^{(d+1)}$ *(with multiplicities): the obstruction of the rank-one paragraph of Section 1.2 recurs in every odd rank, and disappears in every even rank.*

(iii) On the constants, A_j *acts by* $\binom{d+1}{j}_q$ *and the local factor is* $\prod_{i=0}^{d}(1-q^iu)$; *its derivative at* $u=1$ *is* $-\prod_{i=1}^{d}(1-q^i)$. *So the coincidence of Remark 1.6.5 persists in every rank: the first nonzero trace per vertex and the derivative of the trivial local factor at* $u=1$ *are the same number up to sign.*

Proof. (i) Each factor $1-q^i$ has a simple zero at $q=1$. (ii) Each factor is negative for $q \ge 2$. The traces of the powers of a non-negative matrix are non-negative, and padding by zeros does not change them. (iii) $\sum_j (-1)^j q^{\binom{j}{2}}\binom{d+1}{j}_q x^{d+1-j} = \prod_{i=0}^{d}(x-q^i)$ by the q-binomial theorem, and differentiate. □

Remark 1.3.5 (Numerical record). `code/v3p1_ad_trace.py` (numpy, scipy, sympy; about two minutes) checks: the excursion expansion as non-commutative words (W); the identity $T_m = \prod_{i<m}(1-q^i)$ symbolically for $d \le 12$, the order of the zero at $q=1$ and the sign (Q); the chain counts in $\mathbb{F}_q^{d+1}$ by enumeration of all subspaces for $(d,q) = (2,2),(3,2),(2,3),(3,3),(4,2)$, with per-vertex traces 3, -21, 16, -416, 315 (F); the thin $\widetilde{A}_d$ tori $(\mathbb{Z}/m)^{d+1}/\mathbb{Z}(1,\dots,1)$ for $d = 2,3,4$, which satisfy (F_d) for $m \ge d+2$ and give $\mathrm{Tr}(C^{(d+1)})^{d+1} = 0$ there, while for $m = d+1$ the excess walks of increments $(1,\dots,1)$ give $\mathrm{Tr} = (d+1)n$ (T); and Y_{21}, where $651 = 63 + 3\cdot 196$ (Y). No finite thick $\widetilde{A}_3$ complex is constructed; for $d \ge 3$ the thick case rests on Lemma 1.3.2(ii) and on the per-vertex count (F).

Remark 1.3.6 (Corrections to the proposed programme). (1) The proposed $\mathrm{Tr}(C^{(4)})^4 = n(q-1)^3(q+1)(q^2+1)$ is false: the value is $-n(q-1)^3(q+1)(q^2+q+1)$, e.g. $-21n$ instead of $15n$ at $q=2$; the sign is negative and the last factor is $[3]_q$, not q^2+1. (2) The proposed general form $n(q-1)^d\prod_j(q^{j+1}-1)/(q-1)$ is false; the correct product is $(-1)^d n(q-1)^d[d]_q!$. (3) For $d \ge 3$ the trace is not a combination of the cell numbers $|F_k|$ alone: the words $A_{j_1}\cdots A_{j_r}$ count partial flags of every type, which is why the q-multinomials enter. (4) The vertex block of L_E for $d \ge 3$ (Chapter 4, Remark 4.8.1(a)) is a different problem and is not addressed here.

1.4 Garland rigidity by the local spectral gap

Unitary local systems. A rank-N unitary local system $\mathcal{E}$ on a finite simplicial complex Y is given by a Hermitian space $\mathcal{E}_v \cong \mathbb{C}^N$ at each vertex and a unitary transport $g_{uv}\colon \mathcal{E}_v \to \mathcal{E}_u$ for each oriented edge, with $g_{vu} = g_{uv}^{-1}$ and $g_{ab}g_{bc} = g_{ac}$ on every triangle (flatness). Equivalently, it is a representation $\rho\colon \pi_1(Y) \to U(N)$ up to conjugation, and the trivial system $\mathcal{E} = \mathbb{C}$ gives ordinary cohomology. Cochains are

$$C^0 = \{\varphi(v) \in \mathcal{E}_v\}, \quad C^1 = \{f(u,v) \in \mathcal{E}_u : \ f(v,u) = -g_{vu}f(u,v)\},$$

and C^2 consists of alternating functions on oriented triangles with values in the fibre at the first vertex. The differentials are

$$(\delta\varphi)(u,v) = g_{uv}\varphi(v) - \varphi(u), \qquad (\delta f)(a,b,c) = f(a,b) + g_{ab}f(b,c) + g_{ac}f(c,a).$$

Flatness gives $\delta\delta = 0$: $(\delta\delta\varphi)(a,b,c) = g_{ab}g_{bc}\varphi(c) - g_{ac}\varphi(c) = 0$.

Weights. Let $m(e)$ be the number of triangles containing the edge e, and $m(v) = \sum_{w\sim v} m(vw)$. We put

$$\langle\varphi,\psi\rangle = \sum_v m(v)\langle\varphi(v),\psi(v)\rangle, \qquad \langle f,h\rangle = \tfrac{1}{2}\sum_{(u,v)} m(uv)\langle f(u,v),h(u,v)\rangle,$$

where the second sum is over oriented edges. It is well defined because transport is unitary. Under (L), $m(e) = q+1$ and $m(v) = 2(q+1)(q^2+q+1)$ are constant, so these are scalar multiples of the standard inner products, and the harmonic forms below are the standard ones.

Lemma 1.4.1 (Adjoint and Hodge)**.** *$(\delta^* f)(u) = -\frac{1}{m(u)}\sum_{v\sim u} m(uv)f(u,v)$, and $H^1(Y;\mathcal{E}) \cong \ker\delta \cap \ker\delta^* \subseteq C^1$.*

Proof. Start from $\langle\delta\varphi, f\rangle = \frac{1}{2}\sum_{(u,v)} m(uv)\big(\langle g_{uv}\varphi(v), f(u,v)\rangle - \langle\varphi(u), f(u,v)\rangle\big)$. By unitarity and antisymmetry, $\langle g_{uv}\varphi(v), f(u,v)\rangle = \langle\varphi(v), g_{vu}f(u,v)\rangle = -\langle\varphi(v), f(v,u)\rangle$. Relabelling $(u,v) \leftrightarrow (v,u)$ turns the first sum into the second, so $\langle\delta\varphi, f\rangle = -\sum_{(u,v)} m(uv)\langle\varphi(u), f(u,v)\rangle$, which is the

stated adjoint. Since $C^1 = \operatorname{im}\delta \oplus (\operatorname{im}\delta)^\perp$ and $(\operatorname{im}\delta)^\perp = \ker\delta^*$, every class in $\ker\delta/\operatorname{im}\delta$ has a unique representative in $\ker\delta \cap \ker\delta^*$. □

Lemma 1.4.2 (Spectrum of the link)**.** *The adjacency eigenvalues of $\mathcal{I}(q)$ are $\pm(q+1)$, each simple, and $\pm\sqrt{q}$, each with multiplicity q^2+q. The normalized Laplacian $I - \frac{1}{q+1}\mathrm{Adj}$ therefore has smallest positive eigenvalue*

$$\lambda(q) = 1 - \frac{\sqrt{q}}{q+1}, \qquad \text{and} \qquad 2\lambda(q) - 1 = \frac{(\sqrt{q}-1)^2}{q+1}.$$

So $\lambda(q) > \frac{1}{2}$ for every $q \geq 2$ and $\lambda(1) = \frac{1}{2}$. For the Heawood graph, $\lambda(2) = 1 - \sqrt{2}/3 = 0.528595\ldots$

Proof. Let N be the point–line incidence matrix. By Lemma 1.1.1, $NN^t = (q+1)I + (J-I) = qI + J$, since a point lies on $q+1$ lines and two points share one line. Its eigenvalues are $q + (q^2+q+1) = (q+1)^2$ on $\mathbf{1}$ and q on $\mathbf{1}^\perp$. The adjacency $\left(\begin{smallmatrix} 0 & N \\ N^t & 0 \end{smallmatrix}\right)$ has eigenvalues $\pm\sqrt{\cdot}$ of these, and the graph is $(q+1)$-regular. The simplicity of $q+1$ shows that $\mathcal{I}(q)$ is connected. □

For a $\mathcal{E}_v$-valued function ϕ on $\mathrm{Lk}(v)$ put $\|\phi\|_v^2 = \sum_w m(vw)|\phi(w)|^2$ and $\|d\phi\|_v^2 = \sum_{\{w,x\}} |\phi(x) - \phi(w)|^2$, the second sum running over link edges, i.e. over triangles $\{v,w,x\}$. The degree of w in $\mathrm{Lk}(v)$ is $m(vw)$. So if $\sum_w m(vw)\phi(w) = 0$, then $\|d\phi\|_v^2 \geq \lambda(q)\|\phi\|_v^2$, applying Lemma 1.4.2 componentwise in an orthonormal basis of $\mathcal{E}_v$.

Theorem 1.4.3 (Garland vanishing, local form)**.** *Let Y be a finite two-dimensional simplicial complex satisfying* (L) *with $q \geq 2$. No building, free action or flag condition is assumed. Then $H^1(Y;\mathcal{E}) = 0$ for every unitary local system $\mathcal{E}$ on Y. In particular:*

(i) $b_1(Y;\mathbb{R}) = b_1(Y;\mathbb{C}) = 0$, and $H_1(Y;\mathbb{Z})$ is a finite abelian group;

(ii) $\operatorname{coker}_{\mathbb{Z}} \Delta_1 = \mathrm{Jac}_1(Y)$, i.e. the edge critical group is the whole cokernel; the group $K(Y) = \ker\partial_1/\operatorname{im}(\partial_2\partial_2^t)$ is finite and surjects onto $H_1(Y;\mathbb{Z})$; and $b_2 = \chi(Y) - 1$;

(iii) every finite connected cover $Y' \to Y$ satisfies (L)*, hence (i)–(ii).*

Proof. Let $f \in \ker\delta \cap \ker\delta^*$ (Lemma 1.4.1). For each vertex v define the localization $f_v(w) := f(v,w) \in \mathcal{E}_v$ for $w \in \mathrm{Lk}(v)$. All of its values lie in the *single* fibre $\mathcal{E}_v$, so no trivialization is needed. We prove three identities.

(a) Mass. $\sum_v \|f_v\|_v^2 = \sum_v \sum_{w\sim v} m(vw)|f(v,w)|^2 = \sum_{(v,w)} m(vw)|f(v,w)|^2 = 2\|f\|^2$.

(b) Energy. Fix a triangle $\{a,b,c\}$; it is the link edge $\{b,c\}$ of $\mathrm{Lk}(a)$. Since $\delta f = 0$ and $g_{ac}f(c,a) = -f(a,c)$,

$$f_a(c) - f_a(b) = f(a,c) - f(a,b) = g_{ab}f(b,c), \qquad \text{so} \qquad |f_a(c) - f_a(b)| = |f(b,c)|.$$

Summing over the three vertices of every triangle, $\sum_v \|df_v\|_v^2 = \sum_{\{a,b,c\}\in F}(|f(b,c)|^2 + |f(c,a)|^2 + |f(a,b)|^2) = \sum_e m(e)|f(e)|^2 = \|f\|^2$.

(c) Balance. $\sum_w m(vw)f_v(w) = -m(v)(\delta^* f)(v) = 0$.

By (c) and Lemma 1.4.2, $\|df_v\|_v^2 \geq \lambda(q)\|f_v\|_v^2$ for every v. Summing and using (a) and (b),

$$\|f\|^2 = \sum_v \|df_v\|_v^2 \;\geq\; \lambda(q)\sum_v \|f_v\|_v^2 = 2\lambda(q)\|f\|^2, \qquad \text{i.e.} \qquad \frac{(\sqrt{q}-1)^2}{q+1}\,\|f\|^2 \leq 0.$$

Hence $f = 0$ for $q \geq 2$. For (i), take $\mathcal{E}$ trivial: $H^1(Y;\mathbb{C}) = 0$, so $b_1 = 0$, and $H_1(Y;\mathbb{Z})$ is finitely generated of rank 0. For (ii), $\operatorname{coker}\Delta_1 \cong \mathbb{Z}^{b_1} \oplus \mathrm{Jac}_1(Y)$ by Vol. I–II, Proposition 16.2.2. $K(Y)$

has rank $\operatorname{rank}\ker\partial_1 - \operatorname{rank}\partial_2 = b_1 = 0$, and it surjects onto $\ker\partial_1/\operatorname{im}\partial_2 = H_1(Y;\mathbb{Z})$ because $\operatorname{im}(\partial_2\partial_2^t) \subseteq \operatorname{im}\partial_2$. Finally $\chi = b_0 - b_1 + b_2$. For (iii), links are local, and a covering map is a simplicial isomorphism on stars. □

Remark 1.4.4 (Sharpness and provenance). At $q = 1$ the link is the hexagon, $2\lambda(1) - 1 = 0$, and the inequality is empty. The thin tori of Vol. I–II, Construction 16.1.3 indeed have $b_1 = 2$, so the threshold $\lambda > \frac{1}{2}$ is exactly where rank two separates from flat. The argument is Garland's local-to-global method [8], in the form isolated by Ballmann–Świątkowski [9] and Żuk [46]. We have written it for unitary local systems because this costs nothing and gives Proposition 1.5.1. It sharpens [Vol. I–II, Ch. 18, Thm. 18.1.1] in two respects: no group action is used, and the flag condition is not needed. The latter matters because four of the five complexes of [Vol. I–II, Ch. 16] violate it.

Remark 1.4.5 (Correction to [Vol. I–II, Ch. 22, Rem. 22.6.3]). That remark asserts that "no local inspection of a rank-two hyperbolic lattice reveals the obstruction". Theorem 1.4.3 shows the opposite for b_1. The vanishing of b_1 is certified by the vertex links alone, through the single number $\lambda(2) = 1 - \sqrt{2}/3 > \frac{1}{2}$ of the Heawood graph. What is genuinely global is the finite group $H_1(Y;\mathbb{Z})$ itself, e.g. $(\mathbb{Z}/2)^6$ versus $(\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2$ for two complexes with identical links, and property (T) of $\pi_1(Y)$.

Remark 1.4.6 ($\widetilde{A}_d$, $d \geq 3$). Since H^1 depends only on the 2-skeleton, the proof above applies to an $\widetilde{A}_d$ complex as soon as the 1-skeleton of each vertex link (the spherical building of $\mathrm{PG}(d,q)$, weighted by triangle counts) has normalized spectral gap $> \frac{1}{2}$. We have not computed that gap for $d \geq 3$, and we do not claim it. For thick $\widetilde{A}_d$ buildings with $d \geq 3$, $b_1 = 0$ of finite free quotients follows instead from Tits' theorem that such buildings are Bruhat–Tits and from Kazhdan's property (T) for simple groups of rank ≥ 2 over local fields. These are quoted, not proved here, and nothing below uses them.

1.5 What rigidity removes: Brillouin zones, abelian towers, and the non-commutative turn

Proposition 1.5.1 (No continuous character families). *Let Y satisfy* (L) *with $q \geq 2$.*

(i) $\mathrm{Hom}(\pi_1(Y), U(1)) = \mathrm{Hom}(H_1(Y;\mathbb{Z}), U(1))$ *is a finite group, isomorphic to $H_1(Y;\mathbb{Z})$. For every χ in it, $H^1(Y;\mathbb{C}_\chi) = 0$.*

(ii) For every unitary $\rho\colon \pi_1(Y) \to U(N)$, $H^1(\pi_1(Y); \mathfrak{gl}_N(\mathbb{C})_{\mathrm{Ad}\rho}) = 0$. Hence, by Weil's criterion, every first-order deformation of ρ in $\mathrm{Hom}(\pi_1(Y), \mathrm{GL}_N(\mathbb{C}))$ *is tangent to its conjugation orbit: unitary representations are infinitesimally rigid, and the unitary character variety has no positive-dimensional component through any point.*

Proof. (i) $U(1)$ is abelian, $H_1(Y;\mathbb{Z})$ is finite (Theorem 1.4.3), and a finite abelian group is isomorphic to its Pontryagin dual. Then apply Theorem 1.4.3 to the rank-one system $\mathbb{C}_\chi$. (ii) $\mathrm{Ad}\rho$ preserves the Hilbert–Schmidt product $\mathrm{tr}(X^*Y)$, so it defines a unitary local system of rank N^2. In degree one, $H^1(\pi_1(Y); V) = H^1(Y;\mathcal{V})$ for any connected CW complex. Apply Theorem 1.4.3. Weil's criterion [12] identifies the Zariski tangent space of the representation variety at ρ with the cocycles $Z^1(\pi_1(Y); \mathfrak{gl}_N)$, and the tangent space to the orbit with the coboundaries. □

Remark 1.5.2 (Brillouin zones). Bloch–Floquet theory for a $\mathbb{Z}^d$-periodic operator decomposes it over the torus $\widehat{\mathbb{Z}^d} = \mathrm{Hom}(\mathbb{Z}^d, U(1))$, the Brillouin zone. For a periodic operator on the maximal

abelian cover of Y (Hecke operators, the Hodge Laplacians, or any tight-binding Hamiltonian), the same decomposition runs over $\mathrm{Hom}(H_1(Y;\mathbb{Z}), U(1))$. By Proposition 1.5.1(i) this is a *finite* set of $|H_1(Y;\mathbb{Z})|$ points, and the Bloch decomposition is a finite direct sum of twisted operators: 64 of them for Y_{21} and 128 for Y_{42} (Section 1.7). There is no continuous momentum, and therefore no band, group velocity or effective mass. Part (ii) says more: no continuous family of *non-abelian* unitary twists exists either, even infinitesimally. This is the precise content of "the Brillouin zone is erased", and it strengthens [Vol. I–II, Ch. 22, Thm. 22.6.1]. That theorem obtained (iv) from property (T) of the group; here the rigidity is derived from the links.

Proposition 1.5.3 (Galois towers over Y)**.** *Let Y satisfy* (L) *with $q \geq 2$, and let G be the profinite Galois group of a tower of finite connected Galois covers $(Y_U)_U$ of Y, indexed by the open normal subgroups $U \leq G$.*

(i) Every open subgroup $U \leq G$ has finite abelianization $U^{\mathrm{ab}} = U/\overline{[U,U]}$.

(ii) In particular no open subgroup of G surjects onto $\mathbb{Z}_\ell$. There are no $\mathbb{Z}_\ell$-towers, and no $\mathbb{Z}_\ell^r$-towers for any $r \geq 1$, over any finite cover of Y. The commutative Iwasawa theory of [Vol. I–II, Chs. 6, 7] *has no rank-two sector.*

(iii) If G is an infinite ℓ-adic analytic group with Lie algebra $\mathfrak{g}$, then $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$, and $\mathfrak{g}$ has a nonzero semisimple Levi factor. So $\dim G \geq 3$, and the Iwasawa algebra $\mathbb{Z}_\ell[[G]]$ is non-commutative.

Proof. (i) The cover Y_U is a finite connected cover of Y, so $H_1(Y_U;\mathbb{Z})$ is finite by Theorem 1.4.3(iii). The tower gives a homomorphism $\pi_1(Y_U) \to U$ with dense image. Hence the image of $H_1(Y_U;\mathbb{Z}) = \pi_1(Y_U)^{\mathrm{ab}}$ in U^{ab} is dense, and being finite, it is closed; so U^{ab} is finite. (ii) A continuous surjection $U \to \mathbb{Z}_\ell$ would factor through U^{ab}. (iii) Uses Lazard's theory [11, 7], quoted. Suppose $[\mathfrak{g},\mathfrak{g}] \neq \mathfrak{g}$, and choose a nonzero $\mathbb{Q}_\ell$-linear $\mu\colon \mathfrak{g} \to \mathbb{Q}_\ell$ vanishing on $[\mathfrak{g},\mathfrak{g}]$. On a uniform open subgroup U, $\log\colon U \to \mathfrak{g}$ converts the group law into the Baker–Campbell–Hausdorff series. All terms of that series beyond $\log x + \log y$ are iterated brackets, and μ kills them, so $\mu \circ \log$ is a continuous homomorphism $U \to \mathbb{Q}_\ell$. It is nonzero, since $\log U$ spans $\mathfrak{g}$, and its image is a nonzero compact subgroup of $\mathbb{Q}_\ell$, hence infinite. This contradicts (i). A perfect Lie algebra is not solvable unless it is 0, so by Levi's theorem it has a nonzero semisimple part, of dimension ≥ 3. □

Remark 1.5.4 (The non-commutative turn). The proposed programme asks why, without abelian towers, one "must turn to non-linear dynamical modes and non-commutative group algebras". The precise statements are the following. (1) Abelian harmonic analysis over a continuous dual group, which underlies Brillouin zones, Fourier analysis on $\mathbb{Z}_\ell$-towers and the Iwasawa module $\mathfrak{X}_\infty$ over $\mathbb{Z}_\ell[[T]]$, has nothing to act on. By Propositions 1.5.1 and 1.5.3, every infinite Galois structure over Y is non-abelian of dimension ≥ 3, and its natural coefficient ring is a non-commutative Iwasawa algebra $\mathbb{Z}_\ell[[G]]$ in the sense of [1]. On finite levels, the decomposition along irreducible representations of the deck group is the character mechanism of [Vol. I–II, Ch. 18, Thm. 18.2.1]. Whether $\pi_1(Y)$ admits such an infinite ℓ-adic analytic tower depends on its linearity, which we do not decide for the complexes of [Vol. I–II, Ch. 16]. (2) The infinite structures that survive without any group are *dynamical*. They are the symbolic dynamics of $C^{(3)}$ and of its three-step reduction Λ_Q (Vol. I–II, Definition 16.4.4), the geodesic and chamber digraph towers E_k, F_k of [Vol. I–II, Ch. 18], and the Kirchberg algebras $\mathcal{O}(Y, \Delta^{\mathrm{H}})$ of [Vol. I–II, Ch. 16, Thm. 16.5.2]. These are towers of shifts, not of covers. The phrase "non-linear dynamical modes" is not given a mathematical meaning here, and we replace it by (1) and (2).

1.6 The Hecke–Laplacian and the spectrum of the cubic pencil

Recall $P(u) = I - uA_1 + qu^2A_2 - q^3u^3I$ and $\Delta^{\mathrm{H}} = -P(1) = (q^3-1)I + A_1 - qA_2$ (Vol. I–II, Definition 16.3.1). Since A_1 is normal ($A_1A_1^t = A_1^tA_1$), A_1 and $A_2 = A_1^*$ are simultaneously unitarily diagonalizable. The joint eigenpairs are $(a_j, \bar{a}_j)$, $j = 1, \dots, n$, and $P_j(u) = 1 - a_ju + q\bar{a}_ju^2 - q^3u^3$. Put $\kappa := q^2 + q + 1$.

Theorem 1.6.1 (Pencil determinant)**.** *In $\mathbb{Z}[u]$,*

$$\det(I_{3n} - uC^{(3)}) = \det P(u) = \prod_{j=1}^{n} P_j(u),$$

a polynomial of degree exactly $3n$ with leading coefficient $(-q^3)^n$.

Proof. Let $V_u = \begin{pmatrix} I & 0 & 0 \\ uI & I & 0 \\ u^2I & uI & I \end{pmatrix}$, so $\det V_u = 1$. With $I - uC^{(3)} = \begin{pmatrix} I - uA_1 & quA_2 & -q^3uI \\ -uI & I & 0 \\ 0 & -uI & I \end{pmatrix}$, the first column of $(I - uC^{(3)})V_u$ is $\mathrm{col}_1 + u\,\mathrm{col}_2 + u^2\mathrm{col}_3 = (P(u),\, 0,\, 0)^t$. The second and third columns are $(quA_2 - q^3u^2I,\, I,\, -uI)^t$ and $(-q^3uI,\, 0,\, I)^t$. The product is therefore block upper triangular with diagonal blocks $P(u)$ and $\left(\begin{smallmatrix} I & 0 \\ -uI & I \end{smallmatrix}\right)$, so $\det(I - uC^{(3)}) = \det P(u)$. Joint diagonalization gives the product. The u^{3n} coefficient is $\prod_j(-q^3)$. □

Lemma 1.6.2 (Sector lemma)**.** *Let Y be connected with $q \geq 2$. The eigenvalue κ of A_1 is simple, with eigenvector $\mathbf{1}$, and every eigenvalue $\mu_j = -P_j(1) = q^3 - 1 + a_j - q\bar{a}_j$ of Δ^{H} with $a_j \neq \kappa$ satisfies $\mathrm{Re}\,\mu_j > 0$. In particular $P_j(1) \neq 0$ for every nontrivial j, so the hypothesis of Vol. I–II, Theorem 16.3.2(ii) is automatic.*

Proof. $A_1\mathbf{1} = \kappa\mathbf{1}$ and $\mathbf{1}^tA_1 = \kappa\mathbf{1}^t$. If $S \subsetneq V$ were a nonempty set closed under A_1-successors, then all $\kappa|S|$ arcs leaving S would end in S. These account for all $\kappa|S|$ arcs entering S, so no arc enters S from outside, and S would be a union of components of the 1-skeleton. Hence A_1 is irreducible, and by Perron–Frobenius its spectral radius κ is a simple eigenvalue. Write $a_j = x + iy$ with $|a_j| \leq \kappa$. Then $\mathrm{Re}\,\mu_j = q^3 - 1 - (q-1)x \geq q^3 - 1 - (q-1)\kappa = 0$, with equality only if $x = \kappa$, i.e. $a_j = \kappa$. □

Theorem 1.6.3 (The trivial orbit and the exponent of $1 - u^3$)**.** *Let Y be a connected $\widetilde{A}_2$ complex of order $q \geq 2$. Then*

$$\det(I - uC^{(3)}) = (1-u^3)(1-q^3u^3)(1-q^6u^3)\,Q_0(u^3), \qquad Q_0 \in \mathbb{Z}[v],\ \deg Q_0 = n-3,\ Q_0(1) \neq 0.$$

Hence $1 - u^3$ divides $\det(I - uC^{(3)})$ exactly $b_0 = 1$ times. The Betti numbers b_1, b_2 do not appear in this factorization.

Proof. The vectors $\mathbf{1}$, ω^τ and $\omega^{2\tau}$ are joint eigenvectors with pairs (κ, κ), $(\omega\kappa, \omega^2\kappa)$ and $(\omega^2\kappa, \omega\kappa)$. Indeed, $(A_1\omega^{i\tau})(x) = \sum_{x\to y} \omega^{i(\tau(x)+1)} = \omega^i\kappa\,\omega^{i\tau(x)}$, and similarly for A_2. Since $P_{(\omega a, \omega^2\bar{a})}(u) = P_{(a,\bar{a})}(\omega u)$, their product is

$$\prod_{i=0}^{2}(1-\omega^iu)(1-q\omega^iu)(1-q^2\omega^iu) = (1-u^3)(1-q^3u^3)(1-q^6u^3).$$

By Proposition 1.2.1, $\det P(u) = Q(u^3)$ with $Q \in \mathbb{Z}[v]$. Each factor $1 - q^{3i}v$ is primitive, so by Gauss's lemma the quotient Q_0 is integral. If $Q_0(1) = 0$, some other joint eigenpair j would have $P_j(\omega^i) = 0$ for some i. Then the pair $j' = (\omega^ia_j, \omega^{-i}\bar{a}_j)$ is again a joint eigenpair (apply D_1^i), distinct from the trivial one, and it has $P_{j'}(1) = 0$, which contradicts Lemma 1.6.2. □

Theorem 1.6.4 (Derivative identity)**.** *Let Y be a connected $\widetilde{A}_2$ complex of order $q \geq 2$. Then*

$$\frac{d}{du}\det(I - uC^{(3)})\Big|_{u=1} = (-1)^n\, n(q-1)^2(q+1)\,|\operatorname{Jac}^{\mathrm{H}}(Y)|.$$

If Y satisfies the flag condition, then $n(q-1)^2(q+1) = \operatorname{Tr}(C^{(3)})^3$ by Theorem 1.2.3, and

$$\left|\tfrac{d}{du}\det(I - uC^{(3)})\big|_{u=1}\right| = \operatorname{Tr}(C^{(3)})^3 \cdot |\operatorname{Jac}^{\mathrm{H}}(Y)|.$$

Equivalently, $|Q_0(1)| = \dfrac{n(q-1)^2(q+1)}{3(q^3-1)(q^6-1)}\,|\operatorname{Jac}^{\mathrm{H}}(Y)|$.

Proof. By Theorem 1.6.1 and Jacobi's formula, $\frac{d}{du}\det P(u) = \operatorname{tr}\big(\operatorname{adj} P(u)\, P'(u)\big)$, with $P'(1) = -A_1 + 2qA_2 - 3q^3 I$. By Lemma 1.6.2, Δ^{H} has rank $n-1$ with right kernel $\mathbb{Q}\mathbf{1}$ and left kernel $\mathbb{Q}\mathbf{1}^t$ (Vol. I–II, Theorem 16.3.2(i)). Since $\Delta^{\mathrm{H}} \operatorname{adj}\Delta^{\mathrm{H}} = \operatorname{adj}\Delta^{\mathrm{H}}\,\Delta^{\mathrm{H}} = 0$, we get $\operatorname{adj}\Delta^{\mathrm{H}} = c\,\mathbf{1}\mathbf{1}^t$, where c is the common principal cofactor. It has two evaluations:

- nc equals the sum of the principal $(n-1)$-minors, i.e. the product of the nonzero eigenvalues $\prod_{a_j \neq \kappa} \mu_j$. Non-real μ_j occur in conjugate pairs, and real ones are > 0 by Lemma 1.6.2. So $c > 0$.
- $|c| = |\operatorname{Jac}^{\mathrm{H}}(Y)|$. The image of Δ^{H} lies in the sum-zero lattice L, and $\mathbb{Z}^n/L \cong \mathbb{Z}$ splits off. The last column is minus the sum of the others. Dropping the last coordinate identifies L with $\mathbb{Z}^{n-1}$, and the reduced matrix with Δ^{H} minus its last row and column. So $L/\operatorname{im}\Delta^{\mathrm{H}}$ has order $|c|$.

So $\operatorname{adj} P(1) = \operatorname{adj}(-\Delta^{\mathrm{H}}) = (-1)^{n-1}|\operatorname{Jac}^{\mathrm{H}}|\,\mathbf{1}\mathbf{1}^t$, and

$$\tfrac{d}{du}\det P(u)\big|_{u=1} = (-1)^{n-1}|\operatorname{Jac}^{\mathrm{H}}|\,\mathbf{1}^t P'(1)\mathbf{1} = (-1)^{n-1}|\operatorname{Jac}^{\mathrm{H}}| \cdot n(-\kappa + 2q\kappa - 3q^3).$$

Finally $-\kappa + 2q\kappa - 3q^3 = (2q-1)(q^2+q+1) - 3q^3 = -(q^3 - q^2 - q + 1) = -(q-1)^2(q+1)$. The expression for $Q_0(1)$ follows by differentiating the factorization of Theorem 1.6.3 at $u = 1$: $\frac{d}{du}(\cdots)|_1 = -3(1-q^3)(1-q^6)Q_0(1)$. □

Remark 1.6.5 (The same number twice). The integer $n(q-1)^2(q+1)$ plays two roles. It is the chamber-minus-backtrack count of Theorem 1.2.3, and it is n times the product $(1-q)(1-q^2)$ of the nontrivial values at $u = 1$ of the trivial Satake factor $(1-u)(1-qu)(1-q^2u)$. The first reading is combinatorial and needs the flag condition. The second is spectral and does not. The derivative identity is where they meet. On the complexes without the flag condition, the spectral reading survives and the trace does not (Table 1.1).

Remark 1.6.6 (Where Betti numbers do enter). In rank one, Bass's identity reads $\det(I - uA^{\mathrm{nb}}) = (1-u^2)^{m-n}\det(I - uT_p + qu^2)$ [Vol. I–II, Ch. 2]. The factor $(1-u^2)^{b_1-1}$ attaches to the *edge* operator A^{nb}, of size $2m$, not to the vertex companion C, of size $2n$, for which $\det(I-uC) = \det(I - uT_p + qu^2)$ on the nose. Rank two behaves the same way. Kang and Li [10] express the zeta function of an $\widetilde{A}_2$ complex as a quotient with $(1-u^3)^{\chi(Y)}$ in the numerator, and in the denominator $\det P(u)$ together with the determinant of an edge (branch) operator. We quote only this shape and use nothing from it. By Theorem 1.4.3, $\chi(Y) = 1 - b_1 + b_2 = 1 + b_2$, and for $q = 2$, $\chi(Y) = n$ (Vol. I–II, Lemma 16.1.2). The Betti numbers therefore belong to the edge side of the zeta function. The vertex companion $C^{(3)}$ sees only b_0 (Theorem 1.6.3).

1.7 Worked examples: Y_{21}, Y_{24} and Y_{42}

Construction, rebuilt. Take the triangle presentation T of Vol. I–II, Construction 16.1.4: 21 triples over $\mathbb{Z}/7$, with the Singer cycle $\lambda(x) = \{x+1, x+2, x+4\}$; axioms (A)–(C) are re-verified. The type-preserving subgroup Γ^0 has Reidemeister–Schreier generators $\gamma(s,x) = a_0^s a_x a_0^{-(s+1)}$ ($s \in \mathbb{Z}/3$, $x \in \mathbb{Z}/7$; two of them trivial) and 63 relators $\gamma(s,x)\gamma(s{+}1,y)\gamma(s{+}2,z)$ for $(x,y,z) \in T$. The Smith form of the abelianized relator matrix gives

$$H_1(X_3;\mathbb{Z}) = \Gamma^{0,\mathrm{ab}} \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/14,$$

in agreement with [Vol. I–II, Ch. 16]. A homomorphism $\theta\colon \Gamma^0 \to A$ to a finite abelian group defines the voltage cover with vertices $(\alpha, s) \in A \times \mathbb{Z}/3$ and arcs $(\alpha, s) \to (\alpha + \theta\gamma(s,x), s+1)$. Here $\dim \mathrm{Hom}(H_1, \mathbb{F}_7) = 1$ and $\dim \mathrm{Hom}(H_1, \mathbb{F}_2) = 3$. Thus $A = \mathbb{Z}/7$ gives a unique Y_{21}, $A = (\mathbb{Z}/2)^3$ gives Y_{24}, and $A = \mathbb{Z}/14$ gives *seven* covers Y_{42}, one for each nonzero $\mathbb{F}_2$-functional. All seven have identical invariants in every column of Table 1.1.

Y_{21}, step by step ($q = 2$, $n = 21$).

1. *Links.* Every vertex link is the Heawood graph, the incidence graph of the Fano plane $\mathrm{PG}(2,2)$: 7 points $\{i\}$, 7 lines $\{i, i+1, i+3\}$, 21 flags. So every vertex has $7 + 7 = 14$ neighbours, 7 of each foreign type, and 21 triangles.

2. *Edges.* $|E| = \frac{1}{2} \cdot 21 \cdot 14 = 147 = n(q^2+q+1)$.

3. *Chambers.* $|F| = \frac{1}{3} \cdot 21 \cdot 21 = 147 = \frac{1}{3} n(q^2+q+1)(q+1)$. Each edge lies in $3 = q+1$ triangles: $3 \cdot 147 = 441 = 3|F|$ edge–triangle incidences.

4. *Flag part of the trace.* $3|F| - 3q|E| + 3nq^3 = 441 - 882 + 504 = 63 = 21 \cdot 1^2 \cdot 3$.

5. *Actual trace.* Direct computation gives $\mathrm{Tr}\, A_1^3 = 1029 = 441 + 588$. So $s = 196$ closed type-increasing 3-cycles are not triangles; the flag condition fails, as recorded in [Vol. I–II, Ch. 16]. By (1.2.2), $\mathrm{Tr}(C^{(3)})^3 = 1029 - 882 + 504 = 651 = 63 + 588$, confirmed by computing the 63×63 matrix power directly. Also $\mathrm{Tr}(C^{(3)})^k = 0$ for $k = 1, 2, 4, 5$ and $\mathrm{Tr}(C^{(3)})^6 = 15939$.

6. *Garland.* Every link has normalized gap $0.528595 = 1 - \sqrt{2}/3$. Exact ranks give $(b_0, b_1, b_2) = (1, 0, 20)$ and $\chi = 21$; $H_1(Y_{21};\mathbb{Z}) = (\mathbb{Z}/2)^6$, so the "Brillouin zone" is 64 points.

7. *Hecke side.* $\mathrm{Jac}^{\mathrm{H}}(Y_{21}) = (\mathbb{Z}/7)^{17} \oplus \mathbb{Z}/147$, of order $3 \cdot 7^{19}$. The exact derivative is $\frac{d}{du} \det(I - uC^{(3)})|_{u=1} = -2154391190035524027 = (-1)^{21} \cdot 63 \cdot 3 \cdot 7^{19}$, as Theorem 1.6.4 requires, and $Q_0(1) = 7^{18}$. Note that the identity holds with $63 = n(q-1)^2(q+1)$, *not* with the trace 651: here the spectral and combinatorial readings of Remark 1.6.5 part ways.

Y_{42}. $|E| = |F| = 294$, and the flag part is $3 \cdot 294 - 6 \cdot 294 + 3 \cdot 42 \cdot 8 = 882 - 1764 + 1008 = 126$. Then $\mathrm{Tr}\, A_1^3 = 1386 = 882 + 504$ ($s = 168$), so $\mathrm{Tr}(C^{(3)})^3 = 1386 - 1764 + 1008 = 630 = 126 + 504$ and $\mathrm{Tr}(C^{(3)})^6 = 15414$. Further, $(b_0, b_1, b_2) = (1, 0, 41)$, $H_1 = (\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2$ (128 characters), and $\mathrm{Jac}^{\mathrm{H}} = (\mathbb{Z}/7)^{18} \oplus \mathbb{Z}/299593 \oplus \mathbb{Z}/2097151 \oplus \mathbb{Z}/176160684$, as in [Vol. I–II, Ch. 16]. The derivative is $126 \cdot |\mathrm{Jac}^{\mathrm{H}}(Y_{42})| = 22709397752337736246719794532064846488$ (exact, positive since n is even).

Y	n	$\lvert E\rvert = \lvert F\rvert$	flag	$\operatorname{Tr} A_1^3$	$3\lvert F\rvert$	$n(q-1)^2(q+1)$	$\operatorname{Tr}(C^{(3)})^3$	$\operatorname{Tr}(C^{(3)})^6$	(b_0, b_1, b_2)	$H_1(Y;\mathbb{Z})$
Y_{21}	21	147	no	1029	441	63	651	15939	$(1, 0, 20)$	$(\mathbb{Z}/2)^6$
Y_{24}	24	168	no	1008	504	72	576	12840	$(1, 0, 23)$	$(\mathbb{Z}/2)^4 \oplus \mathbb{Z}/14$
Y_{42}	42	294	no	1386	882	126	630	15414	$(1, 0, 41)$	$(\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2$
Y_{168}	168	1176	yes	3528	3528	504	504	—	$(1, 0, 167)$	—

Table 1.1: Exact data. The rows Y_{21}, Y_{24}, Y_{42} were recomputed from scratch by `code/v3p1_verify.py`; the row Y_{168} is quoted from [Vol. I–II, Ch. 16]. The excess $\operatorname{Tr}(C^{(3)})^3 - n(q-1)^2(q+1) = \operatorname{Tr} A_1^3 - 3|F|$ equals $28n$, $21n$ and $12n$, as in Vol. I–II, Construction 16.1.4. For every row the link gap is $1 - \sqrt{2}/3$ and $b_1 = 0$ (Theorem 1.4.3).

check	complexes	result
(P) axioms (A)–(C) of T; $H_1(X_3) = \mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/14$ by Reidemeister–Schreier	X_3	exact
(G) simplicial; $A_1A_2 = A_2A_1$; Heawood links; $\lvert E\rvert = \lvert F\rvert = 7n$	Y_{21}, Y_{24}, seven Y_{42}	exact
(T) $\operatorname{Tr}(C^{(3)})^k$, $k \le 6$, by direct matrix powers; $= 0$ for $3 \nmid k$; (1.2.2)	same	exact
(S) link normalized gap $= 1 - \sqrt{2}/3$ (Lemma 1.4.2)	same, all vertices	numeric 10^{-12}
(B) $b_1 = 0$ from ranks of ∂_1, ∂_2; $H_1(Y;\mathbb{Z}) = \operatorname{Tors}\operatorname{coker}\partial_2$ by Smith form	same	exact
(H) $\operatorname{Jac}^{\mathrm{H}}$ by Smith form; matches [Vol. I–II, Ch. 16, §16.6]	same	exact
(D) $\frac{d}{du}\det P(u)\lvert_1$ by Jacobi's formula, as n Bareiss determinants, against $(-1)^n n(q-1)^2(q+1)\lvert\operatorname{Jac}^{\mathrm{H}}\rvert$	same	exact, equal

Table 1.2: Checks run by `code/v3p1_verify.py`. The derivative in (D) is computed column by column, $\frac{d}{du}\det P = \sum_i \det(P(1)$ with column i replaced by that of $P'(1))$, independently of the adjugate argument used in the proof.

1.8 Verification battery

1.9 Scope, residue, corrections

Remark 1.9.1 (Solved, open). *Solved:* the grading conjugation and the cubic trace identity, with and without the flag condition. The local Garland theorem with unitary coefficients, and its consequences for Brillouin zones and towers. The unconditional form of Vol. I–II, Theorem 16.3.2(ii), via the sector lemma. The exact multiplicity of $1-u^3$, and the derivative identity linking $\operatorname{Tr}(C^{(3)})^3$ to $\lvert\operatorname{Jac}^{\mathrm{H}}\rvert$. The first nonzero trace in every rank, $\operatorname{Tr}(C^{(d+1)})^{d+1} = n\prod_{i=1}^{d}(1-q^i)$ under the flag condition (formerly open problem (a); Theorem 1.3.3). *Open:* (a) a finite thick $\widetilde{A}_d$ complex, $d \ge 3$, on which to evaluate the correction terms $s_{\mathbf{j}}$ of Theorem 1.3.3. (b) The link gap $> \frac{1}{2}$ for $\widetilde{A}_d$, $d \ge 3$ (Remark 1.4.6). (c) Whether $\pi_1(Y_{21})$ is linear over a local field, i.e. whether infinite ℓ-adic analytic towers as in Proposition 1.5.3(iii) exist over the complexes computed here. (d) Structure theorems for $\mathbb{Z}_\ell[[G]]$-modules attached to such towers.

Remark 1.9.2 (Corrections to the proposed programme). (1) The identity $\operatorname{Tr}(C^{(3)})^3 = 3|F| - 3q|E| + 3nq^3 = n(q-1)^2(q+1)$ needs the flag condition. On Y_{21}, which violates it, the exact value is 651, not 63. The general identity is (1.2.2), and 63 is its flag part. (2) "$\operatorname{Jac}_0(Y)$ is pure torsion" is not the content of $b_1 = 0$: Jac_0 is finite for every connected complex, by definition. The content is that $H_1(Y;\mathbb{Z})$, $K(Y)$ and $\operatorname{coker}\Delta_1 = \operatorname{Jac}_1(Y)$ are finite (Theorem 1.4.3(ii)). (3) The factorization $\det(I - uC^{(3)}) = (1-u^3)^{b_2-b_1}P(u)$ is false: the left side has degree $3n$ (Theorem 1.6.1), and the right side has degree $3(b_2-b_1)+3n > 3n$, since $b_2 - b_1 = \chi - 1 > 0$. For Y_{21} that is $123 \ne 63$. The correct statements are Theorems 1.6.1, 1.6.3 and 1.6.4, with Betti numbers entering only on the edge side (Remark 1.6.6). (4) $C^{(3)}$ is the companion of the vertex Hecke pencil, not a Hashimoto-

type edge operator, and "Ihara–Hashimoto–Kang" conflates the two sides of Remark 1.6.6. (5) "Garland–Ballmann–Świątkowski" is proved here in a local form that is stronger in the respects needed (Remark 1.4.4), and it corrects [Vol. I–II, Ch. 22, Rem. 22.6.3] (Remark 1.4.5). (6) "Non-linear dynamical modes" is replaced by the precise dichotomy of Remark 1.5.4. (7) For $\widetilde{A}_d$, $d \geq 3$, the first nonzero trace is proved in Section 1.3; the Garland-type claim of Remark 1.4.6 is quoted, not proved.

Acknowledgement of provenance and caveats

Unrefereed. Every numbered statement is proved from finite linear algebra and earlier chapters, except Proposition 1.5.3(iii) (Lazard, quoted) and the tangent-space identification in Proposition 1.5.1(ii) (Weil, quoted). The Kang–Li formula is cited for its shape only. The complexes of Section 1.7 were rebuilt independently of the script of [Vol. I–II, Ch. 16], and every entry they share with [Vol. I–II, Chs. 16, 18, 22] agrees. The bibliographic data of [11, 7, 12, 46] have been checked against Crossref and zbMATH.

Chapter 2

Simplicial Sandpiles on $\widetilde{A}_2$ Complexes: Sign Obstructions, Hodge Factorization, and Avalanche Statistics

The proposed programme asks for abelian sandpile dynamics on the k-simplices of an $\widetilde{A}_2$ complex, toppled by the Hodge Laplacians Δ_k, for a recurrent-configuration group $\mathrm{Pic}_k(Y)$, and for critical avalanche exponents on buildings. This chapter establishes what can be established and says what cannot.

(1) *The sign obstruction.* A threshold rule $z \mapsto z - Me_i$ is an abelian sandpile in Dhar's sense when M is a Z-matrix, i.e. has non-positive off-diagonal entries. We prove that on every $\widetilde{A}_2$ complex of order $q \geq 2$, none of Δ_1, the up-Laplacian $\partial_2\partial_2^t$, Δ_2 or the Hecke–Laplacian Δ^{H} becomes a Z-matrix under any choice of orientations. The obstructions are local and elementary: any three points of $\mathrm{PG}(2,q)$ (pairwise non-adjacent in the link), one chamber, three chambers on one edge, and the sign of $\Delta^{\mathrm{H}}_{xy}\Delta^{\mathrm{H}}_{yx} = -q$. Consequently the edge and chamber toppling rules are not abelian, which we show both by an explicit two-site example and by randomized stabilizations on Y_{21}. Only the vertex sandpile of the 1-skeleton survives. On thin tori Δ_2 *is* coherent: it is the Laplacian of the dual honeycomb. So it is thickness, not dimension, that destroys the dynamics.

(2) *The Hodge factorization.* The algebraic groups $\operatorname{coker}\Delta_k$ are well defined regardless. We prove $|\operatorname{Jac}_1(Y)| \cdot \operatorname{disc}\ker_{\mathbb{Z}}\Delta_1 = n\,|\operatorname{Jac}_0(Y)|\,|\operatorname{Jac}_2(Y)|\,\operatorname{disc} H_2(Y;\mathbb{Z})$. With Garland rigidity (Chapter 1) this becomes $|\operatorname{Jac}_1| = \det\Delta_1 = n|\operatorname{Jac}_0||\operatorname{Jac}_2|\operatorname{disc} H_2$ on every thick complex. On Y_{21} the discriminant of the 2-cycle lattice is $3 \cdot 7^{29}$.

(3) *Exact avalanche statistics.* For the vertex sandpile with one sink we prove the closed form $\langle s\rangle = n^2 L^+_{ss}/(n-1)$ and the two-sided bound $n/\lambda_{\max} \leq \langle s\rangle \leq n/\lambda_2$ for the mean avalanche size. So Ramanujan families have $\langle s\rangle = \Theta(n)$, against $\Theta(n\log n)$ on two-dimensional tori. This is the rigorous form of the "divergence from the Euclidean lattice", and it is spectral, not topological. The exponents τ and z requested by the proposed programme are not provable on finite complexes. We state the mean-field values as a conjecture, derive them for the approximating branching process, and explain why the dynamic exponent z has no meaning on expanders.

(4) *Example.* The 1-skeleton of Y_{21} is the complete tripartite graph $K_{7,7,7}$, which explains at once its flag defect (343 type-increasing 3-cycles against 147 chambers). Its thresholds are $14, 5, 3, 7$ (vertex, edge, chamber, Hecke); the mean avalanche size is $\langle s\rangle = 29/20$ exactly, and the entropy per vertex is 2.4070.

Status of the results

Every numbered statement is proved, except Conjecture 2.4.2 and the quoted classical inputs: Dhar's recurrent-state theory [2] (Proposition 2.3.1), limit theorems for critical branching processes [5] (Proposition 2.4.1), and Lyons' tree-entropy theorem [6] (Proposition 2.3.4). All finite identities were checked exactly by `code/v3p2_verify.py` (Table 2.2); statements labelled "numeric" are floating point or Monte Carlo.

Notation follows Chapter 1: Y is an $\widetilde{A}_2$ complex of order q with n vertices, edge set E oriented from type t to $t+1$, chamber set F, boundary maps ∂_1, ∂_2, and $\Delta_0 = \partial_1\partial_1^t$, $\Delta_1 = \partial_1^t\partial_1 + \partial_2\partial_2^t$, $\Delta_2 = \partial_2^t\partial_2$, $\mathrm{Jac}_k = \mathrm{Tor}\,\mathrm{coker}_{\mathbb{Z}}\,\Delta_k$ (Vol. I–II, Definition 16.2.1). For a vertex v and an edge $e \ni v$ put $\varepsilon(v,e) = +1$ if v is the head of e and -1 if it is the tail; for a chamber t and an edge $e \subset t$, $[t:e] = \pm 1$ is the incidence number. Hypothesis (L) is that of Chapter 1: every vertex link is the incidence graph of $\mathrm{PG}(2,q)$.

2.1 Toppling matrices and the sign obstruction

Definition 2.1.1 (Threshold dynamics)**.** Let $M \in M_N(\mathbb{Z})$ have positive diagonal. A *configuration* is $z \in \mathbb{Z}^N$; site i is *unstable* if $z_i \geq M_{ii}$, and *toppling* i replaces z by $z - Me_i$. A *stabilization* is a finite sequence of topplings of unstable sites ending in a configuration with no unstable site. The dynamics is *abelian* if every configuration that has a stabilization has exactly one stable result, whatever the order.

The proposed programme's k-simplicial sandpile is this dynamics with $M = \Delta_k$ on C_k. Explicitly:

$k = 0$. $\Delta_0 = 2(q^2+q+1)I - (A_1 + A_2)$ is the Laplacian of the 1-skeleton. A vertex holding at least $2(q^2+q+1)$ grains (14 for $q=2$) topples, sending one grain to each neighbour.

$k = 1$. By Lemma 2.1.2 below, an edge holding at least $q+3$ grains (5 for $q=2$) topples: it loses $q+3$ grains, and each of its $2q(2q+1)$ ($= 20$) neighbours f receives $-\Delta_1(f,e) = -\varepsilon(v,e)\varepsilon(v,f) \in \{\pm 1\}$ grains, where v is the vertex shared by e and f. Edges of a common chamber exchange nothing.

$k = 2$. A chamber holding at least 3 grains topples; each chamber t' sharing an edge e with it receives $-[t:e][t':e] \in \{\pm 1\}$ grains.

Lemma 2.1.2 (Entries of Δ_1)**.** *Assume* (L). *Then* $\Delta_1(e,e) = q+3$. *For* $e \neq f$: $\Delta_1(e,f) = \varepsilon(v,e)\varepsilon(v,f)$ *if* e, f *share a vertex* v *and lie in no common chamber, and* $\Delta_1(e,f) = 0$ *otherwise. Each edge has exactly* $2q(2q+1)$ *neighbours in the support of* Δ_1.

Proof. The diagonal is $\|\partial_1 e\|^2 + \#\{t \supset e\} = 2 + (q+1)$. For $e \neq f$ sharing v, the down term is $\langle \partial_1 e, \partial_1 f\rangle = \varepsilon(v,e)\varepsilon(v,f)$. The up term is $\sum_t [t:e][t:f]$, over the at most one chamber $t = \{v,a,b\}$ containing $e = \{v,a\}$ and $f = \{v,b\}$. Orient t as (v,a,b), so $\partial_2 t = (v,a) + (a,b) + (b,v)$. Then $[t:e] = +1$ exactly when e points $v \to a$, i.e. $[t:e] = -\varepsilon(v,e)$, and $[t:f] = +1$ exactly when f points $b \to v$, i.e. $[t:f] = \varepsilon(v,f)$. The two terms cancel. Edges without a common vertex have neither term. At each endpoint v of e there are $2(q^2+q+1) - 1$ other edges, of which $q+1$ (the link-neighbours of e in $\mathrm{Lk}(v)$) share a chamber with e. That leaves $2q^2+q$ at each end, and $2q(2q+1)$ in total. □

Definition 2.1.3 (Signed support, switching)**.** For a symmetric $M \in M_N(\mathbb{Z})$ the *signed support graph* $\Sigma(M)$ has an edge $\{i, j\}$, $i \neq j$, whenever $M_{ij} \neq 0$, with sign $\operatorname{sgn} M_{ij}$. *Switching* by a signature $S = \operatorname{diag}(s_i)$, $s_i = \pm 1$, replaces M by SMS. Reorienting edges (or chambers) of Y switches Δ_1 (or Δ_2).

Lemma 2.1.4 (Switching criterion)**.** *A symmetric M can be switched to a Z-matrix if and only if every cycle $i_1 i_2 \cdots i_\ell i_1$ of $\Sigma(M)$ has sign product $\prod \operatorname{sgn} M_{i_r i_{r+1}} = (-1)^\ell$.*

Proof. SMS is a Z-matrix iff $s_i s_j M_{ij} < 0$ on every edge of $\Sigma(M)$, i.e. $\operatorname{sgn} M_{ij} = -s_i s_j$. Along a cycle the product of $-s_i s_j$ is $(-1)^\ell \prod s_{i_r}^2 = (-1)^\ell$, so the condition is necessary. Conversely, define s on each component by propagating $s_j = -s_i \operatorname{sgn} M_{ij}$ along a spanning tree; the cycle condition applied to the fundamental cycles makes every non-tree edge consistent. □

Theorem 2.1.5 (Sign obstruction)**.** *Let Y satisfy* (L) *with $q \geq 2$, with any orientations of edges and chambers.*

(i) Δ_0 *is a Z-matrix.*

(ii) The up-Laplacian $U_1 = \partial_2 \partial_2^t$ cannot be switched to a Z-matrix. This holds on every complex with at least one chamber.

(iii) Δ_1 *cannot be switched to a Z-matrix. This holds under* (L) *for every $q \geq 1$.*

(iv) Δ_2 *cannot be switched to a Z-matrix. On the thin tori ($q = 1$) it can: with the coherent orientation it is the Laplacian of the dual honeycomb graph.*

(v) No signature S makes $S\Delta^{\mathrm{H}}S$ a Z-matrix.

Proof. (i) $\Delta_0 = 2(q^2 + q + 1)I - (A_1 + A_2)$.

(ii) For a chamber t with edges e_1, e_2, e_3, the entries $U_1(e_i, e_j) = [t : e_i][t : e_j]$ (no other chamber contains two of the e_i) form a 3-cycle of $\Sigma(U_1)$ with sign product $\prod_{i<j} [t : e_i][t : e_j] = \prod_i [t : e_i]^2 = +1 \neq (-1)^3$.

(iii) Fix a vertex v. Three points p_1, p_2, p_3 of $\mathrm{PG}(2, q)$ are pairwise non-adjacent in the incidence graph $\mathrm{Lk}(v)$, which is bipartite. Hence the edges $e_i = \{v, p_i\}$ pairwise share v and lie in no common chamber. By Lemma 2.1.2 they form a 3-cycle of $\Sigma(\Delta_1)$ with sign product $\prod_{i<j} \varepsilon(v, e_i)\varepsilon(v, e_j) = +1$. For $q = 1$ the three points exist as well, since $\mathrm{PG}(2, 1)$ has 3 points.

(iv) Two distinct chambers share at most one edge, so $\Delta_2(t, t') = [t : e][t' : e]$ for chambers meeting in e, and 0 otherwise. An edge lies in $q + 1 \geq 3$ chambers; three of them, t_1, t_2, t_3, give a 3-cycle with sign product $\prod_i [t_i : e]^2 = +1$. In the thin case, orient every chamber of the planar Coxeter complex counterclockwise. Two chambers sharing an edge then induce opposite orientations on it, so $\Delta_2(t, t') = -1$, and $\Delta_2 = 3I -$ (adjacency of the dual graph). The dual graph is the honeycomb, and the construction descends to the quotient by $m\mathbb{Z}^2$.

(v) For an edge $x \to y$, $\Delta^{\mathrm{H}}_{xy} = A_1(x, y) - qA_2(x, y) = 1$ and $\Delta^{\mathrm{H}}_{yx} = A_1(y, x) - qA_2(y, x) = -q$, so $\Delta^{\mathrm{H}}_{xy}\Delta^{\mathrm{H}}_{yx} = -q < 0$. Switching preserves this product, while in a Z-matrix it is ≥ 0. □

Corollary 2.1.6 (No abelian edge, chamber or Hecke sandpile)**.** *Let M be Δ_1, U_1 or Δ_2 on a complex satisfying* (L)*, $q \geq 2$, with any orientations, or $M = \Delta^{\mathrm{H}}$. Then the threshold dynamics of M is not abelian: some configuration has two stabilizations with different results.*

Proof. Symmetric M. By Theorem 2.1.5, M is not a Z-matrix. All its off-diagonal entries lie in $\{0, \pm 1\}$, so $M_{ef} = M_{fe} = +1$ for some $e \neq f$. Take $z = M_{ee}e_e + M_{ff}e_f$. Toppling e first lowers z_f to $M_{ff} - 1$; toppling f first lowers z_e to $M_{ee} - 1$. Every other site g starts at 0 and receives $-M_{ge} \leq 1$ (resp. $-M_{gf} \leq 1$), which is below $M_{gg} \geq 3$. So $z - Me_e$ and $z - Me_f$ are both stable, and they differ.

$M = \Delta^{\mathrm{H}}$. Take an edge $x \to y$, put $\theta = q^3 - 1$, and let $z = \theta(e_x + e_y)$. A toppling at u adds $-\Delta^{\mathrm{H}}_{wu} \in \{-1, +q\}$ to each neighbour w of u: -1 if $w \to u$, and $+q$ if $u \to w$. Order A: topple y. Then $z_x = \theta - 1$, $z_y = 0$, and every other site holds at most q. The result $z - \Delta^{\mathrm{H}} e_y$ is stable. Order B: topple x. Then $z_y = \theta + q$ is still unstable, so topple y. Now $z_y = q$, $z_x = -1$, and every other site holds at most $2q < \theta$. The result $z - \Delta^{\mathrm{H}} e_x - \Delta^{\mathrm{H}} e_y$ is stable. The two results differ by $\Delta^{\mathrm{H}} e_x \neq 0$. □

Remark 2.1.7 (Numerics on Y_{21}). Take 300 random configurations on the 147 edges of Y_{21} with $z_e \in [0, 4]$, add 5 grains at a random edge, and stabilize twice under Δ_1: once always toppling the lowest-indexed unstable edge, once in random order. Every stabilization terminated. The two results differed in 113 cases, and in 294 cases the first result had negative entries (fixed seed; Table 2.2, line (N)). We have no proof that stabilization always terminates. Termination is irrelevant anyway: without abelianness there is no recurrent class, no stationary measure independent of the protocol, and no recurrent-configuration group.

Remark 2.1.8 (What this means for the proposed programme). The "recurrent configuration group $\mathrm{Pic}_k(Y)$" presupposes Dhar's theory, and that theory needs a Z-matrix: the abelian property follows from the fact that toppling one site can never stabilize another. By Theorem 2.1.5, for $k = 1, 2$ on thick complexes there is no such group; there is no "1-simplex circulation sandpile" and no avalanche of chambers. What survives is (a) the vertex sandpile, a genuine abelian sandpile on a $2(q^2 + q + 1)$-regular graph (Section 2.3), and (b) the *algebraic* groups $\operatorname{coker} \Delta_k$, which need no dynamics (Section 2.2). The Duval–Klivans–Martin group $K(Y)$ [4] is algebraic in the same sense. On thin tori the chamber sandpile is the honeycomb sandpile. The obstruction is exactly the thickness $q + 1 \geq 3$ of the edges.

2.2 Critical groups and the Hodge factorization

For $M \in M_N(\mathbb{Z})$ write $\mathrm{Pic}(M) := \operatorname{coker}_{\mathbb{Z}} M$. For $k = 0$ and connected Y, $\mathrm{Pic}(\Delta_0) \cong \mathbb{Z} \oplus \mathrm{Jac}_0(Y)$, and $\mathrm{Jac}_0(Y) \cong \mathbb{Z}^{n-1}/\operatorname{im} \widetilde{\Delta}_0$ for the reduced Laplacian $\widetilde{\Delta}_0$ (one row and column deleted). The form $\mathbb{Z}^{|S_k|-1}/\operatorname{im}$ proposed in the proposed programme is this *vertex* statement. It does not describe $k = 1$: there $\ker \Delta_1$ has rank b_1, not 1.

Lemma 2.2.1 (Lattice identity). *Let $M \in M_N(\mathbb{Z})$ be symmetric and let $K = \ker_{\mathbb{Z}} M$, a saturated lattice of rank r. Then* $\operatorname{coker}_{\mathbb{Z}} M \cong \mathbb{Z}^r \oplus \operatorname{Tor} \operatorname{coker}_{\mathbb{Z}} M$, *and*

$$|\operatorname{Tor} \operatorname{coker}_{\mathbb{Z}} M| \cdot \operatorname{disc} K = \operatorname{pdet} M,$$

where $\operatorname{disc} K$ *is the Gram determinant of a basis of K and* pdet *is the product of the nonzero eigenvalues.*

Proof. Let $K^{\perp} = \{x \in \mathbb{Z}^N : x \cdot K = 0\}$, which is saturated of rank $N - r$. Symmetry gives $\operatorname{im} M \subseteq K^{\perp}$, and $\mathbb{Z}^N/K^{\perp}$ is free of rank r, so $\operatorname{coker} M \cong \mathbb{Z}^r \oplus K^{\perp}/\operatorname{im} M$. Since K is saturated, $Q = \mathbb{Z}^N/K$ is free, and the dot product identifies $K^{\perp}$ with $\mathrm{Hom}(Q, \mathbb{Z})$. Under this identification M becomes the nondegenerate form $B(\bar{x}, \bar{y}) = x \cdot My$ on Q, and $|K^{\perp}/\operatorname{im} M| = |\det B|$. Choose

lifts $y_1, \dots, y_{N-r}$ of a basis of Q, and let p_i be their orthogonal projections to $V = K_{\mathbb{R}}^{\perp}$. Then $B(\bar{y}_i, \bar{y}_j) = p_i \cdot M p_j$, so $\det B = \det(M|_V) \cdot \det(p_i \cdot p_j)$. The p_i span the projection of $\mathbb{Z}^N$ to V. That lattice is dual to $K^{\perp}$, of covolume $1/\operatorname{covol} K^{\perp} = 1/\operatorname{covol} K$, so $\det(p_i \cdot p_j) = 1/\operatorname{disc} K$. Finally $\det(M|_V) = \operatorname{pdet} M$. □

Theorem 2.2.2 (Hodge factorization of the critical groups)**.** *Let Y be a finite connected simplicial 2-complex with n vertices. Then*

$$|\operatorname{Jac}_1(Y)| \cdot \operatorname{disc}(\ker_{\mathbb{Z}} \Delta_1) = n\,|\operatorname{Jac}_0(Y)| \cdot |\operatorname{Jac}_2(Y)| \cdot \operatorname{disc} H_2(Y;\mathbb{Z}),$$

where $H_2(Y;\mathbb{Z}) = \ker_{\mathbb{Z}} \partial_2$ and $\ker_{\mathbb{Z}} \Delta_1 = \ker \partial_1 \cap \ker \partial_2^t$ is the lattice of integral harmonic 1-chains. If $b_1(Y) = 0$ (in particular under (L) *with $q \geq 2$, by Theorem 1.4.3), then Δ_1 is nonsingular,* $\operatorname{Pic}(\Delta_1) = \operatorname{Jac}_1(Y)$*, and*

$$|\operatorname{Jac}_1(Y)| = \det \Delta_1 = n\,|\operatorname{Jac}_0(Y)| \cdot |\operatorname{Jac}_2(Y)| \cdot \operatorname{disc} H_2(Y;\mathbb{Z}).$$

Proof. Write $D = \partial_1^t \partial_1$ and $U = \partial_2 \partial_2^t$. Since $\partial_1 \partial_2 = 0$, $DU = UD = 0$. Over $\mathbb{R}$, $C_1 = \operatorname{im} \partial_1^t \oplus \operatorname{im} \partial_2 \oplus \ker \Delta_1$ orthogonally (Hodge). D preserves the first summand and kills the other two, and U preserves the second and kills the other two. Hence $\operatorname{pdet} \Delta_1 = \operatorname{pdet} D \cdot \operatorname{pdet} U$. Since AB and BA have the same nonzero eigenvalues, $\operatorname{pdet} D = \operatorname{pdet}(\partial_1 \partial_1^t) = \operatorname{pdet} \Delta_0 = n\,\kappa(Y^{(1)}) = n|\operatorname{Jac}_0|$ (matrix-tree theorem), and $\operatorname{pdet} U = \operatorname{pdet}(\partial_2^t \partial_2) = \operatorname{pdet} \Delta_2$. Apply Lemma 2.2.1 to Δ_1 and Δ_2, using $\ker_{\mathbb{Z}} \Delta_2 = \ker_{\mathbb{Z}} \partial_2$ (because $x^t \Delta_2 x = \|\partial_2 x\|^2$). If $b_1 = 0$, then $\ker \Delta_1 = 0$, the discriminant is 1, and $\operatorname{pdet} = \det > 0$. □

Remark 2.2.3 (An identity of orders, not of groups). Theorem 2.2.2 does not split Jac_1 as $\operatorname{Jac}_0 \oplus \operatorname{Jac}_2$. On Y_{21} the orders differ by the factor $n \cdot \operatorname{disc} H_2 = 21 \cdot 3 \cdot 7^{29}$, and the 2-parts of the three groups have different shapes (Section 2.5). The proposed programme's statement "$|\operatorname{Pic}_1| = \det' \Delta_1$" is true, but under $b_1 = 0$ it is the tautology $\det' = \det$. The content is the factorization: the edge group is built from the vertex group, the chamber group, and the geometry of the 2-cycle lattice. On the thin torus T_6 ($b_1 = 2$) the general form holds with $\operatorname{disc} \ker_{\mathbb{Z}} \Delta_1 = 3888$ and $\operatorname{disc} H_2 = 72$ (Table 2.2, line (H)).

2.3 The vertex sandpile: exact avalanche statistics

Fix a sink $s \in V$. The vertex sandpile has threshold $d = 2(q^2 + q + 1)$ at each $v \neq s$; grains reaching s are lost. Write $\widetilde{\Delta}_0$ for Δ_0 with the row and column of s removed, $G = \widetilde{\Delta}_0^{-1}$, and L^+ for the Moore–Penrose inverse of Δ_0.

Proposition 2.3.1 (Dhar [2])**.** *The recurrent configurations form a group isomorphic to $\mathbb{Z}^{n-1}/\operatorname{im} \widetilde{\Delta}_0 \cong \operatorname{Jac}_0(Y)$. Adding a grain at x acts on it bijectively, so the uniform measure on recurrent configurations is stationary for the dynamics "add a grain at a uniformly chosen $x \neq s$, stabilize".*

Theorem 2.3.2 (Mean avalanche size)**.** *In the stationary state, the expected number of topplings at y caused by adding a grain at x is $G(x,y)$. The mean avalanche size for a uniform addition site is*

$$\langle s \rangle = \frac{\mathbf{1}^t G \mathbf{1}}{n-1} = \frac{n^2}{n-1} L^+_{ss}, \qquad \frac{n}{\lambda_{\max}(\Delta_0)} \;\leq\; \langle s \rangle \;\leq\; \frac{n}{\lambda_2(\Delta_0)}.$$

If $Y^{(1)}$ is vertex-transitive, then $\langle s \rangle = \frac{n}{n-1} \sum_{k \geq 2} \lambda_k(\Delta_0)^{-1}$. Moreover $\sum_y G(x,y) = \mathbb{E}_x[\tau_s]/d$, where τ_s is the hitting time of s by simple random walk.

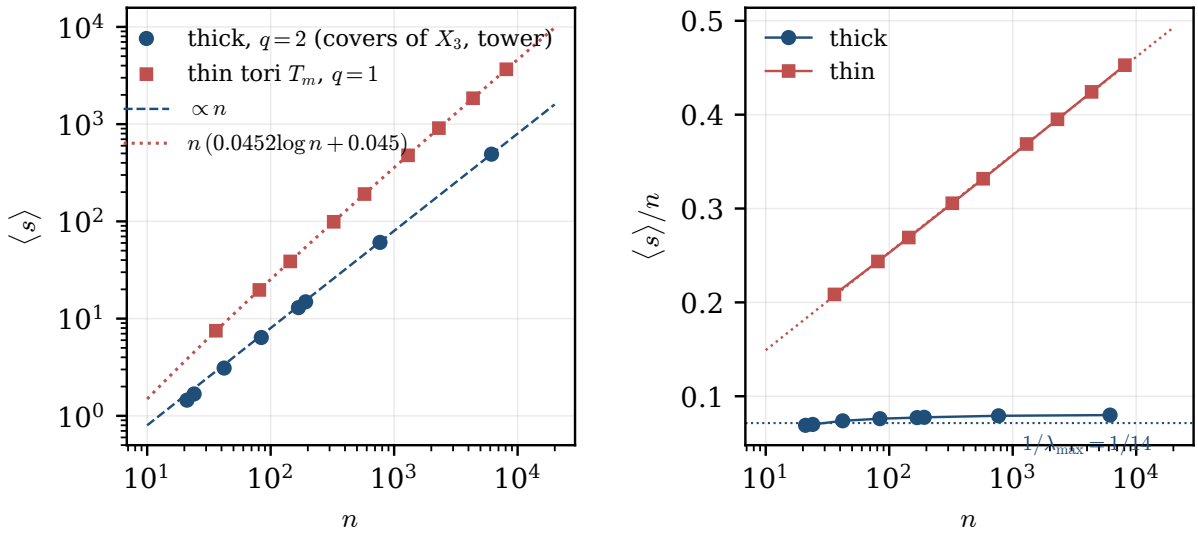


Figure 2.1: Exact mean avalanche size $\langle s\rangle = \mathbf{1}^t G\mathbf{1}/(n-1)$ of the vertex sandpile (sink = vertex 0, Theorem 2.3.2), computed by solving the reduced Laplacian system. Thick, $q=2$: the abelian covers $Y_{21}, Y_{24}, Y_{42}, Y_{84}, Y_{168}$ of X_3 and the levels $n = 192, 768, 6144$ of the canonical 2-tower of Section 6.4. Thin, $q=1$: the tori T_m, $m = 6, \dots, 90$ ($n = m^2$), whose 1-skeleton is the triangular lattice. Left: $\langle s\rangle$ against n on logarithmic axes, with the line $\propto n$ through the thick family and the fit $n(a\log n + c)$ through the thin one. Right: $\langle s\rangle/n$, which stays between 0.069 and 0.080 on the thick family, as Corollary 2.3.3 predicts ($\geq 1/\lambda_{\max} = 1/14$), and grows like $a\log n$ on the thin family, with fitted $a = 0.0452$; the triangular-lattice value is $\sqrt{3}/(12\pi) = 0.0459$, since there $\lambda(k) \approx \frac{3}{2}|k|^2$ and the Brillouin zone has area $8\pi^2/\sqrt{3}$ (the constant $\frac{1}{4\pi}$ of Corollary 2.3.3 is that of the square torus). Drawn by `code/make_figures.py`; $\langle s\rangle(Y_{21}) = 29/20$ is checked there.

Proof. Let z be stationary and $n(x) \in \mathbb{Z}_{\geq 0}^{n-1}$ the toppling vector of the avalanche. The stabilized configuration is $z' = z + e_x - \widetilde{\Delta}_0 n(x)$. Stationarity gives $\mathbb{E}z' = \mathbb{E}z$, so $\widetilde{\Delta}_0\,\mathbb{E}\,n(x) = e_x$, i.e. $\mathbb{E}\,n(x) = Ge_x$, and G is symmetric. For the closed form, put $H(x,y) = L^+_{xy} - L^+_{xs} - L^+_{sy} + L^+_{ss}$. Using $\Delta_0 L^+ = I - \frac{1}{n}J$, $\sum_z (\Delta_0)_{xz} H(z,y) = \delta_{xy} - \delta_{xs}$. Since $H(s,\cdot) = 0$, the restriction of H to $V \setminus s$ inverts $\widetilde{\Delta}_0$, so $G = H|_{V\setminus s}$. Summing, and using $L^+\mathbf{1} = 0$, gives $\mathbf{1}^t G\mathbf{1} = n^2 L^+_{ss}$. With an orthonormal eigenbasis φ_k of Δ_0, $L^+_{ss} = \sum_{k\geq 2} \varphi_k(s)^2/\lambda_k$ and $\sum_{k\geq 2}\varphi_k(s)^2 = 1 - \frac{1}{n}$. This gives the bounds and, by averaging over s, the transitive case. Finally $\widetilde{\Delta}_0 = d(I - \widetilde{P})$, where $\widetilde{P}$ is the walk killed at s, and $\sum_y (I-\widetilde{P})^{-1}_{xy} = \mathbb{E}_x \tau_s$. □

Corollary 2.3.3 (Linear mean avalanches on expanding families)**.** *Suppose* (Y_j) *satisfy* (L) *for a fixed* q *and* $\lambda_2(\Delta_0(Y_j)) \geq c > 0$ *(for instance Ramanujan complexes). Then* $n_j/(4(q^2+q+1)) \leq \langle s\rangle_j \leq n_j/c$*, so* $\langle s\rangle = \Theta(n)$*. On the two-dimensional torus* $(\mathbb{Z}/L)^2$ *with* $n = L^2$*, by contrast,* $\sum_k \lambda_k^{-1} \sim \frac{n\log n}{4\pi}$*, and* $\langle s\rangle = \Theta(n\log n)$*.*

Proof. $\lambda_{\max} \leq 2d$. For the torus, $\lambda_k = 4 - 2\cos\theta_1 - 2\cos\theta_2 \approx |\theta|^2$ at the n Fourier points $\theta \in (2\pi\mathbb{Z}/L)^2$, and $\sum_{\theta\neq 0} |\theta|^{-2} \sim \frac{L^2}{(2\pi)^2}\int_{1/L<|\theta|<1} \frac{d^2\theta}{|\theta|^2} = \frac{L^2}{2\pi}\log L$. □

Figure 2.1 shows the dichotomy on the complexes of this volume.

Proposition 2.3.4 (Lyons [6]). *If finite d-regular graphs G_j converge in the Benjamini–Schramm sense to a transitive graph G, then $\frac{1}{|V(G_j)|}\log|\operatorname{Jac}_0(G_j)| \to h(G) = \log d - \sum_{k\geq 1}\frac{p_k(o,o)}{k}$, where $p_k(o,o)$ is the k-step return probability of simple random walk on G.*

For quotients $\Gamma_j\backslash B$ with injectivity radius tending to infinity, the limit graph is the 1-skeleton of the building. It is transitive, because the Cartwright–Mantero–Steger–Zappa group acts simply transitively on vertices (Vol. I–II, Construction 16.1.4). Its entropy $h(B^{(1)}) < \log(2(q^2+q+1))$ is the "asymptotic principal exponent" of the vertex sandpile groups. We do not compute it. The finite complexes of this volume form a bounded chain, not a Benjamini–Schramm sequence, and their values (Table 2.1) are finite-size data only.

2.4 Scaling: what can and cannot be claimed

Proposition 2.4.1 (Critical branching [5]). *Let (Z_t) be a Galton–Watson process with offspring mean 1 and variance $\sigma^2 \in (0,\infty)$. Then $\mathbb{P}(Z_t > 0) \sim 2/(\sigma^2 t)$. If the offspring law is aperiodic, the total progeny S satisfies $\mathbb{P}(S = s) \sim (2\pi\sigma^2)^{-1/2}s^{-3/2}$. Conditionally on survival to time t, Z_t/t converges to an exponential law (Yaglom). Hence the size exponent is $\tau = 3/2$, the duration exponent is $\tau_T = 2$, and $s \sim T^2$.*

Conjecture 2.4.2 (Mean-field class). *Let (Y_j) be finite quotients of a thick $\widetilde{A}_2$ building of order q with injectivity radius $\to\infty$ and $\lambda_2(\Delta_0)$ bounded below. Then, in the scaling window, the vertex-sandpile avalanche size and duration have the mean-field exponents $\tau = 3/2$ and $\tau_T = 2$, with $s \sim T^2$.*

Heuristic. The local limit is a transitive graph of exponential growth on which simple random walk has spectral radius < 1. A toppling at depth r from the seed reaches, with high probability, sites whose earlier history it has not influenced, so the avalanche is locally a branching process. Criticality of the stationary state (mean offspring 1) then gives Proposition 2.4.1. The same reasoning is known to be exact on the Bethe lattice [3]. The 1-skeleton, however, contains many triangles (the chambers), so the tree approximation is uncontrolled. We do not claim a proof.

Remark 2.4.3 (Corrections concerning exponents). (1) *The dynamic exponent z.* Here z is defined by $T \sim L^z$ with L a linear size. On expanders the diameter is $\Theta(\log n)$, so no power law in L is meaningful. The well-defined replacement is the size–duration exponent γ in $s \sim T^\gamma$, conjecturally $\gamma = 2$. (2) *The comparison value* 1.22. The one-dimensional BTW model has no power-law avalanches at all. The value $\tau \approx 1.22$ is an early numerical estimate for the *two*-dimensional BTW model; later estimates range over roughly 1.2–1.3, and no value is proved. (3) *Topology.* The divergence from Euclidean lattices is not topological. The vertex sandpile does not see $b_1(Y) = 0$: its graph has first Betti number $|E| - n + 1 = 6n + 1$ for $q = 2$. What it sees is expansion, and its rigorous signature is Corollary 2.3.3: $\langle s\rangle = \Theta(n)$ against $\Theta(n\log n)$. (4) *Chambers.* There is no avalanche of chambers to renormalize (Corollary 2.1.6).

2.5 The example Y_{21}

Proposition 2.5.1 (Y_{21} is dense). *The 1-skeleton of Y_{21} is the complete tripartite graph $K_{7,7,7}$. Consequently $\operatorname{Tr} A_1^3 = 3\cdot 7^3 = 1029$, and the $196 = 343 - 147$ non-chamber 3-cycles of Chapter 1 (Section 1.7) are all the type-increasing triangles of $K_{7,7,7}$ that are not chambers.*

Proof. Each vertex has $q^2+q+1=7$ neighbours of each other type, and there are exactly 7 vertices of each type. So every pair of vertices of different types is adjacent, and every choice of one vertex of each type gives a type-increasing 3-cycle. □

Thresholds. The threshold of site i is M_{ii}, and the largest stable height is $M_{ii}-1$:

	vertex Δ_0	edge Δ_1	chamber Δ_2	Hecke Δ^{H}
threshold, general q	$2(q^2+q+1)$	$q+3$	3	q^3-1
threshold, $q=2$	14	5	3	7
neighbours in support	14	20	$3q=6$	14
abelian dynamics	yes	no	no	no

Groups. By Smith forms, recomputed and in agreement with Vol. I–II, Ch. 16:

$$\begin{aligned}
&\mathrm{Jac}_0(Y_{21}) = \mathbb{Z}/2 \oplus (\mathbb{Z}/14)^{15} \oplus \mathbb{Z}/98 \oplus \mathbb{Z}/294, \quad |\,\mathrm{Jac}_0\,| = 21\cdot 14^{18} = 3\cdot 2^{18}\cdot 7^{19},\\
&\mathrm{Jac}_1(Y_{21}) = (\mathbb{Z}/2)^{10} \oplus (\mathbb{Z}/14)^{10} \oplus (\mathbb{Z}/98)^{26} \oplus \mathbb{Z}/2058 \oplus \mathbb{Z}/18522,\\
&\mathrm{Jac}_2(Y_{21}) = (\mathbb{Z}/2)^{13} \oplus (\mathbb{Z}/14)^{15} \oplus \mathbb{Z}/98 \oplus \mathbb{Z}/294.
\end{aligned}$$

Here $|\,\mathrm{Jac}_0\,|$ is the spanning-tree count $N^{k-2}\prod_i (N-n_i)^{n_i-1} = 21\cdot 14^{18}$ of $K_{7,7,7}$. Moreover $H_2(Y_{21};\mathbb{Z}) \cong \mathbb{Z}^{20}$ has discriminant $3\cdot 7^{29}$, and Theorem 2.2.2 reads $|\,\mathrm{Jac}_1\,| = 21\cdot(3\cdot 2^{18}\cdot 7^{19})\cdot |\,\mathrm{Jac}_2\,|\cdot 3\cdot 7^{29} = 2^{48}\cdot 3^4\cdot 7^{68}$. The group exponents are $294 = 2\cdot 3\cdot 7^2$ for Jac_0 and $18522 = 2\cdot 3^3\cdot 7^3$ for Jac_1.

Avalanches. The spectrum of $\Delta_0(K_{7,7,7})$ is 0, $14^{(18)}$, $21^{(2)}$, and $K_{7,7,7}$ is vertex-transitive. By Theorem 2.3.2,

$$\langle s\rangle = \frac{21}{20}\Big(\frac{18}{14}+\frac{2}{21}\Big) = \frac{29}{20} = 1.45,$$

confirmed exactly by solving $\widetilde{\Delta}_0 x = \mathbf{1}$, and by Monte Carlo ($2\cdot 10^5$ additions: 1.45006). This number says nothing about criticality: Y_{21} is too dense (degree 14 on 21 vertices) to host a scaling regime.

Y	n	$\lambda_2(\Delta_0)$	$\lambda_{\max}$	$\langle s\rangle$	$[n/\lambda_{\max},\, n/\lambda_2]$	$\frac{1}{n}\log|\,\mathrm{Jac}_0\,|$
Y_{21}	21	14	21	$29/20 = 1.4500$	[1.00, 1.50]	2.407026
Y_{24}	24	13	21	$7031/4186 = 1.6796$	[1.14, 1.85]	2.426183
Y_{42}	42	8.417424	21	$489107/157768 = 3.1002$	[2.00, 4.99]	2.485136
Y_{84}	84	8.417424	21	6.3933	[4.00, 9.98]	2.532442
Y_{168}	168	8.417424	21	12.9760	[8.00, 19.96]	2.560221

Table 2.1: Vertex sandpiles on the covers of Chapter 1 (sink = vertex 0). The values of $\langle s\rangle$ are exact rationals for $n \le 42$ and floating point otherwise. The entropies for Y_{84}, Y_{168} agree with the Jac_0 data of Vol. I–II, Ch. 16 to six digits. $\lambda_{\max} = 21$ throughout: it comes from the ω-twisted characters ω^τ, with A_1+A_2 eigenvalue $2\,\mathrm{Re}(7\omega) = -7$. For comparison, $\log 14 = 2.639057$.

check	complexes	result
(S) switching to a Z-matrix: Δ_0 yes; Δ_1, U_1 no; Δ_2 no (thick), yes (T_6); $\Delta^{\mathrm{H}}_{xy}\Delta^{\mathrm{H}}_{yx} = -q$	$Y_{21}, Y_{24}, Y_{42}, T_6$	exact
(S') Δ_1: diagonal $q+3$, entries $\{0, \pm 1\}$, 20 neighbours, 0 inside chambers	Y_{21}	exact
(N) two-site non-abelian example; 300 random stabilizations: 0 non-terminating, 113 order-dependent, 294 negative	Y_{21}	exact / numeric (seed 1)
(K) 1-skeleton $= K_{7,7,7}$; $\operatorname{Tr} A_1^3 = 1029$	Y_{21}	exact
(H) $\lvert \operatorname{Jac}_1 \rvert \operatorname{disc} \ker \Delta_1 = n \lvert \operatorname{Jac}_0 \rvert \lvert \operatorname{Jac}_2 \rvert \operatorname{disc} H_2$; $\det \Delta_1 = \lvert \operatorname{Jac}_1 \rvert$ when $b_1 = 0$	Y_{21}, Y_{24}, T_6	exact
(D) $\langle s \rangle$ as exact rationals; Monte Carlo on Y_{21} within 0.02	Y_{21}, Y_{24}, Y_{42}	exact / numeric
(E) $\frac{1}{n} \log \lvert \operatorname{Jac}_0 \rvert$ from spectra; agreement with Vol. I–II, Ch. 16 for Y_{168}	five covers	numeric 10^{-6}

Table 2.2: Checks run by `code/v3p2_verify.py`; the complexes are built by `code/a2_common.py`.

2.6 Verification battery

2.7 Scope, residue, corrections

Remark 2.7.1 (Solved, open). *Solved:* the sign obstruction for every higher toppling operator on thick complexes, and its absence for chambers on thin ones; the Hodge factorization of the orders of the critical groups; exact avalanche means, their spectral bounds, and the $\Theta(n)$ versus $\Theta(n \log n)$ dichotomy; the structure of Y_{21}. *Open:* (a) Conjecture 2.4.2; (b) the entropy $h(B^{(1)})$ of the building's 1-skeleton, computable in principle from return probabilities through the spherical Hecke algebra of (A_1, A_2); (c) whether some non-sign-coherent generalization of chip-firing (a choice of firing cone for a general nonsingular matrix) gives a meaningful dynamical avatar of Jac_1; (d) a structural (not merely numerical) relation between Jac_1, $\operatorname{Jac}_0 \oplus \operatorname{Jac}_2$ and $H_2(Y; \mathbb{Z})$ refining Theorem 2.2.2; (e) the analogue of Theorem 2.1.5 for $\widetilde{A}_d$, where the same 3-cycles exist.

Remark 2.7.2 (Corrections to the proposed programme). (1) The toppling rules built from Δ_1 and Δ_2 are not sandpiles: under every orientation they are not abelian (Theorem 2.1.5, Corollary 2.1.6). The "recurrent configuration group Pic_k" does not exist for $k = 1, 2$, and only the algebraic $\operatorname{coker} \Delta_k$ remains. (2) $\operatorname{Pic}_k \cong \mathbb{Z}^{|S_k|-1} / \operatorname{im} \Delta'_k$ is the vertex statement. For $k = 1$ and $b_1 = 0$, $\operatorname{coker} \Delta_1$ is finite of order $\det \Delta_1$ with no reduction, and "$\lvert \operatorname{Pic}_1 \rvert = \det' \Delta_1$" is a tautology. The substantive statement is Theorem 2.2.2. (3) Avalanche exponents are not provable on finite complexes. $\tau = 3/2$, $\tau_T = 2$, $s \sim T^2$ are proved for the branching approximation and conjectured for the sandpile; z in the usual sense is undefined on expanders. (4) The Euclidean value 1.22 is a numerical estimate for the 2D (not 1D) BTW model. (5) The divergence from Euclidean lattices is spectral (expansion), not topological: it is Corollary 2.3.3, and $b_1(Y) = 0$ plays no role in it. (6) "Local nontrivial toppling thresholds" for Y_{21} are $14, 5, 3, 7$. The "asymptotic principal exponent" is read as the growth rate of $\lvert \operatorname{Jac}_0 \rvert$, which is 2.4070 per vertex on Y_{21}. Its limit along Benjamini–Schramm sequences is the entropy of Proposition 2.3.4, which is not computed here.

Acknowledgement of provenance and caveats

Unrefereed. The classical inputs are Dhar's theory of recurrent states [2], the limit theorems for critical Galton–Watson processes [5], Lyons' entropy theorem [6], and the Bethe-lattice computation [3], which is cited only for the heuristic. Everything else is proved here or in Chapter 1. The complexes are rebuilt by `code/a2_common.py` independently of the scripts of Vol. I–II; where the data overlap, they agree. The bibliographic data of [2, 3, 5] have been checked against Crossref and zbMATH.

Chapter 3

Phase Oscillators on Buildings: Chiral Connections, Frustrated Ground States, and Transverse Bifurcations

The proposed programme asks for a gauge-covariant Kuramoto model on an $\widetilde{A}_2$ complex, with pairwise coupling K_1 through a connection A of chamber flux $2\pi\alpha_0$ and a triadic chamber coupling K_2 with lag Φ. It then asks for proofs of four claims. (a) The flux breaks time reversal and forbids static fixed points, forcing a permanent chiral limit cycle. (b) An Ott–Antonsen reduction closes on the three type classes, and a supercritical Hopf bifurcation produces a chimera with one class coherent and two incoherent. (c) The Berry phase of the synchronized cycle is topological. (d) The frequency shift is fixed by the Euler characteristic and the number of chambers. We prove what is true and correct what is not.

(1) *Chiral connections.* The uniform connection $A^\alpha = \frac{2\pi\alpha}{3}\mathbf{1}_E$ has flux $2\pi\alpha$ on every chamber of every $\widetilde{A}_2$ complex. Chiral connections of a given flux form a torsor under $\mathrm{Hom}(H_1(Y;\mathbb{Z}), U(1))$, which is finite on thick complexes by Garland rigidity (64 classes on Y_{21}). The triadic lag Φ is *not* a gauge field: the only gauge-covariant triadic term is blind to the flux.

(2) *The flux sector is a gradient flow.* For $K_2 = 0$ and identical frequencies, the model is the gradient flow of a frustrated XY energy for every flux. Every trajectory converges to an equilibrium, all Jacobians are symmetric, and there are no limit cycles and no Hopf bifurcations. Claim (a) is therefore false. The ground states are the three *twisted states* $\theta_j = \frac{2\pi m}{3}\tau(j)$; we determine exactly which of them minimize the energy. The flux produces static persistent currents, not rotation.

(3) *The triadic lag.* For all parameters the twisted states are relative equilibria rotating at $\Omega = \omega + K_2 \sin\Phi$. Their Jacobian is $-c_m\Delta_0$, so its spectrum is real, stability changes only through a real, $(n-1)$-fold degenerate crossing, and never through a Hopf bifurcation. On the type-cluster subspace the whole system is a gradient flow modulo rotation. At every cluster equilibrium the Jacobian splits into 3×3 blocks $M(\lambda)$ indexed by the $\mathbb{Z}/3$-orbits of eigenvalues of A_1. On Y_{21} every block is real. On Y_{168} a stable cluster state can lose stability through a complex pair of a transverse block; this happens at about a fifth of the losses of stability found, always with multiplicity 21, and in every case examined the network then freezes into a locked state instead of a limit cycle.

(4) *Ott–Antonsen and chimeras.* The OA manifold is invariant only for $K_2 = 0$: the triadic term is second harmonic in θ_j, and we prove non-closure. For $K_2 = 0$ we derive the closed equations for (r_t, ψ_t) and a Lyapunov function that excludes chimeras and Hopf bifurcations for identical

oscillators. For $K_2 \neq 0$ the chimera of (b) remains unproved. Numerically it is never the primary instability of a stable cluster state, and it did not appear in our runs.

(5) *Phases and frequencies.* A rigid rotation has zero geometric phase. The genuine geometric datum is the monodromy of the twisted states under threading one flux quantum per chamber. It is the cyclic permutation $\Theta_m \mapsto \Theta_{m+1}$, implemented by the grading character of Chapter 1. The frequency shift is $K_2\langle m_\Delta\rangle_F$, a local quantity. The identity $D_2 = 3|F|/\chi(Y)$ holds only for $q = 2$; on the thin torus $\chi = 0$ while the shift is not. Trees fail for two reasons: every connection on a tree is pure gauge, and without 2-cells the dynamics is a gradient flow.

Status of the results

Every numbered statement is proved. The exceptions are the numerical observations in Remark 3.4.4, Theorem 3.8.2 and Remark 3.8.3, which are numerical, and two classical inputs: the Łojasiewicz convergence theorem for analytic gradient flows [16] and LaSalle's invariance principle [15]. The Ott–Antonsen equations are derived here in the form used, together with their failure for $K_2 \neq 0$. All finite identities, and the numerical statements, were checked by `code/v3p3_verify.py` (Table 3.1).

Notation follows Chapters 1–2. Y is a finite $\widetilde{A}_2$ complex satisfying (L) for some $q \geq 1$. For $q = 1$ this means the thin tori, whose links are hexagons. Put $\kappa = q^2 + q + 1$. Then every vertex has κ out-neighbours (type $+1$) and κ in-neighbours (type -1), every edge lies in $q + 1$ chambers, every vertex lies in $D_2 = \kappa(q + 1)$ chambers, and $|E| = \kappa n$, $3|F| = nD_2$. Edges are oriented $j \to k$ with $\tau(k) = \tau(j) + 1$, chambers are oriented $a \to b \to c \to a$, and P_t is the projection onto the class $V_t = \tau^{-1}(t)$, of size $n/3$. As in Vol. I–II, Definition 16.1.1, A_1 is normal: $A_1A_1^t = A_1^tA_1$.

3.1 The model and its gauge structure

Definition 3.1.1 (Connections and flux)**.** A *connection* is a function A on ordered adjacent pairs with $A_{kj} = -A_{jk}$, taken modulo 2π. Equivalently, it is a 1-cochain $A \in C^1(Y; \mathbb{R}/2\pi\mathbb{Z})$ on the oriented edges. Its *flux* is the 2-cochain δA, with $(\delta A)(a, b, c) = A_{ab} + A_{bc} + A_{ca}$. A is *chiral of flux* $2\pi\alpha$ if $\delta A = 2\pi\alpha$ on every (type-increasingly oriented) chamber. A *gauge transformation* $\chi \in C^0(Y; \mathbb{R})$ acts by $\theta_j \mapsto \theta_j + \chi_j$, $A_{jk} \mapsto A_{jk} - (\chi_k - \chi_j)$. It preserves the link variables $\varphi_{jk} = \theta_k - \theta_j + A_{jk}$ and the flux.

Proposition 3.1.2 (Chiral connections)**.** *(i) For every $\alpha \in \mathbb{R}$ the uniform connection A^α, equal to $a = \frac{2\pi\alpha}{3}$ on every oriented edge, is chiral of flux $2\pi\alpha$.*

(ii) $A^{\alpha+1}$ is gauge equivalent to A^α, via the type gauge $\chi = \frac{2\pi}{3}\tau$. In particular A^m is pure gauge for $m \in \mathbb{Z}$.

(iii) The gauge classes of chiral connections of flux $2\pi\alpha$ form a torsor under $\mathrm{Hom}(H_1(Y; \mathbb{Z}), U(1))$. If $q \geq 2$, this group is finite of order $|H_1(Y; \mathbb{Z})|$ (Theorem 1.4.3). On Y_{21}, where $H_1 = (\mathbb{Z}/2)^6$, there are 64 classes.

Proof. (i) Each chamber has three oriented edges. (ii) $\chi_k - \chi_j = \frac{2\pi}{3}$ on every edge $j \to k$. (iii) If A, A' are both chiral of flux $2\pi\alpha$, then $\delta(A - A') = 0$, so $A - A'$ is a flat connection. Modulo gauge, flat connections form $H^1(Y; \mathbb{R}/2\pi\mathbb{Z})$, which equals $\mathrm{Hom}(H_1(Y; \mathbb{Z}), \mathbb{R}/2\pi\mathbb{Z})$ by the universal coefficient theorem, because $\mathbb{R}/2\pi\mathbb{Z}$ is divisible. When $b_1 = 0$ this is the Pontryagin dual of the finite group H_1. □

So the constant flux imposes no quantization. The 2-cochain $\mathbf{1}_F = \frac{1}{3}\delta\mathbf{1}_E$ is exact, and every integral 2-cycle has zero chamber sum.

Definition 3.1.3 (The model)**.** For $\omega \in \mathbb{R}$, $K_1, K_2 \in \mathbb{R}$, $\alpha, \Phi \in \mathbb{R}$ and $a = \frac{2\pi\alpha}{3}$:

$$\dot{\theta}_j = \omega_j + \frac{K_1}{2\kappa} \sum_{k\sim j} \sin(\theta_k - \theta_j + A^{\alpha}_{jk}) + \frac{K_2}{D_2} \sum_{\{j,k,l\}\in F} \sin(\theta_k + \theta_l - 2\theta_j + \Phi). \tag{3.1.1}$$

Unless stated otherwise, $\omega_j = \omega$ for all j. Write φ_e for the link variable on a type-increasing edge e, and for a chamber Δ with corner j write $x^{\Delta}_j = \theta_k + \theta_l - 2\theta_j$ and $m_\Delta = \frac{1}{3}\sum_{j\in\Delta} \sin(x^{\Delta}_j + \Phi)$.

Remark 3.1.4 (The triadic lag is not a gauge field). The pairwise term of (3.1.1) is gauge covariant, but the triadic term with a corner-independent Φ is not: under χ its argument shifts by $\chi_k + \chi_l - 2\chi_j$. The covariant triadic term built from the connection is $\sin(\varphi_{jk} + \varphi_{jl})$. Its three corner arguments sum to $\sum(\varphi_{jk} + \varphi_{kj}) = 0$, whatever the flux, and in the gauge A^α it equals the $\Phi = 0$ term of (3.1.1), because $A_{jk} + A_{jl} = a - a = 0$. Thus the flux α and the lag Φ are independent parameters. Φ is a Sakaguchi-type lag [18] on the triadic coupling of Skardal–Arenas type [20], not a curvature.

3.2 Symmetries and the gradient sector

The model is invariant under the global rotation $\theta \mapsto \theta + c\mathbf{1}$. The conjugation $C\colon \theta \mapsto -\theta$ maps the parameters (ω, α, Φ) to $(-\omega, -\alpha, -\Phi)$. Since α matters only modulo 1 (Proposition 3.1.2(ii)), the flux is C-invariant exactly when $\alpha \in \frac{1}{2}\mathbb{Z}$. This is the precise sense in which the flux is "T-odd".

Proposition 3.2.1 (Frequency sum rule)**.** *For every state θ and arbitrary ω_j,*

$$\frac{1}{n}\sum_j \dot{\theta}_j = \bar{\omega} + K_2 \langle m_\Delta \rangle_F, \qquad \langle m_\Delta \rangle_F = \frac{1}{|F|}\sum_{\Delta\in F} m_\Delta.$$

Hence every frequency-locked state has $|\Omega - \bar{\omega}| \le |K_2|$, and for $K_2 = 0$ its frequency is $\bar{\omega}$ whatever the flux.

Proof. Each edge contributes $\sin\varphi_e - \sin\varphi_e = 0$ to the pairwise part, because all degrees are equal to 2κ. Each chamber contributes $3m_\Delta$ to the triadic part, and $nD_2 = 3|F|$. □

Theorem 3.2.2 (The flux sector is a gradient flow)**.** *Let G be any finite graph and A any connection on it. Suppose $K_2 = 0$ and $\omega_j = \omega$. In the frame rotating at ω, the model $\dot{\theta}_j = \frac{K_1}{\deg j}\sum_{k\sim j}\sin\varphi_{jk}$ is the gradient flow of*

$$V_A(\theta) = -K_1 \sum_e \cos\varphi_e$$

for the metric $\sum_j \deg(j)\, d\theta_j^2$. Consequently:

(i) every trajectory converges to a single equilibrium;

(ii) at every equilibrium the Jacobian is self-adjoint for this metric, so its spectrum is real;

(iii) there is no periodic orbit other than equilibria, and no Hopf bifurcation, for any flux.

In particular, in the model (3.1.1) *with $K_2 = 0$ a nonzero flux does not prevent static fixed points and does not force rotation.*

Proof. $\partial V_A/\partial\theta_j = -K_1 \sum_{k\sim j} \sin\varphi_{jk}$, because $\varphi_{kj} = -\varphi_{jk}$. So $\dot\theta_j = -\deg(j)^{-1}\partial_j V_A$. V_A is real analytic on the compact torus, so by Łojasiewicz [16] every trajectory has finite length and converges. The Jacobian is $-g^{-1}\operatorname{Hess} V_A$, which is self-adjoint for g. Along a non-constant orbit V_A strictly decreases, which excludes periodic orbits. □

Lemma 3.2.3 (Triangle lemma)**.** *Let $S \in [-\pi, \pi]$. On $\{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 \equiv S \bmod 2\pi\}$, $\max \sum_i \cos x_i = 3\cos(S/3)$. For $|S| < \pi$ the maximum is attained only at $x_i \equiv S/3$.*

Proof. At a critical point the Lagrange condition gives $\sin x_1 = \sin x_2 = \sin x_3$, so each $x_i \in \{u, \pi - u\}$ modulo 2π. If all three equal u (or all equal $\pi - u$), the value is $3\cos\frac{S+2\pi k}{3}$ for some k, and it is largest for $k = 0$ because $|S| \le \pi$. If the two values are mixed, then either $(u, u, \pi - u)$ with $u \equiv S - \pi$, of value $\cos u$, or $(u, \pi - u, \pi - u)$ with $u \equiv -S$, of value $-\cos u$; both equal $-\cos S$. Put $c = S/3 \in [-\frac{\pi}{3}, \frac{\pi}{3}]$. Then $3\cos c - (-\cos 3c) = 4\cos^3 c > 0$. □

Theorem 3.2.4 (Frustrated ground states)**.** *Let $K_1 > 0$ and $K_2 = 0$. Write $\alpha = \operatorname{round}(\alpha) + \beta$ with $\beta \in [-\frac{1}{2}, \frac{1}{2}]$, and define the* twisted states *$\Theta_m = \frac{2\pi m}{3}\tau$, $m \in \mathbb{Z}/3$. Then*

$$\min V_{A^\alpha} = -K_1|E|\cos\tfrac{2\pi\beta}{3}.$$

The minimizers are the rotations of Θ_{m_*}, $m_* \equiv -\operatorname{round}(\alpha)$. At a minimizer every type-increasing edge carries the same link variable $\frac{2\pi\beta}{3}$, and hence the static current $K_1 \sin\frac{2\pi\beta}{3}$. For $|\beta| < \frac{1}{2}$ the minimizer is unique up to rotation. For $\alpha \in \frac{1}{2} + \mathbb{Z}$ there are exactly two minimizers, with link variables $\pm\frac{\pi}{3}$, and C exchanges them.

Proof. Around each chamber $\varphi_{ab} + \varphi_{bc} + \varphi_{ca} = 3a = 2\pi\alpha \equiv 2\pi\beta$, because the θ-differences cancel. Each edge lies in $q + 1$ chambers, so by Lemma 3.2.3

$$\sum_e \cos\varphi_e = \frac{1}{q+1}\sum_\Delta \sum_{e\subset\Delta} \cos\varphi_e \le \frac{3|F|}{q+1}\cos\tfrac{2\pi\beta}{3} = |E|\cos\tfrac{2\pi\beta}{3}.$$

On Θ_m every link variable is $\frac{2\pi(m+\alpha)}{3}$, which gives equality for $m = m_*$. If $|\beta| < \frac{1}{2}$, equality forces $\varphi_e \equiv \frac{2\pi\beta}{3}$ on every edge. Then $\theta_k - \theta_j \equiv \frac{2\pi m_*}{3}$ along every edge $j \to k$, so θ is a rotation of Θ_{m_*} (Y is connected). At $|\beta| = \frac{1}{2}$, equality forces each chamber to carry $\frac{\pi}{3}$ on all three edges or $-\frac{\pi}{3}$ on all three. Two chambers sharing an edge agree. Y is gallery connected, because it is connected and its links are connected. So the value is the same everywhere, and θ is a rotation of one of the two twisted states with link variables $\pm\frac{\pi}{3}$. C reverses link variables, so it exchanges them. □

3.3 Twisted states and the triadic lag

Theorem 3.3.1 (Twisted states for all parameters)**.** *For all K_1, K_2, α, Φ and $m \in \mathbb{Z}/3$, the state $\Theta_m + \Omega t$ with $\Omega = \omega + K_2 \sin\Phi$ is a solution of* (3.1.1)*. The Jacobian at Θ_m is*

$$J_m = -c_m\,\Delta_0, \qquad c_m = \frac{K_1\cos\frac{2\pi(m+\alpha)}{3} + 2K_2\cos\Phi}{2\kappa}.$$

Its eigenvalues are $-c_m\lambda_i(\Delta_0)$. Θ_m is linearly stable modulo rotation if and only if $c_m > 0$. When c_m changes sign, all $n - 1$ transverse eigenvalues cross zero together, along the real axis. In particular no Hopf bifurcation ever occurs at a twisted state.

Proof. On Θ_m every link variable is $\pm\frac{2\pi(m+\alpha)}{3}$, and the pairwise contributions from out- and in-neighbours cancel. Every triadic argument is $\frac{2\pi m}{3}(\tau_k+\tau_l-2\tau_j)+\Phi = 2\pi m+\Phi$. Differentiating the pairwise term gives $-\frac{K_1}{2\kappa}\cos\frac{2\pi(m+\alpha)}{3}\Delta_0$. For the triadic term, ∂_{θ_k} of the j-th equation is $\frac{K_2}{D_2}\cos\Phi$ times the number $q+1$ of chambers containing j and k, and ∂_{θ_j} is $-2K_2\cos\Phi$. Since $D_2=\kappa(q+1)$, the triadic part is $\frac{K_2\cos\Phi}{\kappa}(A_1+A_2-2\kappa I) = -\frac{K_2\cos\Phi}{\kappa}\Delta_0$. □

Corollary 3.3.2 (Multistability)**.** *Let $K_2=0$ and $K_1>0$. Θ_m is stable iff the distance from $m+\alpha$ to $3\mathbb{Z}$ is less than $\frac{3}{4}$. Two twisted states are stable exactly when the fractional part of α lies in $(\frac{1}{4},\frac{3}{4})$, and otherwise only the ground state is. Varying α therefore produces hysteresis.*

The type-cluster subspace. If $\omega_j=\omega$, the subspace $\{\theta_j=\psi_{\tau(j)}\}$ is invariant: a vertex of type t sees κ neighbours of types $t\pm1$, and each of its chambers has one vertex of each type. With $d_t=\psi_{t+1}-\psi_t$ (indices mod 3) the reduced equations are

$$\dot\psi_t = \omega + \tfrac{K_1}{2}\big[\sin(d_t+a)-\sin(d_{t-1}+a)\big] + K_2\sin(d_t-d_{t-1}+\Phi). \tag{3.3.1}$$

Theorem 3.3.3 (Gradient structure modulo rotation)**.** *Put $U(\psi) = -\frac{K_1}{2}\sum_t\cos(d_t+a) - \frac{K_2}{3}\sum_t\cos(d_t-d_{t-1}+\Phi)$. Then* (3.3.1) *reads $\dot\psi=\omega\mathbf{1}-\nabla U+\bar m(\psi)\mathbf{1}$ with $\bar m=\frac{K_2}{3}\sum_t\sin(d_t-d_{t-1}+\Phi)$. Modulo rotation it is therefore the gradient flow of U. Every cluster trajectory converges to a relative equilibrium, the reduced Jacobian is symmetric, and there are no periodic orbits or Hopf bifurcations inside the cluster subspace. For the full network,*

$$\dot\theta = \omega\mathbf{1} - \nabla\Big(\tfrac{1}{2\kappa}V_{A^\alpha} + \tfrac{K_2}{D_2}\sum_\Delta W_\Delta\Big) + \tfrac{K_2}{D_2}\sum_\Delta m_\Delta(\theta)\,\mathbf{1}_\Delta, \qquad W_\Delta = -\tfrac{1}{3}\sum_{j\in\Delta}\cos(x_j^\Delta+\Phi),$$

so the only non-gradient part is the chamber drive $\sum_\Delta m_\Delta\mathbf{1}_\Delta$*. It is parallel to $\mathbf{1}$ whenever all m_Δ are equal.*

Proof. For one chamber, $-\partial_{\theta_s}W_\Delta=\sin(x_s^\Delta+\Phi)-m_\Delta$, because $\partial x_j^\Delta/\partial\theta_s$ is -2 for $j=s$ and 1 otherwise. Summing over chambers gives the decomposition. On the cluster subspace all m_Δ equal $\bar m/K_2$, and $\sum_\Delta\mathbf{1}_\Delta=D_2\mathbf{1}$. The reduced flow on T^3/S^1 is then the gradient of U, and the Łojasiewicz argument of Theorem 3.2.2 applies. □

Theorem 3.3.4 (Block decomposition at cluster equilibria)**.** *Let ψ be an equilibrium of* (3.3.1) *and $\theta=\psi\circ\tau$. Put $p_t=\frac{K_1}{2\kappa}\cos(d_t+a)$ and $c_t=\frac{K_2}{\kappa}\cos(d_t-d_{t-1}+\Phi)$, and*

$$\alpha_t = p_t+c_t, \qquad \gamma_t = p_{t-1}+c_t, \qquad \sigma_t=\kappa(\alpha_t+\gamma_t).$$

Then $J=-\sum_t\sigma_tP_t+\sum_tP_t(\alpha_tA_1+\gamma_tA_2)$, and $\mathbb{C}^n=W_0\oplus\bigoplus_{[\lambda]}W_{[\lambda]}$ is a J-invariant decomposition. Here $W_0=\ker A_1$, and $W_{[\lambda]}=E_\lambda\oplus E_{\omega\lambda}\oplus E_{\omega^2\lambda}$ runs over the $\mathbb{Z}/3$-orbits of nonzero eigenvalues of A_1, with $\omega=e^{2\pi i/3}$. On W_0, J acts as $-\sigma_t$ on $W_0\cap\mathbb{C}^{V_t}$. On $W_{[\lambda]}$ it acts as $M(\lambda)\otimes I_{\dim E_\lambda}$, where

$$M(\lambda)=\begin{pmatrix}-\sigma_0 & \alpha_0\lambda & \gamma_0\bar\lambda\\ \gamma_1\bar\lambda & -\sigma_1 & \alpha_1\lambda\\ \alpha_2\lambda & \gamma_2\bar\lambda & -\sigma_2\end{pmatrix}.$$

The orbit of $\lambda=\kappa$ is the tangent space of the cluster subspace, and $M(\kappa)$ is the Jacobian of (3.3.1)*. The rates σ_t are the mean-field transverse rates of the classes.*

Proof. The entries of J follow as in Theorem 3.3.1. Row $j \in V_t$ has α_t at each out-neighbour, γ_t at each in-neighbour, and $-\sigma_t$ on the diagonal, because rows sum to zero. Let $D_1 = \mathrm{diag}(\omega^{-\tau})$. Then $D_1 A_1 D_1^{-1} = \omega A_1$ (Proposition 1.2.1), so D_1 permutes the eigenspaces within an orbit and $P_t = \frac{1}{3}\sum_k \omega^{kt} D_1^k$ preserves $W_{[\lambda]}$ and W_0. A_1 is normal, so $A_2 = A_1^*$ acts as $\bar\lambda$ on E_λ. For $v \in E_\lambda$ we have $A_1 P_s v = P_{s-1} A_1 v = \lambda P_{s-1} v$ and $A_2 P_s v = \bar\lambda P_{s+1} v$. If $P_s v = 0$ for one s, these relations force $v = 0$, since $\lambda \neq 0$. So $\{P_s v_i\}$, for an orthonormal basis (v_i) of E_λ, is a basis of $W_{[\lambda]}$, and $J(P_s v) = -\sigma_s P_s v + \alpha_{s-1}\lambda P_{s-1} v + \gamma_{s+1}\bar\lambda P_{s+1} v$, which is the stated matrix. On W_0 both A_1 and A_2 vanish. □

Corollary 3.3.5. *On Y_{21} the Jacobian at every type-cluster equilibrium has real spectrum. So no Hopf bifurcation occurs at any cluster state of Y_{21}.*

Proof. The 1-skeleton is $K_{7,7,7}$ (Proposition 2.5.1), so $A_1 = C_3 \otimes J_7$, with spectrum $7\{1, \omega, \omega^2\}$ and $0^{(18)}$. The only block is $M(7)$, whose spectrum is that of the reduced cluster Jacobian, and this is real by Theorem 3.3.3. The remaining eigenvalues are the real numbers $-\sigma_t$. □

On complexes where A_1 has nontrivial eigenvalue orbits, $M(\lambda)$ can have complex eigenvalues. On Y_{168} this happened at 6 of the equilibria sampled in line (B) of Table 3.1. A Hopf bifurcation *transverse* to the cluster subspace is therefore not excluded in general. Section 3.8 shows that it occurs on Y_{168}, with multiplicity 21 forced by the symmetry, and that the resulting loss of stability is hard.

3.4 Mean field and Ott–Antonsen reduction

Replace neighbour sums by class averages $Z_t = r_t e^{i\psi_t} = \frac{3}{n}\sum_{j\in V_t} e^{i\theta_j}$, and the pair sum over chambers by $Z_{t+1}Z_{t-1}$. For $j \in V_t$ this gives

$$\dot\theta_j = \omega_j + \mathrm{Im}\big(H_t e^{-i\theta_j}\big) + \mathrm{Im}\big(G_t e^{-2i\theta_j}\big), \quad H_t = \tfrac{K_1}{2}\big(e^{ia} Z_{t+1} + e^{-ia} Z_{t-1}\big), \quad G_t = K_2 e^{i\Phi} Z_{t+1} Z_{t-1}. \tag{3.4.1}$$

On $Y_{21} = K_{7,7,7}$ the pairwise part is *exact*, because every vertex is adjacent to all of the other two classes. The triadic part is not exact: a vertex lies in 21 of the 49 possible cross-class pairs.

For a continuum of oscillators in each class with frequency density g, the Ott–Antonsen ansatz takes the density of class t to be $f_t = \frac{g(\omega)}{2\pi}\big[1 + \sum_{n\geq 1}(z_t^n e^{-in\theta} + \bar z_t^n e^{in\theta})\big]$, with $|z_t| < 1$ and $z_t = z_t(\omega, \cdot)$ [17].

Proposition 3.4.1 (Closure and non-closure)**.** *For the velocity field of* (3.4.1)*, the continuity equation for the Fourier coefficient c_n of $e^{-in\theta}$, evaluated on the ansatz, reads*

$$\begin{aligned} \dot z &= i\omega z + \tfrac{1}{2}(H - \bar H z^2) + \tfrac{1}{2}(G\bar z - \bar G z^3) && (n = 1),\\ \dot z &= i\omega z + \tfrac{1}{2}(H - \bar H z^2) + \tfrac{1}{2}(G z^{-1} - \bar G z^3) && (n \geq 2). \end{aligned}$$

The ansatz is therefore invariant if and only if $G_t \equiv 0$, i.e. $K_2 = 0$. For $K_2 = 0$ and a Lorentzian g of centre ω_0 and width $\Delta \geq 0$, the class order parameters obey

$$\dot r_t = -\Delta r_t + \tfrac{K_1(1-r_t^2)}{4}\big[r_{t+1}\cos(\psi_{t+1} - \psi_t + a) + r_{t-1}\cos(\psi_{t-1} - \psi_t - a)\big],$$

$$\dot\psi_t = \omega_0 + \tfrac{K_1(1+r_t^2)}{4 r_t}\big[r_{t+1}\sin(\psi_{t+1} - \psi_t + a) + r_{t-1}\sin(\psi_{t-1} - \psi_t - a)\big].$$

Proof. $\dot{c}_n = in\omega c_n + \frac{n}{2}(Hc_{n-1} - \bar{H}c_{n+1} + Gc_{n-2} - \bar{G}c_{n+2})$, with $c_0 = 1$, $c_n = z^n$ and $c_{-n} = \bar{z}^n$. For $n = 1$ the term $c_{-1} = \bar{z}$ appears; for $n \geq 2$ it is z^{n-2}. The two equations agree iff $G(\bar{z} - z^{-1}) = 0$, i.e. $G = 0$ or $|z| = 1$. For $G = 0$, closing the contour in the half plane where $z(\omega)$ is analytic gives $Z_t = z_t(\omega_0 + i\Delta)$ and $\dot{Z}_t = (i\omega_0 - \Delta)Z_t + \frac{1}{2}(H_t - \bar{H}_t Z_t^2)$. Taking real and imaginary parts of $\dot{Z}_t e^{-i\psi_t}$ gives the polar form. □

Theorem 3.4.2 (No gauge chimera)**.** *Let $K_2 = 0$, $K_1 \neq 0$ and $\Delta = 0$ (identical oscillators), and let $V(Z) = -\frac{1}{2}\sum_t \operatorname{Re}(\bar{Z}_t H_t)$. Along the reduced system of Proposition 3.4.1,*

$$\dot{V} \leq -\tfrac{1}{2}\sum_t |H_t|^2(1 - r_t^2) \leq 0.$$

Every ω-limit point therefore satisfies, for each t, either $H_t = 0$, or $r_t = 1$ and $H_t\bar{Z}_t \in \mathbb{R}$. Consequently:

(i) no ω-limit point has $r_0 = 1 > r_1, r_2$ (or any rotation of this);

(ii) a class with $r_t < 1$ at an ω-limit point feels zero field, so it does not drift: all limit sets are relative equilibria rotating at ω_0;

(iii) there are no periodic orbits other than rigid rotations, and no Hopf bifurcation.

The same holds for the finite network by Theorem 3.2.2.

Proof. $V = -\frac{1}{2}Z^*WZ$ with W Hermitian, $W_{t,t+1} = \frac{K_1}{2}e^{ia}$. So $\dot{V} = -\operatorname{Re}\sum_t \dot{\bar{Z}}_t H_t$. Substituting $\dot{Z}_t = i\omega_0 Z_t + \frac{1}{2}(H_t - \bar{H}_t Z_t^2)$, the ω_0-term is $\omega_0 \operatorname{Im}(Z^*WZ) = 0$, and $\dot{V} = -\frac{1}{2}\sum_t(|H_t|^2 - \operatorname{Re}((H_t\bar{Z}_t)^2))$, where $|\operatorname{Re}(w^2)| \leq |w|^2$ and $|H_t\bar{Z}_t|^2 = |H_t|^2 r_t^2$. Equality requires, for each t, either $H_t = 0$, or $r_t = 1$ and $(H_t\bar{Z}_t)^2 \geq 0$. The closed polydisc is invariant, since $\frac{d}{dt}|Z_t|^2 = (1 - |Z_t|^2)\operatorname{Re}(\bar{Z}_t H_t)$. LaSalle [15] gives the first claim. For (i): $H_1 = 0$ means $e^{ia}Z_2 = -e^{-ia}Z_0$, so $r_2 = r_0 = 1$. For (ii): if $H_t = 0$ then $\dot{Z}_t = i\omega_0 Z_t$. For (iii): on a periodic orbit V is constant, so the orbit lies in the LaSalle set, where the motion is rigid rotation by (ii). □

Remark 3.4.3 (What is and is not known about chimeras). Chimeras in the sense of [14, 13], a coherent population beside a drifting one, need a non-gradient coupling. With the gauge coupling alone they are impossible (Theorems 3.2.2, 3.4.2). The triadic lag is the only non-gradient ingredient of (3.1.1), and its second harmonic removes the OA closure (Proposition 3.4.1). For $\Delta > 0$ we have no Lyapunov function, and the question is open. Theorem 3.3.4 gives the exact criterion for the *onset* of a type-selective state. At a cluster equilibrium that is stable within the cluster subspace, class t desynchronizes first iff $\sigma_t < 0$ while the other $\sigma_s > 0$ (on Y_{21}; on other complexes the blocks $M(\lambda)$ must also be stable).

Remark 3.4.4 (Numerical observations). (1) We sampled 1500 random parameter sets ($K_1 = \pm 1$, $K_2 \in [-2, 2]$, a, Φ). Among the 2196 cluster equilibria found that are stable within the cluster subspace, none had exactly one class with $\sigma_t > 0$. The pattern of the proposed programme ($r_0 \approx 1$, classes $1, 2$ incoherent) is therefore never the primary instability. The observed pattern is the opposite: one unstable class, in 124 cases. (2) Take $(K_1, K_2, a, \Phi) = (1, -1.99322, -1.01389, -0.06786)$ at the equilibrium $\psi = (0, 1.66481, 2.08094)$, where $\sigma = (3.3251, -0.1916, 3.2556)$. Started there, class 1 breaks up, and the network settles into a *frozen* state with a common frequency: on Y_{21}, $r = (1, 0.7463, 1)$ and $\Omega = 1.18831$; on Y_{168}, $r = (0.9917, 0.9001, 0.9909)$ and $\Omega = 1.29345$. This is not a chimera. (3) An exploratory search on Y_{168} (120 random parameter sets and initial conditions, not part of the battery) found non-locked states only for $|K_2| \gtrsim 1.4$, of two kinds. In

solitary states at most two oscillators drift. In *disordered* states most oscillators of *every* class drift: in the run kept in the battery, $53, 52, 55$ of the 56 per class. We found no state in which one class is coherent while the other two are macroscopically incoherent. (4) Finite populations (3×3000) started on the OA manifold stay on it for $K_2 = 0$ up to sampling error ($|Z_2 - Z_1^2| \le 0.013$). For $K_2 = 1$ they leave it ($|Z_2 - Z_1^2|$ reaches 0.36).

3.5 Frequency shift and geometric phase

Theorem 3.5.1 (Frequency shift)**.** *On every frequency-locked state, $\Omega - \omega = K_2 \langle m_\Delta \rangle_F$. On the twisted states the shift is $\Delta\Omega = K_2 \sin\Phi$, independent of K_1, α, q, n, $|F|$ and $\chi(Y)$. If the triadic sum in* (3.1.1) *is not normalized by D_2, the shift becomes $D_2 K_2 \sin\Phi$ with $D_2 = 3|F|/n = (q+1)(q^2+q+1)$, a count read off the vertex link.*

Proof. Proposition 3.2.1 and Theorem 3.3.1. □

Remark 3.5.2 (The Euler characteristic does not enter). Under (L), $\chi(Y) = n - |E| + |F| = n(1 + \frac{\kappa(q-2)}{3})$. For $q = 2$ this is n ($|E| = |F| = 7n$), so $D_2 = 3|F|/\chi(Y)$. That identity is the only sense in which "the shift is set by χ and the number of chambers" holds, and it is a coincidence of $q = 2$. On the thin torus T_6, $\chi = 0$ while $D_2 = 6$ and $\Delta\Omega = K_2 \sin\Phi \neq 0$. Both $|F|$ and χ are multiplied by the degree under coverings, and $\Delta\Omega$ is a local quantity.

Proposition 3.5.3 (Geometric phase of a rigid rotation)**.** *Let $\theta(t) = \theta^* + \Omega t\mathbf{1}$ be a relative equilibrium, $\Omega \neq 0$, and $u(t) = n^{-1/2}(e^{i\theta_j(t)})_j$. Over one period $T = 2\pi/\Omega$, the total phase $\arg\langle u(0), u(T)\rangle$ equals the dynamical phase $\int_0^T \mathrm{Im}\langle u, \dot{u}\rangle\, dt = 2\pi$. The Aharonov–Anandan geometric phase is therefore* 0.

Proof. $u(t) = e^{i\Omega t}u(0)$, so its projective class is constant and encloses no area in $\mathbb{CP}^{n-1}$. □

Proposition 3.5.4 (Flux-threading monodromy)**.** *Consider the loop $\alpha \mapsto \alpha + 1$ in the circle of chiral fluxes, with fibres identified by the gauge $\chi = \frac{2\pi}{3}\tau$ of Proposition 3.1.2(ii). The continuation of Θ_m over the loop ends at Θ_{m+1}. So the monodromy of the triple $\{\Theta_0, \Theta_1, \Theta_2\}$ is the cyclic permutation, implemented by multiplying $e^{i\theta}$ by the grading character $\omega^\tau = D_1^{-1}$ of Proposition 1.2.1. The three states are pairwise orthogonal ($\sum_j \omega^{\tau(j)} = 0$), so the monodromy is a permutation, not a phase. It has order* 3*. If $K_2 = 0$ and $K_1 > 0$, the branch on which $m + \alpha$ runs from $-\frac{1}{2}$ to $\frac{1}{2}$ stays linearly stable over the whole loop (Corollary 3.3.2).*

Proof. In the gauge A^α, Θ_m does not depend on α. The gauge taking $A^{\alpha+1}$ to A^α maps θ to $\theta + \frac{2\pi}{3}\tau$, and so maps Θ_m to Θ_{m+1}. Along the loop $m + \alpha$ runs over an interval of length 1, from $-\frac{1}{2}$ to $\frac{1}{2}$, where $\cos\frac{2\pi(m+\alpha)}{3} \ge \frac{1}{2} > 0$. □

3.6 Why a tree cannot carry the mechanism

Proposition 3.6.1 (Graphs and trees)**.** *Let G be a finite graph, a* 1*-complex.*

(i) G has no chambers, so there is neither a flux 2*-form nor a triadic term. Every connection is flat, and the gauge classes form the torus $U(1)^{b_1(G)}$ of Aharonov–Bohm phases through* 1*-cycles. On a tree every connection is pure gauge.*

(ii) The dynamics is a gradient flow (Theorem 3.2.2), with no frequency shift, no limit cycle, no Hopf bifurcation, and no chimera for identical frequencies.

(iii) On a tree with $K_1 > 0$, after gauging A away, the equilibria are the states in which every edge difference is 0 or π. The only stable one is the in-phase state.

Proof. (i) and (ii) were shown above. (iii) At a leaf j with neighbour k, the equation $\sin(\theta_k - \theta_j) = 0$ forces a difference of 0 or π. Removing the leaf leaves an equilibrium of the smaller tree, since the removed term vanishes. Now suppose some edge e has difference π. Let v be the indicator of one component of $G \setminus e$. Then $v^t(\operatorname{Hess} V)v = K_1 \cos\pi < 0$, so the equilibrium is a saddle. □

The mechanism needs 2-cells, not thickness. The thin torus T_6 ($q = 1$) carries all of it; Theorems 3.2.4–3.3.4 hold there, and so does line (J) of the battery. Thickness only makes the flat twists finite, through Garland rigidity. On T_6, $b_1 = 2$ and they form a 2-torus.

3.7 The example Y_{21}

On Y_{21}: $\kappa = 7$, the degree is 14, $D_2 = 21$, $|E| = |F| = 147$, and $\chi = 21 = 1 + b_2$. $H_1 = (\mathbb{Z}/2)^6$, so there are 64 chiral connections per flux up to gauge. The spectrum of Δ_0 is $0, 14^{(18)}, 21^{(2)}$. For $K_2 = 0$ the ground-state energy per edge is $-K_1 \cos\frac{2\pi\beta}{3}$, which is $-K_1$, $-0.809017K_1$ and $-\frac{1}{2}K_1$ for $\alpha = 0$, 0.3 and $\frac{1}{2}$, with currents 0, $0.587785K_1$ and $\pm 0.866025K_1$.

Take $K_1 = 1$, $K_2 = 0.2$, $\alpha = 0.3$, $\Phi = \pi/4$. All three twisted states rotate at $\Omega - \omega = 0.2 \sin\frac{\pi}{4} = 0.141421$, and

	c_m	transverse eigenvalues $-14c_m$, $-21c_m$	stable
Θ_0	0.077990	$-1.09186^{(18)}$, $-1.63779^{(2)}$	yes
Θ_1	−0.045050	$0.630703^{(18)}$, $0.946054^{(2)}$	no
Θ_2	0.027669	$-0.387371^{(18)}$, $-0.581057^{(2)}$	yes

Two twisted states are stable, as Corollary 3.3.2 predicts for fractional part $0.3 \in (\frac{1}{4}, \frac{3}{4})$. The K_2 term only shifts every c_m by $2K_2 \cos\Phi/14 = 0.020203$. By Corollary 3.3.5, no parameter change produces a Hopf bifurcation at any cluster state of Y_{21}.

3.8 Transverse Hopf bifurcations on Y_{168}

Theorem 3.3.3 excludes Hopf bifurcations inside the cluster subspace, and Corollary 3.3.5 excludes them on Y_{21}. On Y_{168} the blocks $M(\lambda)$ of Theorem 3.3.4 can have complex spectrum, and the question left open at the end of Section 3.3 is whether a stable cluster equilibrium can lose stability through a complex pair of a transverse block, and what happens then. The answer has three parts: the structure of such a crossing is forced by the symmetry (Proposition 3.8.1); such crossings exist and are common (Theorem 3.8.2); and in every case examined the loss of stability is hard, ending in a frequency-locked state, not on a small limit cycle (Remark 3.8.3). Throughout, $K_1 = 1$, $\omega_j = \omega$, and the numbers come from `code/v3p3_hopf_search.py` (numpy, mpmath; seeded; about 25 minutes; data and figures in `data/`).

Proposition 3.8.1 (Structure of a transverse crossing)**.** *Let ψ be a cluster equilibrium, $\lambda \neq 0$ an eigenvalue of A_1, and $M(\lambda)$ as in Theorem 3.3.4.*

(i) $M(\omega\lambda) = DM(\lambda)D^{-1}$ with $D = \operatorname{diag}(1, \bar{\omega}, \omega)$, and $M(\bar{\lambda}) = \overline{M(\lambda)}$. So the spectrum of $M(\lambda)$ depends only on the orbit $[\lambda]$, and if $\lambda^3 \in \mathbb{R}$ then $M(\lambda)$ is similar to a real matrix.

(ii) Every eigenvalue μ of $M(\lambda)$ is an eigenvalue of the Jacobian J with multiplicity at least $\dim E_\lambda$, and $\bar{\mu}$ is one of $M(\bar{\lambda})$. A crossing of the imaginary axis at $\mu = i\Omega_H$, $\Omega_H \neq 0$, through a transverse block is therefore a Hopf point at which $i\Omega_H$ and $-i\Omega_H$ each have multiplicity at least $\dim E_\lambda$; the conjugate comes from $M(\bar{\lambda})$ if $[\bar{\lambda}] \neq [\lambda]$, and from $M(\lambda)$ itself otherwise.

(iii) On Y_{168} the nonzero eigenvalues of A_1 are 7, the 21 eigenvalues with $\lambda^3 = -1$ (the orbit of $e^{i\pi/3}$, which contains -1; $\dim E_\lambda = 7$), and the 126 eigenvalues with $\lambda^3 = 4(5 \pm \sqrt{-7})$, i.e. $\lambda = 2\sqrt{2}\, e^{\pm i\varphi}\omega^k$ with $\varphi = 0.162232$ ($\dim E_\lambda = 21$ for each sign); 0 has multiplicity 18. Hence a transverse Hopf point through the blocks $2\sqrt{2}\, e^{\pm i\varphi}$ has 21 eigenvalues at $+i\Omega_H$ and 21 at $-i\Omega_H$, and one through the block of $e^{i\pi/3}$ has a conjugate pair of multiplicity 7 each inside the real block $M(-1)$.

Proof. (i) With $d_t = \omega^{-t}$, conjugation by $\mathrm{diag}(d_t)$ multiplies the entry $(t, t+1)$ by $d_t/d_{t+1} = \omega$ and the entry $(t, t-1)$ by $\bar{\omega}$, also across the wrap-around $2 \to 0$, $0 \to 2$, since $\omega^3 = 1$. This turns $\alpha_t\lambda$ into $\alpha_t\omega\lambda$ and $\gamma_t\bar{\lambda}$ into $\gamma_t\overline{\omega\lambda}$. The coefficients $\alpha_t, \gamma_t, \sigma_t$ are real, which gives $M(\bar{\lambda}) = \overline{M(\lambda)}$. If $\lambda^3 \in \mathbb{R}$, then $\bar{\lambda} = \omega^k\lambda$ for some k, so $M(\lambda)$ is similar to its complex conjugate; on the orbit of $e^{i\pi/3}$ it is similar to $M(-1)$, which is real. (ii) J acts on $W_{[\lambda]}$ as $M(\lambda) \otimes I_{\dim E_\lambda}$ (Theorem 3.3.4), and J is real. (iii) The spectrum of A_1 is computed from the 56 character blocks of the deck group $(\mathbb{Z}/2)^3 \times \mathbb{Z}/7$ (line (A)); the identities $\lambda^3 = -1$ and $\lambda^3 = 20 \pm 4\sqrt{-7}$ hold to $3 \cdot 10^{-13}$, and the field $\mathbb{Q}(\sqrt{-7})$ is the one met in Chapter 4, Remark 4.6.8. □

So on Y_{168} no transverse Hopf point is simple, and the classical Hopf theorem, together with its first Lyapunov coefficient, does not apply. The symmetry group of the problem (the deck group extended by the lifted automorphisms of X_3) acts on the 42-dimensional real centre space, and the equivariant Hopf theorem [19] gives a branch of periodic solutions for each $\mathbb{C}$-axial isotropy subgroup; their direction and stability are decided by the equivariant cubic normal form, which we have not computed. The numerics below test the dynamics directly.

Theorem 3.8.2 (Transverse Hopf points exist on Y_{168}; numerical)**.** *(a) Let $K_1 = 1$, $\alpha = 0.55$ and $\Phi = -1.40$. The cluster equilibrium with $d_0 = -1.7681332468697411$, $d_1 = 2.5493466590522488$ undergoes a transverse Hopf bifurcation in the block $\lambda = 2\sqrt{2}\, e^{-i\varphi}$ at*

$$K_2^* = -0.5440732884161930553882 6, \qquad \Omega_H = 0.0074143951488903915028,$$
$$\frac{d\,\mathrm{Re}\,\mu}{dK_2}\Big|_{K_2^*} = 5.0116061224789490.$$

At K_2^* the full 168×168 Jacobian has exactly 21 eigenvalues at $i\Omega_H$, 21 at $-i\Omega_H$, the rotation eigenvalue 0, and all others in $\mathrm{Re} \leq -0.0180$; the equilibrium rotates at $\Omega - \omega = -0.0973938$. Along the branch the rest of the spectrum stays in $\mathrm{Re} < 0$ on $[K_2^* - 0.4, K_2^*]$, and the critical pair crosses once, from left to right. The next crossing is that of a real eigenvalue of the block of $e^{i\pi/3}$, at $\delta_{\mathrm{next}} = K_2 - K_2^* = 0.0035$; for $0 < K_2 - K_2^* < 0.0035$ exactly the 42 Hopf modes are unstable.

(b) Over 150 random slices (α, Φ) with $K_2 \in [-3, 3]$, the 274 losses of stability of stable cluster equilibria (rotated copies counted once) are: 191 real crossings inside the cluster subspace, 52 transverse Hopf points, 23 real crossings of a rate $-\sigma_t$, and 8 real crossings in a transverse block. All 52 Hopf points lie in the blocks $2\sqrt{2}\, e^{\pm i\varphi}$, none in the block of $e^{i\pi/3}$, at $K_2 \in [-2.74, 0.61]$.

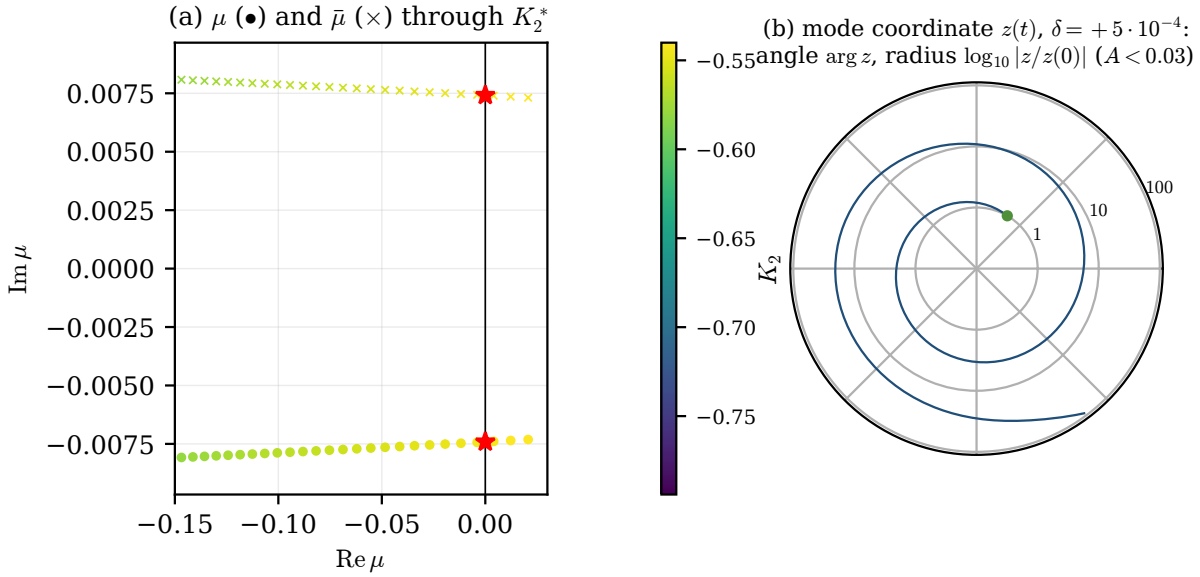


Figure 3.1: The transverse Hopf point of Theorem 3.8.2(a) on Y_{168}, $(K_1, \alpha, \Phi) = (1, 0.55, -1.40)$. (a) Along the branch of cluster equilibria, the leading eigenvalue μ of the block $M(\lambda)$, $\lambda = 2\sqrt{2}\,e^{-i\varphi}$ (dots), and its conjugate partner $\bar{\mu}$ in $M(\bar{\lambda})$ (crosses), coloured by $K_2 \in [K_2^* - 0.25, K_2^* + 0.004]$; they cross the imaginary axis at $\pm i\Omega_H$, $\Omega_H = 0.0074144$ (stars), with speed $d\,\mathrm{Re}\,\mu/dK_2 = 5.0116$, each with multiplicity 21. (b) The full 168-oscillator network at $K_2 = K_2^* + 5\cdot10^{-4}$, started at amplitude 10^{-4} in the 42-dimensional critical space: the critical-mode coordinate $z(t)$ in log-polar form (angle $\arg z$, radius $\log_{10}|z/z(0)|$), shown while the rms deviation $A < 0.03$. It turns at the rate Ω_H and grows at the linear rate $\mathrm{Re}\,\mu$ at first, then faster (Remark 3.8.3(2)), and leaves the neighbourhood without settling on a cycle; the trajectory ends in a frequency-locked state. Drawn by `code/make_figures.py`.

(c) At $\alpha = 0.55$, on 72 values of Φ, there are 26 Hopf points, on two arcs with $1.0 \lesssim |\Phi| \lesssim 2.1$ and $K_2^* \in [-2.4, -0.4]$; there $\Omega_H \in [0.0068, 0.0074]$, and the crossing speed $d\,\mathrm{Re}\,\mu/dK_2$ is positive at every point.

Method. (a) The four real equations "reduced cluster field = 0" and $\det(M(\lambda) - i\omega I) = 0$ in the unknowns (d_0, d_1, K_2, ω) are solved by Newton's method at 40 digits, with λ computed to 40 digits from a character block of A_1; the residual is $3\cdot10^{-41}$, and the slope is a central difference with step 10^{-15}. This is a high-precision numerical solution of an analytic system with nonsingular Jacobian, not an interval certificate. The count of eigenvalues on the axis uses the numerical Jacobian of the 168 oscillators. (b), (c) Equilibria are found by vectorized Newton from 36 seeds at each of 601 values of K_2; each stable one is continued one step and its leading eigenvalue classified; Hopf points are refined by bisection. □

Remark 3.8.3 (Numerical: what happens beyond the crossing). Full-network simulations (RK4, $dt = 0.25$) at the point of Theorem 3.8.2(a), with $\delta = K_2 - K_2^*$:

(1) *Linear regime.* For $\delta = +0.0005$ the linearization gives $\mu = 0.00252 + 0.00740i$. Started at amplitude 10^{-4} in the centre space, the rms deviation A grows at rate 0.00255 and the

critical-mode coordinate turns at rate 0.00743 while $A \in [2 \cdot 10^{-4}, 10^{-3}]$: the instability is the predicted oscillatory one (Figure 3.1).

(2) *No saturation.* The growth then accelerates, to rates 0.0103 for $A \in [3 \cdot 10^{-3}, 10^{-2}]$ and 0.087 for $A \in [10^{-2}, 3 \cdot 10^{-2}]$: the nonlinearity is destabilizing, and there is no small-amplitude cycle to land on. The trajectory ends at a state that is frequency-locked (spread of $\dot{\theta}_j$ below $2 \cdot 10^{-13}$) and stationary in the rotating frame (shape change $4 \cdot 10^{-16}$), with $A = 0.442$ and $r = (0.9994, 0.9965, 0.8155)$: a relative equilibrium off the cluster subspace, in which one class is partly desynchronized.

(3) *All initial directions.* For $\delta = +0.001$ and $+0.002$ (exactly the 42 Hopf modes are unstable there), three generic perturbations in the centre space and three pure character modes (each of the 42 unstable characters of the deck group spans an invariant 3-dimensional subspace) all end in frequency-locked stationary states, with A between 0.40 and 0.52.

(4) *Below threshold.* The equilibrium is linearly stable, but a perturbation of size ε in a fixed generic centre direction escapes to a locked state once $\varepsilon > \varepsilon_c$, with $\varepsilon_c = 0.0053$, 0.0184, 0.0537 at $\delta = -0.002$, -0.008, -0.032. So $\varepsilon_c \to 0$ as $\delta \to 0^-$, as for a subcritical bifurcation; the fitted exponent is 0.83, not the $\frac{1}{2}$ of a generic cubic normal form, so we do not claim a normal-form classification.

(5) *A nearby steady-state crossing.* At every Hopf point examined, a real eigenvalue of the block of $e^{i\pi/3}$ crosses zero shortly after the Hopf pair: $\delta_{\text{next}} = 0.0035$ at the point of Theorem 3.8.2(a), and 0.0005, 0.0020, 0.0020 at three representative points of (b). This is structural rather than accidental: all blocks share the diagonal $-\sigma_t$, so when one block reaches the axis the others are close. The transverse Hopf points of Y_{168} thus sit near a Hopf/steady-state interaction of codimension two.

(6) *Survey.* At those three representative points, each run at $\delta = \delta_{\text{next}}/5$ (so that exactly the 42 Hopf modes are unstable) from a perturbation of size 10^{-3} in the centre space, the outcome is a frequency-locked stationary state in all three cases ($A = 0.27$, 0.22, 0.32, each with one class partly desynchronized). An earlier survey of 12 Hopf points at $\operatorname{Re}\mu = +0.01$ gave the same outcome in all 12 cases, with A from 0.30 to 1.46 (at the largest, $r = (0.45, 0.45, 0.22)$) and $\Omega_H \in [0.0027, 0.0076]$.

(7) *Order parameters.* The critical modes lie in $W_{[\lambda]} \oplus W_{[\bar{\lambda}]}$, which is orthogonal to the class indicators (they span $W_{[\kappa]}$, and A_1 is normal). So the class means of the deviation vanish, and $1 - r_t = \frac{1}{2}\operatorname{Var}_{V_t}(\theta) + O(A^3)$: the oscillation is visible in r_0, r_1, r_2 only at second order, and the time series of the r_t show the growth (with $1 - r_t \propto A^2$) and the final jump, not the rotation. The rotation is seen in the critical-mode coordinate (figure `v3p3_hopf_dynamics.png`).

So transverse Hopf bifurcations exist and account for about a fifth of the losses of stability of stable cluster states on Y_{168}, but no stable limit cycle, chiral or otherwise, was found near any of them: the loss of stability is hard, and the network freezes into a locked state of the kind seen in Remark 3.4.4(2). This happens even where only the Hopf modes are linearly unstable, but the nearby steady-state crossing of (5) means that the relevant normal form is that of a Hopf/steady-state interaction on a $42 + 7$-dimensional centre space, not of an isolated Hopf point. Whether some isotropy branch of the equivariant bifurcation is supercritical is open.

Remark 3.8.4 (Corrections to the proposed programme). (1) The crossing is not a simple Hopf bifurcation: on Y_{168} the pair $\pm i\Omega_H$ has multiplicity 21 each, forced by the deck-group symmetry

(Proposition 3.8.1), so "supercritical" in the sense of a single first Lyapunov coefficient is not defined. (2) The bifurcation point $(K_1^*, K_2^*, \alpha^*, \Phi^*)$ is a curve family, not a point: for fixed (α, Φ) there is a value K_2^*, and Theorem 3.8.2 gives one to 23 digits and the locus at $\alpha = 0.55$. (3) Beyond the crossing ($\mathrm{Re}\,\mu > 0$) the simulations do not produce a stable chiral limit cycle; they produce frequency-locked relative equilibria off the cluster subspace. (4) The class order parameters $r_t(t)$ see the oscillation only at second order and are not a good detector of it. (5) The Hopf points are not isolated codimension-one events: a real crossing in the block of $e^{i\pi/3}$ follows within $\delta_{\mathrm{next}} = 0.0005$–$0.0035$.

3.9 Verification battery

check	complexes	result
(G) $\delta A^\alpha = 2\pi\alpha$; type gauge shifts a by $2\pi/3$; $H_1(Y_{21}) = (\mathbb{Z}/2)^6$, 64 classes	Y_{21}, Y_{24}, T_6	exact
(L) triangle lemma on a 721^2 grid, 200 values of S	—	numeric 10^{-4}
(V) twisted state attains $-\lvert E\rvert \cos\frac{2\pi\beta}{3}$; 30 random descents never lower	Y_{21}, T_6	numeric
(J) $J_m = -c_m \Delta_0$, rotation at $K_2 \sin\Phi$; the Y_{21} example table	Y_{21}, Y_{24}, T_6	numeric 10^{-7}
(F) $\frac{1}{n}\sum \dot\theta_j = \omega + K_2 \langle m_\Delta \rangle_F$ at random states	Y_{21}, Y_{168}, T_6	numeric 10^{-12}
(R) reduced cluster Jacobian symmetric (685 equilibria); field $= -\nabla U + \bar m \mathbf{1}$	—	numeric
(B) spectrum $= \bigcup \mathrm{spec}\, M(\lambda) \cup \{-\sigma_t\}$ at 46 equilibria; real on Y_{21}; complex on Y_{168} in 6	Y_{21}, Y_{168}	numeric 10^{-5}
(O) Lyapunov inequality (2000 points); 60 OA runs without chimera limit; OA defect 0.013 vs 0.36	—	numeric (seeds fixed)
(N) 2196 stable cluster equilibria, none with exactly one stable class; frozen end states; solitary / disordered drift	Y_{21}, Y_{168}	numeric (seeds fixed)
(E) $\chi = n$ for $q = 2$; $\chi(T_6) = 0$	$Y_{21}, Y_{24}, Y_{168}, T_6$	exact

Table 3.1: Checks run by `code/v3p3_verify.py`; the complexes are built by `code/a2_common.py`. Section 3.8 is checked by `code/v3p3_hopf_search.py`.

3.10 Scope, residue, corrections

Remark 3.10.1 (Solved, open). *Solved:* the classification of chiral connections; the gradient structure of the flux sector, with its frustrated ground states and multistability; the twisted states and their exact spectra for all parameters; the gradient structure of the cluster subspace; the block decomposition of the Jacobian at cluster states; OA closure and non-closure; the exclusion of gauge chimeras for identical oscillators; the frequency sum rule; the monodromy of flux threading. *Open:* (a) whether (3.1.1) with $K_2 \neq 0$ has a type-selective chimera on some family of complexes; transverse Hopf bifurcations through a block $M(\lambda)$ do occur on Y_{168} (Theorem 3.8.2), and what remains open is whether any branch of this multiplicity-21 equivariant bifurcation is supercritical (Remark 3.8.3); (b) a low-dimensional closure for the triadic term, perhaps for an OA-compatible variant that is first harmonic in θ_j; (c) the case $\Delta > 0$ of Theorem 3.4.2; (d) the role of the $|H_1|$ flat twists, which break the $\mathbb{Z}/3$ cluster symmetry.

Remark 3.10.2 (Corrections to the proposed programme). (1) A nonzero flux does not remove static fixed points or force a limit cycle. For $K_2 = 0$ the model is a gradient flow for every flux (Theorem 3.2.2). Its ground states are twisted states that carry static persistent currents (Theorem 3.2.4). Breaking the conjugation C is not breaking gradient-ness. (2) Φ_Δ is not part of the gauge field. The gauge-covariant triadic term is blind to the flux (Remark 3.1.4); the only non-gradient ingredient is the triadic lag, through the chamber drive (Theorem 3.3.3). (3) The OA reduction closes only for $K_2 = 0$ (Proposition 3.4.1). (4) There is no Hopf bifurcation at twisted states, anywhere in the cluster subspace, anywhere for $K_2 = 0$, or at any cluster state of Y_{21}

(Theorems 3.3.1, 3.3.3, 3.2.2, Corollary 3.3.5). On Y_{168} transverse Hopf points exist, but they have multiplicity 21 and, numerically, lead to a hard loss of stability (Section 3.8). A supercritical Hopf route to a chimera is not established. (5) The chimera "$r_0 \approx 1$, r_1, r_2 incoherent" is impossible for $K_2 = 0$ and identical oscillators (Theorem 3.4.2). For $K_2 \neq 0$ it is unproved, and it was not seen numerically (Remark 3.4.4). (6) The geometric phase of the synchronized rotation is 0. The geometric content is the $\mathbb{Z}/3$ monodromy of flux threading (Propositions 3.5.3, 3.5.4). (7) $\Delta\Omega = K_2 \sin\Phi$ is local and does not involve $\chi(Y)$ (Theorem 3.5.1, Remark 3.5.2). (8) Trees fail because every connection on a tree is pure gauge and a graph has no 2-cells, hence no triadic term and no non-gradient dynamics (Proposition 3.6.1).

Acknowledgement of provenance and caveats

Unrefereed. The classical inputs are Łojasiewicz's gradient inequality [16] and LaSalle's invariance principle [15]. The Ott–Antonsen ansatz [17] is used only through the computation in Proposition 3.4.1. The chimera and triadic-coupling literature [14, 13, 18, 20] is cited for context and terminology only. The numerical observations of Remark 3.4.4 are exploratory and fixed-seed. The bibliographic data of [16, 15, 17, 13, 18, 20] have been checked against Crossref and zbMATH; [14], which neither indexes, was checked against the reference lists of later work.

Chapter 4

The Geodesic Edge Operator: Unitary Remainders, Ramanujan Equivalence, and Quantum Spectral Chaos

The proposed programme concerns the geodesic edge operator L_E of an $\widetilde{A}_d$ complex. It asks for four claims. (a) After removing the Perron root q^d, the remainder R has spectral radius exactly $q^{(d-1)/2}$ in a suitable inner product, so the Riemann hypothesis for the edge zeta function holds *unconditionally*. (b) Under the $\mathrm{PGL}_{d+1}(\mathbb{F}_q)$ symmetry the degenerate spectrum has Poisson spacings e^{-s}. (c) An arbitrarily small perturbation εV produces level repulsion, with a Brody crossover to GOE ($\beta = 1$) or GUE ($\beta = 2$). (d) All this "unifies" the Ramanujan property with quantum chaos. We prove the correct form of (a) and correct (b)–(d).

(1) *Structure.* L_E is q^2-biregular and block-cyclic in the $\mathbb{Z}/3$ tail type, not block-tridiagonal. It satisfies four intertwining relations with the vertex operators. The subspace $\operatorname{im}\Psi = S^t\mathbb{C}^V + T^t\mathbb{C}^V + Q\mathbb{C}^V$ is invariant, and on it L_E is a quotient of a $3n \times 3n$ matrix whose characteristic polynomial is the Hecke pencil $\det P(u)$ of Chapter 1.

(2) *The unitary remainder, unconditionally.* On $W = (\operatorname{im}\Psi)^\perp = \ker S \cap \ker T \cap \ker Q^t$, L_E^t is exactly $\sqrt{q}$ times an isometry. So the compression of L_E to W is $\sqrt{q}$ times a unitary (in fact real orthogonal) operator, for every complex satisfying (L). The mechanism is local: the point–hyperplane incidence of $\mathrm{PG}(d, q)$ is a 2-design. The same argument works on $\widetilde{A}_d$ with $q^{(d-1)/2}$, which is the circle of the proposed programme. Every eigenvalue of L_E has $|\lambda| \geq \sqrt{q}$, so L_E has no Betti-number factor.

(3) *The Ramanujan equivalence.* $\det(I - uL_E) = \det P(u) \det(I - u\bar{L})/((1-u^3)(1-q^3u^3))$, where $\bar{L}$ is the remainder. This holds whenever no nontrivial vertex eigenvalue lies on an explicit exceptional curve, and in particular for all Ramanujan complexes. The nontrivial spectrum of L_E lies on the two circles $|\lambda| = q$ and $|\lambda| = \sqrt{q}$ *if and only if* Y is Ramanujan. Claim (a) is thus false as stated: $\rho(R) = q$, not $\sqrt{q}$. On Y_{21}, 54 eigenvalues have modulus 2. The Riemann hypothesis is equivalent to the Ramanujan property, not unconditional. On Y_{21}, $\det(I - uL_E) = (1-64u^3)(1-8u^3)^{18}(1-2u^3+8u^6)^9(1+5u^3+8u^6)^6$; this is checked in full numerically and exactly at eight values of u.

(4) *Level statistics.* A spectrum with multiplicities has an atom at $s = 0$, so it is not Poisson. On Y_{168} the 226 remainder phases of one graded piece take only 16 values, and 93% of the spacings are exactly 0. Under a perturbation the spectrum moves continuously, so no arbitrarily small ε gives Wigner–Dyson statistics. Numerically, for $\varepsilon \leq 10^{-2}$, 93% of the spacings are still below a

tenth of the mean. Brody's family is phenomenological: its $b = 1$ is the GOE surmise, but its $b = 2$ is *not* the GUE surmise. The symmetry class is set by antiunitary symmetries: a real V gives GUE-like statistics here. Ramanujan (where the spectrum sits) and chaos (how phases correlate) are logically independent. What links them is the decomposition $\mathbb{C}^E = \operatorname{im}\Psi \oplus W$.

Status of the results

Every numbered statement is proved except the numerical observations in Remark 4.4.4. The classical inputs are analytic perturbation theory [26], which is used only in Proposition 4.4.2, and the definition of Ramanujan complexes [28]. The equivalence of Theorem 4.3.3 is in the spirit of Kang–Li–Wang [25]. We prove it here, and we do not rely on their statement. All identities were checked by `code/v3p4_verify.py` (Table 4.3).

Notation follows Chapters 1–3. Y satisfies (L) with $q \geq 2$, and $\kappa = q^2 + q + 1$. E is the set of type-increasing edges $x \to y$. $S, T\colon \mathbb{C}^E \to \mathbb{C}^V$ sum over edges with a given tail and head: $(Sf)(x) = \sum_{x\to y} f(x \to y)$ and $(Tf)(y) = \sum_{x\to y} f(x \to y)$. For a vertex y, $\mathrm{In}(y)$ and $\mathrm{Out}(y)$ are the edges into and out of y. In $\mathrm{Lk}(y)$ the in-neighbours are lines and the out-neighbours are points of $\mathrm{PG}(2, q)$, and $x \to y \to z$ lie in a chamber iff the point z is on the line x. Inc_y is the line–point incidence matrix, J the all-ones matrix. $P(u) = I - uA_1 + qu^2A_2 - q^3u^3I$ and $P_a(u) = 1 - au + q\bar{a}u^2 - q^3u^3$ for a joint eigenvalue $(a, \bar{a})$ of (A_1, A_2).

4.1 The geodesic edge operator

Definition 4.1.1. $L_E \in M_E(\mathbb{Z})$ has $L_E(x \to y,\, y \to z) = 1$ when x, y, z do *not* span a chamber, and all other entries 0. Define $Q\colon \mathbb{C}^V \to \mathbb{C}^E$ by $(Q\varphi)(x \to y) = \sum_{z:\{x,y,z\}\in F} \varphi(z)$.

Lemma 4.1.2 (Local form)**.** *$L_E = \bigoplus_y M_y$, where $M_y = J - \mathrm{Inc}_y$ maps $\mathbb{C}^{\mathrm{Out}(y)} \to \mathbb{C}^{\mathrm{In}(y)}$. Moreover $M_y\mathbf{1} = q^2\mathbf{1}$, $\mathbf{1}^tM_y = q^2\mathbf{1}^t$, and $M_y^tM_y = M_yM_y^t = qI + (q^2 - q)J$. So on the complements of the constants, M_y is $\sqrt{q}$ times an isometry.*

Proof. The continuations of $x \to y$ are the $\kappa - (q + 1) = q^2$ points of $\mathrm{Lk}(y)$ off the line x. Two distinct points lie on one common line, so $\mathrm{Inc}^t\,\mathrm{Inc} = qI + J$; dually, $\mathrm{Inc}\,\mathrm{Inc}^t = qI + J$. Then $M^tM = \kappa J - 2(q + 1)J + qI + J$. □

Proposition 4.1.3 (Perron root and grading)**.** *L_E has all row and column sums equal to q^2, so $\rho(L_E) = q^2$ with eigenvector $\mathbf{1}_E$. L_E maps edges with tail type t to edges with tail type $t + 1$. So $D_EL_ED_E^{-1} = \omega L_E$ for $D_E = \mathrm{diag}(\omega^{-\tau(\mathrm{tail})})$, and the spectrum is invariant under multiplication by ω. The block structure is* cyclic *of period* 3*, not tridiagonal. On $\widetilde{A}_d$ it is cyclic of period $d + 1$.*

Proposition 4.1.4 (Intertwining relations)**.**

$$TL_E = q^2S, \qquad L_ES^t = q^2T^t, \qquad L_ET^t = T^tA_1 - Q, \qquad L_EQ = qT^tA_2 - qS^t,$$

and $SS^t = TT^t = \kappa I$, $ST^t = A_1$, $SQ = (q + 1)A_2$, $TQ = (q + 1)A_1$.

Proof. The first two relations are the column and row sums of the M_y. For the third, $(L_ET^t\varphi)(x \to y) = \sum_{z\in\mathrm{Out}(y),\, z\notin x} \varphi(z) = (A_1\varphi)(y) - (Q\varphi)(x \to y)$. For the fourth, $(L_EQ\varphi)(x \to y) = \sum_{z\notin x}\sum_{w\ni z} \varphi(w)$, with w ranging over lines of $\mathrm{Lk}(y)$. The line $w = x$ contributes nothing, and each line $w \neq x$ contains exactly q points off x. So the sum is $q\big((A_2\varphi)(y) - \varphi(x)\big)$. The rest are incidence counts: each edge lies in $q + 1$ chambers. □

Let $\Psi\colon \mathbb{C}^V \otimes \mathbb{C}^3 \to \mathbb{C}^E$, $\Psi(\varphi_1,\varphi_2,\varphi_3) = S^t\varphi_1 + T^t\varphi_2 + Q\varphi_3$, and

$$\widehat{N} = \begin{pmatrix} 0 & 0 & -qI \\ q^2 I & A_1 & qA_2 \\ 0 & -I & 0 \end{pmatrix}.$$

Theorem 4.1.5 (The vertex block)**.** *$L_E\Psi = \Psi\widehat{N}$ and $\det(I - u\widehat{N}) = \det P(u)$. So $\operatorname{im}\Psi$ is L_E-invariant, and $\ker\Psi$ is $\widehat{N}$-invariant. On a joint eigenvector with eigenvalue a, $\widehat{N}$ acts by a 3×3 matrix with characteristic polynomial $\lambda^3 - a\lambda^2 + q\bar{a}\lambda - q^3$.*

Proof. The identity is Proposition 4.1.4 read column by column. For the determinant, expand $\det(\lambda I - N(a))$ with $N(a) = \left(\begin{smallmatrix} 0 & 0 & -q \\ q^2 & a & q\bar{a} \\ 0 & -1 & 0 \end{smallmatrix}\right)$ and diagonalize (A_1, A_2) jointly. □

4.2 The unitary remainder

Theorem 4.2.1 (Unitary remainder, unconditional)**.** *Put $W = (\operatorname{im}\Psi)^\perp = \ker S \cap \ker T \cap \ker Q^t$. Then $L_E^t W \subseteq W$ and $\|L_E^t f\| = \sqrt{q}\,\|f\|$ for $f \in W$. Hence the compression $\bar{L} = P_W L_E|_W$ satisfies $\bar{L}^* = L_E^t|_W$, and $\bar{L} = \sqrt{q}\,U$ with U real orthogonal. In particular $\rho(\bar{L}) = \sqrt{q}$, every eigenvalue of $\bar{L}$ lies on $|\lambda| = \sqrt{q}$, and $\bar{L}$ is diagonalizable. No assumption beyond* (L) *is used.*

Proof. W is the orthogonal complement of an L_E-invariant subspace, so it is L_E^t-invariant. $W \subseteq \ker T$ means that $f|_{\mathrm{In}(y)} \perp \mathbf{1}$ for every y. Now $(L_E^t f)|_{\mathrm{Out}(y)} = M_y^t f|_{\mathrm{In}(y)}$, and both families $\{\mathrm{In}(y)\}$ and $\{\mathrm{Out}(y)\}$ partition E. By Lemma 4.1.2, $\|L_E^t f\|^2 = \sum_y \|M_y^t f|_{\mathrm{In}(y)}\|^2 = q\|f\|^2$. The rest is linear algebra: $\langle \bar{L}f, g\rangle = \langle f, L_E^t g\rangle$ for $f, g \in W$, and L_E is real. □

Theorem 4.2.2 (Lower bound)**.** *Every eigenvalue of L_E satisfies $|\lambda| \geq \sqrt{q}$, with equality iff its eigenvectors lie in $\ker S \cap \ker T$. Consequently $1 - u^3$ does not divide $\det(I - uL_E)$, and the Betti numbers of Y do not enter $\det(I - uL_E)$. This is unlike rank one, where the Hashimoto operator has the eigenvalues ± 1 with multiplicity governed by b_1.*

Proof. Let $L_E f = \lambda f$, and write $f|_{\mathrm{Out}(y)} = \sigma(y)\mathbf{1} + g_y$ and $f|_{\mathrm{In}(y)} = \tau(y)\mathbf{1} + h_y$ with $g_y, h_y \perp \mathbf{1}$. By Lemma 4.1.2, $\lambda\tau = q^2\sigma$ and $\|h_y\| = \frac{\sqrt{q}}{|\lambda|}\|g_y\|$. Put $s = \kappa\sum|\sigma|^2$, $G = \sum\|g_y\|^2$ and $\ell = |\lambda|^2$. Computing $\|f\|^2$ in both partitions gives $s + G = \frac{q^4}{\ell}s + \frac{q}{\ell}G$, i.e. $s(\ell - q^4) = G(q - \ell)$. If $\ell < q$, then the left side is ≤ 0 and the right side is ≥ 0, so $s = G = 0$ and $f = 0$. If $\ell = q$, then $s = 0$, i.e. $Sf = 0$, and then $Tf = 0$. □

Remark 4.2.3. This sharpens Remark 1.6.6, which placed the factor $(1-u^3)^{\chi(Y)}$ of the Kang–Li zeta function on the "edge (branch)" side. It cannot come from the edge operator L_E, which has no eigenvalue of modulus 1. It must come from the branch operator.

Theorem 4.2.4 (The $\widetilde{A}_d$ version)**.** *Let Y be a finite complex all of whose vertex links are the flag complex of* $\mathrm{PG}(d,q)$*, with type-increasing edges E. Let L_E continue $x \to y$ to $y \to z$ when $z \not\subset x$ in* $\mathrm{Lk}(y)$*. Let $\mathcal{V}$ be the smallest L_E-invariant subspace containing $T^t\mathbb{C}^V$. Then L_E has Perron root q^d, and L_E^t is $q^{(d-1)/2}$ times an isometry on $\mathcal{V}^\perp$. Every eigenvalue satisfies $|\lambda| \geq q^{(d-1)/2}$. For $d = 2$, $\mathcal{V} = \operatorname{im}\Psi$.*

Proof. Points and hyperplanes of $\mathrm{PG}(d,q)$ form a symmetric 2-design. Each point lies on $r = \frac{q^d-1}{q-1}$ hyperplanes, two points on $\lambda_2 = \frac{q^{d-1}-1}{q-1}$, and $r - \lambda_2 = q^{d-1}$. So $M_y = J - \mathrm{Inc}_y$ has sums q^d, and

on the complements of the constants $M_y^t M_y = M_y M_y^t = q^{d-1} I$. The proofs of Theorems 4.2.1 and 4.2.2 go through verbatim, with (q^2, q) replaced by (q^d, q^{d-1}). For $d = 2$, $\operatorname{im} \Psi$ contains $T^t \mathbb{C}^V$ and is invariant, and $L_E T^t$ and $L_E Q$ generate Q and S^t. □

Remark 4.2.5. The proposed programme attributes the remainder circle to a "global" $\mathrm{PGL}_{d+1}(\mathbb{F}_q)$ symmetry. What is used is local: the double transitivity of $\mathrm{PGL}_{d+1}(\mathbb{F}_q)$ on points of the link makes the incidence matrix a design, and the design makes the remainder an exact isometry. No global group acts on Y for this.

4.3 Factorization and the Ramanujan equivalence

Call a joint eigenvalue a *trivial* if $a \in \{\kappa, \kappa\omega, \kappa\omega^2\}$. These are simple, with eigenvectors $\mathbf{1}, \omega^\tau, \omega^{2\tau}$ (Lemma 1.6.2, Proposition 1.2.1). Y is *Ramanujan* [28] if every nontrivial a is tempered: $a = q(z_1 + z_2 + z_3)$ with $|z_i| = 1$ and $z_1 z_2 z_3 = 1$. Equivalently, every reciprocal root of P_a has modulus q. In general the Satake parameters have the shape $\{re^{i\alpha}, r^{-1}e^{i\alpha}, e^{-2i\alpha}\}$, because $z \mapsto 1/\bar{z}$ permutes them. Define the *exceptional curve*

$$\Gamma_q = \{(q+1)\lambda + q^2\lambda^{-2} : \ |\lambda| = \sqrt{q}\}.$$

Then $a \in \Gamma_q$ iff P_a has a reciprocal root of modulus $\sqrt{q}$, i.e. $r = q^{\pm 1/2}$. In particular Γ_q contains no tempered value.

Lemma 4.3.1 (The kernel of Ψ)**.** $\ker \Psi = K_t \oplus K_n$*, where* K_t *is* 6*-dimensional and lies over the trivial eigenvalues. There* $\widehat{N}$ *has the eigenvalues* ω^k *and* $q\omega^k$*.* $K_n \neq 0$ *only if some nontrivial* a *lies on* Γ_q*.*

Proof. Ψ maps the trivial part $(\bigoplus_{\mathrm{triv}} E_a) \otimes \mathbb{C}^3$ onto the span of $S^t\chi$ for the three trivial characters χ, because $S^t\chi$, $T^t\chi$ and $Q\chi$ are proportional. It maps the nontrivial part into the orthogonal complement of that span: for instance, $\langle S^t\chi, \Psi v\rangle = \langle \chi, S\Psi v\rangle = 0$, since $S\Psi v$ lies in the nontrivial part. Hence $\ker \Psi$ splits, and K_t has dimension $9 - 3 = 6$. On the image, $L_E S^t \chi = q^2 T^t \chi$ has eigenvalue $q^2\omega^k$, so the other two roots $\omega^k, q\omega^k$ of $P_{\kappa\omega^k}$ lie in K_t.

Now let $v = (\varphi_1, \varphi_2, \varphi) \in K_n$ with $\widehat{N} v = \lambda v$. Then $\lambda \neq 0$, $\varphi_1 = -\frac{q}{\lambda}\varphi$, $\varphi_2 = -\lambda\varphi$, and $\Psi v = 0$ reads $(Q\varphi)(x \to y) = \frac{q}{\lambda}\varphi(x) + \lambda\varphi(y)$ on every edge. At a vertex y this says $\mathrm{Inc}_y \varphi_{\mathrm{Out}} = \frac{q}{\lambda}\varphi_{\mathrm{In}} + \lambda\varphi(y)\mathbf{1}$, and on the edges out of y it says $\mathrm{Inc}_y^t \varphi_{\mathrm{In}} = \frac{q}{\lambda}\varphi(y)\mathbf{1} + \lambda\varphi_{\mathrm{Out}}$. Applying Inc^t to the first, and using $\mathrm{Inc}^t \mathrm{Inc} = qI + J$, gives $A_1\varphi = a(\lambda)\varphi$ with $a(\lambda) = (q+1)\lambda + q^2\lambda^{-2}$. Symmetrically, $A_2\varphi = (\lambda^2 + q(q+1)\lambda^{-1})\varphi$. A_1 is normal, so the two values are complex conjugate. Write $\bar{\lambda} = qt/\lambda$ with $t = |\lambda|^2/q$. Then the conjugacy condition becomes $(t-1)\big[q(q+1)\lambda^{-1} - \lambda^2(1+t)t^{-2}\big] = 0$. Either $t = 1$, i.e. $a \in \Gamma_q$; or $\lambda^3 = q(q+1)t^2/(1+t) > 0$, which forces $|\lambda| \in \{1, q\}$ and $a(\lambda)$ trivial. The latter is impossible for φ in the nontrivial part. Since K_n is $\widehat{N}$-invariant, it is 0 unless it contains such an eigenvector. □

Theorem 4.3.2 (Factorization)**.** *If no nontrivial* a *lies on* Γ_q *(in particular, if* Y *is Ramanujan), then* $\dim W = |E| - 3n + 6 = (q-1)(q+2)n + 6$ *and*

$$\det(I - uL_E) = \frac{\det P(u)}{(1-u^3)(1-q^3u^3)} \det(I - u\bar{L}),$$

where all reciprocal roots of $\det(I - u\bar{L})$ *have modulus* $\sqrt{q}$ *(Theorem 4.2.1).*

Proof. With respect to $\mathbb{C}^E = \operatorname{im}\Psi \oplus W$, L_E is block upper triangular with diagonal blocks $L_E|_{\operatorname{im}\Psi}$ and $\bar{L}$. $L_E|_{\operatorname{im}\Psi}$ is conjugate, via Ψ, to the map induced by $\widehat{N}$ on $\mathbb{C}^{3n}/\ker\Psi$. By Lemma 4.3.1, $\ker\Psi = K_t$, which carries $\prod_k(1-\omega^k u)(1-q\omega^k u) = (1-u^3)(1-q^3u^3)$. □

Theorem 4.3.3 (Riemann hypothesis ⇔ Ramanujan)**.** *The eigenvalues of L_E other than $q^2\omega^k$ all lie on the two circles $|\lambda| = q$ and $|\lambda| = \sqrt{q}$ if and only if Y is Ramanujan. Equivalently, the nontrivial poles of $Z_E(u) = \det(I - uL_E)^{-1}$ lie on $|u| = q^{-1}$ or $|u| = q^{-1/2}$. The circle $|u| = q^{-1/2}$ carries the* $\dim W$ *poles of the remainder unconditionally. On a Ramanujan complex the circle $|u| = q^{-1}$ carries $3(n-3)$ poles, and its occupation is* exactly *the Ramanujan condition.*

Proof. (⇐) Theorem 4.3.2: the nontrivial P_a have reciprocal roots qz_i with $|z_i| = 1$, the trivial ones leave only $q^2\omega^k$, and $\bar{L}$ lives on the $\sqrt{q}$-circle. (⇒) Suppose a nontrivial a is not tempered, with $r > 1$. Then $N(a)$ has an eigenvector v over E_a with eigenvalue $\mu = qre^{i\alpha}$, and $|\mu| > q$. If $\Psi v = 0$, Lemma 4.3.1 forces $|\mu| = \sqrt{q}$, which is false. So Ψv is an eigenvector of L_E with $|\mu| \notin \{q, \sqrt{q}\}$. □

Remark 4.3.4 (What the proposed programme's theorem gets right and wrong). The circle $q^{(d-1)/2}$ is right, for the remainder W (Theorems 4.2.1, 4.2.4), and unconditionally so. It is wrong for "L_E minus its Perron root". That operator has spectral radius q on every Ramanujan complex with $n > 3$: 54 eigenvalues of modulus 2 on Y_{21}, and 495 on Y_{168}. Its q-circle is equivalent to the Ramanujan property, which is a global arithmetic input, not a consequence of (L). The five covers of this volume are all Ramanujan (line (F) of Table 4.3). "In a suitable inner product" is not needed: the standard one works, provided one compresses to W rather than restricting (W is L_E^t-invariant, not L_E-invariant).

4.4 Level statistics: what is provable

For a finite spectrum $\{\theta_1, \ldots, \theta_N\}$ on the circle, counted with multiplicity and sorted, the *empirical spacing measure* is $\frac{1}{N}\sum_i \delta_{s_i}$ with $s_i = \frac{N}{2\pi}(\theta_{i+1} - \theta_i)$.

Proposition 4.4.1 (Degenerate spectra are not Poisson)**.** *If the spectrum has $k < N$ distinct values, the spacing measure has an atom of mass $(N-k)/N$ at $s = 0$. So it is not $e^{-s}\,ds$, and a sequence of such spectra with $(N-k)/N$ bounded below cannot converge to it.*

Proof. Exactly $N - k$ consecutive differences vanish. The measure $e^{-s}ds$ has no atoms. □

On Y_{168}, the remainder of one graded piece, U^3 on $W \cap \mathbb{C}^{E_0}$, has $N = 226$ eigenphases with $k = 16$ distinct values. The atom has mass $210/226 = 0.929$. The "arithmetic Poisson spectrum" of the proposed programme is really an arithmetic *degenerate* spectrum. The deck groups of our covers are abelian ($\mathbb{Z}/7$, $(\mathbb{Z}/2)^3$, $\mathbb{Z}/14$, $\mathbb{Z}/2 \oplus \mathbb{Z}/14$, $(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/14$), so they force no degeneracy through irreducible dimensions. Decomposing along the characters χ of the deck group A, one expects pieces of dimension 18 for $\chi = 1$ (the remainder of X_3) and 12 for $\chi \neq 1$. This is consistent with $18 + 12(|A| - 1) = 4n + 6$, but we have not verified it piece by piece. The coincidences of values between pieces are observed, not explained.

Proposition 4.4.2 (Splitting, but not immediately)**.** *Let H_0 be Hermitian with a repeated eigenvalue, and let V be Hermitian with simple spectrum. Then:*

(i) $H_0 + \varepsilon V$ has simple spectrum for all but finitely many $\varepsilon \in \mathbb{R}$;

(ii) an eigenvalue E of H_0 of multiplicity m splits as $E + \varepsilon\mu_i + O(\varepsilon^2)$, where the μ_i are the eigenvalues of $P_E V P_E$ [26];

(iii) for fixed N, as $\varepsilon \to 0$ the spacing measure of $H_0 + \varepsilon V$ converges weakly to that of H_0, which has an atom at 0.

So "any $\varepsilon > 0$ removes the degeneracy" is true in the sense of (i), but "any $\varepsilon > 0$ produces Wigner–Dyson statistics" is false in the sense of (iii). The same holds for the unitary family $U_0 e^{i\varepsilon V}$ on compact ε-intervals.

Proof. (i) The discriminant of $\det(\lambda - H_0 - \varepsilon V)$ is a polynomial in ε. It is not identically zero, because $\varepsilon^{-1}(H_0 + \varepsilon V) \to V$ has simple spectrum. (ii) is first-order degenerate perturbation theory. (iii) Eigenvalues depend continuously on ε. For the unitary family, the discriminant is real analytic in ε. □

Proposition 4.4.3 (Brody is not Dyson)**.** *Let $P_b(s) = (b+1)\alpha_b s^b \exp(-\alpha_b s^{b+1})$ with $\alpha_b = \Gamma\big(\frac{b+2}{b+1}\big)^{b+1}$, normalized to unit mass and mean. Then $P_0 = e^{-s}$, and $P_1 = \frac{\pi}{2} s e^{-\pi s^2/4}$ is the GOE Wigner surmise. But $P_2 \neq \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$, the GUE surmise: their logarithms decay like $-s^3$ and $-s^2$ respectively. The Brody parameter is therefore not Dyson's index. The family is an interpolation [23], not a consequence of perturbation theory, and no "crossover formula" P_β with $\beta = 2$ for GUE follows from it.*

Remark 4.4.4 (Numerical observations). Let U_3 be the orthogonal 226×226 remainder of Y_{168} above. Take $U_\varepsilon = U_3 e^{i\varepsilon V}$, with $V = (A + A^*)/(2\sqrt{N})$ and A a complex ("GUE") or real ("GOE") Gaussian matrix, and 12 samples per ε (seed fixed). We use the unfolding-free ratio $\langle r \rangle$ of consecutive spacings; the reference values are 0.386 (Poisson), 0.531 (GOE), 0.600 (GUE) [21].

ε	10^{-3}	10^{-2}	10^{-1}	1	3
GUE-type V: $\langle r \rangle$	0.509	0.513	0.527	0.595	0.593
spacings $< 0.1\times$mean	0.929	0.929	0.134	0.001	0.001
GOE-type V: $\langle r \rangle$	0.523	0.510	0.538	0.601	0.595
spacings $< 0.1\times$mean	0.929	0.929	0.261	0.003	0.001

(1) For $\varepsilon \leq 10^{-2}$ the spectrum is still 93% clustered. The moderate $\langle r \rangle$ there is an artefact: the ratios inside a split cluster follow $P_E V P_E$ (Proposition 4.4.2(ii)). (2) The crossover occurs around $\varepsilon \sim 0.1$, the scale at which the splittings reach the mean spacing $2\pi/226$. (3) At saturation both kinds of V give GUE-like values. U_ε has no antiunitary symmetry even for real V, since $\overline{U_\varepsilon} = U_3 e^{-i\varepsilon V}$. GOE statistics would require a time-reversal symmetry of the perturbed operator, not "saturation".

Remark 4.4.5 (Ramanujan and chaos). The two properties are about different things. Ramanujan is a statement about *where* the nontrivial spectrum lies: the q-circle, Theorem 4.3.3. Chaos is about *how* the phases on the $\sqrt{q}$-circle correlate. The five covers of this volume are all Ramanujan, and their remainders are maximally degenerate. What is true is structural. The decomposition $\mathbb{C}^E = \operatorname{im}\Psi \oplus W$ separates the two questions. On $\operatorname{im}\Psi$ the vertex pencil lives, and there sits the Ramanujan condition. On W lives an exact unitary quantum map U, in the sense of quantum graphs [27], whose eigenphase statistics is the natural object for a Bohigas–Giannoni–Schmit type question [22, 24]. We do not prove any universality statement for U.

4.5 The example Y_{21}

On Y_{21}, $|E| = 147$. The joint eigenvalues are $7\omega^k$ (trivial) and $a = 0$ with multiplicity 18 (Corollary 3.3.5), and 0 is tempered ($z = (1, \omega, \omega^2)$). $\dim W = 90$ and $\operatorname{rank}\Psi = 57$. The

characteristic polynomial is

$$\det(I - uL_E) = (1 - 64u^3)\,(1 - 8u^3)^{18}\,(1 - 2u^3 + 8u^6)^9\,(1 + 5u^3 + 8u^6)^6,$$

The eigenvalues on both sides agree numerically, and the two integer polynomials agree exactly at $u = 1, 2, 3, -1, -2, 5, 7, -3$ (Bareiss determinants). Eight points do not determine a polynomial of degree 147, so the identity is checked, not proved. The factors $1 - 64u^3$ and $(1 - 8u^3)^{18}$ are proved by Theorem 4.3.2. The first factor is the Perron orbit. The second is $P_0(u) = 1 - 8u^3$, giving 54 eigenvalues of modulus 2. The last two are the remainder: $\lambda^3 \in \{1 \pm i\sqrt{7},\ (-5 \pm i\sqrt{7})/2\}$, all of modulus $2^{3/2}$. The spectrum is therefore highly degenerate: only 4 distinct values of λ^3 on W. For the other covers, see Table 4.3.

4.6 Breaking the arithmetic degeneracy: the crossover to GUE

This section takes up open problem (d) of Remark 4.8.1, and on the way settles most of (b). Throughout, $Y = Y_{168}$, $q = 2$, and U_3 denotes the compression of U^3 to $W \cap \mathbb{C}^{E_0}$, the real orthogonal $N \times N$ matrix of Remark 4.4.4, $N = 226$. Its 16 distinct eigenphases have multiplicities $21^8, 9^2, 7^4, 6^2$, the spacing measure has the atom 210/226 at $s = 0$ (Proposition 4.4.1), and the mean spacing is $2\pi/N = 0.0278$. We perturb along one-parameter families

$$U_\varepsilon = U_3\, e^{i\varepsilon V}, \qquad V = V^*, \qquad \operatorname{Tr} V^2 = N,$$

so that the eigenvalues of V have root mean square 1, and we ask three questions. At which ε does the degeneracy give way to level repulsion, and is that scale predicted by first-order perturbation theory? Which Wigner–Dyson class appears at saturation, and what decides it? Which perturbations respect the symmetry that produced the degeneracy, and what do they leave behind? Every numerical statement below is produced by `code/v3p4_quantum_chaos.py` (seed 2026, numpy and scipy, about six minutes) and is labelled as numerical; the propositions are proved.

Four kinds of perturbation

Let $W_0 \colon \mathbb{C}^N \to \mathbb{C}^E$ be an orthonormal basis of $W \cap \mathbb{C}^{E_0}$.

(G) *GUE-type:* $V = (A + A^*)/2$, A a complex Gaussian matrix, rescaled.

(O) *GOE-type:* the same with A real.

(S) *On-site (gauge-type):* d_e independent and uniform in $(-\pi, \pi)$ on the edges, and $V = W_0^t \operatorname{diag}(d) W_0$. The diagonal operator $\operatorname{diag}(d)$ does not preserve W, so $e^{i\varepsilon \operatorname{diag}(d)}$ is not a map of W; its compression V is Hermitian (real symmetric), and $e^{i\varepsilon V}$ is unitary on $W \cap \mathbb{C}^{E_0}$. This is a random $U(1)$ potential on the edges, seen from the remainder.

(L) *Local* $0/\pm 1$*:* M real symmetric on $\mathbb{C}^{E_0}$, with independent entries in $\{-1, 0, 1\}$ on the 2352 pairs of type-0 edges that share their tail or their head, and on the diagonal; $V = W_0^t M W_0$.

(S) and (L) are built from the geometry, and (L) is sparse; (G) and (O) are not. All four are rescaled to $\operatorname{Tr} V^2 = N$, so that the first-order scales of Proposition 4.6.1 are comparable.

The first-order scale

Proposition 4.6.1 (First-order splitting of a unitary family)**.** *Let U_0 be unitary, V Hermitian, and $U_\varepsilon = U_0 e^{i\varepsilon V}$. Let $e^{i\theta}$ be an eigenvalue of U_0 of multiplicity m, with orthogonal eigenprojection P, and let $\mu_1 \le \cdots \le \mu_m$ be the eigenvalues of PVP on* $\operatorname{ran} P$*. Then for small ε the m eigenvalues of U_ε near $e^{i\theta}$ have the phases*

$$\theta + \varepsilon\mu_j + o(\varepsilon), \qquad j = 1, \dots, m.$$

In particular the $m-1$ spacings inside the cluster are $\varepsilon(\mu_{j+1} - \mu_j) + o(\varepsilon)$.

Proof. $U_\varepsilon = U_0 + \varepsilon\, iU_0V + O(\varepsilon^2)$ is analytic in ε, and $e^{i\theta}$ is a semisimple eigenvalue of U_0, because U_0 is normal. By the reduction process of analytic perturbation theory [26], the eigenvalues of U_ε in the $e^{i\theta}$-group are $e^{i\theta} + \varepsilon\nu_j + o(\varepsilon)$, where the ν_j are the eigenvalues of $P(iU_0V)P = ie^{i\theta}PVP$ on $\operatorname{ran} P$; so $\nu_j = ie^{i\theta}\mu_j$, and $e^{i\theta}(1 + i\varepsilon\mu_j) + o(\varepsilon) = e^{i(\theta+\varepsilon\mu_j)} + o(\varepsilon)$. Since the eigenvalues of U_ε lie on the unit circle, their phases are $\theta + \varepsilon\mu_j + o(\varepsilon)$. □

Write $\delta\mu$ for the collection of the within-cluster first-order gaps $\mu_{j+1} - \mu_j$ over the 16 clusters (210 gaps), and let $f(\varepsilon)$ be the fraction of spacings of U_ε below one tenth of the mean. Proposition 4.6.1 gives two predictions. First, $f(\varepsilon) = f_1(\varepsilon) + o(1)$ with $f_1(\varepsilon) = N^{-1}\#\{\delta\mu : \varepsilon\,\delta\mu < 0.2\pi/N\}$, because the spacings between distinct clusters stay above 1.6 mean spacings; let $\varepsilon_{1/2}$ be the solution of $f_1(\varepsilon_{1/2}) = \frac{1}{2}$. Second, the first-order splittings reach the mean spacing at

$$\varepsilon_1 = \frac{2\pi/N}{\overline{\delta\mu}}.$$

For (G) the matrix PVP on an m-dimensional eigenspace is an $m \times m$ GUE matrix with entries of variance $1/N$, so its eigenvalues fill a semicircle of radius $2\sqrt{m/N}$ and $\overline{\delta\mu} \approx 4\sqrt{m/N}/(m-1)$; for $m = 21$ this is 0.061, and

$$\varepsilon_1 \approx \frac{\pi(m-1)}{2\sqrt{mN}} = 0.46.$$

This is sixteen times the "naive" scale $2\pi/N = 0.028$ of a splitting of order ε: inside a 21-fold cluster the first-order spread $4\sqrt{m/N} \approx 1.2$ is divided among 20 gaps. The naive scale happens to agree with $\varepsilon_{1/2} \approx 0.1\,\varepsilon_1$ within a factor of two, but for the wrong reason.

Remark 4.6.2 (Numerical: the crossover window). Table 4.1 reports $\langle r \rangle$ (the ratio statistic of Remark 4.4.4; Poisson 0.3863, GOE 0.5307, GUE 0.5996 [21]) and $f(\varepsilon)$ on a grid of 26 values $\varepsilon = 10^{-4}, \dots, 10$ (ratio $10^{0.2}$), averaged over 20 samples of V, for the four families (G), (O), (S), (L), for the time-reversal symmetric family (T) of Proposition 4.6.4 and for the deck-equivariant family (B) of Proposition 4.6.10. The observed $\varepsilon_{1/2}$ is the log-linear crossing of f through $\frac{1}{2}$; $\varepsilon_{\mathrm{sat}}$ is the first grid value at which $\langle r \rangle$ comes within 0.01 of its saturation value (the mean over $\varepsilon \ge 2$; the sampling error of one grid value is about 0.004). The findings are:

(1) For $\varepsilon \le 10^{-2}$ the spectrum is still 93% clustered, for every family.

(2) The first-order prediction $\varepsilon_{1/2}$ of Proposition 4.6.1, computed from the same samples of V, agrees with the observed value to within 10% in all six families (0.047–0.050 against 0.048–0.051), far inside the factor 3 one might have asked for. The scale is the same for all four non-symmetric families, including the sparse geometric ones, because the normalization $\operatorname{Tr} V^2 = N$ fixes $\overline{\delta\mu}$ to 0.059–0.062.

(3) Saturation of $\langle r\rangle$ needs splittings of a few mean spacings: $\varepsilon_{\rm sat}/\varepsilon_1$ lies between 1.5 and 3.5 (the grid resolves a factor $10^{0.2}$). The crossover window therefore spans more than a decade, from $\varepsilon_{1/2}\approx 0.05$, where the clusters have opened, to $\varepsilon_{\rm sat}\approx 1$, where the correlations between former clusters are those of the Wigner–Dyson ensemble. At $\varepsilon = 0.1$ only 13% of the spacings are small, but $\langle r\rangle = 0.537$ is still far from 0.60.

(4) At saturation all four non-symmetric families give the GUE value, 0.599–0.602, the symmetric family gives the GOE value 0.532, and the deck-equivariant family gives the Poisson value 0.385.

family	$\varepsilon_{1/2}$ obs.	$\varepsilon_{1/2}$ 1st order	$\overline{\delta\mu}$	ε_1	$\varepsilon_{\rm sat}$	$\varepsilon_{\rm sat}/\varepsilon_1$	$\langle r\rangle_{\rm sat}$	class
(G) GUE-type	0.0485	0.0470	0.0615	0.452	1.00	2.21	0.599	GUE
(O) GOE-type	0.0482	0.0473	0.0620	0.449	1.58	3.53	0.599	GUE
(S) on-site	0.0507	0.0498	0.0585	0.475	1.00	2.10	0.600	GUE
(L) local $0/\pm 1$	0.0488	0.0481	0.0608	0.458	1.58	3.46	0.602	GUE
(T) symmetric, real V	0.0488	0.0476	0.0631	0.441	1.00	2.27	0.532	GOE
(B) deck-equivariant	0.0343	0.0314	0.1403	0.198	0.10	0.50	0.385	Poisson

Table 4.1: Crossover scales (numerical; $N = 226$, 20 samples per ε, $\operatorname{Tr} V^2 = N$). The first-order columns are computed from the eigenvalues of PVP on the 16 eigenspaces of U_3 (for (T), of S_0); the reference scale $2\pi/N$ is 0.0278.

Which class, and why

Proposition 4.6.3 (No antiunitary time reversal for generic V)**.** *Let U_0 be unitary and V Hermitian, and $U_\varepsilon = U_0 e^{i\varepsilon V}$.*

(i) An antiunitary T satisfies $TU_\varepsilon T^{-1} = U_\varepsilon^{-1}$ for all real ε if and only if $TU_0T^{-1} = U_0^{-1}$ and $TVT^{-1} = U_0VU_0^{-1}$.

(ii) If such a T exists, then for all integers b, c

$$\kappa_{b,c}(V) := \operatorname{Tr}(VU_0^bVU_0^cV) - \operatorname{Tr}(VU_0^cVU_0^bV) = 0.$$

(iii) $\kappa_{b,c}$ is a cubic polynomial in the entries of V. Hence if $\kappa_{b,c}(V_0)\neq 0$ for one Hermitian (resp. real symmetric) V_0, then no such T exists for V outside a proper algebraic subset of the Hermitian (resp. real symmetric) matrices, which has Lebesgue measure zero.

(iv) For $U_0 = U_3$: numerically $|\kappa_{1,2}(V_0)| = 0.38$ for a sample of the real symmetric family (O). So for generic real or complex V the family $U_3e^{i\varepsilon V}$ has no antiunitary time reversal, and the real structure of U_3 does not help: complex conjugation K commutes with U_3 instead of inverting it, and $KU_\varepsilon K = U_3e^{-i\varepsilon\bar V}$ is $U_{-\varepsilon}$ for real V, not a symmetry of U_ε.

Proof. (i) If the two relations hold, then $Te^{i\varepsilon V}T^{-1} = e^{-i\varepsilon TVT^{-1}} = U_0e^{-i\varepsilon V}U_0^{-1}$, so $TU_\varepsilon T^{-1} = U_0^{-1}U_0e^{-i\varepsilon V}U_0^{-1} = U_\varepsilon^{-1}$. Conversely, $\varepsilon = 0$ gives the first relation. T is real-linear and continuous, so it commutes with $d/d\varepsilon$; differentiating at $\varepsilon = 0$ gives $T(iU_0V)T^{-1} = -iU_0^{-1}TVT^{-1}$ on the left and $\frac{d}{d\varepsilon}e^{-i\varepsilon V}U_0^{-1}|_0 = -iVU_0^{-1}$ on the right, whence $U_0^{-1}TVT^{-1} = VU_0^{-1}$.

(ii) For antiunitary T and any A, $\mathrm{Tr}(TAT^{-1}) = \overline{\mathrm{Tr}\,A}$: in an orthonormal basis $T = ZK$ with Z unitary, and $TAT^{-1} = Z\bar{A}Z^*$. Apply this to $A = VU_0^bVU_0^cV$. By (i), $TAT^{-1} = (U_0VU_0^{-1})U_0^{-b}(U_0VU_0^{-1})U_0^{-c}(U_0VU_0^{-1}) = U_0VU_0^{-b}VU_0^{-c}VU_0^{-1}$, whose trace is $\mathrm{Tr}(VU_0^{-b}VU_0^{-c}V)$. On the other hand $\overline{\mathrm{Tr}\,A} = \mathrm{Tr}\,A^* = \mathrm{Tr}(VU_0^{-c}VU_0^{-b}V)$. Equating and replacing (b,c) by $(-b,-c)$ gives $\kappa_{b,c}(V) = 0$.

(iii) is clear, and (iv) is the numerical certificate; the check also confirms $\kappa_{1,2} = 0$ to machine precision for the two symmetric constructions of Proposition 4.6.4, as (ii) requires. □

So the GUE value in Remark 4.4.4 and in Table 4.1 is not a property of the perturbations: it is forced by the absence of an antiunitary symmetry compatible with the perturbation. The converse is a construction. U_3 is far from symmetric ($\max|U_3 - U_3^t| = 0.51$), and it cannot be made symmetric without changing it: a real orthogonal symmetric matrix is an involution. But U_3 has symmetric unitary partners with the same spectrum.

Proposition 4.6.4 (Time-reversal symmetric families with the spectrum of U_3)**.** *Let $U_3 = ORO^t$ be a real Schur form, O real orthogonal and R block diagonal with 2×2 rotation blocks $\left(\begin{smallmatrix} a & b \\ -b & a\end{smallmatrix}\right)$ (U_3 is normal and has no real eigenvalue). Let Λ be the diagonal matrix with the entries $a \pm ib$ in the place of each block, and D the diagonal matrix with the entries $1, -1$ in the place of each block.*

(a) $S_0 = O\Lambda O^t$ is a symmetric unitary matrix with the spectrum of U_3. For real symmetric V the family $U'_\varepsilon = e^{i\varepsilon V/2}S_0e^{i\varepsilon V/2}$ consists of symmetric unitaries, has the time reversal K ($KU'_\varepsilon K = U'^{-1}_\varepsilon$), and is unitarily equivalent to $S_0e^{i\varepsilon V}$.

(b) $Z = ODO^t$ is a real symmetric involution with $ZU_3Z = U_3^{-1}$, so $T = ZK$ is an antiunitary involution with $TU_3T^{-1} = U_3^{-1}$. For a real symmetric V the family $U_3e^{i\varepsilon V}$ has the time reversal T if and only if $V = OMO^t$ with $M = YMY$, where $Y = DR$ is a real symmetric involution; these are the matrices $O(A + YAY)O^t$, A real symmetric.

Proof. (a) S_0 is unitary and symmetric since O is real and Λ is diagonal unitary, and $S_0 = O\Lambda O^{-1}$ has the eigenvalues of Λ, which are those of R. For $V^t = V$, $(e^{i\varepsilon V/2}S_0e^{i\varepsilon V/2})^t = e^{i\varepsilon V/2}S_0e^{i\varepsilon V/2}$. A symmetric unitary U' satisfies $KU'K = \bar{U}' = (U'^*)^t = (U'^{-1})^t = U'^{-1}$. Finally $e^{-i\varepsilon V/2}U'_\varepsilon e^{i\varepsilon V/2} = S_0e^{i\varepsilon V}$.

(b) A reflection conjugates a rotation to its inverse, so $DRD = R^t$ blockwise, and $ZU_3Z = ODRDO^t = OR^tO^t = U_3^t = U_3^{-1}$; Z is real symmetric with $Z^2 = 1$, so $T^2 = Z\bar{Z} = 1$. By Proposition 4.6.3(i) the family has the time reversal T iff $Z\bar{V}Z = U_3VU_3^{-1}$. For real $V = OMO^t$ this reads $DMD = RMR^t$, i.e. $M = DRMR^tD$. From $DRD = R^t$ one gets $R^tD = DR$, so $Y = DR$ is symmetric and $Y^2 = DRDR = R^tR = 1$, and the condition is $M = YMY$. The map $A \mapsto (A + YAY)/2$ is the orthogonal projection onto this subspace. □

Remark 4.6.5 (Numerical: symmetric families are GOE). Both constructions were run with real symmetric Gaussian V (rescaled), the time-reversal relations were verified to 10^{-10}, and at saturation ($\varepsilon = 3$, 40 samples): $\langle r\rangle = 0.534$ for (a) and 0.528 for (b), against the COE/GOE value 0.531 and a Haar-COE reference of 0.532 at $N = 226$. The non-symmetric families give 0.600–0.604, the Haar-CUE reference 0.602. Note that (a) and (b) have the *same* unperturbed spectrum as (G)–(L); what differs is the position of the real structure relative to the unperturbed operator: K inverts S_0 but commutes with U_3. The reality of U_3, which comes from the integrality of L_E, is a symmetry of the type "spectrum closed under conjugation", not a time reversal; Haar-random real orthogonal matrices have GUE bulk statistics (the Weyl density of $O(N)$ is a $\beta = 2$ ensemble in $\cos\theta$), and numerically their eigenphases in $(0,\pi)$ give $\langle r\rangle = 0.600$ at $N = 226$.

Dyson's classification applies to the pair (operator, antiunitary symmetry), never to the operator alone.

The saturated spacing distribution, and Brody once more

Remark 4.6.6 (Numerical: unfolded spacings at saturation). On the circle the density of eigenphases is uniform, so unfolding is the rescaling $s_i = \frac{N}{2\pi}(\theta_{i+1} - \theta_i)$, which has mean exactly 1 in every sample. For $\varepsilon = 3$ and 40 samples (9040 spacings per family) Table 4.2 gives the maximum-likelihood Brody parameter b of Proposition 4.4.3, the Kolmogorov distance $\sup_s |F_{\mathrm{emp}} - F|$ to the Poisson, GOE and GUE surmises and to the Brody curves with $b = 2$ and with the fitted b, and the Kullback–Leibler divergence $\sum_i p_i \log(p_i/q_i)$ on bins of width 0.1. Three Haar ensembles at $N = 226$ calibrate the finite-N and finite-sample effects.

(1) For the four non-symmetric families the empirical distribution is the GUE surmise to within the sampling resolution: Kolmogorov distance 0.006–0.009, the same as for Haar CUE (0.007), while the distance to the Brody $b = 2$ curve is 0.044–0.050, which is the exact distance $\sup |F_{P_2} - F_{\mathrm{GUE}}| = 0.043$ between the two curves. The Kullback–Leibler divergence to GUE is 0.001–0.002 against 0.044–0.050 to GOE.

(2) The fitted Brody parameter of a GUE spectrum is $b = 1.5 \pm 0.05$, both here and for Haar CUE. This is Proposition 4.4.3 seen from the data: $\log P_b(s)$ decays like $-\alpha_b s^{b+1}$, i.e. like $-s^3$ for $b = 2$, whereas the GUE surmise decays like $-4s^2/\pi$; the likelihood balances the small-s behaviour s^2 (which wants $b = 2$) against the Gaussian tail (which wants $b = 1$) and settles in between. A fitted $b \approx 2$ would be evidence *against* GUE, not for it. For the symmetric families $b = 0.93$–0.95, as for Haar COE (0.94), and the Kolmogorov distance to the GOE surmise is 0.008–0.014.

(3) For the deck-equivariant family (B) the fitted b is 0.02 and the Kolmogorov distance to the Poisson law is 0.013: a superposition of 56 independent spectra of sizes 4 and 6 is Poisson to this resolution.

The deck group, its blocks, and what explains the coincidences

The remaining question is what the degeneracy of U_3 is made of, and which perturbations respect it. $Y_{168} \to X_3$ is the voltage cover of `a2_common` with group $A = (\mathbb{Z}/2)^3 \times \mathbb{Z}/7$: its vertices are the pairs (a, s), $a \in A$, $s \in \mathbb{Z}/3$ the type, and the edge with label $x \in \mathbb{Z}/7$ out of (a, s) ends at $(a + \theta(s, x), s + 1)$, where $\theta\colon \{0, 1, 2\} \times \mathbb{Z}/7 \to A$ is the voltage assignment.

Proposition 4.6.7 (Deck equivariance and the character blocks)**.** *A acts on Y_{168} by $g \cdot (a, s) = (a + g, s)$, and this action is by type-preserving simplicial automorphisms. Consequently the induced permutations P_g of E commute with L_E, S, T, Q, preserve W and each graded piece $W \cap \mathbb{C}^{E_t}$, and their compressions D_g to $W \cap \mathbb{C}^{E_0}$ are orthogonal and commute with U_3. Hence*

$$W \cap \mathbb{C}^{E_0} = \bigoplus_{\chi \in \widehat{A}} W_\chi, \qquad U_3 W_\chi \subseteq W_\chi,$$

over the 56 characters of A. Numerically, $\dim W_\chi = 6$ for $\chi = 1$ (the type-0 remainder of X_3) and 4 for $\chi \neq 1$, so $6 + 55 \cdot 4 = 226$; this is the type-0 third of the count $18 + 12 \cdot 55 = 678$ anticipated in Section 4.4.

	$\langle r\rangle$	b	d_{Poi}	d_{GOE}	d_{GUE}	d_{P_2}	d_{P_b}	$\mathrm{KL}_{\mathrm{GOE}}/\mathrm{KL}_{\mathrm{GUE}}$
(G) GUE-type	0.603	1.51	0.283	0.070	0.006	0.048	0.015	0.047 / 0.001
(O) GOE-type	0.602	1.51	0.284	0.071	0.009	0.049	0.017	0.049 / 0.002
(S) on-site	0.600	1.48	0.280	0.069	0.007	0.050	0.017	0.044 / 0.002
(L) local $0/\pm 1$	0.604	1.54	0.286	0.073	0.006	0.044	0.016	0.050 / 0.002
(T_a) symmetric, S_0	0.534	0.95	0.211	0.008	0.076	0.111	0.006	0.002 / 0.078
(T_b) T-real V	0.528	0.93	0.209	0.014	0.080	0.118	0.007	0.003 / 0.083
(B) deck-equivariant	0.386	0.02	0.013	0.207	0.271	0.303	0.006	0.565 / 1.251
(P) pull-back on-site	0.359	0.01	0.071	0.247	0.311	0.337	0.071	0.852 / 1.731
Haar CUE	0.602	1.52	0.283	0.073	0.007	0.043	0.016	0.049 / 0.002
Haar COE	0.532	0.94	0.211	0.014	0.078	0.115	0.008	0.003 / 0.081
Haar $O(N)$, $(0,\pi)$	0.600	1.52	0.284	0.070	0.008	0.043	0.018	0.049 / 0.004

Table 4.2: Saturated regime, $\varepsilon = 3$, 40 samples, 9040 unfolded spacings per row (numerical). b is the maximum-likelihood Brody parameter, d_F the Kolmogorov distance of the empirical distribution to F, and P_2, P_b the Brody curves with $b = 2$ and with the fitted b. (T_a) is $e^{i\varepsilon V/2} S_0 e^{i\varepsilon V/2}$ and (T_b) is $U_3 e^{i\varepsilon V}$ with $V = O(A + YAY)O^t$ (Proposition 4.6.4); for Haar $O(N)$ the eigenphases in $(0,\pi)$ are used.

Proof. Translation by g commutes with the step maps $(a,s) \mapsto (a + \theta(s,x), s+1)$, so it maps the edge with label x out of (a,s) to the edge with the same label out of $(a+g,s)$, and the chambers, which are the closed label paths (x,y,z) of the triangle presentation, to chambers. Types are preserved by definition. L_E, S, T, Q are defined from edges, types and chambers alone, so they commute with P_g; then $W = \ker S \cap \ker T \cap \ker Q^t$ and the graded pieces are invariant, and the compression of $(L_E^t)^3$ commutes with that of P_g. A is abelian, so $\mathbb{C}^{E_0}$ is the direct sum of the isotypic subspaces of its characters, each D_g-invariant subspace splits accordingly, and U_3 preserves the pieces since it commutes with all D_g. The dimensions are counted by the script (the χ-isotypic part of $\mathbb{C}^{E_0}$ is 7-dimensional, since A acts freely on E_0 with 7 orbits, and W_χ is cut out by the three χ-twisted constraints S, T, Q^t). □

Remark 4.6.8 (Numerical: inside and across the blocks). The block spectra are as follows (phases of U_3 in degrees; $\mu = 2^{3/2}\lambda$ is the corresponding eigenvalue of $(L_E^t)^3$).

order of χ	blocks	phases	block polynomial in μ
1	1	152.11^3, 207.89^3	$(\mu^2 + 5\mu + 8)^3$
7, $\chi(1) \in \{\zeta, \zeta^2, \zeta^4\}$	3	69.30^3, 152.11	$(\mu - 2\pi)^3(\mu - \bar{\pi}^3)$
7, $\chi(1) \in \{\zeta^3, \zeta^5, \zeta^6\}$	3	207.89, 290.71^3	$(\mu - 2\bar{\pi})^3(\mu - \pi^3)$
2	7	66.72, 113.28, 246.72, 293.28	$\mu^4 + 11\mu^2 + 64$
14, $\chi\vert_{\mathbb{Z}/7}$ a square	21	59.32, 178.57, 201.67, 280.44	
14, $\chi\vert_{\mathbb{Z}/7}$ a non-square	21	79.56, 158.33, 181.43, 300.68	

Here $\zeta = e^{2\pi i/7}$ and $\pi = \frac{1+\sqrt{-7}}{2}$, the prime above 2 in $\mathbb{Q}(\sqrt{-7})$; the phases of the $\mathbb{Z}/7$ sector are $\pm\varphi$ and $\pm 3\varphi$ with $\varphi = \arg\pi = \arctan\sqrt{7} = 69.295°$, and the four values of λ^3 there are $2^{-3/2}$ times the four elements $2\pi, 2\bar{\pi}, \pi^3, \bar{\pi}^3$ of norm 8 in $\mathbb{Z}[\pi]$. The $\mathbb{Z}/7$ sector (characters trivial on $(\mathbb{Z}/2)^3$) is the pull-back of the type-0 remainder of Y_{21}, of dimension $6 + 6 \cdot 4 = 30 = 90/3$, and the

exponents 9 and 6 of the polynomial of Section 4.5 are visible as the multiplicities $9 = 3 \cdot 3$ of $2\pi, 2\bar{\pi}$ and $6 = 3 + 3 \cdot 1$ of $\pi^3, \bar{\pi}^3$. The block polynomials were verified to 10^{-6}.

(i) *Within a block.* The 49 blocks of characters of order 2 and 14 have simple spectrum, with a smallest gap of 0.40 rad, i.e. 14.5 mean spacings. The 7 blocks of the $\mathbb{Z}/7$ sector are not simple: they carry triple eigenvalues. (ii) *Across blocks.* Of the 1866 coincident pairs of eigenphases, 24 lie inside one block and 1842 across blocks; every one of the 16 values occurs in at least 3 blocks. The 56 block spectra form exactly the 6 multisets of the table.

The six classes are not a coincidence. The triangle presentation of `a2_common` has the chambers $(c, c+1, c+3)$, $c \in \mathbb{Z}/7$ (cyclic triples): the translates of the Singer difference set $\{0, 1, 3\}$ of $\mathrm{PG}(2, 2)$.

Proposition 4.6.9 (Automorphisms of X_3 explain the cross-block coincidences)**.** *(i) The affine maps $x \mapsto ax + b$ of $\mathbb{Z}/7$ with $a \in \{1, 2, 4\}$ preserve the chamber list, hence act on X_3 as type-preserving automorphisms fixing the three vertices; they form the Frobenius group $F_{21} = \mathbb{Z}/7 \rtimes \mathbb{Z}/3$.*

(ii) Every automorphism of X_3 lifts to Y_{168}, and the lift normalizes A; it acts on A by an automorphism ϕ and maps W_χ isometrically onto $W_{\chi\circ\phi^{-1}}$, commuting with U_3. Blocks in one orbit of this action are isospectral.

(iii) (Numerical.) The lifts of $x \mapsto x + 1$ and $x \mapsto 2x$ were constructed explicitly (vertex maps $(a, s) \mapsto (\phi(a) + c_s, s)$; the lift of $x \mapsto x + 1$ acts trivially on $\mathbb{Z}/7$ and by an element of order 7 of $\mathrm{GL}_3(\mathbb{F}_2)$ on $(\mathbb{Z}/2)^3$, the lift of $x \mapsto 2x$ by multiplication by 2 on $\mathbb{Z}/7$ and by an element of order 3). Their orbits on the 56 characters have sizes $1, 3, 3, 7, 21, 21$ and coincide with the six spectral classes. So 1824 of the 1842 cross-block coincidences are between blocks in one orbit, and are explained.

Proof. (i) Translation is obvious. Doubling maps the cyclic triple $(c, c+1, c+3)$ to $(2c, 2c+2, 2c+6)$, which is the cyclic rotation $(d, d+1, d+3)$ with $d = 2c + 6$. The maps generate F_{21}, and the chamber condition and the types are preserved, so these are automorphisms of X_3. (ii) Y_{168} is the cover of X_3 defined by the kernel of $\pi_1(X_3) \to H_1(X_3; \mathbb{Z}) \to H_1(X_3; \mathbb{Z}) \otimes \mathbb{Z}/14 = A$, since the voltages are a basis of the null spaces of the relator matrix modulo 2 and modulo 7. This kernel is characteristic, so every automorphism of X_3 lifts, and the lift normalizes the deck group and induces an automorphism ϕ of A. A lift τ is a type-preserving automorphism of Y_{168}, so it commutes with L_E, S, T, Q, preserves $W \cap \mathbb{C}^{E_0}$ and commutes with U_3 there; and $\tau D_g \tau^{-1} = D_{\phi(g)}$ gives $\tau W_\chi = W_{\chi\circ\phi^{-1}}$. (iii) is checked by the script: the lifts are found by solving $\phi(\theta(s, x)) - \theta(s, \pi(x)) = c_s - c_{s+1}$ over $\phi \in \mathrm{GL}_3(\mathbb{F}_2) \times (\mathbb{Z}/7)^\times$ (one solution each), the resulting edge permutations commute with L_E, S, T, Q and with U_3 and normalize A, and the orbit count is $1, 3, 3, 7, 21, 21$. □

What is *not* explained by A and $\mathrm{Aut}(X_3)$ is confined to the $\mathbb{Z}/7$ sector: the 24 coincidences inside a block (the triple 2π in each order-7 block, the triples $\pi^3, \bar{\pi}^3$ in the trivial block) and the 18 coincidences between the trivial block and the order-7 blocks. These are the exponents 9 and 6 of the Y_{21} polynomial, and the identification of the values with the norm-8 elements of $\mathbb{Z}[\pi]$, $\mathbb{Q}(\sqrt{-7})$ being the field of the quadratic Gauss sum of $\mathbb{Z}/7$, is as far as we go. This is what remains of open problem (b) for Y_{168}.

Proposition 4.6.10 (Deck-equivariant perturbations)**.** *Let V be Hermitian and commute with all D_g, i.e. $V = \bigoplus_\chi V_\chi$. Then $U_\varepsilon = U_3 e^{i\varepsilon V}$ preserves every W_χ, and $\det(\lambda - U_\varepsilon) = \prod_\chi \det(\lambda - U_{\chi,\varepsilon})$.*

Along such a family, eigenphases of different blocks cross: their coincidence is one real equation in ε, so for generic V it happens at isolated ε and transversally, without repulsion. Eigenphases of one block do not cross for generic V: the unitary $d \times d$ matrices with a repeated eigenvalue form a set of real codimension 3. Consequently the saturated statistics of the family is that of a superposition of 56 independent spectra of sizes 4 and 6, and in particular not Wigner–Dyson.

Proof. $e^{i\varepsilon V} = \bigoplus e^{i\varepsilon V_\chi}$ and $U_3 = \bigoplus U_\chi$ by Proposition 4.6.7, so the blocks are preserved and the characteristic polynomial factors. Two eigenphases from different blocks are two analytic functions of ε with independent perturbations, and their difference vanishes on a discrete set for generic V. For the codimension count, a unitary $d \times d$ matrix with exactly one double eigenvalue is $W \operatorname{diag}(\lambda_1, \lambda_1, \lambda_3, \dots, \lambda_d) W^*$, and the stabilizer of the diagonal matrix is $U(2) \times U(1)^{d-2}$ of dimension $d + 2$, so these matrices form a set of dimension $(d-1) + d^2 - (d+2) = d^2 - 3$. Levels of distinct blocks are therefore uncorrelated, and levels within a block repel; this is a superposition. □

Remark 4.6.11 (Numerical: deck-equivariant and pull-back perturbations). (B) *Independent blocks.* $V = \bigoplus_\chi B_\chi H_\chi B_\chi^*$ with independent GUE blocks H_χ of sizes 4 and 6, rescaled. The commutation with the D_g and the block invariance of U_ε were verified; a generic V of type (G) does not commute. At saturation every exact coincidence is lifted (no zero spacings in 40 samples), but $\langle r \rangle = 0.386$ (Poisson 0.3863), $f = 0.09$ (Poisson $1 - e^{-0.1} = 0.095$), and the Kolmogorov distance to the Poisson law is 0.013: the cross-block degeneracy is replaced by the absence of cross-block repulsion, which is what Proposition 4.6.10 predicts. The first-order scale is larger ($\overline{\delta\mu} = 0.14$, since a 4×4 block has few, wide gaps) and the crossover is correspondingly earlier ($\varepsilon_{1/2} = 0.034$, again within 10% of the first-order prediction 0.031).

(P) *Pull-backs from X_3.* An on-site potential constant on the 21 deck orbits of edges (the pull-back of an on-site potential on X_3; 7 free parameters on the type-0 edges) is deck-equivariant, and it keeps an exact degeneracy: in every sample, at every ε tested, the three blocks of each class $\{\chi, \chi^2, \chi^4\}$ of characters of order 7 remain isospectral, so the spectrum has 8 triple eigenphases (16 zero spacings per sample, 640 in 40), and $\langle r \rangle = 0.359$. The lifted automorphism $x \mapsto 2x$ of Proposition 4.6.9 maps W_χ to W_{χ^2} but conjugates the potential d to $d \circ (2x)^{-1}$, so it does not explain this; the entrywise moduli of the orthogonal projections onto W_χ and W_{χ^2} agree to 10^{-16}, while the obvious gauge $\operatorname{diag}(\chi(a_e))$ does not map W_χ to W_{χ^2}. We record the observation without a proof.

So the "discrete subgroup symmetry" statement is this: the degeneracy of U_3 is the A-block structure together with the F_{21} permutation of the blocks and the arithmetic of $\mathbb{Q}(\sqrt{-7})$ inside the $\mathbb{Z}/7$ sector. A perturbation that commutes with A keeps the blocks and produces Poisson statistics at best; the GUE statistics of Table 4.2 require a perturbation that mixes the blocks, and the GOE statistics require in addition an antiunitary symmetry of the perturbed family, which the real structure of U_3 does not provide.

The full remainder on W and the window $\varepsilon \leq 1/\dim W$

The question is sometimes posed for the remainder U itself, on all of W ($\dim W = 4n + 6 = 678$), and for the window $0 \leq \varepsilon \leq 1/\dim W$: does $\langle r \rangle$ cross over to the GUE value there? It does not, and the reason is a limit that can be computed exactly. Everything below is produced by `code/v3p4_gue_transition.py` (seed 2027, about 25 minutes; data and figures in `data/`).

Let $U_0 = W^t L_E W/\sqrt{q}$ on W, $N = \dim W = 678$. It is real orthogonal, and the grading $J = \omega^{\text{type}}$ ($\omega = e^{2\pi i/3}$) satisfies $JU_0J^{-1} = \omega^{-1}U_0$, because L_E raises the type by one. Hence

$\operatorname{spec} U_0$ is invariant under rotation by $2\pi/3$ and consists of the cube roots of $\operatorname{spec} U_3$: 48 distinct eigenphases with multiplicities $21^{24}, 9^6, 7^{12}, 6^6$, and the same atom $630/678 = 0.929$ at $s = 0$. Every eigenvalue is non-real, so every eigenspace P_k is a complex subspace on which $P_k V P_k$ is a complex Hermitian matrix even for real V. Besides (G) and (O) we use the geometric phase perturbation (S): a random $U(1)$ potential $d_e \in (-\pi, \pi)$ on the edges, $V = W^t \operatorname{diag}(d) W$. It commutes with J, so $U_0 e^{i\varepsilon V}$ stays $\mathbb{Z}/3$-covariant; for all three families $\kappa_{1,2}(V) \neq 0$, so no family has an antiunitary time reversal (Proposition 4.6.3).

Proposition 4.6.12 (The first-order plateau)**.** *Let U_0 be unitary on $\mathbb{C}^N$ with distinct eigenvalues $e^{i\theta_k}$ of multiplicities $m_k \geq 2$ and eigenprojections P_k, let V be Hermitian such that every $P_k V P_k$ has simple spectrum $\mu_{k,1} < \cdots < \mu_{k,m_k}$ on $\operatorname{ran} P_k$, and put $g_{k,j} = \mu_{k,j+1} - \mu_{k,j}$. Let $\langle r \rangle(\varepsilon)$ be the mean over the N consecutive pairs of spacings of $U_\varepsilon = U_0 e^{i\varepsilon V}$ on the circle of $\min(s_i, s_{i+1}) / \max(s_i, s_{i+1})$. Then*

$$\lim_{\varepsilon \to 0^+} \langle r \rangle(\varepsilon) = \langle r \rangle_1 := \frac{1}{N} \sum_k \sum_{j=1}^{m_k - 2} \frac{\min(g_{k,j}, g_{k,j+1})}{\max(g_{k,j}, g_{k,j+1})} \;\leq\; \frac{N - 2K}{N},$$

K the number of clusters, while $\langle r \rangle(0)$ is undefined (pairs $0/0$).

Proof. By Proposition 4.6.1 the eigenphases near θ_k are $\theta_k + \varepsilon \mu_{k,j} + o(\varepsilon)$, so for small $\varepsilon > 0$ they stay ordered as the $\mu_{k,j}$, the $m_k - 1$ spacings inside the cluster are $\varepsilon g_{k,j} + o(\varepsilon)$, and the K spacings between neighbouring clusters tend to the positive gaps of $\operatorname{spec} U_0$. Among the N pairs of consecutive spacings, $m_k - 2$ lie inside the k-th cluster, and their ratios tend to $\min(g_{k,j}, g_{k,j+1}) / \max(g_{k,j}, g_{k,j+1})$ because the ratio is homogeneous of degree 0; the remaining $2K$ pairs consist of one spacing $O(\varepsilon)$ and one bounded below, so their ratio tends to 0. Each ratio is at most 1. □

So the ratio statistic does not see the size of the splitting at all, only the shape of the first-order spectra $P_k V P_k$: it jumps at $\varepsilon = 0^+$ to $\langle r \rangle_1$ and stays there as long as the clusters have not met. For Y_{168}, $N - 2K = 582$, and if each $P_k V P_k$ is a GUE-like matrix the internal ratios average about 0.59, which predicts $\langle r \rangle_1 \approx 0.858 \cdot 0.59 \approx 0.51$, between Poisson and GOE and *not* the GUE value.

Remark 4.6.13 (Numerical: the window and beyond). (1) *Window* $0 \leq \varepsilon \leq 1/N = 1.475 \cdot 10^{-3}$, 21 values, 16 samples. For all three families $\langle r \rangle$ is flat on the whole window and equal to the first-order plateau computed from the same V:

family	$\varepsilon = \frac{1}{20N}$	$\frac{1}{4N}$	$\frac{1}{2N}$	$\frac{1}{N}$	$\langle r \rangle_1$ (Prop. 4.6.12)
(G) GUE-type	0.5083	0.5084	0.5085	0.5087	0.5083 ± 0.0030
(S) geometric phase	0.5082	0.5082	0.5083	0.5085	0.5082 ± 0.0072
(O) GOE-type	0.5096	0.5096	0.5097	0.5099	0.5095 ± 0.0031

At $\varepsilon = 1/N$ the fraction of spacings below 0.1 is still 0.929, the atom itself, and the first-order spread of a 21-fold cluster is 0.11 mean spacings. The real family (O) gives the same plateau as (G) because the eigenspaces of U_0 are complex.

(2) *Extended scan* $\varepsilon = 10^{-4}, \ldots, 10$ (26 values, 8 samples; Figure 4.1). $\langle r \rangle$ stays on the plateau (0.50–0.52 with 8 samples) up to $\varepsilon \approx 10^{-2}$, rises through the GOE value near $\varepsilon \approx 0.03$, and reaches the GUE value; above $\varepsilon \approx 0.4$ it fluctuates around 0.60. The scales are $\varepsilon_{1/2} = 0.028$ (half of the spacings have opened), $\varepsilon_1 = 0.26$ (mean first-order splitting = mean spacing; the formula $\pi(m-1)/(2\sqrt{mN})$ with $m = 21$ gives 0.263), and $\varepsilon_{\text{sat}} = 0.40$ for (G), (O) and 0.63 for (S), i.e. 270–430 times $1/N$; $\langle r \rangle_{\text{sat}} = 0.601$, 0.600, 0.596.

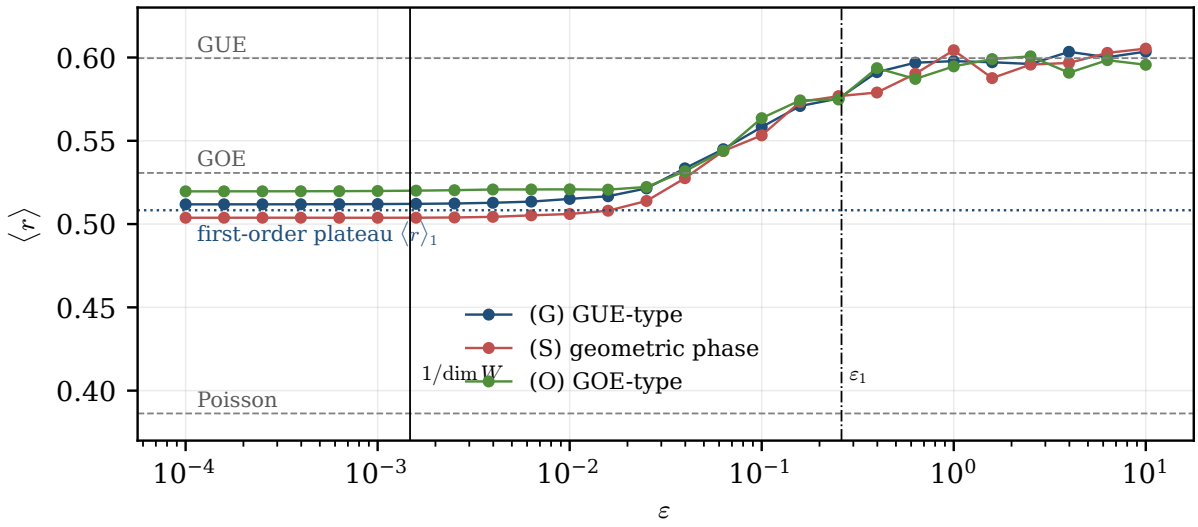


Figure 4.1: The ratio statistic $\langle r\rangle$ of $U_0 e^{i\varepsilon V}$ on the full space W of Y_{168} ($\dim W = 678$) against the perturbation strength, $\varepsilon = 10^{-4}, \dots, 10$ (26 values, 8 samples each), for a GUE-type perturbation (G), the geometric phase perturbation (S) and a GOE-type perturbation (O), all with $\operatorname{Tr} V^2 = \dim W$. The dotted line is the first-order plateau $\langle r\rangle_1 = 0.5083$ of Proposition 4.6.12; the sample-to-sample scatter of the plateau is visible below $\varepsilon \approx 10^{-2}$. The solid vertical line marks the window edge $\varepsilon = 1/\dim W$, deep inside the plateau, and the dash-dotted line the first-order scale $\varepsilon_1 = 0.26$ at which the mean splitting reaches the mean spacing. Saturation at the GUE value 0.600 occurs for $\varepsilon \gtrsim 0.4$. Data of `code/v3p4_gue_transition.py`, drawn by `code/make_figures.py`.

(3) *Spacing distributions* (24 samples, 16272 spacings each). At $\varepsilon = 1/N$ the empirical $P(s)$ is the atom plus a small continuous part; the maximum-likelihood Brody parameter sits at the boundary $b = 0$ of the family, which cannot represent an atom, and the Kolmogorov distances to the Poisson, GOE and GUE laws are 0.91–0.93. At $\varepsilon = \varepsilon_1$: $\langle r\rangle = 0.577$ and 0.580, still $b = 0$ because of the excess of small spacings. At $\varepsilon = 3$: $\langle r\rangle = 0.601$ (G), 0.597 (S), $b = 1.53$ and 1.51 (the GUE value of the estimator, Remark 4.6.6), Kolmogorov distance 0.005 and 0.010 to the GUE surmise against 0.069 and 0.072 to GOE.

So the answer to the question is negative in the window and positive outside it: on $0 < \varepsilon \le 1/\dim W$ the statistic is pinned at the plateau $\langle r\rangle_1 \approx 0.51$ of Proposition 4.6.12, and the GUE value 0.600 is reached only for $\varepsilon \gtrsim 0.4$, where the first-order splittings exceed the mean spacing; the scale is $\varepsilon_1 \propto (m-1)/\sqrt{mN}$, not $1/N$.

Remark 4.6.14 (Scope). Proved: Propositions 4.6.1, 4.6.3(i)–(iii), 4.6.4, 4.6.7 except the block dimensions, 4.6.9(i)–(ii), 4.6.10, 4.6.12. Numerical, with the code as the record: Tables 4.1 and 4.2, Remark 4.6.13, the block dimensions and spectra, the lifts and their orbits, the certificate $\kappa_{1,2}(V_0) \neq 0$, and the persistence of the Galois triples under pull-back potentials. Not proved: any universality statement for the family U_ε as $n \to \infty$; the numbers here concern one complex, Y_{168}, and what they establish is that on this complex the crossover scale is the first-order splitting scale, and the symmetry class is decided by the antiunitary symmetries of the perturbed family, in the sense of [22, 27], not by the arithmetic that produced the degeneracy.

4.7 Verification battery

check	complexes	result
(R) sums q^2; block-cyclic; the relations of Proposition 4.1.4	five covers	exact
(U) W is L_E^t-invariant, $\|L_E^t f\| = \sqrt{2}\|f\|$; $\operatorname{rank}\Psi = 3n-6$, $\dim W = 4n+6$	five covers	numeric 10^{-10}
(F) spec L_E as in Theorem 4.3.2; $\|\lambda\| \geq \sqrt{2}$; Ramanujan; circle counts $(q, \sqrt{q})$: Y_{21} $(54, 90)$, Y_{24} $(63, 102)$, Y_{42} $(117, 174)$, Y_{84} $(243, 342)$, Y_{168} $(495, 678)$	five covers	numeric 10^{-6}
(X) the Y_{21} product formula at $u = 1, 2, 3, -1, -2, 5, 7, -3$	Y_{21}	exact
(K) $\ker\Psi$ 6-dimensional, trivial characters; Γ_q property and disjointness from tempered a $(q \leq 5)$	five covers	numeric
(D) $J - \operatorname{Inc}(\mathrm{PG}(d,q))$: sums q^d, q^{d-1} on $\mathbf{1}^\perp$, $(d,q) \in \{(2,2),(2,3),(3,2),(3,3),(4,2)\}$	—	numeric 10^{-8}
(S) distinct λ^3 on W: $4, 6, 16, 16, 16$; atom 210/226; perturbation table	Y_{168}	numeric (seed 1)
(B) Brody $b = 1$ is the GOE surmise; $b = 2$ differs from GUE by 0.162 (sup)	—	numeric

Table 4.3: Checks run by `code/v3p4_verify.py`. Section 4.6 is checked by `code/v3p4_quantum_chaos.py` and `code/v3p4_gue_transition.py`.

4.8 Scope, residue, corrections

Remark 4.8.1 (Solved, open). *Solved:* the local form and intertwining relations of L_E; the unconditional unitary remainder, for all d; the lower bound $|\lambda| \geq q^{(d-1)/2}$; the factorization and the Ramanujan equivalence for $d = 2$; the exact kernel of Ψ; the exact polynomial for Y_{21}. *Open:* (a) the vertex block for $d \geq 3$, which needs one operator like Q for each intermediate type; (b) an explanation of the few distinct remainder values on abelian covers; (c) a (L)-complex with $q \geq 2$ that is not Ramanujan, or a proof that none exists among quotients of a given building; (d) any universality statement for the eigenphases of U along families.

Remark 4.8.2 (Corrections to the proposed programme). (1) The block structure of L_E is cyclic in the type grading, not tridiagonal. (2) The unconditional theorem is the unitary remainder on W, with $\rho = q^{(d-1)/2}$. The spectral radius of L_E away from the Perron orbit is q, not $q^{(d-1)/2}$, and the full Riemann hypothesis for $\det(I - uL_E)$, in its two-circle form, is *equivalent* to the Ramanujan property (Theorem 4.3.3). (3) Degenerate spectra have an atom at 0 and are not Poisson (Proposition 4.4.1). $\mathrm{PGL}_{d+1}(\mathbb{F}_q)$ acts on links, not on Y. (4) L_E is not Hermitian, so $L_E + \varepsilon V$ is not a Hamiltonian. The natural unitary object is U on W. A "nonlinear" perturbation has no spectrum; we use linear ones. (5) Small ε splits degeneracies but does not produce Wigner–Dyson statistics (Proposition 4.4.2); the crossover requires splittings of the order of the mean spacing. (6) Brody's parameter is not Dyson's β: $b = 2$ is not GUE (Proposition 4.4.3). GOE versus GUE is decided by antiunitary symmetries (Remark 4.4.4(3)). (7) The "unification" is the decomposition $\operatorname{im}\Psi \oplus W$, not an implication between Ramanujan and chaos (Remark 4.4.5).

Acknowledgement of provenance and caveats

Unrefereed. The definition of Ramanujan complexes is that of [28]. The two-circle characterization is in the spirit of [25], and it is proved here independently. The perturbation-theoretic and random-matrix references [26, 23, 21, 27, 22, 24] are cited for standard facts and terminology. The bibliographic data of these references have been checked against Crossref and zbMATH.

Chapter 5

Chamber Games and Higher-Order Interactions: Pairwise Reductions, Interfaces, and Entrainment

The proposed programme places a three-player public-goods game on every chamber of an $\widetilde{A}_2$ complex, with strategies updated by the Fermi rule. It asks for four claims. (a) A "topological shielding cage" built from the $\mathrm{PG}(2,q)$ flags lowers the critical synergy to $r_c = 1 + 2/(q+1)$. (b) This is far below the value $r_c > 3$ of "traditional regular networks", a "topological catalysis" that grows with the thickness q. (c) A $\mathbb{Z}/3$-chiral rock–paper–scissors variant has a torus attractor and persistent oscillations in the replicator dynamics. (d) There is a global phase diagram in $|F|/|V|$. We prove the following.

(1) *Pairwise reduction.* The public-goods payoff is linear in the strategies. On *every* 2-complex the chamber game is therefore the additive pairwise game $\pi_i = (\frac{r}{3} - 1)D_2(i)s_i + \frac{r}{3}\sum_j w_{ij}s_j$ on the 1-skeleton, where w_{ij} is the number of chambers on the edge ij. Under (L), $w \equiv q+1$. After normalization by D_2, the flag geometry disappears entirely, and only the degree 2κ of the 1-skeleton remains. There is no higher-order mechanism to exploit.

(2) *Thresholds.* Defection is strictly dominant in every configuration for $r < 3$. The mean-field (replicator/Fermi) threshold is exactly $r_c = 3$, for every q and every $|F|/|V|$. At zero noise, a cooperator cluster A is stable iff $r > 3\kappa/(\kappa + \delta(A))$, where $\delta(A)$ is an integer boundary margin. So $5/3 = 1 + 2/(q+1)$ at $q = 2$ is not the threshold of any cluster on any $q = 2$ complex. A single mutant needs $r > 3\kappa/(\kappa - 1)$ ($= 3.5$ for $q = 2$). The best natural cage, the vertex star of Y_{168}, gives $7/3$. On $Y_{21} = K_{7,7,7}$ no cluster resists below $r = 3$.

(3) *No catalysis.* In Monte Carlo runs at noise $K = 0.1$, Y_{168} behaves like a random 14-regular graph: cooperators die at $r = 2.7$, and the transition is near 3. The *thin* complex, the triangular lattice T_{12}, keeps 40% cooperators at $r = 2.7$. Thickness makes the complex an expander, and expanders are mean-field. The effect claimed in (b) is reversed.

(4) *Cyclic dominance.* For the three-strategy replicator dynamics, $\frac{d}{dt}\log(x_1x_2x_3) = (a-b)(1-3\sigma)$. So there is global convergence ($a > b$), a heteroclinic attractor ($a < b$), or a neutral family of closed orbits ($a = b$). There are never limit cycles, and never a torus, since the phase space is a triangle. With mutation rate μ a Hopf bifurcation occurs at $\mu_H = (b-a)/18$. Below it there is a stable limit cycle (numerically supercritical): this is the correct form of "persistent oscillations".

(5) *Nonlinear and chiral chamber games (Sections 5.5, 5.6).* A chamber game is pairwise iff its synergy is affine. Symmetric chamber games are exact potential games, so best-response learning

selects a state that does not depend on the geometry. Role-dependent (chiral) games break the potential. In the supply-chain game with hold-up, logit learning in the mean field has a Hopf bifurcation at noise $K_H = \frac{1}{4}$ and periodic orbits below it. On large sparse complexes, however, the oscillation of class densities decays with size. Under imitation, flat walls of the thin lattice give the threshold $\frac{18}{11}$, while thick stars give $\frac{63}{19}$. We find no spectral-gap law, and we show that the economic "iff" proposed for dominant voluntary cooperation is false.

Status of the results

Every numbered statement is proved, with two exceptions. The finite-noise thresholds are Monte Carlo observations (Remark 5.3.1). In Theorem 5.4.2 the uniqueness of the interior equilibrium and the supercriticality are verified numerically. All checks are in `code/v3p5_verify.py` (Table 5.1). Notation follows Chapters 1–4: Y satisfies (L), $\kappa = q^2 + q + 1$, the 1-skeleton is 2κ-regular, every vertex lies in $D_2 = \kappa(q+1)$ chambers, and $|F|/|V| = \kappa(q+1)/3$.

5.1 The model and its pairwise reduction

Definition 5.1.1 (Chamber public-goods game)**.** A configuration is $s \in \{0,1\}^V$, with $1 = C$. In a chamber Δ each cooperator pays 1; the pool is multiplied by r and shared equally among the three members. The payoff of i is $\pi_i(s) = \sum_{\Delta \ni i}\big(\frac{r}{3}\, n_C(\Delta) - s_i\big)$, and the normalized payoff is $\hat{\pi}_i = \pi_i / D_2(i)$. Under the *Fermi rule* [32] a uniformly random vertex i picks a uniformly random neighbour j and adopts s_j with probability $f(\hat{\pi}_i - \hat{\pi}_j)$, where $f(z) = 1/(1+e^{z/K})$. The master equation is

$$\frac{d}{dt}P(s) = \sum_i \big[w_i(s^{(i)})P(s^{(i)}) - w_i(s)P(s)\big], \qquad w_i(s) = \frac{1}{n \deg i} \sum_{j\sim i,\ s_j \neq s_i} f(\hat{\pi}_i - \hat{\pi}_j),$$

where $s^{(i)}$ is s with s_i flipped.

Theorem 5.1.2 (Pairwise reduction)**.** *On any finite 2-complex,*

$$\pi_i = \Big(\frac{r}{3} - 1\Big) D_2(i)\, s_i + \frac{r}{3} \sum_j w_{ij} s_j, \qquad w_{ij} = \#\{\Delta \in F :\ i,j \in \Delta\}.$$

So the chamber game is an additive pairwise game on the weighted 1-skeleton. Under (L), $w_{ij} = (q+1)[i \sim j]$, $D_2 = \kappa(q+1)$, and

$$\hat{\pi}_i = \Big(\frac{r}{3} - 1\Big) s_i + \frac{r}{3\kappa} N_C(i), \qquad N_C(i) = \#\{j \sim i : s_j = 1\}.$$

The normalized dynamics depends on the complex only through its 1-skeleton.

Proof. $n_C(\Delta) = \sum_{j\in\Delta} s_j$. Summing $\frac{r}{3} n_C(\Delta)$ over the chambers at i counts s_i $D_2(i)$ times and s_j w_{ij} times. Under (L), each edge lies in $q+1$ chambers. □

Remark 5.1.3. The "shielding cage from the $\mathrm{PG}(2,q)$ flags" has no separate existence. The flags enter only through the constant $q+1$, which cancels after normalization. Any two complexes with the same 1-skeleton and constant w have identical dynamics, whether or not they satisfy the flag condition. This is a structural feature of public goods, which are linear. A nonlinear group payoff (threshold or volunteer games) would be genuinely higher-order.

Proposition 5.1.4 (Dominance and the mean-field threshold)**.** *(i) In every configuration, switching i from D to C changes π_i by $D_2(r/3-1)$. So D is strictly dominant for $r<3$ and C for $r>3$.*

(ii) In the well-mixed (mean-field) limit the Fermi dynamics gives $\dot{x} = x(1-x)\tanh(\frac{r/3-1}{2K})$, and the replicator dynamics gives $\dot{x} = x(1-x)(r/3-1)$. Both have the threshold $r_c = 3$, independent of q and $|F|/|V|$.

Proof. (i) is read off Theorem 5.1.2. (ii) With independent neighbours, $\mathbb{E}(\hat{\pi}_C - \hat{\pi}_D) = r/3-1$. The rates $D \to C$ and $C \to D$ are $x(1-x)f(-\Delta)$ and $x(1-x)f(\Delta)$, and $f(-\Delta) - f(\Delta) = \tanh(\Delta/2K)$. □

5.2 Zero noise: pairs, clusters, cages

As $K \to 0$ the Fermi rule becomes "imitate a strictly better neighbour".

Lemma 5.2.1 (Pair criterion)**.** *For adjacent i (C) and j (D), $\hat{\pi}_i > \hat{\pi}_j$ iff $\kappa + N_C(i) - N_C(j) > 0$ and $r > 3\kappa/(\kappa + N_C(i) - N_C(j))$. Moreover $N_C(i) - N_C(j) \le 2\kappa - 2 - c_{ij} \le 2\kappa - q - 3$, where $c_{ij} \ge q+1$ is the number of common neighbours.*

Proof. $\hat{\pi}_i - \hat{\pi}_j = \frac{r}{3} - 1 + \frac{r}{3\kappa}(N_C(i) - N_C(j))$. Let C_{ij} be the set of common neighbours and m the number of cooperators in it. Since $s_j = 0$, $N_C(i) \le m + (2\kappa - 1 - c_{ij})$. Since $s_i = 1$, $N_C(j) \ge m+1$. The edge ij lies in $q+1$ chambers, whose third vertices are common neighbours. □

For a set $A \subsetneq V$ of cooperators, define the *boundary margin* $\delta(A) = \min\{N_A(i) - N_A(j) : i \in A,\ j \notin A,\ i \sim j\}$, where $N_A(x)$ is the number of neighbours of x in A.

Theorem 5.2.2 (Cluster stability at zero noise)**.** *Under the zero-noise rule, no cooperator of A ever imitates a defector iff $\kappa + \delta(A) > 0$ and $r > r_A := 3\kappa/(\kappa + \delta(A))$. In particular:*

(i) a cluster can resist for some $r < 3$ only if $\delta(A) > 0$;

(ii) a single defector in a sea of cooperators, and a single cooperator in a sea of defectors, both change sides at $r_* = 3\kappa/(\kappa-1) = 3 + \frac{3}{q(q+1)}$. This is 3.5 for $q=2$, and it is above *the mean-field value;*

(iii) every zero-noise cluster threshold on a $q=2$ complex has the form $21/(7+\delta)$ with $\delta \in \mathbb{Z}$. Since $21/(7+\delta) = 5/3$ would need $\delta = 5.6$, the value $1 + 2/(q+1) = 5/3$ is not the threshold of any cluster.

Proof. Apply Lemma 5.2.1 to every boundary pair. For (ii), use the sets $V \setminus \{v\}$ ($\delta = -1$) and $\{v\}$: a lone C beats a D neighbour with one cooperating neighbour iff $\frac{r}{3} - 1 > \frac{r}{3\kappa}$. □

Proposition 5.2.3 (Y_{21} has no cages)**.** *On Y_{21} every proper nonempty A has $\delta(A) \le 0$, with equality iff $|A \cap V_t|$ is the same for all three types. So $r_A \ge 3$, and no cluster resists below the mean-field threshold.*

Proof. The 1-skeleton is $K_{7,7,7}$ (Proposition 2.5.1), so $N_A(x) = |A| - a_{\tau(x)}$ with $a_t = |A \cap V_t|$, and $\delta(A) = \min(a_{t'} - a_t)$ over $t \ne t'$ with $a_t > 0$, $a_{t'} < 7$. Let $a_{t_0} = \max a_t \ge 1$. If some $a_{t'} < a_{t_0}$, the pair (t_0, t') gives ≤ -1. Otherwise all a_t are equal and < 7, and every pair gives 0. □

Remark 5.2.4 (Natural cages on the covers). Computed margins: vertex stars have $\delta = -6, 0, 2$ on Y_{21}, Y_{84}, Y_{168}; chambers have $\delta = 0$ and edges $\delta = -1$ everywhere. The star of Y_{168} is stable exactly for $r > 21/9 = 7/3$, which is checked directly. On the thin torus T_{12} ($\kappa = 3$), stars have $\delta = 1$ ($r > 9/4$) and a half-torus strip has $\delta = 2$ ($r > 9/5$). These are statements about zero noise only.

5.3 Finite noise: no topological catalysis

Remark 5.3.1 (Monte Carlo). We ran the Fermi dynamics with normalized payoffs, starting from 50% cooperators, with 60 replicas and 400 Monte Carlo steps. The table gives the final cooperator fraction.

K	structure	$r = 2.4$	2.7	2.9	3.0	3.1	3.3
0.5	Y_{168}	0.00	0.00	0.00	0.29	0.95	1.00
	Y_{84}	0.00	0.00	0.05	0.27	0.80	1.00
	random 14-regular	0.00	0.00	0.00	0.17	0.96	1.00
	T_{12} (thin)	0.00	0.00	0.09	0.64	0.93	1.00
0.1	Y_{168}	0.00	0.00	0.31	0.80	1.00	1.00
	Y_{84}	0.00	0.00	0.00	0.50	0.97	1.00
	random 14-regular	0.00	0.00	0.01	0.76	1.00	1.00
	T_{12} (thin)	0.00	0.40	0.69	0.79	0.89	1.00

The rows for $K = 0.1$ and Y_{168}, T_{12} and the random graph are reproduced by the battery. The $K = 0.5$ rows and Y_{84} come from the same code in an exploratory run. The thick complexes are indistinguishable from a random 14-regular graph, and their transition is near the mean-field value 3. The thin (Euclidean) complex supports cooperation at lower r: this is the classical network reciprocity of lattices [30, 31]. The "topological catalysis" of the proposed programme is reversed. The 1-skeleton of a thick complex is an expander (Chapter 1), its neighbourhoods are large ($2\kappa = 14$), and the dynamics is close to mean field. The sizes are small ($n \leq 168$), and we claim no critical values beyond this table.

Remark 5.3.2 (The comparison value "$r_c > 3$"). For groups of size G the well-mixed threshold is $r = G$, here 3. On lattices with overlapping groups, network reciprocity gives $r_c < G$; for example $r_c/G \approx 0.75$ on the square lattice with $G = 5$ [31]. "$r_c > 3$ on traditional regular networks" is not the correct baseline: our random 14-regular graph has its transition near 3.

The "global phase diagram". Under (L), $|F|/|V| = \kappa(q+1)/3$ is a function of q (7 for $q = 2$). The proved thresholds are the mean-field $r_c = 3$ for all q and the single-mutant $r_* = 3+3/(q(q+1))$; both come from the table in line (E) of the battery. The normalized dynamics does not contain $|F|/|V|$ at all (Theorem 5.1.2). A phase diagram in $|F|/|V|$ therefore carries no information beyond q, and no finite-noise phase boundary is proved.

5.4 Cyclic dominance and oscillations

Give each vertex a strategy in $\mathbb{Z}/3$. In every chamber it plays the rock–paper–scissors matrix $A = \begin{pmatrix} 0 & a & -b \\ -b & 0 & a \\ a & -b & 0 \end{pmatrix}$ against the two other members: k beats $k+1$ (gain a) and loses to $k-1$ (loss b). The chiral asymmetry is $a \neq b$. By the argument of Theorem 5.1.2 the chamber game is again

$(q+1)$ times a pairwise game, and its mean field is the replicator equation $\dot{x}_i = x_i((Ax)_i - x^tAx)$ on the triangle Δ^2.

Theorem 5.4.1 (No limit cycles, no torus)**.** *Let $V = x_1x_2x_3$ and $\sigma = x_1x_2 + x_2x_3 + x_3x_1 \le \frac{1}{3}$. Along the replicator flow, $\frac{d}{dt}\log V = (a-b)(1-3\sigma)$. Hence:*

(i) if $a > b$, every interior orbit converges to the barycentre;

(ii) if $a < b$, every interior orbit other than the barycentre converges to the heteroclinic boundary cycle, and its oscillations slow down without bound;

(iii) if $a = b$, V is conserved and interior orbits are closed. This family is neutral, not an attractor.

In no case is there an isolated periodic orbit. Since the phase space is two-dimensional, there is no invariant torus.

Proof. $\frac{d}{dt}\log V = \sum_i (Ax)_i - 3x^tAx$. The column sums of A are $a-b$, and $x^tAx = (a-b)\sigma$. Also $\sigma \le (\sum x_i)^2/3$, with equality only at the barycentre. In (i) and (ii), V is a strict Lyapunov function on the interior minus the barycentre; LaSalle [15] gives the limits. In (iii) the level sets of V are closed curves. A torus does not embed in the plane. □

Theorem 5.4.2 (Mutation creates a limit cycle)**.** *Consider $\dot{x}_i = x_i((Ax)_i - x^tAx) + \mu(1-3x_i)$ with $\mu > 0$. At the barycentre the linearization has eigenvalues $\frac{b-a}{6} - 3\mu \pm i\frac{\sqrt{3}(a+b)}{6}$. For $a < b$ a Hopf bifurcation occurs at $\mu_H = (b-a)/18$. For $0 < \mu < \mu_H$, the triangle is forward invariant and the barycentre is a repelling focus. So, if the barycentre is the only equilibrium, every other orbit accumulates on a periodic orbit by Poincaré–Bendixson.*

Proof. On $\{\sum \xi = 0\}$ the linearization is $\frac{1}{3}A - 3\mu I$, because $\delta(x^tAx) = 0$ there. The eigenvalues of the circulant A on that plane are $a\omega - b\omega^2$ and its conjugate, i.e. $\frac{b-a}{2} \pm i\frac{\sqrt{3}(a+b)}{2}$. On the face $x_i = 0$ we have $\dot{x}_i = \mu > 0$. □

Numerically, for $(a,b) = (1,2)$ the barycentre is the only equilibrium, on a 400×400 grid and for four parameter sets. The cycle is reached from inside and from near the boundary, with amplitudes $0.397, 0.160, 0.048$ at $\mu/\mu_H = 0.5, 0.9, 0.99$. The squared amplitude scales like $1-\mu/\mu_H$, which indicates a supercritical bifurcation, and the cycle is gone at $1.05\mu_H$. This agrees with [29]. It is a limit cycle, a one-dimensional attractor, not a torus. A quasi-periodic torus would need at least three dimensions, for example the three type classes as separate populations; we do not treat that case.

5.5 Nonlinear chamber games

This section treats open problem (d) of Remark 5.9.1. Let $f\colon \{0,1,2,3\} \to \mathbb{R}$ with $f(0) = 0$ be the synergy of a chamber. Every chamber pays each member $f(n_C)/3$ and charges each cooperator 1, so

$$\pi_i(s) = \sum_{\Delta \ni i} \Big(\tfrac{1}{3} f(n_C(\Delta)) - s_i \Big).$$

The examples are the linear game $f = r(0,1,2,3)$, the threshold game $f = r(0,0,2,3)$ (so $f(0) = f(1) = 0$, $f(2) = 2r$, $f(3) = 3r$), and the quadratic game $f(n) = rn^2$. On $\{0,1\}^3$,

$$f(s_1+s_2+s_3) = c_1(s_1+s_2+s_3) + c_2(s_1s_2 + s_1s_3 + s_2s_3) + c_3 s_1s_2s_3,$$

with $c_1 = f(1)$, $c_2 = f(2) - 2f(1)$ and $c_3 = f(3) - 3f(2) + 3f(1)$. The threshold game has $(c_1, c_2, c_3) = (0, 2r, -3r)$.

Theorem 5.5.1 (Irreducibility)**.** *Call a game* pairwise *if every π_i is a sum of a function of s_i and functions of (s_i, s_j) over the neighbours j of i. On a 2-complex in which every vertex lies in a chamber, the chamber game is pairwise iff $c_2 = c_3 = 0$, i.e. iff f is affine on $\{0, 1, 2, 3\}$. Otherwise π_i contains the monomials $\frac{c_2}{3} s_j s_k$ and $\frac{c_3}{3} s_i s_j s_k$ for every chamber $\{i, j, k\}$, which are genuine three-body terms.*

Proof. Multilinear expansions of functions on $\{0,1\}^V$ are unique. A pairwise payoff contains only the monomials 1, s_i, s_j and $s_i s_j$ with $j \sim i$. The monomial $s_j s_k$, for $\{i, j, k\} \in F$, arises only from that chamber, with coefficient $c_2/3$. Likewise $s_i s_j s_k$ arises only from that chamber, with coefficient $c_3/3$. For affine f we are back in Theorem 5.1.2. □

Theorem 5.5.2 (Exact potential)**.** *For every f and every 2-complex, $\Phi(s) = \sum_\Delta \big(\frac{1}{3} f(n_C(\Delta)) - n_C(\Delta)\big)$ is an exact potential: switching i changes π_i and Φ by the same amount. Hence:*

(i) the Nash equilibria are the local maxima of Φ under single switches;

(ii) logit (best-response) dynamics is reversible, with Gibbs measure $\propto e^{\Phi/K}$;

(iii) if $\psi(n) = \frac{1}{3} f(n) - n$ is maximized over $\{0, 1, 2, 3\}$ at $n \in \{0, 3\}$ (true for convex f and for the threshold game), then as $K \to 0$ the stochastically stable state is all-D if $f(3) < 9$ and all-C if $f(3) > 9$, on every complex.

In particular the threshold game switches at $r = 3$, and $f = rn^\gamma$ switches at $r = 3^{2-\gamma}$, whatever the geometry.

Proof. The change of π_i is $\sum_{\Delta \ni i} \big(\frac{1}{3}(f(k_\Delta + 1) - f(k_\Delta)) - 1\big)$, where k_Δ counts the other cooperators, and this is also the change of Φ. (ii) is standard for exact potential games. (iii) Every chamber contributes at most $\max \psi$, and the uniform states attain that bound in every chamber simultaneously. For the threshold game, $\psi = (0, -1, \frac{2r}{3} - 2, r - 3)$, and $\psi(2) \le \max(0, r - 3)$. □

Proposition 5.5.3 (Type-stratified equilibria)**.** *On any complex in which every chamber has one vertex of each type ($\widetilde{A}_2$ complexes, thick or thin), the state "D exactly on one type class" puts two cooperators in every chamber. It is a Nash equilibrium iff $f(2) - f(1) \ge 3 \ge f(3) - f(2)$. For the threshold game this is $\frac{3}{2} \le r \le 3$. For $r \in (\frac{3}{2}, 3)$ it is a strict local maximum of Φ, but not the global one, so it is metastable under logit dynamics. Under imitation it is unstable: a cooperator and a neighbouring defector see identical chambers, and the defector earns D_2 more.*

Proof. A cooperator's other chamber members are one C and one D, so switching changes its payoff by $D_2(1 - \frac{1}{3}(f(2) - f(1)))$. A defector's other members are two C's, so switching changes its payoff by $D_2(\frac{1}{3}(f(3) - f(2)) - 1)$. □

This is the one place where the 2-cells matter for equilibria. What it uses is partiteness (the $\mathbb{Z}/3$ grading), not q and not the flag structure.

Imitation: mean field. For the Fermi rule in the well-mixed limit, $\dot{x} = x(1-x)\tanh(\Delta(x)/2K)$ with $\Delta(x) = \frac{1}{3}\sum_{k=0}^{2}\binom{2}{k}x^k(1-x)^{2-k}(f(k+1)-f(k))-1$. For the threshold game, $\Delta(x) = \frac{r}{3}(4x-3x^2)-1$. All-$D$ is always stable. For $r > \frac{9}{4}$ a stable mixed state appears, and all-C is stable iff $r > 3$. So nonlinearity lowers the mean-field threshold from 3 to $\frac{9}{4}$. This effect does not depend on the geometry.

Proposition 5.5.4 (Star cages under imitation, $q \in \{1,2\}$)**.** *Let A be the star of a vertex in a complex whose radius-3 balls are those of an $\widetilde{A}_2$ building of order $q \in \{1,2\}$. Put $\varphi = f/r$. Then A loses no cooperator at zero noise iff $r > r_{\mathrm{star}}$, where*

$$r_{\mathrm{star}}(q) = \frac{3(q+1)(q^2+q+1)}{(q+1)\big(\varphi(3)+q\varphi(2)+q^2\varphi(1)\big)-\varphi(2)-2q\varphi(1)}.$$

This gives $\frac{7}{3}, \frac{63}{19}, \frac{63}{55}$ for $q = 2$ (linear, threshold, quadratic) and $\frac{9}{4}, \frac{9}{4}, \frac{9}{11}$ for $q = 1$.

Proof. A boundary vertex u of the star has $q+1$ chambers through v (three cooperators), $q(q+1)$ chambers with exactly one other cooperator, and $q^2(q+1)$ chambers with none. This follows from the link: its neighbours in the star are the $q+1$ link-neighbours of u in $\mathrm{Lk}(v)$, and these are pairwise non-adjacent. The most dangerous outside neighbour j is the third vertex of a chamber over a boundary edge. It has one chamber with two cooperators and $2q$ chambers with one. Then apply the zero-noise pair criterion. For $q = 2$ the count at j, and the maximality of that j, are verified exactly in the radius-3 ball of the building (a level-12 cover of the canonical tower of Chapter 6, whose ball sizes $1, 14, 98, 560$ agree with those of the building) and on Y_{168}. For $q = 1$ they are verified on T_{12} (line (NL4)). □

For large q the displayed formula grows like $\frac{3}{2}q$ for the threshold game and tends to 3 in the other two cases, assuming the same counts. A linear 3-uniform hypertree with the same chamber degree D_2 gives $3D_2/(\varphi(3)+(D_2-2)\varphi(1))$ for the star, i.e. 21 for the threshold game when $D_2 = 21$. So the building's star beats the hypertree's star. The mechanism is the $(q+1)$-fold overlap of chambers along edges, not the incidence "two points on one line". The star does not beat the thin lattice, and it gets worse as q grows.

Remark 5.5.5 (Exhaustive comparison on 21 vertices). For hypergraphs on 21 vertices with $D_2 = 21$ we enumerated all $2^{21}-2$ cooperator sets and computed the best zero-noise cage threshold $\min_A r_A$ (line (NL3)).

hypergraph	linear	threshold	quadratic
Y_{21} (chambers)	3	$\frac{9}{4}$	0.677
random tripartite, 3 seeds	2.739	2.10–2.17	0.670–0.685
random 3-uniform, 3 seeds	2.86–3	2.17–2.25	0.663–0.685

Y_{21} is never better than the random models; for the linear game it is worse. The battery reproduces seed 0 of each random model. The quadratic cages are almost full cooperation. They sit near the mean-field stability value $\frac{3}{5}$ of all-C, where $\Delta(1) = \frac{5}{3}r - 1$.

Remark 5.5.6 (Monte Carlo). Threshold game, Fermi rule, $K = 0.1$, 50% initial cooperators, 60 replicas, 300 sweeps. Final cooperator fraction:

structure	$r = 2.25$	2.5	2.75	3.0	3.25	3.5
Y_{21}	0.00	0.00	0.50	0.77	0.85	0.90
Y_{168}	0.00	0.59	0.94	1.00	1.00	1.00
random, $n = 168$, $D_2 = 21$	0.00	0.60	0.94	1.00	1.00	1.00
T_{12} (thin)	0.09	0.87	0.99	1.00	1.00	1.00
random, $n = 144$, $D_2 = 6$	0.00	0.48	0.90	0.98	1.00	1.00

For comparison, the linear game on the same structures (exploratory run) switches between 3.0 and 3.25. The exceptions are T_{12}, which already has 0.50 at 2.75, and Y_{21}, which still has only 0.27 at 3.25. The battery reproduces the columns 2.25–3.0 for Y_{21}, T_{12}, Y_{168} and the $D_2 = 21$ random hypergraph. Three facts stand out. Nonlinearity lowers the transition from about 3 to about 2.4–2.5, in line with the mean-field value $\frac{9}{4}$. The thick complex Y_{168} is indistinguishable from a random hypergraph with the same chamber degree. The thin lattice is the most favourable structure, and the dense $Y_{21} = K_{7,7,7}$ the least.

Remark 5.5.7 (Answer to the deepening request). (1) "Pairwise iff linear" is true (Theorem 5.5.1); nonlinear chamber games are genuinely three-body. (2) The incidence fact that is actually used is "each edge lies in $q + 1$ chambers". Two edges at a vertex lie in at most one common chamber, not "exactly one". (3) Under best-response learning the geometry is irrelevant for the selected state (Theorem 5.5.2). The one structural effect, the type-stratified equilibria of Proposition 5.5.3, comes from partiteness and is metastable. (4) Under imitation, the $\widetilde{A}_2$ star cage is much better than a hypertree star, but its threshold grows with q and exceeds the thin-lattice value (Proposition 5.5.4). Exhaustively, Y_{21} is no better than random hypergraphs with the same parameters (Remark 5.5.5). At finite noise the thick complex matches a random hypergraph (Remark 5.5.6). A "topological shielding threshold significantly below that of random regular hypergraphs" is therefore not established, and the data contradict it. What nonlinearity does lower, uniformly, is the mean-field threshold, from 3 to $\frac{9}{4}$.

5.6 Chiral games, interfaces, and an economic reading

Every chamber has one vertex of each type. Let the vertex of type ρ play *role* ρ in all its chambers, so that roles are ordered cyclically, $0 \to 1 \to 2 \to 0$. A *chiral chamber game* gives role ρ the per-chamber payoff $u_\rho(s_0, s_1, s_2) = s_\rho\, g_\rho(s_{\rho-1}, s_{\rho+1})$. The gain g_ρ may depend on the upstream member $\rho - 1$ and the downstream member $\rho + 1$ differently. A vertex's payoff is the average of $u_{\tau(i)}$ over its D_2 chambers. The *supply-chain game* has roles technology (0), manufacturing (1) and channel (2), with

$$g_0 = 1 - 2s_2, \qquad g_1 = 2s_0 - 1, \qquad g_2 = 2s_1 - 1.$$

Manufacturing's investment pays only if technology invested; the channel's pays only if manufacturing did; technology is held up once the channel invests. The *three-body variant* replaces g_2 by $4s_0s_1 - 1$: the channel's return is activated only when both upstream partners invest.

Theorem 5.6.1 (Potential criterion). *Let Y satisfy* (L)*, and for $\rho \neq \sigma$ let $D_{\rho\sigma}u_\rho$ be the mixed difference of u_ρ in (s_ρ, s_σ), a function of the third strategy s_κ. The network game has an exact potential iff $D_{\rho\sigma}u_\rho \equiv D_{\rho\sigma}u_\sigma$ for all $\rho \neq \sigma$. It has a weighted potential iff there are $w_\rho > 0$ with $w_\rho D_{\rho\sigma}u_\rho \equiv w_\sigma D_{\rho\sigma}u_\sigma$. The symmetric games of Section 5.5 satisfy the exact criterion. The supply-chain game and its three-body variant satisfy neither. Their Monderer–Shapley circulation on an upstream edge is $\pm(q+1)\cdot 2/D_2$ ($\pm 2/7$ on Y_{21}), and a single chamber has a 6-cycle of strict*

improvements $(0,0,0) \to (1,0,0) \to (1,1,0) \to (1,1,1) \to (0,1,1) \to (0,0,1) \to (0,0,0)$. *So there is not even an ordinal potential.*

Proof. A finite game has an exact potential iff every 2×2 subgame has zero circulation, i.e. $D_{ij}\pi_i = D_{ij}\pi_j$ for all players i, j and all profiles of the others [43]. Players interact only along edges. For an edge ij with roles (ρ, σ), the $q+1$ chambers containing it all assign the same roles, and their third vertices are distinct. So $D_{ij}\pi_i = \frac{1}{D_2}\sum_{\Delta \supset ij} D_{\rho\sigma}u_\rho(s_{k_\Delta})$. Setting all third vertices to 0, or all to 1, gives necessity of the per-chamber identity; sufficiency is clear. The weighted case is the same computation with $w_i\pi_i$. For the supply-chain game, $D_{10}u_1 = 2$ while $D_{10}u_0 = 0$, which cannot be balanced by any positive weights. The improvement cycle is read off the signs of g_ρ. □

In the well-mixed limit each type class is a population with cooperator fraction x_ρ, and logit learning at noise K gives

$$\dot{x}_\rho = \sigma\big(G_\rho(x)/K\big) - x_\rho, \qquad \sigma(z) = \frac{1}{1+e^{-z}}, \qquad G_\rho(x) = \mathbb{E}\, g_\rho. \tag{5.6.1}$$

Proposition 5.6.2 (Potential mean fields converge)**.** *If $w_\rho G_\rho = \partial\Phi/\partial x_\rho$ for some Φ and weights $w_\rho > 0$, then $L = \Phi/K + \sum_\rho w_\rho H(x_\rho)$, with $H(x) = -x\log x - (1-x)\log(1-x)$, increases strictly along* (5.6.1) *away from equilibria. So there are no periodic orbits.*

Proof. $\dot{L} = \sum_\rho w_\rho\big(G_\rho/K - \operatorname{logit} x_\rho\big)\big(\sigma(G_\rho/K) - x_\rho\big) \geq 0$, because σ is increasing. Equality holds iff every factor vanishes. □

Theorem 5.6.3 (Endogenous cycles in cyclic chiral games)**.** *Suppose $G_\rho = G_\rho(x_{\rho-1})$ is strictly monotone with sign ε_ρ. Then* (5.6.1) *is a monotone cyclic feedback system on $[0,1]^3$.*

(i) If $\varepsilon_0\varepsilon_1\varepsilon_2 = +1$, the system is cooperative and irreducible, and no nontrivial periodic orbit is asymptotically stable [45].

(ii) If $\varepsilon_0\varepsilon_1\varepsilon_2 = -1$, the equilibrium x^* is unique. Its linearization has eigenvalues $-1 + |P|^{1/3}e^{i\pi(2k+1)/3}$, $k = 0,1,2$, where $P = \prod_\rho \sigma'(G_\rho(x^*_{\rho-1})/K)G'_\rho(x^*_{\rho-1})/K < 0$. So x^* is stable iff $|P| < 8$. At $|P| = 8$ a pair of eigenvalues crosses the imaginary axis at $\pm i\sqrt{3}$. For $|P| > 8$ a nonconstant periodic orbit exists (Mallet-Paret–Smith [44]).

For the supply-chain game $x^* = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ and $|P| = (2K)^{-3}$. The Hopf point is $K_H = \frac{1}{4}$, and the period there is $2\pi/\sqrt{3} = 3.628$.

Proof. (i) With positive feedback the off-diagonal Jacobian entries are ≥ 0, and they form an irreducible cyclic pattern in the interior. (ii) An equilibrium solves $x_\rho = F_\rho(x_{\rho-1})$ with $F_\rho = \sigma \circ (G_\rho/K)$ monotone of sign ε_ρ. The composite $F_0 \circ F_2 \circ F_1$ is decreasing, so it has exactly one fixed point. The Jacobian is $-I + C$, where C is the cyclic matrix with entries $\sigma' G'_\rho/K$, and C has eigenvalues $P^{1/3}$. The real part of $-1 + |P|^{1/3}e^{\pm i\pi/3}$ is $-1 + |P|^{1/3}/2$. For $|P| > 8$ the equilibrium is unique and has a two-dimensional unstable manifold. The Poincaré–Bendixson theory for monotone cyclic feedback systems [44] (quoted) then gives a nonconstant periodic orbit. □

Remark 5.6.4 (Mean-field numerics)*.* For the supply-chain game the cycle amplitude of x_0 is $0.294, 0.287, 0.263, 0.206, 0.098$ at $K = 0.05, 0.10, 0.15, 0.20, 0.24$, and 0 for $K \geq 0.26$. The period is $3.00, 3.26, 3.47, 3.59, 3.63$, approaching $2\pi/\sqrt{3}$. The squared amplitude scales like $K_H - K$, which indicates a supercritical bifurcation. The three-body variant is not a cyclic feedback system, since

x_2 responds to both x_0 and x_1. Its Jacobian at $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is $-I + \frac{1}{4K}M$, where M has characteristic polynomial $\lambda^3 + 4\lambda + 8$. So its Hopf point is $K_H = 0.170582$, a quarter of the real part of the complex root. Numerically the cycle exists below K_H (amplitude 0.186 at $K = 0.05$) and not above. Its existence beyond the local Hopf regime is not proved. Line (NC4) checks both Hopf points and the amplitudes.

Remark 5.6.5 (Networks: the cycle is a mean-field phenomenon). Logit dynamics of the supply-chain game on the complexes themselves, at $K = 0.1$ (below K_H), gives the following standard deviation of the class density $x_0(t)$ (exploratory runs over 120 time units):

	Y_{24}	Y_{768}	Y_{6144}	T_{12}	T_{24}	T_{48}
n	24	768	6144	144	576	2304
std x_0, $K = 0.1$	0.262	0.091	0.023	0.094	0.044	0.025
std x_0, $K = 0.4$	0.224	0.041	0.015	0.074	0.036	0.021

Here Y_{768} and Y_{6144} are levels 4 and 5 of the canonical 2-tower of Chapter 6. In the mean field the standard deviation would be about 0.2. On both thick and thin complexes it decays with n. The single-site autocorrelation has only a weak negative lobe (-0.06 on Y_{768}, -0.09 on T_{24} at lag 3), so local oscillations are weak and incoherent. The limit cycle of Theorem 5.6.3 describes well-mixed populations; on $Y_{21} = K_{7,7,7}$ the drift is exactly mean-field. No coherent network rhythm was found on large sparse complexes, thick or thin. Line (NC6) reproduces the decay for $Y_{24} \to Y_{168}$ and $T_{12} \to T_{48}$.

Proposition 5.6.6 (Flat interfaces). *In the thin complex T_m ($3 \mid m$), let A be a half-torus bounded by two walls parallel to a lattice direction. At zero noise under imitation, A loses no cooperator iff*

$$r > \frac{18}{3\varphi(3) + \varphi(2) - \varphi(1)},$$

i.e. $r > \frac{9}{5}$, $\frac{18}{11}$, $\frac{3}{5}$ for the linear, threshold and quadratic games.

Proof. A boundary cooperator has 6 chambers: three with three cooperators, two with two, and one with one. Its defecting neighbour across the wall has one chamber with two cooperators, two with one, and three with none. So the gap in normalized payoffs is $r(3\varphi(3) + \varphi(2) - \varphi(1))/18 - 1$, with $D_2 = 6$. The two boundary vertices of each type are equivalent by translation. □

The flat wall is the exact mechanism behind the advantage of the thin lattice (Remark 5.5.6). A boundary cooperator on a wall has only one "lonely" chamber out of 6. At the boundary of a thick star it has $q^2(q+1) = 12$ out of 21 (Proposition 5.5.4). Enlarging the cluster to balls of radius 2 and 3 in the thick building gives exactly the star thresholds again, as we computed in the level-14 cover. Thick buildings have no flat interfaces: their spheres grow exponentially.

Remark 5.6.7 (Why no analytic law $r_c(q+1, \lambda_2)$). A finite-noise critical value is not sharply defined on finite complexes, where the dynamics is absorbing. On the infinite building, the existence of a phase transition for this non-reversible particle system is itself open. The data also rule out λ_2 as the controlling variable. With normalized gaps $\lambda_2/\deg = 0.089$ (T_{12}), 0.441 (random, $D_2 = 6$), 0.601 (Y_{168}), 0.697 (random, $D_2 = 21$) and 1.000 (Y_{21}), the threshold-game transitions of Remark 5.5.6 lie at about 2.38, 2.51, 2.48, 2.48 and 2.75. The ordering is roughly monotone, but Y_{168} sits below a hypergraph with a smaller gap. Interfaces are what the proved statements identify (Propositions 5.5.4, 5.6.6). A chamber-cluster approximation could be written down, but it would be an uncontrolled closure, and we do not present its output as a law.

An economic reading. Firms (vertices) join three-party projects (chambers). Each project has one firm of each role, and each firm has D_2 projects. Axioms:

(A1) a firm chooses one policy, contribute (1) or not (0), for all its projects;

(A2) contributing costs 1 per project;

(A3) a project's output depends on its contributors through a technology, which is symmetric, $f(n_C)$ shared equally, or role-dependent, u_ρ;

(A4) payoffs are averaged over projects.

An R&D consortium is the symmetric case, with a threshold technology when a breakthrough needs at least two partners. A syndicated credit line is the same with "lend" for "contribute". A supply chain with hold-up is the chiral case.

Proposition 5.6.8 (Free riding, equilibrium and selection)**.** *In the symmetric case:*

(i) contributing is a dominant strategy iff $\min_{k=0,1,2}\big(f(k+1)-f(k)\big)\geq 3$. *In particular it is* never *dominant under a threshold technology (*$f(1)=f(0)$*);*

(ii) universal contribution is a Nash equilibrium iff $f(3)-f(2)\geq 3$*;*

(iii) under logit learning, universal contribution is selected as $K\to 0$ *iff* $f(3)>9$ *(when* ψ *peaks at the ends),* whatever the contract network.

The network matters only for non-equilibrium social learning (imitation), through its interfaces (Proposition 5.6.6).

Proof. (i) A firm's gain from contributing in a project with k other contributors is $\frac{1}{3}(f(k+1)-f(k))-1$; dominance must hold for every k in every project. (ii) is the case $k=2$. (iii) is Theorem 5.5.2(iii). □

The proposed programme asks to prove that voluntary cooperation is a globally dominant equilibrium *iff* the technology has a threshold and the network is moderately local. This is false in both directions. A threshold technology rules out dominance (i). Equilibrium selection under best-response learning ignores the network (iii). What locality does is help *imitation* spread cooperation across flat interfaces. In the chiral (supply-chain) case, hold-up creates negative feedback and endogenous investment cycles for low noise. These are coherent in a well-mixed market (Theorem 5.6.3) but not across a large sparse contract network (Remark 5.6.5).

5.7 Clocks, entrainment and anchors

The two obstacles found above are the decay of network oscillations (Remark 5.6.5) and the absence of flat interfaces in thick buildings (Proposition 5.6.6). This section tests the two remedies proposed in the programme. The first is a settlement clock carried by the gauge field of Chapter 3. The second is a committed core ("chain leader"). The first produces coherence, but by entrainment rather than endogenously. The second cannot overcome the geometry.

The clock. Each vertex carries a phase θ_i evolving by the flux sector of Chapter 3: identical frequencies ω, coupling $K_1 > 0$, flux $2\pi\alpha$. Its settlement cycle enters the supply-chain gain as $g_{\tau(i)} + \varepsilon \cos\theta_i$. The coupling is one-way: strategies do not act on phases.

Proposition 5.7.1 (The clock locks with a flux-selected twist)**.** *The phases converge to an equilibrium of the flux sector in the rotating frame (Theorem 3.2.2). The ground states are the twisted states $\theta_i = \omega t + c + \frac{2\pi m^*}{3}\tau(i)$ with $m^* \equiv -\operatorname{round}(\alpha)$ (Theorem 3.2.4). So a synchronized clock forces the three roles with a relative phase lag of $\frac{2\pi m^*}{3}$, and the flux selects this chirality. From random initial phases the clock reached the ground state, with order > 0.999, on Y_{24} and Y_{168} for $\alpha = 0.3, 1.2, -0.8$ (so $m^* = 0, 2, 1$), and on Y_{768}. On the thin torus T_{48} it stopped at order 0.41, trapped in a non-ground equilibrium. Expansion, which hurts cooperation, helps the clock.*

Theorem 5.7.2 (Entrained cycle: linear response)**.** *Let $K > K_H = \frac{1}{4}$, so that the mean-field equilibrium $x^* = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ of the supply-chain game is a stable focus (Theorem 5.6.3). Force it with a clock of twist m, i.e. $\dot{x}_\rho = \sigma\big((G_\rho(x) + \varepsilon\cos(\omega t + \frac{2\pi m\rho}{3}))/K\big) - x_\rho$. For small ε this has a unique attracting periodic solution $x^* + \varepsilon\,\mathrm{Re}\big(e^{i\omega t}X_m(\omega)\big) + O(\varepsilon^2)$, with*

$$X_m(\omega) = \frac{1}{4K}\big(i\omega I + I - C\big)^{-1} f_m, \qquad f_m = \big(e^{2\pi i m\rho/3}\big)_\rho,$$

where C is the cyclic matrix with entries $C_{02} = -\frac{1}{2K}$ and $C_{10} = C_{21} = \frac{1}{2K}$. The response peaks near the endogenous frequency $\sqrt{3}/(4K)$. At $K = 0.4$ and resonance the x_0-amplitude per unit ε is 0.94, 0.78, 1.20 for $m = 0, 1, 2$. If the clock is only partly synchronized, with class order parameters $r_\rho = |\langle e^{i\theta_i}\rangle_{V_\rho}|$, the first-order forcing is multiplied by r_ρ. So a macroscopic response exists at first order iff the clock is synchronized ($r > 0$).

Proof. The linearization at x^* is $-I + C$, which is hyperbolic and stable for $K > \frac{1}{4}$. The forcing enters through $\sigma'(0)/K = 1/(4K)$. The periodic solution and its expansion follow from the implicit function theorem for periodic forcing of a hyperbolic equilibrium. For a partly synchronized clock, the class average of $\cos\theta_i$ equals $r_\rho \cos\psi_\rho$ to first order. □

Remark 5.7.3 (Coherence on large complexes). On the complexes themselves, the class-density standard deviation of the forced logit dynamics ($\varepsilon = 0.3$, clock from the flux sector with $\alpha = 0.3$) is:

	clock	Y_{24}	Y_{768}	Y_{6144}
$K = 0.4$, $\omega = 1.083$	synchronized	0.215	0.110	0.103
	random offsets	0.201	0.037	0.013
	none	0.222	0.035	0.013
$K = 0.1$, $\omega = 1.927$	synchronized	0.233	0.156	0.148
	random offsets	0.239	0.041	0.022
	none	0.247	0.069	0.028

With a synchronized clock the macroscopic oscillation no longer decays with n, both above and below K_H. With random offsets it decays like the unforced dynamics. Line (NA3) checks the Y_{168} case. What produces coherence is the *common clock*. The cycle is entrained, i.e. imposed by an external rhythm whose phases were synchronized separately. It is not an endogenous macroscopic cycle of the game. The game contributes the resonance and the chirality dependence of Theorem 5.7.2. The "iff" of the proposed programme holds in this sense: coherent macroscopic response iff the clock order parameter is positive. For heterogeneous clock frequencies that is the classical Kuramoto synchronization condition, which concerns the clock and not the game.

Anchors. A *committed set* Z never changes strategy; it cooperates always. A chain leader enforcing cooperation in its own chambers is the case $Z =$ the star of a vertex.

Proposition 5.7.4 (Frontier principle)**.** *Under zero-noise imitation, a committed set changes only the inequalities at its own boundary. In the thick building ($q = 2$, computed in the level-14 cover, whose balls of radius 4 have $1, 14, 98, 560, 2912$ vertices), we find the following.*

(i) From an anchored ball of radius 1 (the leader's star), cooperation spreads to the next sphere iff $r > \frac{7}{3}, \frac{63}{19}, \frac{63}{55}$ (linear, threshold, quadratic).

(ii) From an anchored ball of radius 2, it spreads iff $r > \frac{21}{10}, 3, \frac{21}{20}$. But the newly reached ball of radius 3 is not anchored, and it is stable only for $r > \frac{7}{3}, \frac{63}{19}, \frac{63}{55}$.

So sustained growth beyond an anchor of radius ≤ 2 needs $r > \frac{63}{19}$ in the threshold game, whatever the anchor. In the thin lattice an anchored wall spreads, and every new front is again a flat wall, stable for $r > \frac{9}{5}, \frac{18}{11}, \frac{3}{5}$.

Proof. A defector j adjacent to the cooperating set converts iff some cooperating neighbour earns more. This gives the thresholds $\min_i D_2/(\Phi_i - \Phi_j)$, maximized over the frontier. An unanchored cooperator keeps its strategy iff it earns more than every defecting neighbour (Theorem 5.2.2). The values are exact computations over the ball (line (NA4)). □

The mechanism-design reading is negative. A leader's enforced contract protects its own neighbourhood, but the moving front has the geometry of spheres, and spheres in a thick building have the star thresholds. The absence of flat interfaces is a property of the geometry, and no finite anchor removes it. Only an *infinite* anchored set shaped like a wall would help, and thick buildings have no sets with the small boundary of a wall.

Remark 5.7.5 (Anchors at finite noise). Threshold game, Fermi rule, $K = 0.1$. Cooperator fraction among the non-committed vertices:

	$r = 2.0$	2.25	2.4	2.5
Y_{168}, no anchor	0.00	0.00	0.27	0.57
Y_{168}, leader star (15 committed)	0.00	0.00	0.65	0.82
Y_{168}, 15 random committed	0.00	0.00	0.68	0.80
T_{12}, no anchor	0.00	0.04	0.72	0.89
T_{12}, committed row (12)	0.01	0.64	0.87	0.95
T_{12}, 12 random committed	0.00	0.60	0.87	0.92

About 9% of committed vertices raises cooperation near the transition on both geometries. The leader's star does no better than random commitment. On the thick complex cooperation is still absent at $r = 2.25$, where the thin lattice with a committed row already has 0.64. Anchors shift the transition slightly; they do not pin it low. Line (NA5) checks the $r = 2.25$ and 2.5 columns.

5.8 Verification battery

5.9 Scope, residue, corrections

Remark 5.9.1 (Solved, open). *Solved:* the pairwise reduction; dominance and the mean-field threshold; the exact zero-noise cluster criterion, with its integrality consequence and the absence

check	complexes	result
(R) chamber payoff = reduced additive payoff; $w = q + 1$ on edges, 0 elsewhere; $D_2 = \kappa(q + 1)$	Y_{21}, Y_{168}, T_9	numeric 10^{-9}
(Dm) own switch changes own payoff by $D_2(r/3 - 1)$	same	numeric
(Z) $r_* = 3.5$ both ways; margins of stars, chambers, edges; star of Y_{168} at 7/3; $5/3 \notin \{21/(7 + \delta)\}$	$Y_{21}, Y_{84}, Y_{168}, T_{12}$	exact
(MC) Fermi, $K = 0.1$: Y_{168}, random 14-regular graph, T_{12}	—	Monte Carlo (seeded)
(P) $\frac{d}{dt} \log V$ identity; μ_H and frequency; cycle amplitudes; unique equilibrium	—	numeric
(E) $\lvert F\rvert/\lvert V\rvert$, D_2, r_* for $q = 2, 3, 4, 5, 7, 8$	—	exact
(NL1) multilinear coefficients; pairwise iff f affine; $s_j s_k$ coefficient $c_2/3$	Y_{21}	exact
(NL2) exact potential; type-stratified Nash window $[\frac{3}{2}, 3]$	Y_{21}, Y_{168}, T_{12}	exact
(NL3) best cages by enumeration of 2^{21} sets; $\max \Phi$ switches at $r = 3$	Y_{21}, random	exact
(NL4) star cages = closed form	Y_{168}, T_{12}	exact
(NL5) threshold game, Fermi, $K = 0.1$	Y_{21}, T_{12}, Y_{168}, random	Monte Carlo (seeded)
(NC1)–(NC2) circulation $\pm 2/7$; strict improvement 6-cycle	Y_{21}	exact
(NC3) Lyapunov function of potential mean fields	—	numeric
(NC4) Hopf points $\frac{1}{4}$ and 0.170582; amplitudes; period $2\pi/\sqrt{3}$; unique equilibrium	—	numeric
(NC5) flat interfaces $18/(3\varphi_3 + \varphi_2 - \varphi_1)$	T_{12}	exact
(NC6) class-density oscillation shrinks with n	$Y_{24}, Y_{168}, T_{12}, T_{48}$	Monte Carlo (seeded)
(NA1) gauge clock reaches $m^* = -\operatorname{round}(\alpha)$ with order > 0.999	Y_{24}, Y_{168}	numeric
(NA2) linear response $X_m(\omega)$ vs forced ODE (3%); resonance	—	numeric
(NA3) synchronized vs random clock vs none	Y_{168}	Monte Carlo (seeded)
(NA4) spreading from anchored balls; stability of the radius-3 ball; anchored wall	building (level 14), T_{24}	exact
(NA5) committed star vs random vs none; committed row	Y_{168}, T_{12}	Monte Carlo (seeded)

Table 5.1: Checks run by `code/v3p5_verify.py`.

of cages on Y_{21}; the trichotomy for cyclic replicator dynamics; the Hopf point with mutation. *Open:* (a) finite-noise thresholds on families of thick complexes of growing size; (b) a pair or cluster approximation adapted to the $\widetilde{A}_2$ 1-skeleton; (c) a proof that the interior equilibrium with mutation is unique, and the sign of the first Lyapunov coefficient; (d) genuinely higher-order (nonlinear) chamber games, treated in Section 5.5. What remains open there is the star formula for $q \geq 3$, the optimal cages on large thick complexes, and finite-noise thresholds along growing families; (e) three-population versions with a three-dimensional mean field, where tori are possible.

Remark 5.9.2 (Corrections to the proposed programme). (1) The chamber public-goods game is a pairwise game on the 1-skeleton (Theorem 5.1.2). The $\mathrm{PG}(2, q)$ flags enter only through the constant $q + 1$, and there is no separate "shielding cage". (2) $r_c = 1 + 2/(q + 1)$ is not derived by any of the available notions. The mean-field threshold is 3 (Proposition 5.1.4), and zero-noise cluster thresholds are $3\kappa/(\kappa + \delta)$, which never equals 5/3 for $q = 2$ (Theorem 5.2.2). The best cage found, the star of Y_{168}, needs $r > 7/3$, and Y_{21} has none (Proposition 5.2.3). (3) There is no topological catalysis. Thick complexes behave like random regular graphs, and the thin lattice supports cooperation better (Remark 5.3.1). "$r_c > 3$ on regular networks" is not the correct baseline (Remark 5.3.2). (4) The three-strategy replicator dynamics has no limit cycle and no torus (Theorem 5.4.1). Persistent oscillations arise as a limit cycle once mutation is added (Theorem 5.4.2). (5) $\lvert F\rvert/\lvert V\rvert$ is a function of q and does not enter the normalized dynamics.

Acknowledgement of provenance and caveats

Unrefereed. The Fermi rule is that of [32]. Lattice reciprocity and the r_c/G values are quoted from [30, 31], and the mutation-driven cycle is compared with [29]. LaSalle's principle [15] is classical. The bibliographic data of [32, 30, 31, 29] have been checked against Crossref and zbMATH.

Chapter 6

Hecke Algebras on Chambers, Iwasawa Modules, and the 2-adic Valuation Tower

The proposed programme asks for three things. (a) The operators of this volume — Δ^{H}, L_E and the chiral Kuramoto derivative — are to be realized in representations of the degenerate affine Hecke algebra $\mathbb{H}_d$. (b) A non-commutative p-adic L-element $L_p(Y) \in K_1(\Sigma^{-1}\Lambda)$ is to be built from $\det(I - uC^{(d+1)})$, where $\Lambda = \mathbb{Z}_p[[G]]$. (c) A "main conjecture" $\partial L_p(Y) = [\mathrm{Sel}(Y/G)^\vee]$ is to be proved by a five-step scheme. This closing chapter states what is true.

(1) *Hecke algebras.* We give $\mathbb{H}_d$ and its polynomial representation, checked exactly. Every operator L is x_1 in the $\mathbb{H}_d$-module $s_i \mapsto 1$, $x_i \mapsto L + (i-1)c$, so "realized in a representation of $\mathbb{H}_d$" has no content. The algebra that acts naturally is the Iwahori–Hecke algebra $H_q(\widetilde{A}_2)$ with parameter q, acting on functions on chambers. We prove that the quadratic and braid relations hold on every complex satisfying (L), including Y_{21}, which fails the flag condition. In this language $\Delta_2 = 3 + T_0 + T_1 + T_2$, $A_1 = (q+1)^{-1}\sum_t \pi_t \pi_{t+1}^t$, and $L_E = (q(q+1))^{-1}\, p\, T_t T_{t+2} T_t\, p^t$ blockwise. The vertex operators are images of symmetric polynomials under the Satake parameters: $A_1 = qe_1$, $A_2 = qe_2$, $e_3 = 1$. The Kuramoto derivative is nonlinear and is not an algebra representation. Its linearization at the twisted states is $-c_m\Delta_0$, which is central in this sense.

(2) *Localization.* We state Cohn localization and its K-theory sequence correctly, and note that the "$\to 0$" holds when $K_0(\Lambda) \to K_0(\Sigma^{-1}\Lambda)$ is injective. Garland rigidity (Chapter 1) implies that every open subgroup of a tower group has *finite* abelianization. So no quotient $G/H \cong \mathbb{Z}_p$ exists, and the Ore set of Coates–Fukaya–Kato–Sujatha–Venjakob, with its category $\mathfrak{M}_H(G)$, is *not available* for any tower over a thick complex. One must localize at all nonzero elements of the domain Λ.

(3) *The theorem that replaces the conjecture.* For every infinite pro-p tower, the equivariant Hecke–Laplacian $\Delta^{\mathrm{H}}_\Lambda \in M_n(\Lambda)$ is injective. If the tower group is moreover p-adic analytic, then $X = \operatorname{coker}\Delta^{\mathrm{H}}_\Lambda$ is a torsion module of projective dimension ≤ 1, and $\mathcal{L} = [\Delta^{\mathrm{H}}_\Lambda] \in K_1(Q(\Lambda))$ satisfies $\partial\mathcal{L} = [X]$. In all cases descent gives $X_U \cong \mathbb{Z}_p \oplus \mathrm{Jac}^{\mathrm{H}}(Y_U) \otimes \mathbb{Z}_p$. For every nontrivial Artin representation ρ, $\rho(\mathcal{L}) = \det \Delta^{\mathrm{H}}_\rho \neq 0$, and there is an analytic class number formula $\prod_{\rho\neq 1} \det(\Delta^{\mathrm{H}}_\rho)^{\dim\rho} = [G : U]\,|\,\mathrm{Jac}^{\mathrm{H}}(Y_U)|/|\,\mathrm{Jac}^{\mathrm{H}}(Y)|$. This formula is checked exactly on the abelian covers of X_3; on Y_{21} the product is 7^{18}. The "Selmer group" has no definition for a complex. Replacing it by X makes $\partial\mathcal{L} = [X]$ a theorem, whose content is the descent and the class number formula. Of the five proposed steps, one is used in a different role, one is not needed,

and three do not apply (Section 6.6).

(4) *The canonical tower (Sections 6.4–6.5).* The triples of the Vol. I–II presentation satisfy $x+2y+4z\equiv 0 \pmod 7$. So $a_x \mapsto 1+\alpha^x\varphi$ embeds Γ in the cyclic division algebra $(\mathbb{F}_8(t)/\mathbb{F}_2(t), \mathrm{Frob}, t)$. The building is therefore the Bruhat–Tits building of $\mathrm{PGL}_3(\mathbb{F}_2((s)))$, $s=1+t$; it is not exotic. The closure of $\pi_1(X_3)$ in the units of the maximal order at $t=0$ is a pro-2 group P with $d(P)=3$ and *infinite rank*. This gives an unconditional exhausting pro-2 tower, whose level 2 is Y_{24}. Its group is *not* p-adic analytic: the proposed "uniform pro-p group with Lie algebra $\mathfrak{pgl}_3$" does not exist here. Γ is discrete in $\mathrm{PGL}_3(\mathbb{F}_2((s)))$, so no congruence map exists at that place. Along any exhausting tower $\frac{1}{n}\log|\,\mathrm{Jac}^{\mathrm{H}}\,| \to h_H(2) = 2.063693$, a Plancherel integral; the tower gives 1.8629, 2.0416, 2.0544, 2.0625. The growth of the 2-part stays open.

(5) *Retrospect and outlook.* We summarize the six chapters of Volume III, and we list the open problems that Volume IV inherits.

Status of the results

Every numbered statement is proved, apart from the classical inputs, which are marked where they are used: the PBW theorem and the center of $\mathbb{H}_d$ [36], Neumann's theorem that $\mathbb{Z}_p[[G]]$ is a domain [37], and the localization sequences [38]. Parts (i′) and (ii) of Theorem 6.3.1 need a p-adic analytic tower (Hypothesis (T)); the other parts need only an infinite pro-p tower (T_p). Such a tower exists over X_3 for $p=2$ (Theorem 6.4.2), which answers Chapter 1, Remark 1.9.1(c), for the Vol. I–II complexes. Further quoted inputs are Lazard's rank criterion [7], Oppenheim's trickling-down theorem [41], Macdonald's Plancherel measure [39], and, only in Remark 6.4.4, Margulis [40]. The identities are checked by `code/v3p6_verify.py` and `code/v3p6_deep_verify.py` (Table 6.2).

6.1 Hecke algebras

Definition 6.1.1 (Degenerate affine Hecke algebra [34, 36])**.** Let $c\in\mathbb{C}$. $\mathbb{H}_d$ is generated by $s_1,\dots,s_d$ and $x_1,\dots,x_{d+1}$ subject to the relations: the s_i satisfy the Coxeter relations of S_{d+1}; $x_ix_j = x_jx_i$; $s_ix_j = x_js_i$ for $j\notin\{i,i+1\}$; and $s_ix_{i+1} - x_is_i = c$. By the PBW theorem, $\mathbb{H}_d = \mathbb{C}[S_{d+1}]\otimes\mathbb{C}[x_1,\dots,x_{d+1}]$ as vector spaces. Its center is $\mathbb{C}[x]^{S_{d+1}}$, and its finite-dimensional irreducible modules have dimension at most $(d+1)!$ [36].

Proposition 6.1.2 (Two representations)**.** *(i) On $\mathbb{C}[y_1,\dots,y_{d+1}]$, the assignment $x_i\mapsto y_i$ and $s_i\mapsto \sigma_i - c\,\partial_i$, where $\partial_i f = (f-\sigma_i f)/(y_i-y_{i+1})$, is a representation.*

(ii) For every *endomorphism L of a vector space M, the assignment $s_i\mapsto 1$, $x_i\mapsto L+(i-1)c$ is a representation on M with $x_1 = L$.*

In particular, "L is realized in a representation of $\mathbb{H}_d$" holds for every L, and it says nothing about L.

Proof. (i) We have $\partial_i\sigma_i = -\partial_i$, $\sigma_i\partial_i = \partial_i$ and $\partial_i^2 = 0$, so $(\sigma_i - c\partial_i)^2 = 1$. Also $\partial_i(y_{i+1}f) - y_i\partial_i f = -f$, which gives the cross relation. The braid relation is checked on all polynomials of degree ≤ 4 in line (Dg). (ii) The x_i commute, and $x_{i+1}-x_i = c$. □

The algebra that acts naturally is the one-parameter Iwahori–Hecke algebra of the affine Weyl group of type $\widetilde{A}_2$ [35], acting on functions on chambers. For $i\in\mathbb{Z}/3$, the *panel of cotype i* of a chamber is its edge that misses type i. Define T_i on $\mathbb{C}^F$ by $(T_if)(c) = \sum f(c')$, the sum over chambers $c'\neq c$ with the same cotype-i panel.

Theorem 6.1.3 (Iwahori–Hecke action)**.** *If Y satisfies* (L)*, then $T_i^2 = (q-1)T_i + q$ and $T_iT_jT_i = T_jT_iT_j$ for $i \neq j$. So $\mathbb{C}^F$ is a module over $H_q(\widetilde{A}_2) = \langle T_0, T_1, T_2\rangle$. The flag condition is not needed.*

Proof. Each panel lies in $q+1$ chambers, which gives the quadratic relation. A gallery of type (i,j,i) or (j,i,j) pivots around the vertex v of c of the third type k. By (L), the chambers containing v are the flags of $\mathrm{PG}(2,q)$. Write $c = (P,\ell)$. A non-stuttering gallery of type (i,j,i) runs $(P,\ell) \to (P',\ell) \to (P',\ell') \to (P'',\ell')$, and it ends at a flag opposite to c: $P'' \notin \ell$ and $P \notin \ell'$. Conversely, each opposite flag is reached by exactly one such gallery, with $P' = \ell \cap \ell'$. The same holds for (j,i,j). So both products equal the "opposite flag" matrix of the star. □

Let $\pi_t \colon \mathbb{C}^F \to \mathbb{C}^{V_t}$ sum over the chambers at a vertex of type t. For each cotype i, let $p_i \colon \mathbb{C}^F \to \mathbb{C}^{E_i}$ sum over the chambers on an edge of that cotype.

Theorem 6.1.4 (The operators of Volume III in Hecke language)**.** *Assume* (L)*. Then:*

(i) $\Delta_2 = 3 + T_0 + T_1 + T_2$*, with the type-increasing orientation of chambers;*

(ii) $A_1 = \frac{1}{q+1}\sum_t \pi_t\pi_{t+1}^t$*, so* $\Delta^{\mathrm{H}} = (q^3-1) + A_1 - qA_2$ *and* $\Delta_0 = 2\kappa - A_1 - A_2$*;*

(iii) L_E *maps* $\mathbb{C}^{E_t}$ *to* $\mathbb{C}^{E_{t+2}}$*, and on this block* $L_E = \frac{1}{q(q+1)}\, p_{t+2}\, T_tT_{t+2}T_t\, p_t^t$*;*

(iv) on the joint eigenspace with Satake parameters z (where $z_1z_2z_3 = 1$), $A_1 = qe_1(z)$ and $A_2 = qe_2(z)$. So Δ^{H}, Δ_0 and $\det P(u) = \prod_i(1 - qx_iu)$ are images of symmetric polynomials, i.e. of central elements of $\mathbb{H}_2$ or of H_q modulo $e_3 - 1$. The Kuramoto field of Chapter 3 is nonlinear. Its linearization at the twisted states, $-c_m\Delta_0$, is of this central form.

Proof. (i) All incidences $[t : e]$ are $+1$, and distinct chambers share at most one panel. (ii) $\pi_t\pi_{t+1}^t(v,w)$ counts the chambers containing both v and w, which is $q+1$ for an edge $v \to w$. (iii) An edge $e = x \to y$ has cotype $\tau(y)+1$, and $f = y \to z$ has cotype $\tau(y)-1$. The product $T_tT_{t+2}T_t$ connects exactly the opposite flags at y (proof of Theorem 6.1.3). For $z \notin x$ there are $q(q+1)$ opposite pairs: $q+1$ lines through z, and q points of x off each such line. For $z \in x$ there are none. (iv) If $P_a(u) = \prod(1 - qz_iu)$, then $a = qe_1(z)$ and $q\bar{a} = q^2e_2(z)$. □

Remark 6.1.5. The degenerate algebra $\mathbb{H}_d$ is Lusztig's graded version of H_q, obtained by filtering at a central character [36]. It models the representations of H_q near a given Satake parameter. It does not act on the chain spaces of Y, and nothing in this volume needs it. The spectral data of Chapters 1–4 are central characters, i.e. Satake parameters; the non-central operators (L_E, Δ_2) live in H_q and its parabolic modules.

6.2 Localization and the absence of a $\mathbb{Z}_p$-direction

Definition 6.2.1 (Cohn localization [33])**.** Let Σ be a set of maps between finitely generated projective Λ-modules. A ring map $\Lambda \to \Sigma^{-1}\Lambda$ is the *Cohn localization* if it makes every $\sigma \in \Sigma$ invertible and is universal with that property. If $\Lambda \to \Sigma^{-1}\Lambda$ is injective and stably flat (for instance, an Ore localization), there is an exact sequence [38]

$$K_1(\Lambda) \to K_1(\Sigma^{-1}\Lambda) \xrightarrow{\partial} K_0(\Lambda,\Sigma) \to K_0(\Lambda) \to K_0(\Sigma^{-1}\Lambda).$$

Here $K_0(\Lambda,\Sigma)$ is the K_0 of Σ-torsion modules of projective dimension ≤ 1, and $\partial[\alpha] = [\operatorname{coker}\alpha]$ for injective $\alpha \colon \Lambda^n \to \Lambda^n$ in Σ (this is our sign convention). ∂ is surjective when $K_0(\Lambda) \to K_0(\Sigma^{-1}\Lambda)$ is injective, e.g. when Λ is local and $\Sigma^{-1}\Lambda$ is a skew field.

The Ore set of [1] is $S = \{s \in \Lambda(G) : \Lambda(G)/\Lambda(G)s \text{ is finitely generated over } \Lambda(H)\}$. It requires a closed normal subgroup H with $G/H \cong \mathbb{Z}_p$, and its torsion category is $\mathfrak{M}_H(G)$. The proposed programme's Σ ("$M/M[p^\infty]$ finite, pseudo-null") is not this set.

Theorem 6.2.2 (No $\mathbb{Z}_p$-direction)**.** *Let Y satisfy* (L) *with $q \geq 2$. Let G be a profinite group that receives $\pi_1(Y)$ with dense image, so that its open subgroups U correspond to finite covers Y_U. Then every open $U \leq G$ has finite abelianization. In particular no open subgroup of G has a quotient $\mathbb{Z}_p$, and the Ore set S and the category $\mathfrak{M}_H(G)$ of [1] do not exist for any open subgroup of G.*

Proof. This is Proposition 1.5.3(i),(ii). $H_1(Y_U;\mathbb{Z})$ is finite by Garland rigidity, and it surjects onto a dense subgroup of U^{ab}. □

So the only natural localization is the Goldie quotient. Under the following hypothesis, Λ is a Noetherian domain [11, 37], and the nonzero elements form an Ore set with skew field of fractions $Q(\Lambda)$.

(T_p) $\pi_1(Y) \to G$ has dense image, where G is an infinite pro-p group. Put $\Lambda = \mathbb{Z}_p[[G]]$.

(T) In addition, G is a torsion-free p-adic Lie group. (Such G has dimension ≥ 3 and perfect Lie algebra, by Proposition 1.5.3(iii).)

Section 6.4 shows that (T_2) holds *unconditionally* over X_3, for a canonical tower. It also shows that this tower does *not* satisfy (T).

6.3 The Iwasawa module of Hecke Jacobians

Under (T_p), choose lifts of the vertices of Y to the tower. Each arc $e\colon v \to w$ of Y acquires a voltage $g_e \in G$. Define $\Delta^{\mathrm{H}}_\Lambda \in M_n(\Lambda)$ by

$$(\Delta^{\mathrm{H}}_\Lambda)_{vw} = (q^3-1)\delta_{vw} + \sum_{e:v\to w} g_e - q\sum_{e:w\to v} g_e^{-1},$$

and put $X(Y/G) = \operatorname{coker}\Delta^{\mathrm{H}}_\Lambda$. For U open normal, $\mathbb{Z}_p[G/U]\otimes_\Lambda \Delta^{\mathrm{H}}_\Lambda$ is $\Delta^{\mathrm{H}}(Y_U)\otimes\mathbb{Z}_p$, after identifying $\mathbb{Z}_p[G/U]^n$ with $\mathbb{Z}_p^{V(Y_U)}$.

Theorem 6.3.1 (Main theorem, replacing the main conjecture)**.** *Assume* (L)*, $q \geq 2$, and* (T_p)*. Then* (i)*,* (iii)*,* (iv) *and* (v) *hold. If moreover* (T) *holds, so do* (i′) *and* (ii)*.*

(i) $\Delta^{\mathrm{H}}_\Lambda$ *is injective, and* $0 \to \Lambda^n \to \Lambda^n \to X \to 0$ *is exact.*

(i′) $\Delta^{\mathrm{H}}_\Lambda$ *is invertible over $Q(\Lambda)$, and X is a finitely generated torsion Λ-module of projective dimension ≤ 1.*

(ii) $\mathcal{L}(Y/G) := [\Delta^{\mathrm{H}}_\Lambda] \in K_1(Q(\Lambda))$ *satisfies $\partial\mathcal{L}(Y/G) = [X(Y/G)]$, and every characteristic element of X differs from $\mathcal{L}$ by an element of the image of $K_1(\Lambda)$.*

(iii) Descent: $\mathbb{Z}_p[G/U]\otimes_\Lambda X \cong \mathbb{Z}_p \oplus \big(\mathrm{Jac}^{\mathrm{H}}(Y_U)\otimes\mathbb{Z}_p\big)$*.*

(iv) Interpolation: for every irreducible representation $\rho \neq 1$ of G/U, $\rho(\Delta^{\mathrm{H}}_\Lambda) = \Delta^{\mathrm{H}}_\rho$ and $\det\Delta^{\mathrm{H}}_\rho \neq 0$. Moreover

$$\prod_{\rho\neq 1}\det(\Delta^{\mathrm{H}}_\rho)^{\dim\rho} = [G:U]\,\frac{|\,\mathrm{Jac}^{\mathrm{H}}(Y_U)|}{|\,\mathrm{Jac}^{\mathrm{H}}(Y)|}.$$

(v) In the variable u: $\det P_{Y_U}(u) = \prod_\rho \det P_\rho(u)^{\dim \rho}$*, and the trivial factor is governed by the derivative identity of Theorem 1.6.4.*

Proof. (i) Let $x \in \Lambda^n$ with $\Delta^{\mathrm{H}}_\Lambda x = 0$, and let x_U be its image at level U. Then $\Delta^{\mathrm{H}}(Y_U)x_U = 0$, so x_U is constant (Lemma 1.6.2 applied to Y_U). That is, $x_U = c_U N_U \mathbf{1}$, where $N_U = \sum_{g \in G/U} g$. Compatibility under $U' \subset U$ gives $c_U = [U : U']c_{U'}$. Since G is infinite pro-p, the indices $[U : U']$ are unbounded powers of p, so $c_U = 0$ and $x = 0$. Only (T_p) has been used so far. (i′) Under (T), Λ is an Ore domain, and an injective square matrix stays injective over $Q(\Lambda)$ (clear denominators), hence is invertible. Its cokernel has rank 0, and $0 \to \Lambda^n \to \Lambda^n \to X \to 0$ is a free resolution. (ii) is the definition of ∂. Two characteristic elements have the same image under ∂, so by exactness they differ by an element of the image of $K_1(\Lambda)$. (iii) Tensor products are right exact. In the proof of Theorem 1.6.4, $\operatorname{coker} \Delta^{\mathrm{H}}(Y_U) \cong \mathbb{Z} \oplus \operatorname{Jac}^{\mathrm{H}}(Y_U)$. (iv) Over $\mathbb{C}$, $\Delta^{\mathrm{H}}(Y_U) \cong \bigoplus_\rho (\Delta^{\mathrm{H}}_\rho)^{\oplus \dim \rho}$. The kernel is $\mathbb{C}\mathbf{1}$, in the trivial summand. The other eigenvalues have positive real part by the sector lemma. The product of the nonzero eigenvalues of $\Delta^{\mathrm{H}}(Y_U)$ is $n_U |\operatorname{Jac}^{\mathrm{H}}(Y_U)|$ (proof of Theorem 1.6.4), and similarly for Y. Divide, using $n_U/n_Y = [G : U]$. (v) The same decomposition applies to $P(u)$. □

Remark 6.3.2 (Finite abelian check). Statement (iv) is a finite-level identity, and it holds for every finite regular cover. For the abelian covers of X_3 it is checked exactly (line (CN)): $|\operatorname{Jac}^{\mathrm{H}}(X_3)| = 147$, and $\prod_{\chi \neq 1} \det \Delta^{\mathrm{H}}_\chi$ equals $7 \cdot 3 \cdot 7^{19}/147 = 7^{18}$ for Y_{21}, $8|\operatorname{Jac}^{\mathrm{H}}(Y_{24})|/147$ for Y_{24}, and $14|\operatorname{Jac}^{\mathrm{H}}(Y_{42})|/147$ for Y_{42}. On Y_{21} the value 7^{18} is also $Q_0(1)$ of Chapter 1: two readings of the same determinant. By Theorem 6.2.2 no infinite abelian tower exists. Y_{21} and Y_{42} illustrate (iv) only, while Y_{24} is level 2 of the canonical 2-tower of Section 6.4.

Remark 6.3.3 (What happened to the Selmer group). "Sel(Y/G)" has no definition for a simplicial complex. The natural candidate is the Iwasawa module $X(Y/G)$ of Hecke Jacobians, with $\mathbb{Z}_p$-coinvariants the p-parts of the $\operatorname{Jac}^{\mathrm{H}}(Y_U)$. With this choice $\partial\mathcal{L} = [X]$ holds by construction. The non-formal content is the descent (iii) and the class number formula (iv), in the spirit of Iwasawa theory for graphs. Because G^{ab} is finite, there is no Iwasawa μ–λ formula along a $\mathbb{Z}_p$-direction. The total order $|\operatorname{Jac}^{\mathrm{H}}(Y_U)|$ grows like $e^{h_H n_U}$ (Theorem 6.5.1). The growth of the p-part is open (Remark 6.5.2).

6.4 The Cartwright–Steger model and the canonical 2-tower

Let $E = \mathbb{F}_8 = \mathbb{F}_2(\alpha)$ with $\alpha^3 = \alpha + 1$, and let $\sigma(a) = a^2$. Put $K = \mathbb{F}_2(t)$ and let D be the cyclic algebra $(E(t)/K, \sigma, t)$: $D = \bigoplus_{j<3} E(t)\varphi^j$, $\varphi a = \sigma(a)\varphi$, $\varphi^3 = t$. It is a division algebra of degree 3, because t is not a norm from $E(t)$ (v_t of a norm is divisible by 3). Let Γ be the group of the triangle presentation of Vol. I–II, Construction 16.1.4, with generators a_x ($x \in \mathbb{Z}/7$) and relations $a_x a_y a_z = 1$ for $(x, y, z) \in T$. It acts simply transitively on the vertices of an $\widetilde{A}_2$ building B, and $\Gamma^0 = \pi_1(X_3)$ is its type-preserving subgroup of index 3.

Theorem 6.4.1 (The Cartwright–Steger model, in the style of [42])**.** *(i) The assignment $a_x \mapsto u_x := 1 + \alpha^x \varphi$ defines a homomorphism $\rho\colon \Gamma \to D^\times/K^\times$. Indeed $u_x u_y u_z = 1 + t$ for every $(x, y, z) \in T$.*

(ii) Put $s = 1 + t$. Then $D \otimes \mathbb{F}_2((s)) \cong M_3(\mathbb{F}_2((s)))$. The group $\rho(\Gamma)$ acts on the Bruhat–Tits building B_s of $\mathrm{PGL}_3(\mathbb{F}_2((s)))$, and the $\rho(a_x)v_0$ are the 7 neighbours of one type of the base vertex v_0.

(iii) *ρ is injective, $\rho(\Gamma)$ acts simply transitively on the vertices of B_s, and $B \cong B_s$. In particular the building of Vol. I–II is Bruhat–Tits, not exotic, and $X_3, Y_{21}, \dots, Y_{168}$ are quotients of the building of* $\mathrm{PGL}_3(\mathbb{F}_2((s)))$.

Proof. (i) For $\beta_i \in E$, $(1+\beta_1\varphi)(1+\beta_2\varphi)(1+\beta_3\varphi) = 1 + (\beta_1+\beta_2+\beta_3)\varphi + (\beta_1\sigma\beta_2 + \beta_1\sigma\beta_3 + \beta_2\sigma\beta_3)\varphi^2 + \beta_1\sigma(\beta_2)\sigma^2(\beta_3)\,t$. Take $\beta = (\alpha^x, \alpha^y, \alpha^z)$ with $(x,y,z) \in T$. The triples of T satisfy $x + 2y + 4z \equiv 0 \pmod 7$, so the last coefficient is 1, and the other two coefficients vanish (line (CS); the φ-coefficient vanishes because $\{x,y,z\}$ is a line of $\mathrm{PG}(2,2) = \mathbb{F}_8^\times/\mathbb{F}_2^\times$).

(ii) At s, $E \otimes \mathbb{F}_2((s)) = \mathbb{F}_8((s))$ is unramified and $t = 1+s$ is a unit. So $t = N(e)$ for some unit $e \in \mathbb{F}_8[[s]]$. Then $\varphi' = e^{-1}\varphi$ satisfies $\varphi'^3 = 1$, so $D_s \cong (\mathbb{F}_8((s)), \sigma, 1) = \mathrm{End}_{\mathbb{F}_2((s))}(\mathbb{F}_8((s)))$, with φ' acting as Frobenius. Let v_0 be the class of the lattice $L_0 = \mathbb{F}_8[[s]]$. Since $(\alpha^x\varphi)^3 = t$, the reduced norm is $\mathrm{Nrd}(u_x) = 1 + t = s$. Also $u_x = 1 + \alpha^x e \varphi'$ is integral, so $sL_0 \subset u_xL_0 \subset L_0$ with $u_xL_0/sL_0 = \operatorname{im}\bar{u}_x$, where $\bar{u}_x(y) = y + \beta_x y^2$, $\beta_x = \alpha^x\bar{e}$. Similarly $su_y^{-1}L_0/sL_0 = \ker\bar{u}_y = \{0, \beta_y^{-1}\}$. Line (NB) checks, for every $\bar{e} \neq 0$, that the seven images are the seven hyperplanes of $\mathbb{F}_8$, that the seven kernels are the seven lines, and that $\ker\bar{u}_y \subset \operatorname{im}\bar{u}_x$ iff $(x,z,y) \in T$ for some z. So the link of v_0 in B_s is matched with the link of the base vertex of B, incidence for incidence.

(iii) Define $f\colon B \to B_s$ on vertices by $f(\gamma v) = \rho(\gamma)v_0$. This is well defined because Γ is simply transitive on the vertices of B. It is simplicial and equivariant, and by (ii) it is an isomorphism from each vertex star onto the image star. Hence f is a covering map. B_s is contractible, and B is connected, so f is an isomorphism. If $\rho(\gamma) = 1$, then $f(\gamma v) = f(v)$, so $\gamma v = v$ and $\gamma = 1$. □

At $t = 0$ the algebra $D_0 = D \otimes \mathbb{F}_2((t))$ is a division algebra, with maximal order $\mathcal{O} = \mathbb{F}_8[[\varphi;\sigma]]$ and uniformizer φ. Put

$$P = (1+\varphi\mathcal{O})/(1+t\mathbb{F}_2[[t]]), \qquad P_m = \text{image of } 1+\varphi^m\mathcal{O}.$$

Theorem 6.4.2 (The canonical 2-tower)**.** *(i) $\rho(\Gamma^0) \subset P$, and its closure is all of P.*

(ii) $\Phi(P) = P_2$, so $d(P) = 3$, and $|P/P_m| = 2^{3(m-1)-\lfloor(m-1)/3\rfloor}$.

(iii) $d(P_k) \geq \dim_{\mathbb{F}_2} P_k/P_{2k} \geq 2k$. So P has infinite rank, and it is not a p-adic analytic group for any p.

(iv) The groups $U_m = \Gamma^0 \cap \rho^{-1}(P_m)$ are normal in Γ^0, with $\Gamma^0/U_m \cong P/P_m$ and $\bigcap_m U_m = 1$. The covers $Y_m = U_m\backslash B$ satisfy (L)*, $Y_1 = X_3$, and $Y_2 = Y_{24}$.*

Proof. (i) Each $u_x \equiv 1 \bmod \varphi$, and $\mathrm{Nrd}(u_x) = 1+t$ is a unit at $t = 0$, so $\rho(\Gamma) \subset \mathcal{O}^\times/\mathbb{F}_2[[t]]^\times$ and $\rho(\Gamma^0) \subset P$. The Schreier generators $u_0^s u_x u_0^{-((s+1) \bmod 3)}$ reduce modulo P_2 to $\alpha^x + 1$ or α^x, and these span $P/P_2 \cong \mathbb{F}_8$ (line (FR)). By the Burnside basis theorem for pro-p groups and (ii), $\overline{\rho(\Gamma^0)} = P$.

(ii) The level-k quotient $(1+\varphi^k\mathcal{O})/(1+\varphi^{k+1}\mathcal{O})$ is $\mathbb{F}_8$, via $1 + c\varphi^k \mapsto c$. In P the centre removes $\mathbb{F}_2$ when $3 \mid k$. We have $[1+a\varphi^i, 1+b\varphi^j] \equiv 1 + (a\sigma^i(b) + b\sigma^j(a))\varphi^{i+j}$ and $(1+a\varphi^i)^2 = 1 + a\sigma^i(a)\varphi^{2i}$. For $k \geq 2$ these values span the level-k quotient of P. This depends only on k mod 6, and it is checked for $k = 2, \dots, 13$ in line (FR). By successive approximation in the complete filtered group, $P_2 \subset \Phi(P)$. The reverse inclusion holds because P/P_2 is elementary abelian. The order formula counts 3 bits per level, minus one at each level divisible by 3.

(iii) In characteristic 2, $(1+x)^2 = 1 + x^2$, and commutators of elements of P_k lie in P_{2k}. So $\Phi(P_k) \subset P_{2k}$ and $d(P_k) \geq \dim P_k/P_{2k}$. A pro-$p$ group is p-adic analytic iff it has finite rank [7]. An infinite pro-2 group is not p-adic analytic for odd p, because its open pro-p subgroups would be trivial.

(iv) The U_m are preimages of normal subgroups, and they surject by density. Their intersection is $\ker\rho\cap\Gamma^0=1$ (Theorem 6.4.1(iii)). Covers are local isomorphisms, so (L) is inherited. $P/P_2\cong(\mathbb{Z}/2)^3$ is the maximal elementary abelian 2-quotient of Γ^0, since $\dim H_1(X_3;\mathbb{F}_2)=3$. So Y_2 is the unique such cover, Y_{24}, and its Hecke Jacobian agrees (line (L2)). □

Corollary 6.4.3 (Status of the hypotheses)**.** *Over X_3 (and each Y_m), (T_2) holds unconditionally, with $G=P$. Hence Theorem 6.3.1(i),(iii),(iv),(v) hold unconditionally along the canonical 2-tower. The tower does not satisfy (T). Parts (i′) and (ii) remain open for it, because we do not know whether $\mathbb{Z}_2[[P]]$ is Noetherian or has a skew field of fractions.*

Remark 6.4.4 (Analytic towers). Γ is a cocompact lattice in $\mathrm{PGL}_3(\mathbb{F}_2((s)))$, a rank-2 group over a local field of characteristic 2, and it is S-arithmetic in D. Congruence towers at the other places of $\mathbb{F}_2(t)$ are again characteristic-2 groups of infinite rank. By Margulis' superrigidity and arithmeticity [40], one expects every representation of Γ over a field of characteristic 0 to have finite image (we have not re-checked the precise statement). If so, no infinite p-adic Lie tower exists over X_3 for any p, and (T) is false for every complex of Vol. I–II. The request to make (T) unconditional is therefore answered by Theorem 6.4.2 and Corollary 6.4.3: the natural tower is pro-2 and not analytic, and the theory has to be based on (T_2), not on (T).

6.5 Growth of Hecke Jacobians

Let ν_q be the spectral measure of A_1 at a vertex of an $\widetilde{A}_2$ building of order q. It depends only on q, because closed-walk counts do. For the Bruhat–Tits building it is Macdonald's spherical Plancherel measure [39]: $a=q(z_1+z_2+z_3)$ on $\{|z_i|=1,\ z_1z_2z_3=1\}$, with density proportional to $\prod_{i<j}|1-z_j/z_i|^2/|1-q^{-1}z_j/z_i|^2$.

Theorem 6.5.1 (Exponential growth of the Hecke Jacobian)**.** *Let Y_m be an exhausting tower of normal covers of a complex satisfying* (L) *with $q\geq 2$, i.e. U_m normal, decreasing, and $\bigcap U_m=1$. Then*

$$\lim_{m\to\infty}\frac{\log|\operatorname{Jac}^{\mathrm{H}}(Y_m)|}{n_m}=h_H(q):=\int\log|q^3-1+a-q\bar{a}|\,d\nu_q(a).$$

For $q=2$, $h_H(2)=2.063693$. Along the canonical 2-tower the values are 1.862909, 2.041580 *and* 2.054414 *at $m=2,3,4$ (line (J)), and* 2.062520 *at $m=5$, where $n=6144$ (exploratory run).*

Proof. Normality and $\bigcap U_m=1$ make the injectivity radius tend to infinity. Hence the moments $\frac{1}{n_m}\operatorname{Tr}(A_1^jA_2^k)$ converge to those of ν_q, and the empirical measures of the joint eigenvalues converge weakly. They live on a compact subset of $\mathbb{C}$, where polynomials in a and $\bar{a}$ are dense. By Theorem 1.6.4, $\log|\operatorname{Jac}^{\mathrm{H}}(Y_m)|=\sum_{a_i\neq\kappa}\log|\mu_i|-\log n_m$. The only zero of $a\mapsto q^3-1+a-q\bar{a}$ is $a=\kappa$. Trickling down [41] from the link gap gives a uniform bound: every nontrivial eigenvalue of the normalized 1-skeleton walk is at most $\sqrt{q}/(q+1-\sqrt{q})$. Hence $\operatorname{Re}\mu_i\geq(q^3-1)(\sqrt{q}-1)^2/(q+1-\sqrt{q})>0$ for all levels and all $a_i\neq\kappa$. So $\log|\mu|$ is bounded and continuous on a set carrying all the measures, and the integrals converge. □

Remark 6.5.2 (The 2-part). Along the canonical tower, $v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)|=0,18,99,289$ for $m=1,2,3,4$, with $n=3,24,192,768$ ($|\operatorname{Jac}^{\mathrm{H}}(X_3)|=147$ is odd). Per vertex this is $0,0.75,0.52,0.38$; per unit of index it is $0,2.25,1.55,1.13$. No asymptotic law is proved. P is not analytic, so the growth theorems of analytic Iwasawa theory do not apply, and "pseudo-null" and "characteristic divisor" have no established meaning over $\mathbb{Z}_2[[P]]$. The total order grows like e^{h_Hn} (Theorem 6.5.1). Whether v_2 grows linearly in n is open.

6.6 The five-step scheme

proposed step	status
① Garland excludes H^1 poles	True fact (Theorem 1.4.3), but it plays another role. It makes every U^{ab} finite and so *removes* the $\mathbb{Z}_p$-direction (Theorem 6.2.2). The only degenerate factor of $\mathcal{L}$ is the trivial representation, which is handled in (iv).
② unitary remainder fixes Euler factors	Not used. The remainder of L_E (Chapter 4) lives on edges and does not enter the vertex Jacobians.
③ $\mathbb{Z}/3$ chirality cancels odd obstructions	Does not apply. What is true is $\det P(u) \in \mathbb{Z}[u^3]$ and $\mathrm{Tr}(C^{(3)})^k = 0$ for $3 \nmid k$ (Proposition 1.2.1). No "odd obstruction" is defined.
④ spectral rigidity: determinant = Fitting ideal	Does not apply in the stated form. Over commutative $\mathbb{Z}_p[G/U]$, $\mathrm{Fitt}_0(\mathrm{coker}\, M) = (\det M)$ for square M, which is a tautology. Over non-commutative Λ, Fitting ideals are not defined in general, and K_1 replaces them.
⑤ spectrum = class field invariants	Does not apply. No class field theory is involved. What is proved is descent and the class number formula (Theorem 6.3.1(iii),(iv)).

6.7 Volume III in retrospect, and Volume IV

What Volume III established. Chapter 1 proved local Garland rigidity from the link gap alone, the cubic trace identity with and without the flag condition, and the exact Hecke pencil. Chapter 2 showed that no higher simplicial sandpile is abelian, and proved the Hodge factorization of critical groups and exact avalanche means. Chapter 3 showed that the gauge sector of chiral Kuramoto dynamics is a gradient flow, and derived the frustrated ground states and the block spectra. Chapter 4 proved the unconditional unitary remainder of the geodesic edge operator, and its Riemann hypothesis equivalent to Ramanujan. Chapter 5 showed that chamber public goods reduce to a pairwise game, and that thickness gives no catalysis. Chapter 6 described the Hecke action on chambers, the absence of a $\mathbb{Z}_p$-direction, and the Iwasawa module of Hecke Jacobians. The recurring lesson is that thick $\widetilde{A}_2$ complexes are *rigid* (Garland) and *expanding*, and most proposed higher-order effects either collapse to the 1-skeleton or are obstructed by rigidity. The surviving structure is spectral and local: links, Satake parameters, Hecke operators.

What Volume IV inherits. Precise open problems. Problems (a) and (b) of the first version of this list are now settled as far as stated in Sections 6.4 and 6.5. The complexes of Vol. I–II are Bruhat–Tits quotients in characteristic 2 with an unconditional non-analytic pro-2 tower, and $|\,\mathrm{Jac}^{\mathrm{H}}\,|$ grows like $e^{2.063693\,n}$. What remains of them: (a) whether $\mathbb{Z}_2[[P]]$ admits a localization in which Theorem 6.3.1(ii) holds (Section 6.9 proves the finite-level form unconditionally and reduces the question to whether $\mathbb{Z}_2[[P]]$ is an Ore domain), and the characteristic-0 statement of Remark 6.4.4; (b) the growth of $v_2|\,\mathrm{Jac}^{\mathrm{H}}(Y_m)|$ along the canonical tower; (c) the vertex block of L_E and the trace identities for $\widetilde{A}_d$, $d \geq 3$; (d) universality, or its failure, for the eigenphases of the unitary remainder; (e) finite-noise thresholds and genuinely nonlinear chamber games; (f) chimeras and transverse Hopf bifurcations for the triadic lag. The first plan for Volume IV (affine Kac–Moody algebras and cosmology) was replaced by the authors with topological economics and higher-order games, which continue Chapter 5. Nothing proved here points to cosmology. For the geometric problems above, the natural continuation remains Kac–Moody buildings and their lattices, where Garland-type rigidity may fail.

6.8 The 2-part along the canonical tower

This section takes up Remark 6.5.2. We write $G_m = P/P_m$, $n_m = 3|G_m|$, $N_m = P_m/P_{m+1}$ and $d_m = \log_2|N_m|$, so $d_m = 2$ if $3 \mid m$ and $d_m = 3$ otherwise (Theorem 6.4.2(ii)). The vertices of Y_m are the pairs (g,s), $g \in G_m$, $s \in \mathbb{Z}/3$, and the type-increasing arcs are $(g,s) \to (g\theta_{s,x}, s+1)$, $x \in \mathbb{Z}/7$, with the Schreier generators $\theta_{s,x} = \rho(a_0^s a_x a_0^{-(s+1)})$ of the proof of Theorem 6.4.2. The group N_m is central in G_{m+1} and elementary abelian (the commutator and square formulas in that proof land in P_{m+1} and P_{2m}), and $Y_{m+1} \to Y_m$ is a regular cover with group N_m. Since $\operatorname{coker}\Delta^{\mathrm{H}}(Y_m) = \mathbb{Z} \oplus \operatorname{Jac}^{\mathrm{H}}(Y_m)$, the Smith form of $\Delta^{\mathrm{H}}(Y_m)$ has one zero invariant factor; let e_m be the number of its *even* nonzero invariant factors, i.e. the number of cyclic factors of even order in the invariant-factor decomposition of $\operatorname{Jac}^{\mathrm{H}}(Y_m)$, and let κ_m be the largest 2-adic valuation of an invariant factor. Trivially $e_m \le v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| \le \kappa_m e_m$.

Proposition 6.8.1 (Block decomposition of a layer)**.** *Fix $m \ge 1$ and representatives $\tilde g \in G_{m+1}$ of the cosets of N_m. Every arc $e\colon v \to w$ of Y_m, $v = (\bar g, s)$ with label x, acquires a voltage $c_e \in N_m$ by $\tilde g_v \theta_{s,x} = c_e \tilde g_w$. For a character χ of N_m (all of them are real, with values ±1) put $A_\chi = \sum_e \chi(c_e) E_{vw}$ and $D_\chi = 7I + A_\chi - 2A_\chi^t \in M_{n_m}(\mathbb{Z})$.*

(i) $D_1 = \Delta^{\mathrm{H}}(Y_m)$ *and* $D_\chi \equiv \Delta^{\mathrm{H}}(Y_m) \pmod 2$ *for every χ. With $H = (\chi(c))_{\chi,c}$, which satisfies $HH^t = 2^{d_m} I$,*

$$(H \otimes I)\,\Delta^{\mathrm{H}}(Y_{m+1})\,(H \otimes I)^{-1} = \bigoplus_\chi D_\chi .$$

The conjugation is defined over $\mathbb{Z}[\frac12]$, not over $\mathbb{Z}_2$.

(ii) $\det D_\chi \ne 0$ *for $\chi \ne 1$, and*

$$|\operatorname{Jac}^{\mathrm{H}}(Y_{m+1})| = 2^{-d_m}\,|\operatorname{Jac}^{\mathrm{H}}(Y_m)| \prod_{\chi\ne1} |\det D_\chi|,$$

$$v_2|\operatorname{Jac}^{\mathrm{H}}(Y_{m+1})| = v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| - d_m + \sum_{\chi\ne1} v_2 \det D_\chi .$$

(iii) Each D_χ, $\chi \ne 1$, has exactly $e_m + 1$ even invariant factors. Hence $\sum_{\chi\ne1} v_2 \det D_\chi \ge (2^{d_m}-1)(e_m+1)$ and

$$v_2|\operatorname{Jac}^{\mathrm{H}}(Y_{m+1})| \;\ge\; v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| - d_m + (2^{d_m}-1)(e_m+1).$$

(iv) The maps $a_x \mapsto a_{2x}$ and $a_x \mapsto a_{x+1}$ are automorphisms of Γ which extend to the tower. They act on $N_m = \{1 + c\varphi^m\}$ by $c \mapsto c^2$ and $c \mapsto \alpha^{2^m-1}c$, and they permute the nontrivial characters of N_m transitively. Consequently all D_χ, $\chi \ne 1$, have the same invariant factors over $\mathbb{Z}_2$.

Proof. The lifts of $v = (\bar g, s)$ are $(c\tilde g_v, s)$, $c \in N_m$, and since N_m is central the arc of label x at $c\tilde g_v$ ends at $c\tilde g_v\theta_{s,x} = c\,c_e\tilde g_w$. In the basis indexed by (c,v), $A_1(Y_{m+1}) = \sum_e R(c_e) \otimes E_{vw}$, with R the regular representation of N_m. The characters diagonalize all $R(c)$ at once, $HR(c)H^{-1} = \operatorname{diag}(\chi(c))$; $H \otimes I$ is real, its inverse is $2^{-d_m}(H^t \otimes I)$, and transposition commutes with this conjugation. This gives the decomposition of $\Delta^{\mathrm{H}} = 7 + A_1 - 2A_1^t$, and $D_1 = \Delta^{\mathrm{H}}(Y_m)$ because the voltages of the trivial character are 1. Between two vertices of Y_m there is at most one arc, so A_χ has entries ±1 exactly on the arcs, and $D_\chi \equiv \Delta^{\mathrm{H}}(Y_m)$ modulo 2. Finally $\det H = \pm 2^{d_m 2^{d_m-1}}$, so H is not invertible over $\mathbb{Z}_2$. This proves (i).

(ii) The spectrum of $\Delta^{\mathrm{H}}(Y_{m+1})$ is the union of the spectra of the D_χ. Its kernel is $\mathbb{C}\mathbf{1}$ by the sector lemma (Lemma 1.6.2), and it lies in the trivial block, so the D_χ, $\chi \neq 1$, are nonsingular. The product of the nonzero eigenvalues of $\Delta^{\mathrm{H}}(Y)$ is $\pm n_Y |\operatorname{Jac}^{\mathrm{H}}(Y)|$ (proof of Theorem 1.6.4; this is the step used in Theorem 6.3.1(iv)). Dividing the identities for Y_{m+1} and Y_m, with $n_{m+1}/n_m = 2^{d_m}$, gives the product formula, and v_2 of it gives the valuation identity. This is the class number formula of Theorem 6.3.1(iv) for the layer Y_{m+1}/Y_m, where it is unconditional (Remark 6.3.2).

(iii) For a nonsingular integer matrix, the number of even invariant factors is its corank over $\mathbb{F}_2$. By (i) the $\mathbb{F}_2$-corank of D_χ equals that of $\Delta^{\mathrm{H}}(Y_m)$, which is $e_m + 1$ because of the zero invariant factor. Each even invariant factor contributes at least 1 to $v_2 \det D_\chi$, and there are $2^{d_m} - 1$ nontrivial characters.

(iv) The set T is invariant under $x \mapsto 2x$ and $x \mapsto x + 1$ (line (SY)), so both maps are automorphisms of Γ, and they preserve Γ^0. On D the first is induced by the Frobenius σ applied to the coefficients of $\mathbb{F}_8[[\varphi;\sigma]]$, which is a ring automorphism because σ commutes with itself; the second by conjugation with α^{-1}, since $\alpha^{-1}(\alpha^x\varphi)\alpha = \alpha^{x-1}\sigma(\alpha)\varphi = \alpha^{x+1}\varphi$. Both fix the centre and the filtration $1 + \varphi^m\mathcal{O}$, so they preserve every $U_m = \Gamma^0 \cap \rho^{-1}(P_m)$ and induce type-preserving automorphisms ψ_m of the Y_m that commute with the projections (checked on the vertex model for $m \leq 5$ in line (SY)). On N_m: $\sigma(1 + c\varphi^m) = 1 + c^2\varphi^m$ and $\alpha^{-1}(1 + c\varphi^m)\alpha = 1 + \alpha^{-1}c\,\sigma^m(\alpha)\varphi^m = 1 + \alpha^{2^m-1}c\,\varphi^m$. If $3 \nmid m$, then $2^m \not\equiv 1 \pmod 7$, so α^{2^m-1} generates $\mathbb{F}_8^\times$, and multiplication by it is transitive on the nontrivial characters of $\mathbb{F}_8$ (identified with $\mathbb{F}_8^\times$ through the trace form). If $3 \mid m$, then $N_m = \mathbb{F}_8/\mathbb{F}_2$, on which σ has order 3 and no nonzero fixed point, so it permutes the three nontrivial characters cyclically. Now let ψ be one of these automorphisms of the layer, with $\psi(c \cdot \tilde{y}) = f(c) \cdot \psi(\tilde{y})$ for $c \in N_m$. Writing $\psi(\tilde{g}_v) = b_v \tilde{g}_{\psi v}$ with $b_v \in N_m$, the voltage of the image arc is $c_{\psi e} = f(c_e)\, b_w b_v^{-1}$. Hence A_χ, after the vertex permutation ψ, equals $B\, A_{\chi\circ f}\, B^{-1}$ with $B = \operatorname{diag}(\chi(b_v))$: the blocks of χ and $\chi \circ f$ are conjugate by a signed permutation matrix, which is unimodular, and they have the same Smith form. The orbits of f on the nontrivial characters are transitive, which proves (iv). The common valuations are 12, 64 and 208 for the layers $m = 2, 3, 4$ (line (V2)). □

Theorem 6.8.2 (The 2-part at level 5)**.** $v_2|\operatorname{Jac}^{\mathrm{H}}(Y_5)| = 1742$*, with* $n_5 = 6144$*. The invariant factors of* $\Delta^{\mathrm{H}}(Y_5)$ *have the* 2*-adic profile of Table 6.1, with* $e_5 = 546$ *even factors and* $\kappa_5 = 11$*.*

Proof. Proposition 6.8.1(ii) for the layer $Y_5 \to Y_4$ ($d_4 = 3$, seven twisted blocks of size 768) gives $v_2|\operatorname{Jac}^{\mathrm{H}}(Y_5)| = 289 - 3 + 7 \cdot 208 = 1742$. The valuations $v_2 \det D_\chi$ are computed by 2-adic elimination modulo 2^{30}, which never reaches the cap (every block ends with no zero invariant factor, and $\Delta^{\mathrm{H}}(Y_m)$ with exactly one). The same procedure recovers $99 = 18 - 3 + 7 \cdot 12$ for $Y_3 \to Y_2$ and $289 = 99 - 2 + 3 \cdot 64$ for $Y_4 \to Y_3$ from the values of Remark 6.5.2, which are themselves obtained by direct elimination (line (V2)); the exact Smith forms at levels 2 and 3 agree with the eliminations (line (SM)), and the traces $\operatorname{Tr} A_1(Y_{m+1})^k$, $k = 3, 6$, equal the sums of the block traces (line (BL)). Independently, the direct 2-adic elimination of the 6144×6144 matrix gives 1742 again, together with the profile of the table (line (FD), about three minutes). The block decomposition also gives $\frac{1}{n_5}\log|\operatorname{Jac}^{\mathrm{H}}(Y_5)| = 2.062520$, the value of Theorem 6.5.1 that was obtained there from the full spectrum (line (TO)). □

The number e_m is an invariant of the group ring $\mathbb{F}_2[G_m]$.

Proposition 6.8.3 (Even invariant factors and the Schur complement)**.** *Let* $\bar{f}_s = \sum_x \theta_{s,x}^{-1} \in \mathbb{F}_2[G_m]$ *and* $F_m = 1 + \bar{f}_1\bar{f}_0\bar{f}_2$*, and let* $C_m = \mathbb{F}_2[G_m]/\mathbb{F}_2[G_m]F_m$*.*

(i) $\ker_{\mathbb{F}_2}(\Delta^{\mathrm{H}}(Y_m) \bmod 2) = \ker(I + A_1) \cong \{v \in \mathbb{F}_2[G_m] : vF_m = 0\}$*, and* $e_m + 1 = \dim_{\mathbb{F}_2} C_m$*.*

(ii) F_m *lies in the augmentation ideal* I *of* $\mathbb{F}_2[G_m]$*, the projection* $\mathbb{F}_2[G_{m+1}] \to \mathbb{F}_2[G_m]$ *maps* F_{m+1} *to* F_m*, and* $C_{m+1}/I_{N_m}C_{m+1} \cong C_m$*, where* I_{N_m} *is the augmentation ideal of* $\mathbb{F}_2[N_m]$*.*

(iii) $e_{m+1} \geq e_m$*, and* $(e_{m+1}+1)/n_{m+1} \leq (e_m+1)/n_m$*. Hence the limit* $c_e = \lim_m (e_m+1)/n_m$ *exists.*

(iv) For $m = 2, \dots, 5$*:* $e_m = 6, 36, 99, 546$*,* $\dim C_m = 7, 37, 100, 547$*, and* $F_m \in I^3 \setminus I^4$ *(the Loewy lengths of* $\mathbb{F}_2[G_m]$ *are* $4, 10, 16, 28$*). The resulting bound* $e_m + 1 \geq \dim \mathbb{F}_2[G_m]/I^3 = 10$ *(*$m \geq 3$*) does not grow with* m*.*

Proof. (i) Identify $\mathbb{F}_2^{V(Y_m)} = \bigoplus_s \mathbb{F}_2[G_m]$ by $v \mapsto (v_s)_s$, $v_s = \sum_g v(g,s)\, g$. Then $(A_1 v)(g,s) = \sum_x v(g\theta_{s,x}, s+1)$, i.e. $(A_1 v)_s = v_{s+1}\bar{f}_s$ (right multiplication, which commutes with the left action of G_m). In characteristic 2, $v \in \ker(I + A_1)$ iff $v_s = v_{s+1}\bar{f}_s$ for all s, i.e. $v_1 = v_2\bar{f}_1$, $v_0 = v_2\bar{f}_1\bar{f}_0$ and $v_2 = v_0\bar{f}_2 = v_2\bar{f}_1\bar{f}_0\bar{f}_2$. So v is determined by v_2, and v_2 is admissible iff $v_2 F_m = 0$. The left annihilator of F_m is the kernel of the right multiplication R_{F_m} on $\mathbb{F}_2[G_m]$, whose dimension equals $\dim \operatorname{coker} R_{F_m} = \dim C_m$. Finally $\dim \ker_{\mathbb{F}_2} \Delta^{\mathrm{H}}(Y_m)$ is the number of invariant factors divisible by 2, which is $e_m + 1$ (line (E) computes both sides).

(ii) The augmentation of F_m is $1 + 7^3 = 344 \equiv 0$. The projection is a ring map sending $\theta_{s,x}$ to $\theta_{s,x}$, with kernel $I_{N_m}\mathbb{F}_2[G_{m+1}]$ (a two-sided ideal, N_m being normal). C_{m+1} is a left $\mathbb{F}_2[G_{m+1}]$-module, since left multiplication commutes with $R_{F_{m+1}}$, and $C_{m+1}/I_{N_m}C_{m+1} = \mathbb{F}_2[G_m]/\mathbb{F}_2[G_m]F_m = C_m$.

(iii) C_m is a quotient of C_{m+1}, so $e_{m+1} \geq e_m$. The ring $\mathbb{F}_2[N_m]$ is local with nilpotent maximal ideal I_{N_m} (N_m is a 2-group). By Nakayama's lemma, lifts of a basis of $C_m = C_{m+1}/I_{N_m}C_{m+1}$ generate C_{m+1} over $\mathbb{F}_2[N_m]$, so $\dim C_{m+1} \leq |N_m| \dim C_m$, i.e. $(e_{m+1}+1)/n_{m+1} \leq (e_m+1)/n_m$. A non-increasing sequence of nonnegative numbers converges.

(iv) Lines (E) and (AU): I^k is computed as $\sum_g I^{k-1}(g-1)$ over three generators of G_m, and F_m is reduced against a basis of I^k. Since the projections map I^k onto I^k, the depth of F_m is non-increasing in m, so $F_m \notin I^4$ for all $m \geq 2$. □

Theorem 6.8.4 (Bounds)**.** *For* $m \geq 2$*:*

(i) $e_m \leq v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| \leq \kappa_m e_m$*, and* $v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| \leq \log_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| = (h_H(2)/\log 2 + o(1))\, n_m$ *with* $h_H(2)/\log 2 = 2.977$*.*

(ii) $$\frac{v_2|\operatorname{Jac}^{\mathrm{H}}(Y_{m+1})|}{n_{m+1}} \;\geq\; (1 - 2^{-d_m})\,\frac{e_m+1}{n_m} \;\geq\; \tfrac{3}{4}\, c_e.$$

(iii) If $c_e > 0$*, then* $v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)|$ *grows linearly in* n_m*:*

$$\liminf_{m\to\infty} \frac{v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)|}{n_m} \;\geq\; \tfrac{3}{4}\, c_e.$$

If $v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| = o(n_m)$*, then* $c_e = 0$*, that is,* $e_m = o(n_m)$*. If moreover* $\kappa_m = O(\log n_m)$*, then* $c_e = 0$ *implies* $v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| = o(n_m \log n_m)$*.*

Proof. (i) is the definition of e_m and κ_m, and Theorem 6.5.1. (ii) By Proposition 6.8.1(iii), $v_2|\operatorname{Jac}^{\mathrm{H}}(Y_{m+1})| \geq v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| - d_m + (2^{d_m} - 1)(e_m + 1)$, and $v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| \geq e_m \geq e_2 = 6 \geq d_m$ by Proposition 6.8.3(iii). Since $2^{d_m} - 1 = (n_{m+1} - n_m)/n_m$, dividing by n_{m+1} gives the first inequality, and $(e_m+1)/n_m \geq c_e$ for every m gives the second. (iii) follows from (i) and (ii). □

Remark 6.8.5 (The data, and what they support). Table 6.1 and line (GR) give the following.

m	n_m	$v_2\lvert\operatorname{Jac}^{\mathrm{H}}(Y_m)\rvert$	e_m	v_2/n_m	e_m/n_m	κ_m	2-adic profile of the invariant factors
1	3	0	0	0	0	0	— ($\lvert\operatorname{Jac}^{\mathrm{H}}(X_3)\rvert = 147$)
2	24	18	6	0.7500	0.2500	3	$2^3\colon 6$
3	192	99	36	0.5156	0.1875	6	$2^1\colon 9,\ 2^2\colon 9,\ 2^3\colon 12,\ 2^6\colon 6$
4	768	289	99	0.3763	0.1289	8	$2^1\colon 24,\ 2^2\colon 23,\ 2^3\colon 25,\ 2^4\colon 9,\ 2^5\colon 12,\ 2^8\colon 6$
5	6144	1742	546	0.2835	0.0889	11	$2^1\colon 132,\ 2^2\colon 83,\ 2^3\colon 111,\ 2^4\colon 104,\ 2^5\colon 64,$ $2^6\colon 25,\ 2^7\colon 9,\ 2^8\colon 12,\ 2^{11}\colon 6$

Table 6.1: The 2-part of the Hecke Jacobian along the canonical tower. "$2^k\colon c$" means c invariant factors of 2-adic valuation exactly k. Lines (SM), (V2), (E), (FD), (TB) of `code/v3p6_v2_growth.py`.

(a) $v_2/n_m = 0.7500, 0.5156, 0.3763, 0.2835$ for $m = 2, \dots, 5$, decreasing at every level. The ratios $v_2(m)/v_2(m-1)$ are 5.500, 2.919, 6.028 against the indices 8, 4, 8, i.e. the local exponents $\log(v_2 \text{ ratio})/\log(\text{index})$ are 0.820, 0.773, 0.864, and the least-squares fit $v_2 \sim cn^\beta$ over $m = 2, \dots, 5$ gives $\beta = 0.822$ (0.830 over $m = 3, \dots, 5$). Per unit of index, $v_2/(n_m/3) = 2.250$, 1.547, 1.129, 0.851.

(b) $(e_m+1)/n_m = 0.2917, 0.1927, 0.1302, 0.0890$, non-increasing as Proposition 6.8.3(iii) requires, with successive ratios 0.661, 0.676, 0.684. The lower bound of Theorem 6.8.4(ii) is 0.2552, 0.1445, 0.1139 at $m = 3, 4, 5$, against the actual 0.5156, 0.3763, 0.2835; the layer sums $\sum_{\chi\neq 1} v_2 \det D_\chi = 84$, 192, 1456 exceed the bounds $(2^{d_m} - 1)(e_m + 1) = 49$, 111, 700 by the factors 1.71, 1.73, 2.08.

(c) At every computed level the largest exponent is $\kappa_m = \log_2 |G_m| = \log_2(n_m/3)$ $(3, 6, 8, 11)$, attained by exactly six invariant factors: the 2-part of $\operatorname{Jac}^{\mathrm{H}}(Y_m)$ has exponent $|G_m|$ and contains $(\mathbb{Z}/|G_m|)^6$. We have no explanation. The invariant factors of the twisted blocks (largest exponent 5 at the layer $Y_5 \to Y_4$) are not those of $\Delta^{\mathrm{H}}(Y_5)$, as Proposition 6.8.1(i) warns.

Verdict. The conjecture that $\lim v_2\lvert\operatorname{Jac}^{\mathrm{H}}(Y_m)\rvert/n_m$ exists and is positive is *not supported by the data*: on the five computed levels the ratio decreases monotonically from 0.75 to 0.28, every local exponent is below 1, and $(e_m+1)/n_m$ decays geometrically by a factor of about 0.68 per level. The data support *sublinear* growth, $v_2\lvert\operatorname{Jac}^{\mathrm{H}}(Y_m)\rvert \asymp n_m^\beta$ with $\beta \approx 0.82$, and, through Theorem 6.8.4(iii), they suggest $c_e = 0$. This is a statement about five levels only; the levels 6–8 of Theorem 6.8.7 and Remark 6.8.9 weaken it (the decay rates are slowing down), and Theorem 6.8.8 improves the constant $\frac{3}{4}$ below to 1. We have no proof of an asymptotic law in either direction. What is proved is Theorem 6.8.4: linear growth follows from $c_e > 0$, sublinear growth forces $c_e = 0$, and c_e is the limit of the normalized kernel dimensions of the single element F_m of $\mathbb{F}_2[G_m]$, a mod-2 analogue of a von Neumann dimension along the tower, whose vanishing is the real open question. The augmentation filtration does not decide it: F_m has depth exactly 3 for all computed m, which gives only $e_m + 1 \geq 10$. Because P has infinite rank (Theorem 6.4.2(iii)), $\mathbb{F}_2[[P]]$ is not known to be Noetherian, and the growth bounds for torsion modules over the Iwasawa algebras of p-adic analytic groups are not available. Open problem (b) of Section 6.7 is therefore sharpened to: decide whether

$$c_e = \lim_{m\to\infty} \frac{\dim_{\mathbb{F}_2}(\mathbb{F}_2[G_m]/\mathbb{F}_2[G_m]F_m)}{n_m}$$

vanishes.

Induced characters and levels 6–8

The layer decomposition of Proposition 6.8.1 uses the central subgroup N_m, so each new level costs $2^{d_m}-1$ blocks of the size of the previous level. Going down to Y_k with $k \approx m/2$ is much cheaper, because the whole kernel P_k/P_m is still abelian.

Proposition 6.8.6 (Induced characters of P_k/P_m)**.** *Let $2 \le k < m \le 2k$ and $A = P_k/P_m \subset G_m$.*

(i) A is an elementary abelian normal 2-subgroup, and the coefficients of $\varphi^k, \dots, \varphi^{m-1}$ (without the $\mathbb{F}_2$-part of the coefficients of φ^{3i}) identify it with $\mathbb{F}_2^d$, $2^d = n_m/n_k$. The left action of A on Y_m is free, and $A\backslash Y_m = Y_k$.

(ii) With coset representatives $r_c \in G_m$ ($c \in G_k$), every arc $c \to c'$ of Y_k with label x has the voltage $a_e = r_c \theta_{s,x} r_{c'}^{-1} \in A$. For $\chi \in \widehat{A}$ put $A_\chi = \sum_e \chi(a_e) E_{cc'}$ and $D_\chi = 7I + A_\chi - 2A_\chi^t$. Then $\Delta^{\mathrm{H}}(Y_m)$ is conjugate over $\mathbb{Z}[\frac12]$ to $\bigoplus_\chi D_\chi$, $D_1 = \Delta^{\mathrm{H}}(Y_k)$, $D_\chi \equiv \Delta^{\mathrm{H}}(Y_k) \pmod 2$, and

$$v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| = v_2|\operatorname{Jac}^{\mathrm{H}}(Y_k)| - \log_2 \frac{n_m}{n_k} + \sum_{\chi \neq 1} v_2 \det D_\chi .$$

(iii) (Artin–Clifford.) D_χ is the Hecke operator of Y_k twisted by $\operatorname{Ind}_A^{G_m} \chi$. G_m acts on $\widehat{A}$ by conjugation; blocks in one orbit O are conjugate by signed permutation matrices, hence have the same Smith form over $\mathbb{Z}_2$, and

$$\prod_{\chi \in O} \det D_\chi = \prod_{\rho \in \operatorname{Irr}(G_m) \text{ over } O} \det(\Delta_\rho^H)^{\dim \rho}.$$

Proof. (i) A is the kernel of the projection $G_m \to G_k$, hence normal. For $x, y \in \varphi^k\mathcal{O}$, $(1+x)(1+y) = 1+x+y+xy$ and $xy \in \varphi^{2k}\mathcal{O} \subseteq \varphi^m\mathcal{O}$, so A is abelian, the multiplication is addition of coefficients, and every element has order 2. The central units $1+t^i$ used to normalize only change the $\mathbb{F}_2$-part of the coefficient of φ^{3i}, and they also act additively, since the cross terms again lie in $\varphi^{2k}\mathcal{O}$. The deck group G_m acts on Y_m by left multiplication and the arcs are right multiplications, so the left action of any subgroup is free and commutes with the arcs. (ii) The arc of label x at ar_c ends at $ar_c\theta_{s,x} = (aa_e)r_{c'}$. In the basis (a,c), $A_1(Y_m) = \sum_e R(a_e) \otimes E_{cc'}$, and the proof of Proposition 6.8.1(i)–(ii) applies word for word with A in place of N_m; it never used that N_m is central, only that it is abelian and acts on the left. (iii) $\mathbb{C}[G_m] = \operatorname{Ind}_A^{G_m} \mathbb{C}[A] = \bigoplus_\chi \operatorname{Ind}_A^{G_m} \chi$, and the χ-summand is the χ-isotypic part for the left A-action, on which Δ^{H} acts as D_χ. For $g \in G_m$, left multiplication by g maps ar_c to $(gag^{-1})(gr_c)$ and $gr_c = b_c r_{\bar g c}$ with $b_c \in A$; it carries the χ-isotypic part to the $\chi \circ \operatorname{Ad}(g)^{-1}$-isotypic part, and in the bases above it is the permutation $c \mapsto \bar g c$ times the signs $\chi(b_c)$. By Clifford's theorem every irreducible ρ of G_m lies over exactly one orbit, and $\bigoplus_{\chi \in O} \operatorname{Ind} \chi = \bigoplus_{\rho \text{ over } O} \rho^{\dim \rho}$ as summands of the regular representation; take determinants of Δ^{H} on both sides. □

Theorem 6.8.7 (Levels 6, 7, 8)**.** *The 2-parts and the total orders of the Hecke Jacobians of $Y_5, \dots, Y_8$ are:*

m	n_m	$v_2\lvert\operatorname{Jac}^{\mathrm{H}}(Y_m)\rvert$	v_2/n_m	$\frac{1}{n_m}\log\lvert\operatorname{Jac}^{\mathrm{H}}(Y_m)\rvert$
5	6144	1742	0.28353	2.062520
6	49152	11595	0.23590	2.063511
7	196608	36937	0.18787	2.063639
8	1572864	254662	0.16191	2.063684

Proof. Proposition 6.8.6 with $(m,k) = (6,3)$: 255 blocks of size 192 in 24 conjugation orbits; $(6,4)$: 63 blocks of size 768 in 21 orbits; $(7,4)$: 255 blocks in 24 orbits; $(8,4)$: 2047 blocks of size 768 in 52 orbits (sizes $1^7, 4^{14}, 64^{31}$). The block valuations are computed by the 2-adic elimination of Theorem 6.8.2 (cap 2^{30}, never reached), on one representative of each orbit, and checked to be constant on the orbit (all members for $m \le 6$, two further members of every orbit for $m = 7, 8$). The two routes to level 6 give $99 + 7 - 15 + 11504 = 289 + 9 - 15 + 11312 = 11595$, and the two routes to level 7, over Y_4 and over Y_5 ($(m,k) = (7,5)$: 31 blocks of size 6144 in 10 orbits, valuations 1408 and 1056), give $289 + 9 - 17 + 36656 = 1742 + 12 - 17 + 35200 = 36937$. The same procedure recovers 99, 289 and 1742 from Y_2 and Y_3, i.e. Theorem 6.8.2 by a route independent of the central layers. The total orders are the sums of $\log|\det D_\chi|$ (floating point). □

The column of total orders converges to $h_H = 2.063693$ from below, the last value being $9 \cdot 10^{-6}$ short: the Plancherel limit of Theorem 6.5.1 is visible to five digits at $n = 1.57 \cdot 10^6$. The 2-part does not converge at that rate, if at all.

Theorem 6.8.8 (A sharper lower bound)**.** *For $2 \le k < m \le 2k$,*

$$v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)| \;\ge\; v_2|\operatorname{Jac}^{\mathrm{H}}(Y_k)| - \log_2 \frac{n_m}{n_k} + \Big(\frac{n_m}{n_k} - 1\Big)(e_k + 1).$$

Consequently $\liminf_m v_2|\operatorname{Jac}^{\mathrm{H}}(Y_m)|/n_m \ge c_e$*, improving the constant* $\frac{3}{4}$ *of Theorem 6.8.4(iii) to* 1*. Moreover* $e_6 = 3169$*, so* $c_e \le (e_6+1)/n_6 = 0.06449$*.*

Proof. By Proposition 6.8.6(ii) each of the $n_m/n_k - 1$ nontrivial blocks is congruent to $\Delta^{\mathrm{H}}(Y_k)$ modulo 2, hence has $\mathbb{F}_2$-corank $e_k + 1$ and $v_2 \det D_\chi \ge e_k + 1$ (as in Proposition 6.8.1(iii)). For the limit take $k = \lceil m/2 \rceil$: then $n_m/n_k \to \infty$, $\log_2(n_m/n_k) = O(m) = o(n_m)$, $v_2|\operatorname{Jac}^{\mathrm{H}}(Y_k)| \ge 0$, and $(e_k+1)/n_k \ge c_e$ by Proposition 6.8.3(iii). The value of e_6 is $\dim \mathbb{F}_2[G_6]/\mathbb{F}_2[G_6]F_6 - 1$ by Proposition 6.8.3(i), computed by Gaussian elimination over $\mathbb{F}_2$ on the 16384×16384 matrix of right multiplication by F_6 ($|\operatorname{supp} F_6| = 154$). □

Remark 6.8.9 (The data at levels 2–8). The numbers below are produced by lines (FT) and (E) of the script `code/v3p6_level5_v2.py`.

(a) The sequences, with the local exponent $\beta_m = \log(v_2(m)/v_2(m-1))/\log(n_m/n_{m-1})$ and the ratio $\rho_m = ((e_m+1)/n_m)/((e_{m-1}+1)/n_{m-1})$:

m	2	3	4	5	6	7	8
n_m/n_{m-1}	—	8	4	8	8	4	8
v_2/n_m	0.750	0.516	0.376	0.284	0.236	0.188	0.162
β_m	—	0.820	0.773	0.864	0.912	0.836	0.928
$(e_m+1)/n_m$	0.292	0.193	0.130	0.089	0.064		
ρ_m	—	0.661	0.676	0.684	0.724		

v_2/n_m still decreases at every level, and every β_m is below 1. But the exponents are increasing; along the layers of index 8 they are 0.820, 0.864, 0.912, 0.928. Likewise $(e_m+1)/n_m$ decreases with increasing ratios ρ_m, so the geometric decay by a constant factor suggested by Remark 6.8.5(b) is not borne out (Figure 6.1).

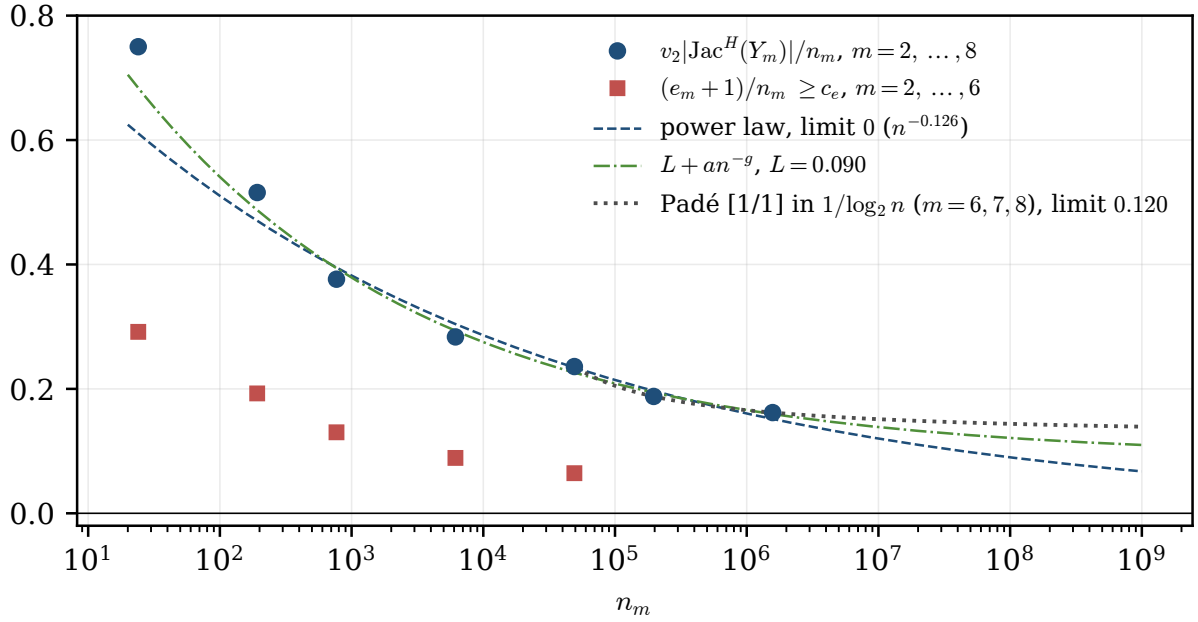


Figure 6.1: The 2-part of the Hecke Jacobian along the canonical 2-tower: $v_2|\,\mathrm{Jac}^{\mathrm{H}}(Y_m)|/n_m$ for $m = 2,\ldots,8$ (Theorems 6.8.2, 6.8.7) and the normalized $\mathbb{F}_2$-kernel dimensions $(e_m+1)/n_m$ for $m = 2,\ldots,6$, which decrease to the limit c_e (Proposition 6.8.3, Theorem 6.8.8). Three extrapolations of v_2/n_m that fit the data about equally well: a power law $n^{-0.126}$ with limit 0 (fitted on $m = 3,\ldots,8$); the least-squares optimum $0.090 + an^{-g}$ of Remark 6.8.9(b); and the Padé approximant [1/1] in $1/\log_2 n$ through $m = 6,7,8$, with limit 0.120, drawn only on the range of its data and beyond. The data do not decide between a positive limit and slow decay to 0; what is proved is $\liminf v_2/n_m \geq c_e$. Data of `code/v3p6_level5_v2.py`, drawn by `code/make_figures.py`.

(b) *Fits.* With limit $L = 0$ the best power law is $v_2 \sim n^{0.874}$. The three-parameter model $v_2/n = L + an^{-g}$ fitted on $m = 3,\ldots,8$ has its least-squares optimum at $L = 0.090$, but the residual at $L = 0$ is only 1.09 times the optimum, and the leave-one-out optima range over $[0, 0.125]$. Exact fits through three consecutive levels jump with the period-3 pattern of the indices $(8,4,8,8,4,8)$ and are not used. The data do not distinguish a positive limit of order 0.1 from slow decay to 0.

(c) *Rigorous interval.* $c_e \leq \liminf v_2/n_m \leq \limsup v_2/n_m \leq h_H/\log 2 = 2.977$, with $0 \leq c_e \leq 0.0645$. The bound of Theorem 6.8.8 is far from sharp at finite level: over Y_4 it gives $v_2/n_m \geq 0.134, 0.131, 0.130$ for $m = 6,7,8$, against the actual $0.236, 0.188, 0.162$.

(d) *The ratio to c_e.* At $m = 2k$ the observed quotient $(v_2|\,\mathrm{Jac}^{\mathrm{H}}(Y_{2k})|/n_{2k})/((e_k+1)/n_k)$ is $1.290, 1.224, 1.243$ for $k = 2,3,4$: on average a twisted block has about 1.25 times the minimal valuation $e_k + 1$. If this quotient stays bounded, which would follow from a uniform bound on the 2-exponents of the blocks (they are at most 10 at levels 6–8), then $v_2|\,\mathrm{Jac}^{\mathrm{H}}(Y_m)|/n_m$ and $(e_{\lceil m/2\rceil}+1)/n_{\lceil m/2\rceil}$ have the same order, and $\lim v_2/n_m > 0$ iff $c_e > 0$.

Verdict. Strict positivity of the slope is not decided by levels 2–8. What is proved is that a positive mod-2 kernel density c_e forces slope at least c_e; the numbers agree with slope $\asymp c_e$, and both

sequences are still decreasing, but more slowly at every level. The open problem remains whether $c_e = \lim \dim_{\mathbb{F}_2}(\mathbb{F}_2[G_m]/\mathbb{F}_2[G_m]F_m)/n_m$ vanishes.

Remark 6.8.10 (Verification). The script `code/v3p6_v2_growth.py` uses numpy and the sparse module of scipy. It runs in about five minutes (one minute with `-quick`, which skips line (FD)). It builds levels 2–5 (line (LV)), checks the central elementary abelian layers and the voltages, the trace identities and the Hadamard decomposition (BL), the valuation identity at the layers $2 \to 3$, $3 \to 4$ against the direct values and its application at $4 \to 5$ (V2), the agreement of the 2-adic eliminations with the exact Smith forms at levels $2, 3$ (SM), e_m from the $\mathbb{F}_2$-rank of $I + A_1$ and from $\dim C_m$ (E), the depth of F_m (AU), the two automorphisms and their action on the characters (SY), the total order and $v_2 \le \log_2 |\operatorname{Jac}^{\mathrm{H}}|$ (TO), the direct elimination of the 6144×6144 matrix (FD), and the table and growth statistics (TB), (GR).

The script `code/v3p6_level5_v2.py` (numpy, matplotlib; about three minutes, plus about 70 minutes with `-check7`) verifies Proposition 6.8.6 and Theorems 6.8.7, 6.8.8: the quotient models and the elementary abelian structure of P_k/P_m (QT), the conjugation orbits on the characters (CJ), the re-derivation of $18, 99, 289$ by direct elimination and of $99, 289, 1742$ through induced characters (VA), the new levels with the two routes to level 6 (and, with `-check7`, the route to level 7 over Y_5) (NW), $e_m + 1$ for $m \le 6$ and the sharpened bound (E), the total orders (TO), and the fits (FT). Data and the figure are written to `data/`.

6.9 Localization over the non-analytic group $\mathbb{Z}_2[[P]]$

Corollary 6.4.3 leaves parts (i′) and (ii) of Theorem 6.3.1 open for the canonical tower, because nothing is known about $\Lambda = \mathbb{Z}_2[[P]]$ beyond what holds for every pro-2 group. This section separates what can be proved about Λ from what cannot, and proves an unconditional substitute for (ii): a compatible system of characteristic elements at the finite levels, whose boundaries are the finite parts of the Iwasawa module and whose reduced norms are the Artin factors of the class number formula. Throughout, $G_m = P/P_m$, $n_m = 3|G_m|$, $N_m = \sum_{g \in G_m} g \in \mathbb{Z}_2[G_m]$, and $\Delta^{\mathrm{H}}_\Lambda \in M_3(\Lambda)$ is the equivariant operator of Section 6.3 for the tower $Y_m \to X_3$, so that $\mathbb{Z}_2[G_m] \otimes_\Lambda \Delta^{\mathrm{H}}_\Lambda = \Delta^{\mathrm{H}}(Y_m) \otimes \mathbb{Z}_2$ under $\mathbb{Z}_2[G_m]^3 = \mathbb{Z}_2^{V(Y_m)}$.

What is known about Λ

Proposition 6.9.1 (The group P and its ring)**.** *(i) P is a torsion-free pro-2 group, topologically generated by 3 elements, of infinite rank; $d(P_k) \ge 2k$.*

(ii) $\Lambda = \varprojlim_m \mathbb{Z}_2[G_m]$ is a compact local ring with maximal ideal $(2, I_P)$, I_P the augmentation ideal, and every finitely generated compact Λ-module M satisfies $M = \varprojlim_m \mathbb{Z}_2[G_m] \otimes_\Lambda M$.

(iii) The necessary condition for Λ to be a domain holds: $\mathbb{Z}_2[[P]]$ has no zero divisor of the form $(g-1)\nu$ with g of finite order and ν its norm element.

Proof. (i) Generation, rank and $d(P_k)$ are Theorem 6.4.2. A pro-2 group has no torsion of odd order, so let $x \in 1 + \varphi\mathcal{O}$ with $x^{2^k} \in K^\times$, $K = \mathbb{F}_2((t))$ the centre of D_0. The field $K(x) \subseteq D_0$ has degree 1 or 3 over K, since D_0 is a division algebra of degree 3. Over $K(x)$, in characteristic 2, $y^{2^k} - x^{2^k} = (y-x)^{2^k}$, and the minimal polynomial f of x over K divides it, so $f = (y-x)^j$ with $j = [K(x) : K] \in \{1, 3\}$. If $j = 3$, then $f = y^3 + xy^2 + x^2y + x^3$ has its coefficients in K, so $x \in K$, a contradiction. Hence $j = 1$, $x \in K^\times \cap (1 + \varphi\mathcal{O}) = 1 + t\mathbb{F}_2[[t]]$, and x is trivial in P. (ii) is standard for pro-p groups: $\Lambda/(2, I_P) = \mathbb{F}_2$, and every element outside $(2, I_P)$ is a unit, being a

unit in every $\mathbb{Z}_2[G_m]$, which is local because G_m is a 2-group; the limit statement holds because M is a quotient of Λ^r and inverse limits of compact modules are exact. (iii) If g has order p, then $(g-1)(1+g+\cdots+g^{p-1})=0$; by (i) there is no such $g\neq 1$. □

The ring-theoretic questions that (i′) and (ii) need are open for Λ.

Remark 6.9.2 (What is not known). (a) *Domain.* Neumann's theorem [37] says that $\mathbb{Z}_p[[G]]$ is a domain when G is a torsion-free *p-adic analytic* pro-p group; its proof goes through Lazard's filtration and the fact that the associated graded ring is a polynomial ring [11]. P is not analytic (Theorem 6.4.2(iii)), so the theorem does not apply. Proposition 6.9.1(i) supplies the necessary condition, torsion-freeness, but we know of no theorem that makes $\mathbb{Z}_2[[P]]$ a domain. Lazard's argument would go through if the graded ring of $\mathbb{F}_2[[P]]$ for the filtration induced by the P_m were a domain. The P_m form a restricted N-series, so by Jennings and Quillen this graded ring is the restricted enveloping algebra of the graded restricted Lie algebra $\operatorname{gr} P$. In $\operatorname{gr} P$ the squaring map is injective on every homogeneous piece: the square of $1+a\varphi^k$ is $1+a\sigma^k(a)\varphi^{2k}$, and $a\sigma^k(a)$ is nonzero modulo the centre. So the obvious source of zero divisors, $x^{[2]}=0$, is absent, but whether the enveloping algebra is a domain is not known to us.

(b) *Fractions.* Even for a domain, the nonzero elements need not form an Ore set. Goldie's theorem gives the skew field $Q(\Lambda)$ for Noetherian domains, and whether $\mathbb{Z}_2[[P]]$ is Noetherian is open (Corollary 6.4.3); to our knowledge no pro-p group of infinite rank is known to have a Noetherian completed group ring. The Cohn localization at $\Sigma=\Lambda\setminus\{0\}$ exists by its universal property, but the localization sequence of Definition 6.2.1 needs $\Lambda\to\Sigma^{-1}\Lambda$ to be injective and stably flat, which we cannot verify.

(c) *Consequence.* The element $\mathcal{L}=[\Delta_\Lambda^{\mathrm{H}}]\in K_1(Q(\Lambda))$ and the identity $\partial\mathcal{L}=[X(Y/P)]$ are available only conditionally. Theorem 6.3.1(i) does hold unconditionally: $\Delta_\Lambda^{\mathrm{H}}$ is injective on Λ^3.

An unconditional substitute: the tower of finite characteristic elements

The trivial representation is the only obstruction at a finite level, so we remove it. Put

$$A_m=\mathbb{Z}_2[G_m]/\mathbb{Z}_2N_m,\qquad \alpha_m=\text{image of }\Delta_\Lambda^{\mathrm{H}}\text{ in }M_3(A_m),\qquad X_m^{\mathrm{nt}}=\operatorname{coker}\bigl(\alpha_m\colon A_m^3\to A_m^3\bigr).$$

A_m is a ring, because $\mathbb{Z}_2N_m$ is a two-sided ideal ($gN_m=N_mg=N_m$). It is a $\mathbb{Z}_2$-order in the semisimple algebra $\mathbb{Q}_2A_m=(1-e_m)\mathbb{Q}_2[G_m]=\bigoplus_{\rho\neq 1}$, where $e_m=N_m/|G_m|$ is the trivial idempotent; in particular A_m is Noetherian, free of rank $|G_m|-1$ over $\mathbb{Z}_2$, and $\mathbb{Q}_2A_m=A_m[\frac12]$ is its localization at the central multiplicative set $\{2^k\}$. For such an order the localization sequence is classical and unconditional:

$$K_1(A_m)\to K_1(\mathbb{Q}_2A_m)\xrightarrow{\partial}K_0(A_m,\mathbb{Q}_2)\to K_0(A_m)\to K_0(\mathbb{Q}_2A_m),$$

where $K_0(A_m,\mathbb{Q}_2)$ is the Grothendieck group of finite A_m-modules of projective dimension ≤ 1 and $\partial[\alpha]=[\operatorname{coker}\alpha]$ for injective $\alpha\in M_r(A_m)$, with the sign convention of Definition 6.2.1.

Lemma 6.9.3 (The rings A_m form the tower)**.** *The projections $G_{m+1}\to G_m$ induce surjective ring maps $\pi_m\colon A_{m+1}\to A_m$ with $\pi_m(\alpha_{m+1})=\alpha_m$, and $\Lambda\to\varprojlim_m A_m$ is an isomorphism of topological rings.*

Proof. $N_{m+1} \mapsto |P_m/P_{m+1}|\, N_m = 2^{d_m} N_m$, so the ideal $\mathbb{Z}_2 N_{m+1}$ maps into $\mathbb{Z}_2 N_m$, and π_m is well defined and surjective. Both α_{m+1} and α_m are images of $\Delta^{\mathrm{H}}_\Lambda$. If $\lambda \in \Lambda$ maps to 0 in every A_m, its images in $\mathbb{Z}_2[G_m]$ are $c_m N_m$ with $c_m = 2^{d_m} c_{m+1}$, so c_m is divisible by every power of 2, and $\lambda = 0$. Surjectivity follows from compactness, since all transition maps are surjective. □

Theorem 6.9.4 (Finite characteristic elements along the canonical tower)**.** *For every $m \geq 1$, unconditionally:*

(i) α_m is invertible in $M_3(\mathbb{Q}_2 A_m)$, the sequence $0 \to A_m^3 \xrightarrow{\alpha_m} A_m^3 \to X_m^{\mathrm{nt}} \to 0$ is exact, and

$$X_m^{\mathrm{nt}} = \operatorname{coker}\big[\Delta^{\mathrm{H}}(Y_m) \otimes \mathbb{Z}_2 \;\big|\; \mathbf{1}_{V_0}, \mathbf{1}_{V_1}, \mathbf{1}_{V_2}\big]$$

is a finite A_m-module of projective dimension ≤ 1, where $\mathbf{1}_{V_s}$ is the indicator of the vertices of type s.

(ii) $\mathcal{L}_m := [\alpha_m] \in K_1(\mathbb{Q}_2 A_m)$ satisfies $\partial \mathcal{L}_m = [X_m^{\mathrm{nt}}]$, and any $\mathcal{L}' \in K_1(\mathbb{Q}_2 A_m)$ with $\partial \mathcal{L}' = [X_m^{\mathrm{nt}}]$ differs from $\mathcal{L}_m$ by an element of the image of $K_1(A_m)$.

(iii) (Compatibility.) $\pi_{m*} \mathcal{L}_{m+1} = \mathcal{L}_m$ and $A_m \otimes_{A_{m+1}} X_{m+1}^{\mathrm{nt}} \cong X_m^{\mathrm{nt}}$ with $\operatorname{Tor}_1^{A_{m+1}}(A_m, X_{m+1}^{\mathrm{nt}}) = 0$, so the boundaries are compatible: $\pi_{m*} \partial \mathcal{L}_{m+1} = \partial \mathcal{L}_m$. Hence

$$\mathcal{L}_\infty := (\mathcal{L}_m)_m \in \varprojlim_m K_1(\mathbb{Q}_2 A_m), \qquad X(Y/P) = \operatorname{coker} \Delta^{\mathrm{H}}_\Lambda \cong \varprojlim_m X_m^{\mathrm{nt}}.$$

(iv) (Artin descent.) Over $\mathbb{C}$, $\mathbb{Q}_2 A_m \otimes \mathbb{C} = \bigoplus_{\rho \neq 1} M_{\dim \rho}(\mathbb{C})$ and the reduced norm of $\mathcal{L}_m$ has components $\det \Delta^{\mathrm{H}}_\rho$, $\rho \neq 1$. The order of the boundary is the class number formula:

$$\log_2 |X_m^{\mathrm{nt}}| = v_2 \prod_{\rho \neq 1} \det(\Delta^{\mathrm{H}}_\rho)^{\dim \rho} = v_2\Big([P : P_m]\, \frac{|\operatorname{Jac}^{\mathrm{H}}(Y_m)|}{|\operatorname{Jac}^{\mathrm{H}}(X_3)|}\Big) = \log_2 |G_m| + v_2 |\operatorname{Jac}^{\mathrm{H}}(Y_m)|.$$

Proof. (i) $\mathbb{Q}_2 A_m \otimes \mathbb{C}$ is the sum of the nontrivial isotypic components of $\mathbb{C}[G_m]$, and on the ρ-component α_m acts as Δ^{H}_ρ (with multiplicity $\dim \rho$), which is nonsingular by Theorem 6.3.1(iv). So α_m is invertible over $\mathbb{Q}_2 A_m$ and injective on the lattice A_m^3, whose cokernel is then finite; the exact sequence is a resolution of length 1 by free modules. Under $\mathbb{Z}_2[G_m]^3 = \mathbb{Z}_2^{V(Y_m)}$ the element $N_m e_s$ is $\mathbf{1}_{V_s}$, so $A_m^3 = \mathbb{Z}_2^{V(Y_m)}/\langle \mathbf{1}_{V_0}, \mathbf{1}_{V_1}, \mathbf{1}_{V_2} \rangle$ and $X_m^{\mathrm{nt}} = \mathbb{Z}_2^{V(Y_m)}/(\operatorname{im} \Delta^{\mathrm{H}}(Y_m) + \langle \mathbf{1}_{V_s} \rangle)$.

(ii) is the definition of ∂ and the exactness of the localization sequence at $K_1(\mathbb{Q}_2 A_m)$.

(iii) $\pi_m(\alpha_{m+1}) = \alpha_m$ by Lemma 6.9.3. Applying $A_m \otimes_{A_{m+1}} -$ to the resolution of (i) gives $A_m^3 \xrightarrow{\alpha_m} A_m^3 \to A_m \otimes X_{m+1}^{\mathrm{nt}} \to 0$, and α_m is injective, so the Tor term vanishes and the base change is X_m^{nt}. The localization sequences are natural for the ring map π_m, which sends 2 to 2, so ∂ commutes with π_{m*} (on $K_0(-, \mathbb{Q}_2)$ the map is $[M] \mapsto [A_m \otimes^{\mathbf{L}} M]$). For the limit, $X = \varprojlim X_m$ with $X_m = \mathbb{Z}_2[G_m] \otimes_\Lambda X = \operatorname{coker} \Delta^{\mathrm{H}}(Y_m) \otimes \mathbb{Z}_2$ (Proposition 6.9.1(ii)), and $X_m \to X_m^{\mathrm{nt}}$ is surjective with kernel K_m, the $\mathbb{Z}_2$-span of the images of the $\mathbf{1}_{V_s}$. The transition maps send $\mathbf{1}_{V_s}(Y_{m+1}) \mapsto 2^{d_m} \mathbf{1}_{V_s}(Y_m)$, so a compatible family in (K_m) lies in $\bigcap_j 2^j K_m = 0$ (Krull, K_m being a finitely generated $\mathbb{Z}_2$-module); hence $\varprojlim K_m = 0$, and $\varprojlim X_m \cong \varprojlim X_m^{\mathrm{nt}}$ because $\varprojlim$ is exact on compact modules.

(iv) The first statement is the decomposition used in (i). For the order: A_m^3 is a free $\mathbb{Z}_2$-module of rank $3(|G_m| - 1)$, and for an injective endomorphism of a lattice $|\operatorname{coker} \alpha_m| = |\det_{\mathbb{Q}_2} \alpha_m|_2^{-1}$. The $\mathbb{Q}_2$-determinant of α_m on $\mathbb{Q}_2 A_m^3$ is the determinant of $\Delta^{\mathrm{H}}(Y_m)$ on the nontrivial isotypic part, i.e. $\prod_{\rho \neq 1} \det(\Delta^{\mathrm{H}}_\rho)^{\dim \rho}$, which equals $[P : P_m] |\operatorname{Jac}^{\mathrm{H}}(Y_m)|/|\operatorname{Jac}^{\mathrm{H}}(X_3)|$ by Theorem 6.3.1(iv). Finally $|\operatorname{Jac}^{\mathrm{H}}(X_3)| = 147$ is odd. □

Remark 6.9.5 (The orders, and the checks). With Theorems 6.8.2 and 6.8.7, (iv) gives $\log_2|X_m^{\mathrm{nt}}| = 3+18 = 21,\ 6+99 = 105,\ 8+289 = 297,\ 11+1742 = 1753,\ 14+11595 = 11609,\ 16+36937 = 36953$ and $19+254662 = 254681$ for $m = 2,\dots,8$. Over $\mathbb{Z}$ the same construction gives $|X_2^{\mathrm{nt}}| = 8\cdot|\operatorname{Jac}^{\mathrm{H}}(Y_{24})|/147 = 2^{21}7^{14}$, which is the product of the seven twisted determinants of the layer $Y_2 \to Y_1$. The script `code/v3p6_pro2_localization.py` (numpy; about three minutes) checks the ingredients at the levels $m \le 5$: the orders $|G_m|$ and the central elementary abelian layers (OR); the push-forward and pull-back identities $\Pi\,\Delta^{\mathrm{H}}(Y_{m+1}) = \Delta^{\mathrm{H}}(Y_m)\,\Pi$ and $\Delta^{\mathrm{H}}(Y_{m+1})\Pi^t = \Pi^t\Delta^{\mathrm{H}}(Y_m)$ behind Lemma 6.9.3 (PF); the block decomposition of each layer (BL); the sector lemma for all twisted blocks (SE); the Smith forms $\operatorname{coker}\Delta^{\mathrm{H}}(Y_m) = \mathbb{Z}\oplus\operatorname{Jac}^{\mathrm{H}}(Y_m)$ and the order $|X_m^{\mathrm{nt}}| = |G_m||\operatorname{Jac}^{\mathrm{H}}(Y_m)|/147$ over $\mathbb{Z}$ for $m = 2,3$ (SM); the relative class number formula of the layers Y_2/Y_1, Y_3/Y_2 and Y_4/Y_3 with exact integers, $|\operatorname{Jac}^{\mathrm{H}}(Y_4)|$ being reconstructed by Chinese remaindering (CN); and the surjectivity of $\operatorname{Jac}^{\mathrm{H}}(Y_{m+1}) \to \operatorname{Jac}^{\mathrm{H}}(Y_m)$ with kernel of order $\prod_{\chi\neq1}\det\Delta_\chi^{\mathrm{H}}/|H_m|$ (KR).

Remark 6.9.6 (How far this goes). Theorem 6.9.4 is the form of Theorem 6.3.1(i′),(ii) that holds without knowing anything about $\mathbb{Z}_2[[P]]$: the characteristic element lives in the inverse limit $\varprojlim K_1(\mathbb{Q}_2A_m)$ instead of $K_1(Q(\Lambda))$, its boundary is the system of finite modules X_m^{nt} whose limit is $X(Y/P)$, and its reduced norms are the Artin factors. This is the shape of the finite-level formulations of non-commutative Iwasawa theory, where K_1 of a localized Iwasawa algebra is approached through the finite group rings. If Λ turns out to be an Ore domain, then $\mathcal{L} = [\Delta_\Lambda^{\mathrm{H}}] \in K_1(Q(\Lambda))$ exists and $\partial\mathcal{L} = [X]$ by Theorem 6.3.1(ii); we do not claim a map relating $\mathcal{L}$ to $\mathcal{L}_\infty$, because a skew field of fractions of Λ need not map to the finite-level algebras $\mathbb{Q}_2A_m$. Open problem (a) of Section 6.7 is therefore sharpened to: is $\mathbb{Z}_2[[P]]$ a Noetherian domain, or at least an Ore domain?

Remark 6.9.7 (Corrections to the proposed programme). (1) Neumann's theorem does not say that $\mathbb{Z}_2[[P]]$ is a domain: it requires a p-adic analytic group, and P is not one. What is proved is that P is torsion-free (Proposition 6.9.1), which is necessary, not sufficient. (2) The Ore localization at $\Lambda\setminus\{0\}$, i.e. the skew field $Q(\Lambda)$, is not known to exist: Goldie's theorem needs a Noetherian ring (Remark 6.9.2). (3) Therefore $\partial[\Delta_\Lambda^{\mathrm{H}}] = [X(Y/P)]$ in $K_1(Q(\Lambda))$ is not proved unconditionally. What holds unconditionally is Theorem 6.9.4, the same identity at every finite level, compatibly along the tower, with $X(Y/P)$ recovered as the limit. (4) The descent formula with denominator $|\operatorname{Jac}^{\mathrm{H}}(X_3)|$ is Theorem 6.3.1(iv); in Theorem 6.9.4(iv) it computes the order of the boundary. (5) $d(P_k) \ge 2k$ and the absence of the Ore set of [1] are Theorems 6.4.2(iii) and 6.2.2, as stated in the proposed programme.

6.10 Verification battery

6.11 Corrections

Remark 6.11.1 (Corrections to the proposed programme). (1) "Strictly realized as a representation of $\mathbb{H}_d$" is vacuous (Proposition 6.1.2(ii)). The natural structure is H_q on chambers (Theorems 6.1.3, 6.1.4), and the Kuramoto field is not linear. (2) The localization sequence ends in $\to 0$ only under an injectivity condition on K_0. The set Σ of the proposed programme is not the Ore set of [1], and that Ore set does not exist over thick complexes (Theorem 6.2.2). (3) An "unconditional" L-element requires a tower, which is Hypothesis (T); the element is then $[\Delta_\Lambda^{\mathrm{H}}]$, and $\det(I-uC^{(3)})$ enters through its specializations (Theorem 6.3.1(v)). (4) $\mathrm{Sel}(Y/G)$ is undefined. With $X(Y/G)$ in its place the "main conjecture" is a theorem, and its content is descent and the class number

check	complexes	result
(Dg) polynomial representation of $\mathbb{H}_2$ (degree ≤ 4); trivial module with $x_1 = L_E(Y_{21})$	—	exact / numeric
(Hq) $T_i^2 = (q-1)T_i + q$, braid relations	$Y_{21}, Y_{24}, Y_{168}, T_6$	exact
(Op) $\Delta_2 = 3 + \sum T_i$; $A_1 = (q+1)^{-1}\sum \pi_t \pi_{t+1}^t$; $L_E = p\,T_t T_{t+2} T_t\, p^t/(q(q+1))$	Y_{21}, Y_{168}	exact
(Sa) $a = qe_1(z)$, $\bar{a} = qe_2(z)$, $e_3 = 1$; Δ^{H} as symmetric polynomial	Y_{42}, Y_{168}	numeric 10^{-8}
(CN) $\prod_{\chi\neq 1} \det \Delta_\chi^{\mathrm{H}} = \lvert A\rvert \lvert \mathrm{Jac}^{\mathrm{H}}(Y_A)\rvert / \lvert \mathrm{Jac}^{\mathrm{H}}(X_3)\rvert$	Y_{21}, Y_{24}, Y_{42}	numeric vs exact Smith forms
(B1) $b_1 = 0$, hence no $\mathbb{Z}_p$-quotient	five covers	exact
`code/v3p6_deep_verify.py`		
(CS) $x + 2y + 4z \equiv 0$ on T; $u_x u_y u_z = 1 + t$	—	exact
(NB) the 7 hyperplanes, 7 lines and their incidence = link of the CMSZ building, for every $\bar{e}$	—	exact
(FR) generators span $P/P_2 = \mathbb{F}_8$; squares and commutators fill levels 2..13	—	exact
(IM) image of $\Gamma^0 = P/P_m$ for $m \le 6$; lower bounds for $d(P_k)$	—	exact
(L2) level 2 has $\mathrm{Jac}^{\mathrm{H}}(Y_{24})$	Y_2	exact
(J) $v_2\lvert \mathrm{Jac}^{\mathrm{H}}\rvert = 18, 99, 289$; $\frac{1}{n}\log\lvert \mathrm{Jac}^{\mathrm{H}}\rvert = 1.8629, 2.0416, 2.0544$; Ramanujan; $\min \mathrm{Re}\,\mu > 0$	Y_2, Y_3, Y_4	exact / numeric
(PL) Plancherel moments $\lvert a\rvert^2 \mapsto 7$, $a^3 \mapsto 21$; $h_H(2) = 2.063693$	—	numeric 10^{-6}

Table 6.2: Checks run by `code/v3p6_verify.py`.

formula. (5) The five-step scheme is not a proof (Section 6.6). (6) Nothing in Volume III supports a cosmological reading. (7) (Deepening request.) Γ is discrete in $\mathrm{PGL}_3(\mathbb{F}_2((s)))$, so the proposed map $\Gamma \to \mathrm{PGL}_3(\mathcal{O}_K/\mathfrak{p}^m)$ at that place does not exist. Congruence towers come from other places of $\mathbb{F}_2(t)$. In characteristic 2 their groups have infinite rank, so they are not "uniform pro-p with Lie algebra $\mathfrak{pgl}_3$". What is unconditional is (T_2) and the parts of Theorem 6.3.1 that need only it (Corollary 6.4.3). An "exact asymptotic formula" is proved for the total order (Theorem 6.5.1), not for the 2-part.

Acknowledgement of provenance and caveats

Unrefereed. The inputs quoted without proof are the structure theory of $\mathbb{H}_d$ [34, 36], the Iwahori–Hecke algebra [35], Cohn localization and its K-theory [33, 38], Lazard's and Neumann's theorems on $\mathbb{Z}_p[[G]]$ [11, 37], and the framework of [1], which is used only to explain why it does not apply. The bibliographic data of [34, 36, 35, 33, 38, 37] have been checked against Crossref and zbMATH.

State of Volume III

- **Rank-two topology, proved locally** (Ch. 1): the general cubic trace identity $\operatorname{Tr}(C^{(3)})^3 = \operatorname{Tr} A_1^3 - 3q|E| + 3nq^3$, which equals $n(q-1)^2(q+1)$ *only under the flag condition* — on Y_{21} the true trace is 651, of which 63 is the flag part and 588 comes from non-simplex 3-cycles (Thm. 1.2.3); Garland vanishing $H^1(Y;\mathcal{E}) = 0$ for every unitary local system, from the link gap $1 - \sqrt{q}/(q+1) > \frac{1}{2}$ alone, with no building, group action or flag condition (Thm. 1.4.3); $\det(I - uC^{(3)}) = \det P(u)$ with $1 - u^3$ of multiplicity exactly $b_0 = 1$, the Betti numbers b_1, b_2 entering only on the edge (branch) side of the Kang–Li zeta function (Thms. 1.6.1, 1.6.3, Rem. 1.6.6); the sector lemma removing the hypothesis of Vol. I–II, Ch. 16, Thm. 16.3.2(ii); and the derivative identity $\frac{d}{du}\det(I - uC^{(3)})|_{u=1} = (-1)^n n(q-1)^2(q+1)|\operatorname{Jac}^{\mathrm{H}}(Y)|$ (Thm. 1.6.4).

- **Simplicial sandpiles** (Ch. 2): on thick complexes none of Δ_1, $\partial_2\partial_2^t$, Δ_2, Δ^{H} can be switched to a Z-matrix, so edge, chamber and Hecke toppling are not abelian and have no recurrent group; only the vertex sandpile survives, and on thin tori the chamber sandpile is the honeycomb one (Thm. 2.1.5, Cor. 2.1.6). The Hodge factorization $|\operatorname{Jac}_1|\operatorname{disc}\ker_{\mathbb{Z}}\Delta_1 = n|\operatorname{Jac}_0||\operatorname{Jac}_2|\operatorname{disc} H_2(Y;\mathbb{Z})$, which reads $|\operatorname{Jac}_1| = \det\Delta_1$ on thick complexes (Thm. 2.2.2). Exact mean avalanche size $\langle s\rangle = n^2 L^+_{ss}/(n-1) \in [n/\lambda_{\max}, n/\lambda_2]$, hence $\Theta(n)$ on expanding families against $\Theta(n\log n)$ on 2D tori (Thm. 2.3.2, Cor. 2.3.3); mean-field exponents conjectural (Conj. 2.4.2). The 1-skeleton of Y_{21} is $K_{7,7,7}$ (Prop. 2.5.1).

- **Phase oscillators** (Ch. 3): chiral connections of any flux exist on every $\widetilde{A}_2$ complex and form a torsor under $\operatorname{Hom}(H_1, U(1))$, which has 64 elements on Y_{21} (Prop. 3.1.2). The flux sector is a gradient flow for every flux, so there are no limit cycles and no Hopf bifurcation. Its ground states are twisted states carrying static currents (Thms. 3.2.2, 3.2.4). For all parameters the twisted states rotate at $\omega + K_2\sin\Phi$ with Jacobian $-c_m\Delta_0$ (Thm. 3.3.1). The cluster subspace is gradient modulo rotation, and the Jacobian at cluster states splits into 3×3 blocks, all real on Y_{21} (Thms. 3.3.3, 3.3.4). The Ott–Antonsen reduction closes only for $K_2 = 0$, and then excludes chimeras for identical oscillators (Prop. 3.4.1, Thm. 3.4.2); chimeras for $K_2 \neq 0$ remain open. The geometric phase of a rigid rotation is 0; flux threading permutes the twisted states cyclically; and $\Delta\Omega$ does not involve $\chi(Y)$ (Prop. 3.5.4, Rem. 3.5.2).

- **Geodesic edge operator** (Ch. 4): L_E is q^2-biregular and block-cyclic. The subspace $S^t\mathbb{C}^V + T^t\mathbb{C}^V + Q\mathbb{C}^V$ is invariant and carries the Hecke pencil (Thm. 4.1.5). On its complement W, L_E^t is exactly $\sqrt{q}$ times an isometry, unconditionally and for all $\widetilde{A}_d$ with $q^{(d-1)/2}$ (Thms. 4.2.1, 4.2.4), and every eigenvalue has $|\lambda| \geq \sqrt{q}$ (Thm. 4.2.2). $\det(I - uL_E) = \det P(u)\det(I - u\bar{L})/((1-u^3)(1-q^3u^3))$ off an explicit exceptional curve, and the two-circle Riemann hypothesis is equivalent to the Ramanujan property (Thms. 4.3.2, 4.3.3).

Degenerate spectra are not Poisson, small perturbations do not give Wigner–Dyson statistics, and Brody $b = 2$ is not GUE (Props. 4.4.1–4.4.3).

- **Chamber public-goods games** (Ch. 5): the chamber game is an additive pairwise game on the 1-skeleton, with the flags entering only through $q + 1$ (Thm. 5.1.2). The mean-field threshold is $r_c = 3$ for every q (Prop. 5.1.4). Zero-noise cluster thresholds are $3\kappa/(\kappa + \delta)$ with integer δ, so $5/3$ never occurs for $q = 2$, and Y_{21} has no cages (Thm. 5.2.2, Prop. 5.2.3). In Monte Carlo runs thick complexes behave like random regular graphs, while the thin lattice favours cooperation (Rem. 5.3.1). Cyclic replicator dynamics has no limit cycle and no torus, and mutation produces a Hopf cycle at $\mu_H = (b - a)/18$ (Thms. 5.4.1, 5.4.2).

- **Hecke algebras and Iwasawa theory** (Ch. 6): realizing an operator in a representation of the degenerate affine Hecke algebra is vacuous. The Iwahori–Hecke algebra $H_q(\widetilde{A}_2)$ acts on chambers under (L), and Δ_2, A_1 and L_E are expressed through it (Prop. 6.1.2, Thms. 6.1.3, 6.1.4). Garland rigidity removes every $\mathbb{Z}_p$-direction, so the Ore set of Coates et al. is not available (Thm. 6.2.2). Under a tower hypothesis the equivariant Hecke–Laplacian is injective, $\partial[\Delta^{\mathrm{H}}_\Lambda] = [X]$, descent recovers the $\mathrm{Jac}^{\mathrm{H}}(Y_U)$, and the class number formula $\prod_{\rho\neq 1} \det(\Delta^{\mathrm{H}}_\rho)^{\dim\rho} = [G : U]\,|\,\mathrm{Jac}^{\mathrm{H}}(Y_U)|/|\,\mathrm{Jac}^{\mathrm{H}}(Y)|$ holds; it equals 7^{18} on Y_{21} (Thm. 6.3.1). The presentation embeds in the cyclic division algebra $(\mathbb{F}_8(t)/\mathbb{F}_2(t), \mathrm{Frob}, t)$ via $a_x \mapsto 1+\alpha^x\varphi$, so the building is that of $\mathrm{PGL}_3(\mathbb{F}_2((s)))$ (Thm. 6.4.1). The resulting canonical pro-2 tower over X_3 is unconditional, starts with Y_{24}, and has a group of infinite rank that is not p-adic analytic (Thm. 6.4.2, Cor. 6.4.3). $\frac{1}{n}\log|\,\mathrm{Jac}^{\mathrm{H}}\,| \to h_H(2) = 2.063693$ along any exhausting tower (Thm. 6.5.1).

Status at the close of Volume III

The open problems carried by Chapter 6 (Section 6.7) stand as follows.

(a) *Towers.* Settled in the form in which it can be settled. The group of Vol. I–II is the Cartwright–Steger lattice of $\mathrm{PGL}_3(\mathbb{F}_2((s)))$, and its building is Bruhat–Tits. The canonical pro-2 tower over X_3 exists unconditionally, starts with Y_{24}, and satisfies (T_2). Hence Theorem 6.3.1(i),(iii),(iv),(v) are unconditional. The tower is *not* p-adic analytic, and (T) is expected to fail for every tower (Theorems 6.4.1, 6.4.2, Corollary 6.4.3, Remark 6.4.4). P is torsion-free, and the characteristic element exists unconditionally as a compatible system $\mathcal{L}_\infty \in \varprojlim K_1(\mathbb{Q}_2 A_m)$ with boundaries the finite modules X^{nt}_m, whose limit is $X(Y/P)$ (Theorem 6.9.4). *Still open:* whether $\mathbb{Z}_2[[P]]$ is an Ore domain, i.e. a localization in which Theorem 6.3.1(ii) holds.

(b) *Growth of Hecke Jacobians.* Settled for the total order: along every exhausting tower, $\frac{1}{n}\log|\,\mathrm{Jac}^{\mathrm{H}}\,| \to h_H(2) = 2.063693$, a Plancherel integral (Theorem 6.5.1). For the 2-part, $v_2|\,\mathrm{Jac}^{\mathrm{H}}(Y_m)|$ is known exactly up to $n = 1572864$ (Theorem 6.8.7), and $\liminf v_2/n \geq c_e$ with $0 \leq c_e \leq 0.0645$ (Theorem 6.8.8). *Still open:* whether the slope is positive, i.e. essentially whether $c_e > 0$ (Remark 6.8.9).

(e) *Finite-noise thresholds and nonlinear chamber games.* Answered in Chapter 5, Sections 5.5–5.7. Nonlinearity is exactly what makes a chamber game irreducible. Symmetric games are potential games, so their selected state does not depend on geometry. Chiral games break the potential and cycle in the mean field; on large networks the cycle is coherent only when entrained by a synchronized clock. The thin-lattice advantage is the flat interface, and finite

anchors cannot remove the frontier threshold of thick buildings. No "topological catalysis" by thickness exists. *Still open:* the star formula for $q \geq 3$, and finite-noise phase transitions on infinite complexes.

Problems (c), (d) and (f) of that list remain open: the vertex block of L_E for $\widetilde{A}_d$, $d \geq 3$ (the trace identity of (c) is settled: $\mathrm{Tr}(C^{(d+1)})^{d+1} = n \prod_{i=1}^{d}(1-q^i)$ under the flag condition, Theorem 1.3.3); universality for the remainder eigenphases (on Y_{168} the crossover to GUE and its first-order scale are settled, Section 4.6); and chimeras and transverse Hopf bifurcations for the triadic lag. Of (f), the transverse Hopf bifurcations are found on Y_{168}: they have multiplicity 21, and in every case examined the loss of stability is hard and ends in a locked state (Section 3.8); whether some equivariant branch is supercritical, and the chimeras, remain open. Volume III closes with these results and these open problems.

Bibliography

[1] J. Coates, T. Fukaya, K. Kato, R. Sujatha and O. Venjakob, *The* GL_2 *main conjecture for elliptic curves without complex multiplication*, Publ. Math. Inst. Hautes Études Sci. **101** (2005), 163–208. doi:10.1007/s10240-004-0029-3

[2] D. Dhar, *Self-organized critical state of sandpile automaton models*, Phys. Rev. Lett. **64** (1990), 1613–1616. doi:10.1103/PhysRevLett.64.1613

[3] D. Dhar and S. N. Majumdar, *Abelian sandpile model on the Bethe lattice*, J. Phys. A **23** (1990), 4333–4350. doi:10.1088/0305-4470/23/19/018

[4] A. M. Duval, C. J. Klivans and J. L. Martin, *Critical groups of simplicial complexes*, Ann. Comb. **17** (2013), 53–70. doi:10.1007/s00026-012-0168-z

[5] T. E. Harris, *The Theory of Branching Processes*, Grundlehren Math. Wiss. **119**, Springer, Berlin, 1963. doi:10.1007/978-3-642-51866-9

[6] R. Lyons, *Asymptotic enumeration of spanning trees*, Combin. Probab. Comput. **14** (2005), 491–522. doi:10.1017/S096354830500684X

[7] J. D. Dixon, M. P. F. du Sautoy, A. Mann and D. Segal, *Analytic Pro-p Groups*, 2nd ed., Cambridge Stud. Adv. Math. **61**, Cambridge Univ. Press, Cambridge, 1999. doi:10.1017/CBO9780511470882

[8] H. Garland, *p-adic curvature and the cohomology of discrete subgroups of p-adic groups*, Ann. of Math. (2) **97** (1973), 375–423. doi:10.2307/1970829

[9] W. Ballmann and J. Świątkowski, *On* L^2*-cohomology and property (T) for automorphism groups of polyhedral cell complexes*, Geom. Funct. Anal. **7** (1997), 615–645. doi:10.1007/s000390050022

[10] M.-H. Kang and W.-C. W. Li, *Zeta functions of complexes arising from* PGL(3), Adv. Math. **256** (2014), 46–103. doi:10.1016/j.aim.2013.12.033

[11] M. Lazard, *Groupes analytiques p-adiques*, Publ. Math. Inst. Hautes Études Sci. **26** (1965), 5–219. doi:10.1007/BF02684303

[12] A. Weil, *Remarks on the cohomology of groups*, Ann. of Math. (2) **80** (1964), 149–157. doi:10.2307/1970495

[13] D. M. Abrams, R. Mirollo, S. H. Strogatz and D. A. Wiley, *Solvable model for chimera states of coupled oscillators*, Phys. Rev. Lett. **101** (2008), 084103. doi:10.1103/PhysRevLett.101.084103

[14] Y. Kuramoto and D. Battogtokh, *Coexistence of coherence and incoherence in nonlocally coupled phase oscillators*, Nonlinear Phenom. Complex Syst. **5** (2002), 380–385.

[15] J. P. LaSalle, *Some extensions of Liapunov's second method*, IRE Trans. Circuit Theory **7** (1960), 520–527. doi:10.1109/TCT.1960.1086720

[16] S. Łojasiewicz, *Une propriété topologique des sous-ensembles analytiques réels*, in: Les Équations aux Dérivées Partielles (Paris, 1962), Colloques Internat. CNRS **117**, Éditions du CNRS, Paris, 1963, 87–89.

[17] E. Ott and T. M. Antonsen, *Low dimensional behavior of large systems of globally coupled oscillators*, Chaos **18** (2008), 037113. doi:10.1063/1.2930766

[18] H. Sakaguchi and Y. Kuramoto, *A soluble active rotator model showing phase transitions via mutual entrainment*, Progr. Theoret. Phys. **76** (1986), 576–581. doi:10.1143/PTP.76.576

[19] M. Golubitsky, I. Stewart and D. G. Schaeffer, *Singularities and Groups in Bifurcation Theory, Vol. II*, Appl. Math. Sci. **69**, Springer, New York, 1988. doi:10.1007/978-1-4612-4574-2

[20] P. S. Skardal and A. Arenas, *Abrupt desynchronization and extensive multistability in globally coupled oscillator simplexes*, Phys. Rev. Lett. **122** (2019), 248301. doi:10.1103/PhysRevLett.122.248301

[21] Y. Y. Atas, E. Bogomolny, O. Giraud and G. Roux, *Distribution of the ratio of consecutive level spacings in random matrix ensembles*, Phys. Rev. Lett. **110** (2013), 084101. doi:10.1103/PhysRevLett.110.084101

[22] O. Bohigas, M.-J. Giannoni and C. Schmit, *Characterization of chaotic quantum spectra and universality of level fluctuation laws*, Phys. Rev. Lett. **52** (1984), 1–4. doi:10.1103/PhysRevLett.52.1

[23] T. A. Brody, *A statistical measure for the repulsion of energy levels*, Lett. Nuovo Cimento **7** (1973), 482–484. doi:10.1007/BF02727859

[24] D. Jakobson, S. D. Miller, I. Rivin and Z. Rudnick, *Eigenvalue spacings for regular graphs*, in: Emerging Applications of Number Theory, IMA Vol. Math. Appl. **109**, Springer, New York, 1999, 317–327. doi:10.1007/978-1-4612-1544-8_12

[25] M.-H. Kang, W.-C. W. Li and C.-J. Wang, *The zeta functions of complexes from* PGL(3)*: a representation-theoretic approach*, Israel J. Math. **177** (2010), 335–348. doi:10.1007/s11856-010-0049-2

[26] T. Kato, *Perturbation Theory for Linear Operators*, Grundlehren Math. Wiss. **132**, Springer, Berlin, 1966; reprint of the 2nd ed., Classics in Mathematics, Springer, Berlin, 1995. doi:10.1007/978-3-642-66282-9

[27] T. Kottos and U. Smilansky, *Quantum chaos on graphs*, Phys. Rev. Lett. **79** (1997), 4794–4797. doi:10.1103/PhysRevLett.79.4794

[28] A. Lubotzky, B. Samuels and U. Vishne, *Ramanujan complexes of type* $\tilde{A}_d$, Israel J. Math. **149** (2005), 267–299. doi:10.1007/BF02772543

[29] M. Mobilia, *Oscillatory dynamics in rock–paper–scissors games with mutations*, J. Theoret. Biol. **264** (2010), 1–10. doi:10.1016/j.jtbi.2010.01.008

[30] M. A. Nowak and R. M. May, *Evolutionary games and spatial chaos*, Nature **359** (1992), 826–829. doi:10.1038/359826a0

[31] A. Szolnoki, M. Perc and G. Szabó, *Topology-independent impact of noise on cooperation in spatial public goods games*, Phys. Rev. E **80** (2009), 056109. doi:10.1103/PhysRevE.80.056109

[32] G. Szabó and C. Tőke, *Evolutionary prisoner's dilemma game on a square lattice*, Phys. Rev. E **58** (1998), 69–73. doi:10.1103/PhysRevE.58.69

[33] P. M. Cohn, *Free Rings and Their Relations*, 2nd ed., London Math. Soc. Monogr. **19**, Academic Press, London, 1985.

[34] V. G. Drinfeld, *Degenerate affine Hecke algebras and Yangians*, Funct. Anal. Appl. **20** (1986), 58–60. doi:10.1007/BF01077318

[35] N. Iwahori and H. Matsumoto, *On some Bruhat decomposition and the structure of the Hecke rings of p-adic Chevalley groups*, Publ. Math. Inst. Hautes Études Sci. **25** (1965), 5–48. doi:10.1007/BF02684396

[36] G. Lusztig, *Affine Hecke algebras and their graded version*, J. Amer. Math. Soc. **2** (1989), 599–635. doi:10.1090/S0894-0347-1989-0991016-9

[37] A. Neumann, *Completed group algebras without zero divisors*, Arch. Math. **51** (1988), 496–499. doi:10.1007/BF01261969

[38] A. Neeman and A. Ranicki, *Noncommutative localisation in algebraic K-theory I*, Geom. Topol. **8** (2004), 1385–1425. doi:10.2140/gt.2004.8.1385

[39] I. G. Macdonald, *Spherical Functions on a Group of p-adic Type*, Publ. Ramanujan Inst. **2**, Univ. of Madras, Madras, 1971.

[40] G. A. Margulis, *Discrete Subgroups of Semisimple Lie Groups*, Ergeb. Math. Grenzgeb. (3) **17**, Springer, Berlin, 1991. doi:10.1007/978-3-642-51445-6

[41] I. Oppenheim, *Local spectral expansion approach to high dimensional expanders. Part I: Descent of spectral gaps*, Discrete Comput. Geom. **59** (2018), 293–330. doi:10.1007/s00454-017-9948-x

[42] D. I. Cartwright and T. Steger, *A family of $\tilde{A}_n$-groups*, Israel J. Math. **103** (1998), 125–140. doi:10.1007/BF02762271

[43] D. Monderer and L. S. Shapley, *Potential games*, Games Econom. Behav. **14** (1996), 124–143. doi:10.1006/game.1996.0044

[44] J. Mallet-Paret and H. L. Smith, *The Poincaré–Bendixson theorem for monotone cyclic feedback systems*, J. Dynam. Differential Equations **2** (1990), 367–421. doi:10.1007/BF01054041

[45] H. L. Smith, *Monotone Dynamical Systems: An Introduction to the Theory of Competitive and Cooperative Systems*, Math. Surveys Monogr. **41**, Amer. Math. Soc., Providence, RI, 1995. doi:10.1090/surv/041

[46] A. Żuk, *La propriété (T) de Kazhdan pour les groupes agissant sur les polyèdres*, C. R. Acad. Sci. Paris Sér. I Math. **323** (1996), 453–458.